Gasoline Profiles and Impact of Ethanol Blending in Latina America

August 2025

U.S. GRAINS & BIOPRODUCTS COUNCIL

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Ethanol Blending in Latin America

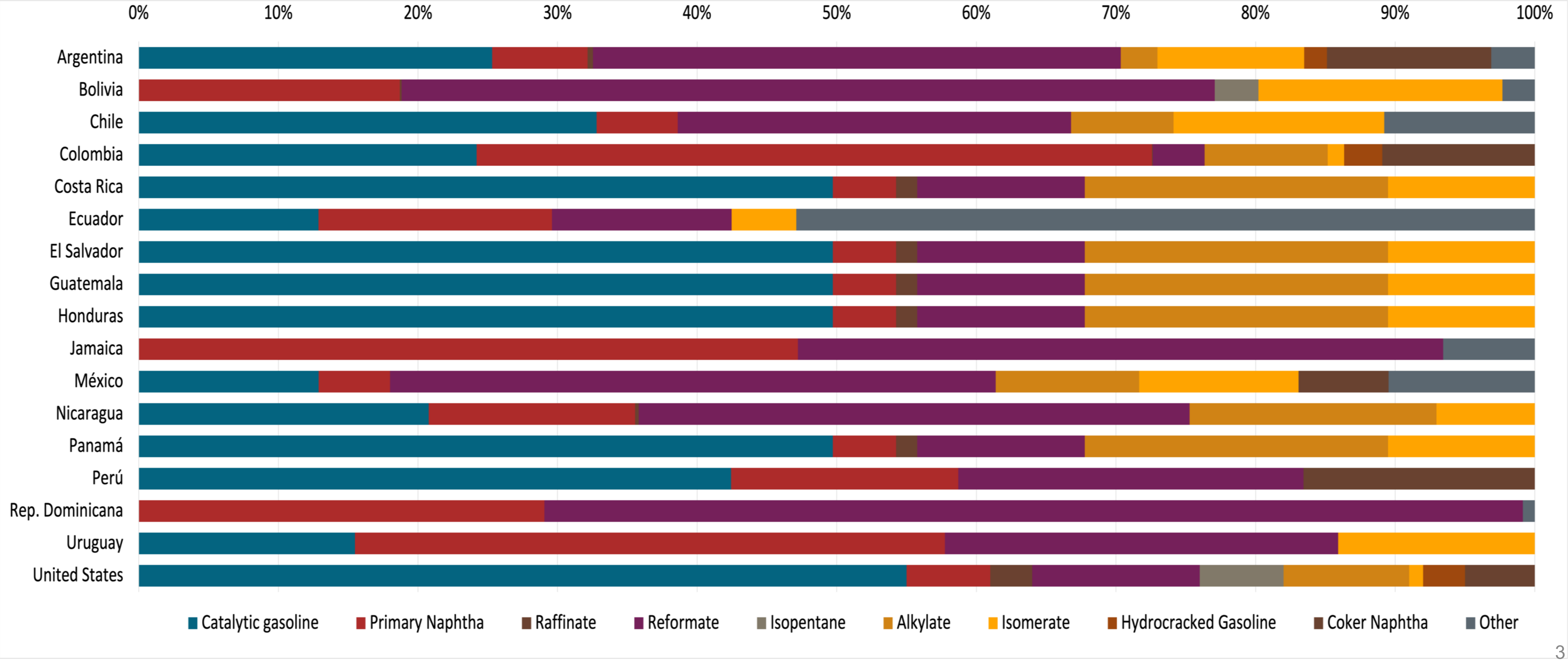
There are important fuel quality and environmental impact of vehicle emission challenges in the Region.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Latin America to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.

Sixteen countries with potential and additional use of ethanol were studied: 1) gasoline market profiles; 2) Optimization of gasoline blends with ethanol and 3) Environmental impact of gasolines blended with ethanol.

Gasoline Component Blending in Latin America

Gasoline is a blend of a base gasoline and other components. This blending is usually done at blending terminals as only 30% of the world's finished gasoline is distributed directly from refineries. Each component provides different properties to the final blend, for example, isomerates, alkylates and butanes increase the octane. The components commonly used in Latin America are:



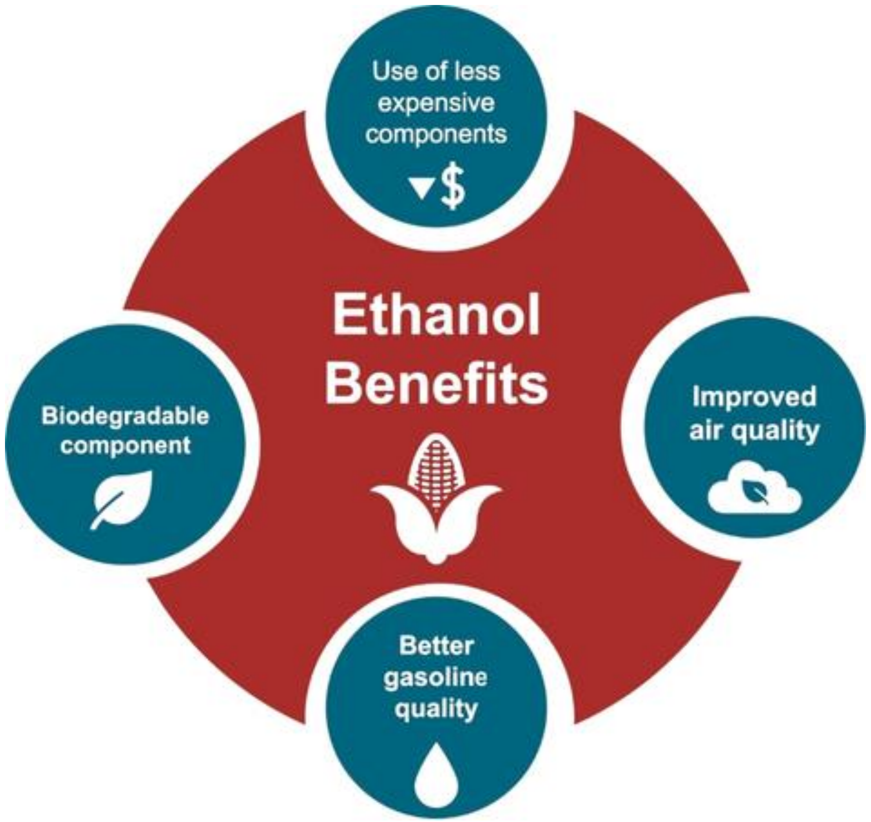
Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component.The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives.To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

The blending model uses gasoline component spot average prices 2024 and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.



Vehicle Emission Impact for Ethanol Gasoline Blending

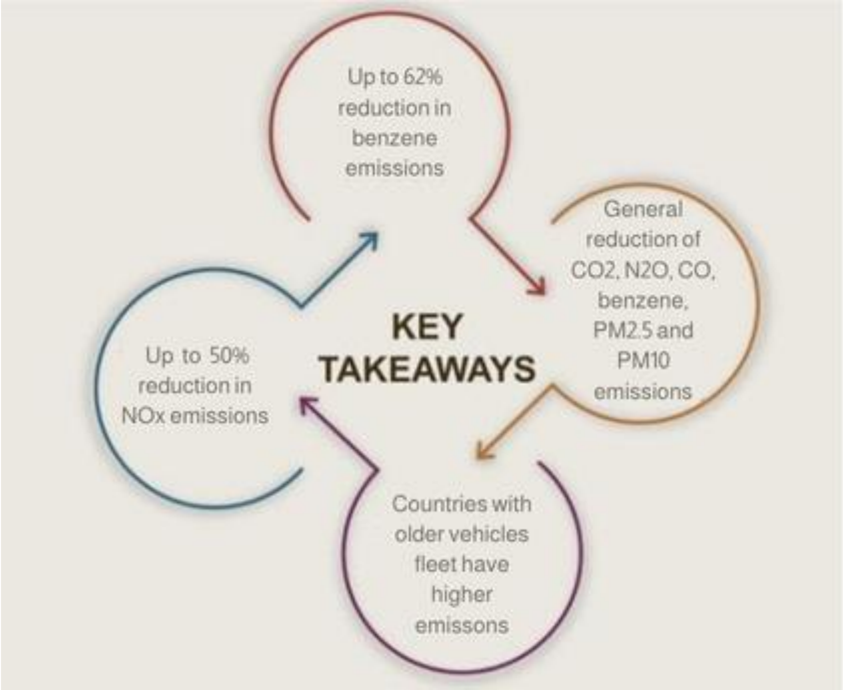
The model used in this analysis takes as a reference the **International Vehicle Emissions Model (IVE).**
The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

Main Results



**Source: Bureau of transportation statistics.*

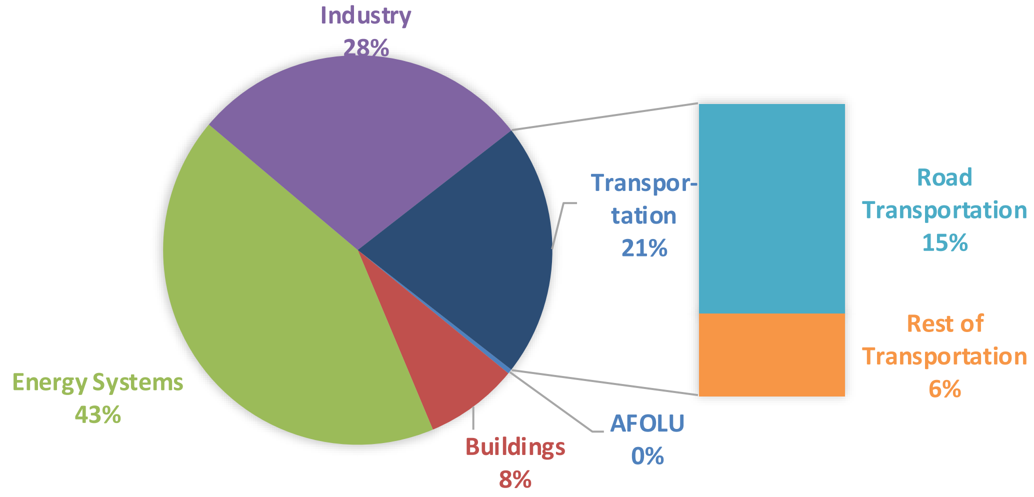
Life Cycle GHG Emissions

The Global Agenda for Climate Action calls for countries, cities and regions, businesses, and members of civil society around the world to to achieve a low-carbon and climate-resilient economies in support of the Paris Agreement. Road Transportation accounts for 15% of total GHG emissions (IPCC,2023).

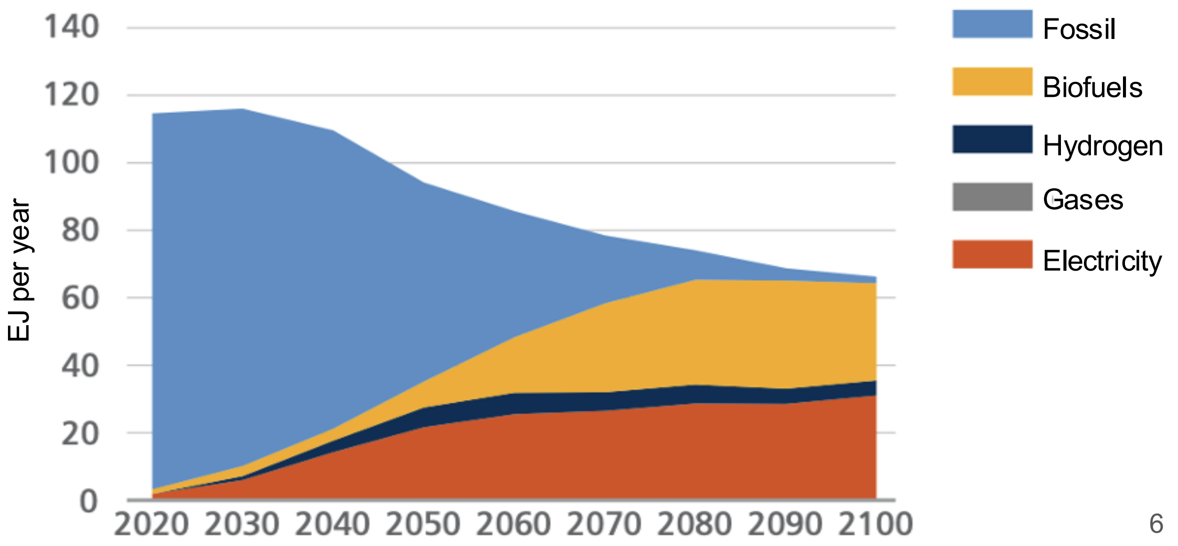
Ethanol gasoline blends are an internationally adopted practice that offer significant short-term benefits. Biofuels are an important solution to achieve reduction targets in the transportation sector.

Country GHG emission reductions are calculated using **RED II/III** and **GREET** life cycle emission methodologies and the **International Vehicular Emissions Model (IVE).** Results are compared to the current country targets for 2035. Current 2024 country emission are those published by **IPCC**.

Current GHG Global Emissions Distribution



IPCC Transport Sector Sustainable Development Goals and climate policies Scenario, 2023



Case Studies

1. Argentina
2. Bolivia
3. Chile
4. Colombia
5. Costa Rica
6. Dominican Republic
7. Ecuador
8. El Salvador
9. Guatemala
10. Honduras
11. Jamaica
12. México
13. Nicaragua
14. Perú
15. Panamá
16. Uruguay

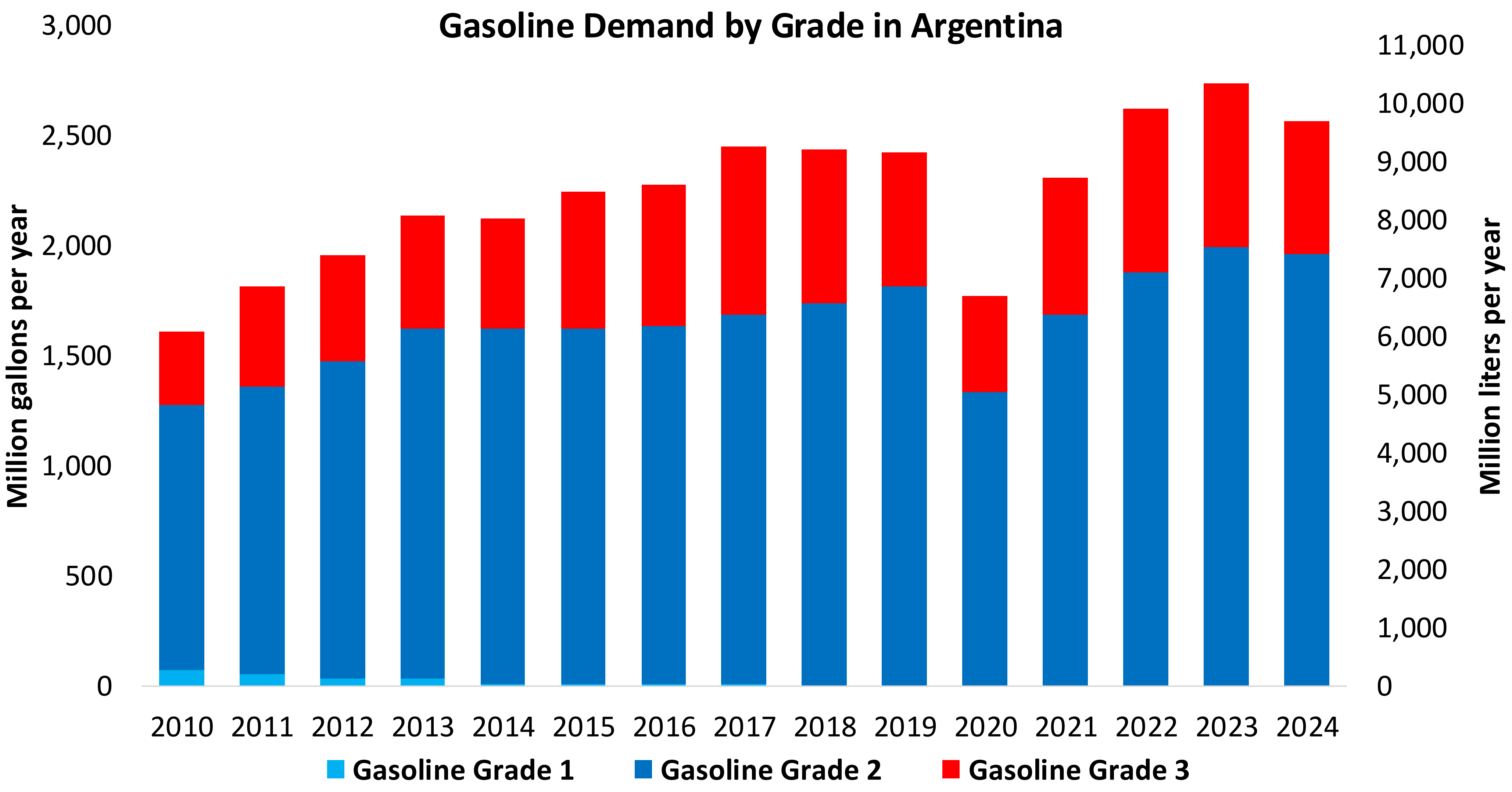
Ethanol Blending in Gasoline - Argentina

Argentina's total gasoline demand was 2,566 million gallons per year in 2024 (9,712 million liters per year), with a market share of approximately 76% as Nafta Súper / Grado 2 (RON 93 min – AKI 87) and 24% as Nafta Ultra / Grado 3 (RON 97 min – AKI 92). Octane delivered to market is usually higher than the minimum required. The demand for gasoline is supplied by local production, requiring imports of less than 75 million liters per year.

Argentina's policy is self-sufficiency and production and supply of ethanol for gasoline blend. Ethanol concentration varies according to local production. Currently, the concentration limit is 12% v/v. (Law 27.640) and a future E15 requirement is under discussion. Ethanol production is equal to consumption with some exports and imports from other countries.

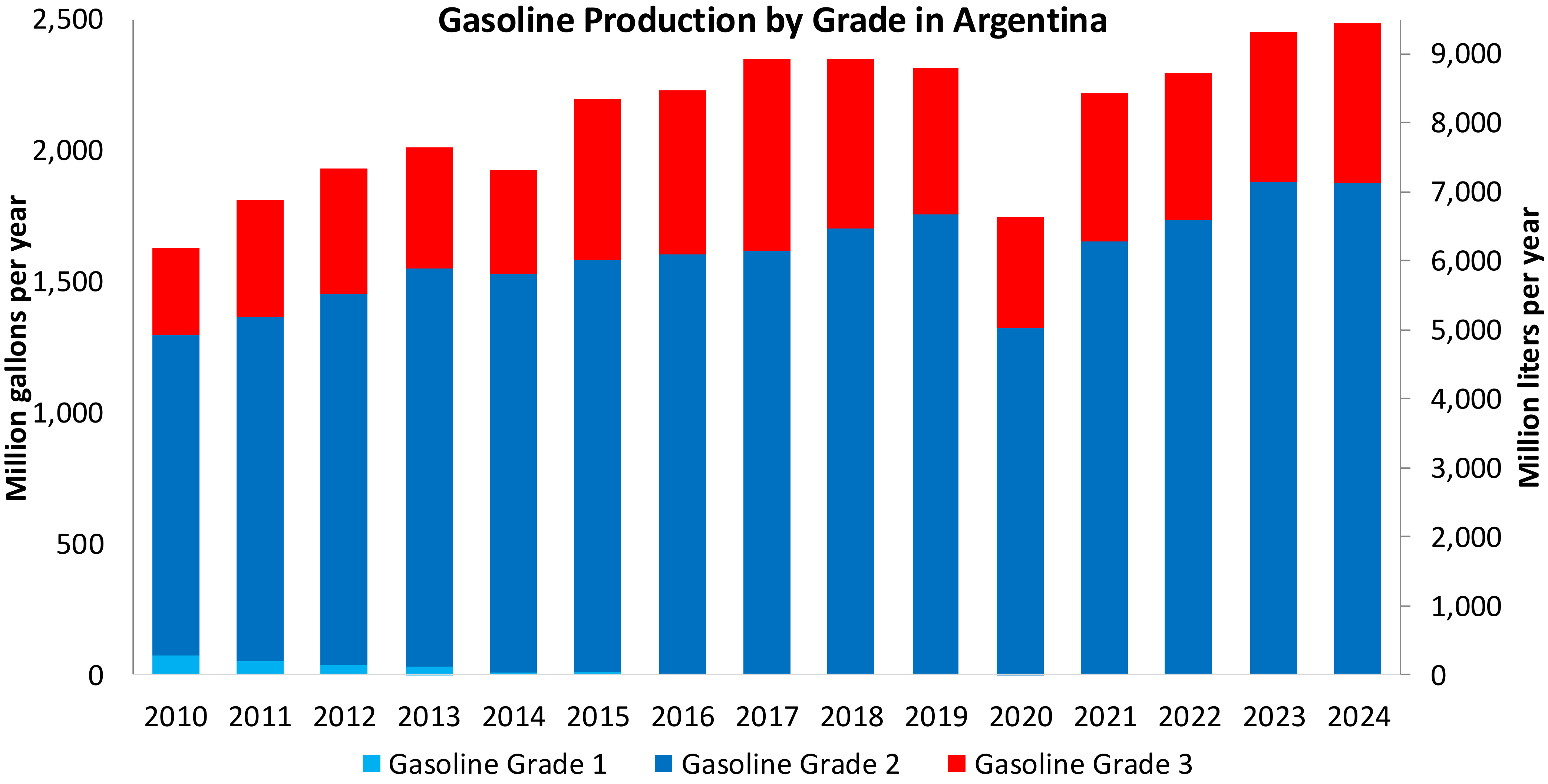
Source: Ministerio de Economía, 2025

Gasoline Demand in Argentina



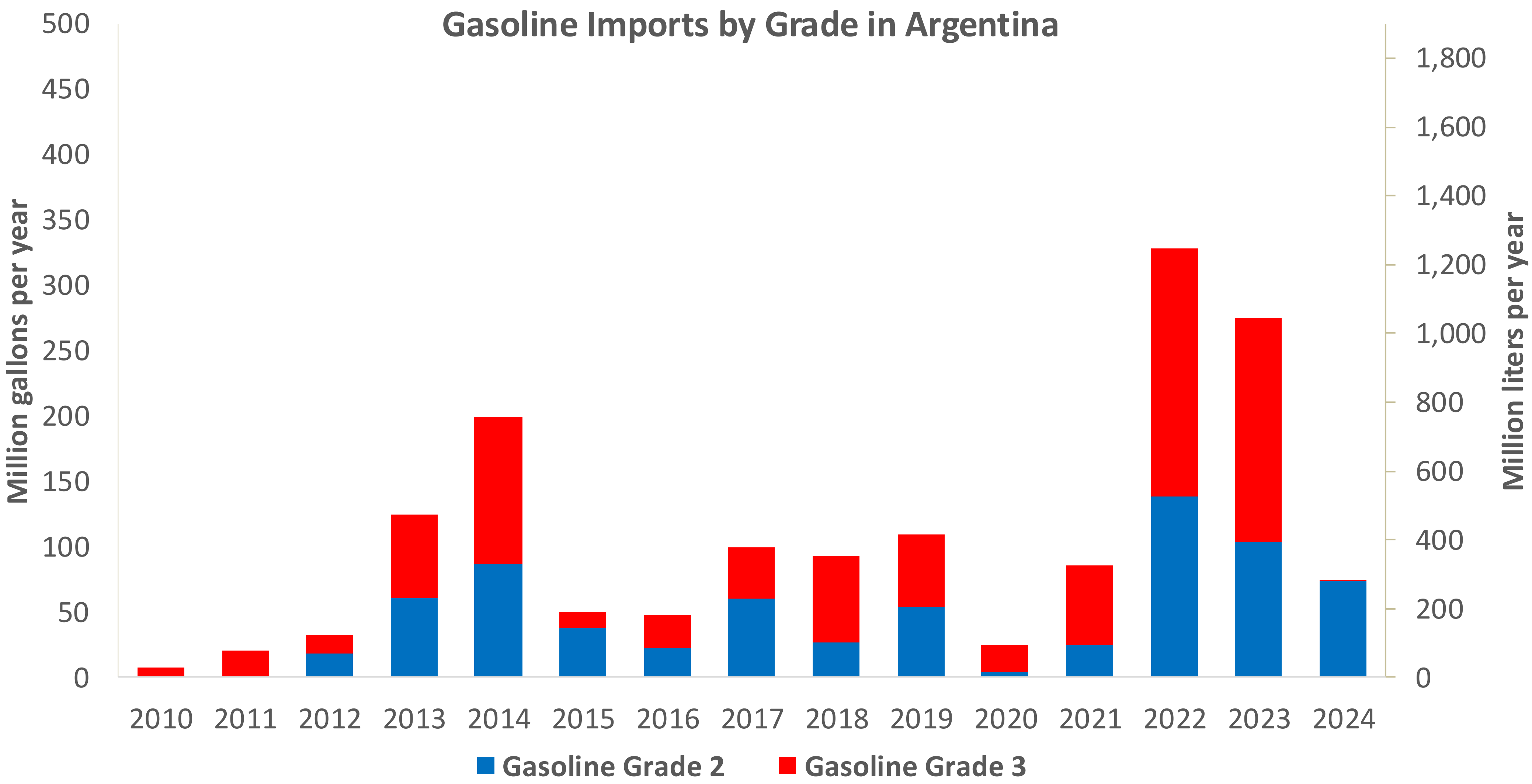
Source: Ministerio de Economía, 2025

Gasoline Production in Argentina



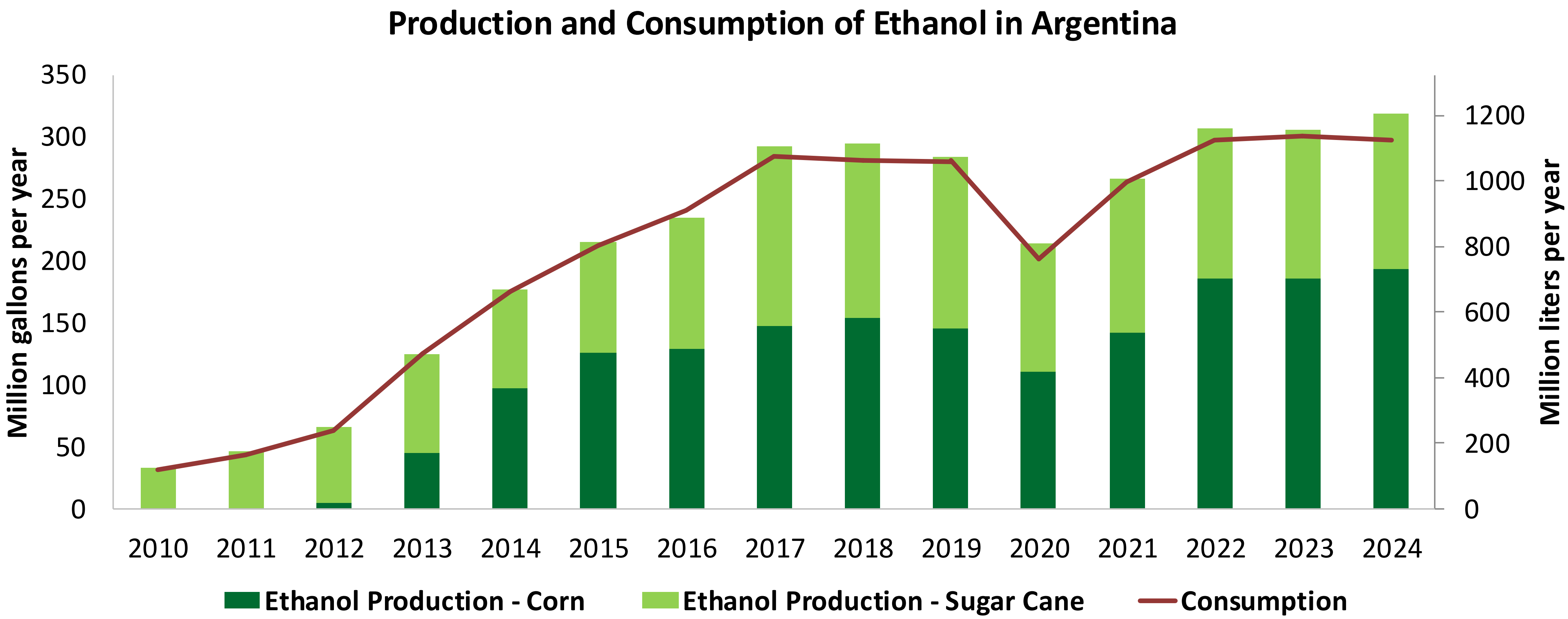
Source: Ministerio de Economía, 2025

Gasoline Imports in Argentina



Source: Ministerio de Economía, 2025

Ethanol Balance in Argentina



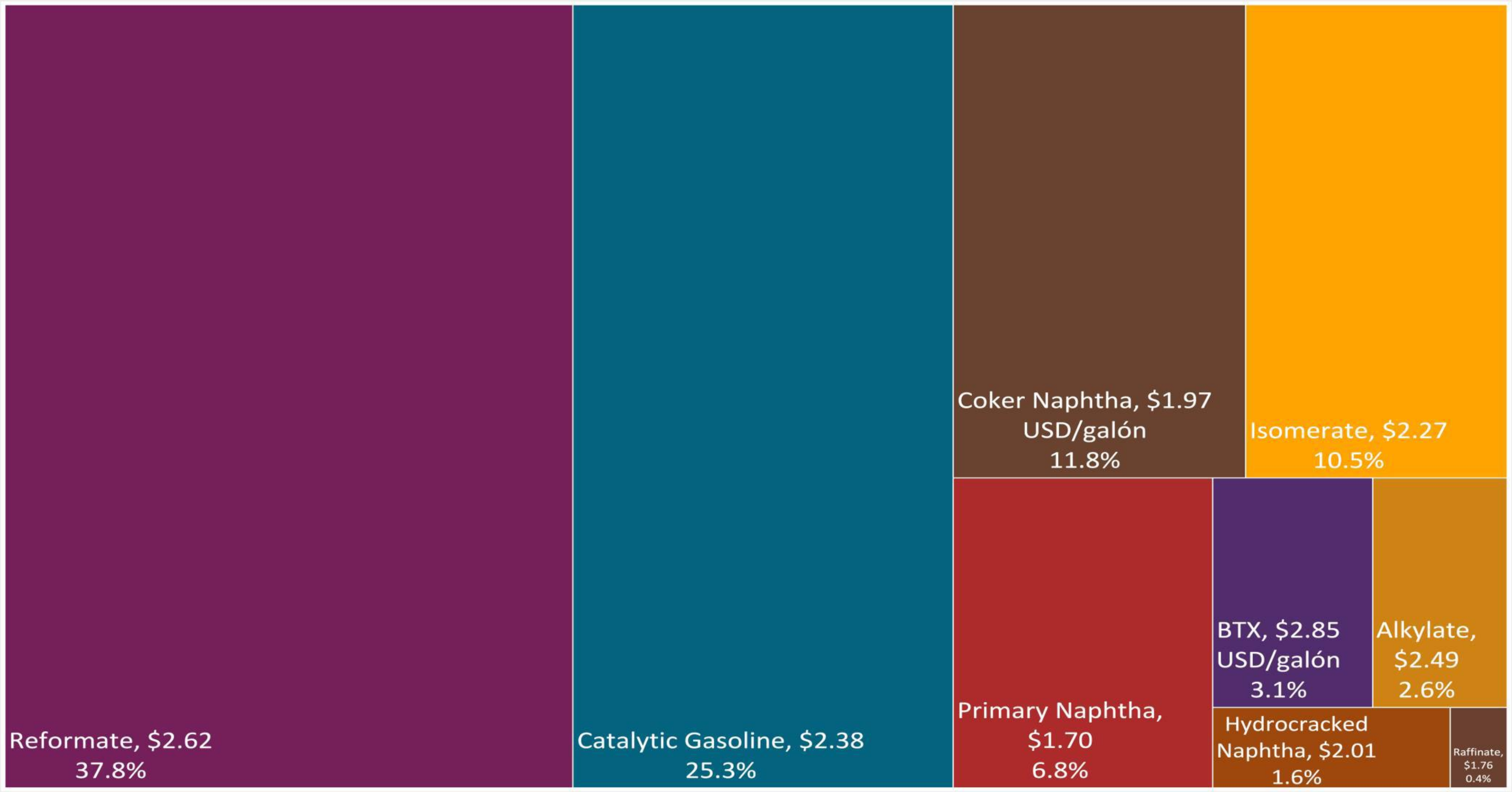
Source: Ministerio de Economía, 2025

Gasoline Quality in Argentina

Name	Resolution 576/2019 and Resolution 492/2023						EN 228:2012 + A1:2017 (Euro 6 enabling)			
Implementation Date	2019						2017			
Applicability	North Zone		Central Zone		South Zone		All countries			
Selected Grade	Gasoline Grade 2	Gasoline Grade 3	Gasoline Grade 2	Gasoline Grade 3	Gasoline Grade 2	Gasoline Grade 3	RON 95 E5	RON 95 E10	RON 98 E5	RON 98 E10
Benzene Content	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v
Aromatics	< 40 %v/v	< 40 %v/v	< 40 %v/v	< 40 %v/v	< 40 %v/v	< 40 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v
Olefins	-	-	-	-	-	-	< 18 %v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v
Lead Content	< 5 mg/l	< 5 mg/l	< 5 mg/l	< 5 mg/l	< 5 mg/l	< 5 mg/l	< 5 mg/l	< 5 mg/l	< 5 mg/l	< 5 mg/l
Manganese	< 2.5 mg/l	< 2.5 mg/l	< 2.5 mg/l	< 2.5 mg/l	< 2.5 mg/l	< 2.5 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l
RON	> 93	> 97	> 93	> 97	> 93	> 97	> 95	> 95	> 98	> 98
MON	> 83	> 85	> 83	> 85	> 83	> 85	> 85	> 88	> 85	> 88
AKI										
Sulfur Content	< 150 mg/kg	< 10 mg/kg	< 150 mg/kg	< 10 mg/kg	< 150 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg
Oxygen Content	< 4.5 %m/m	< 4.5 %m/m	< 4.5 %m/m	< 4.5 %m/m	< 4.5 %m/m	< 4.5 %m/m	<2,7 % m/m	<3,7 % m/m	<2,7 % m/m	<3,7 % m/m
Ethanol (EtOH)	<> 12 - 14 %v/v	<> 12 - 14 %v/v	<> 12 - 14 %v/v	<> 12 - 14 %v/v	<> 12 - 14 %v/v	<> 12 - 14 %v/v	<5 %v/v	<10 %v/v	<5 %v/v	<10 %v/v
RVP 37.8°C (Summer)	<> 35 - 70 kPa	<> 35 - 70 kPa	<> 35 - 70 kPa	<> 35 - 70 kPa	<> 45 - 80 kPa	<> 45 - 80 kPa	<> 60 - 70 kPa *Depends on the country, RVP is regulated in the EU Fuel Quality Directive			
RVP 37.8 °C(Winter)	<> 45 - 80 kPa	<> 45 - 80 kPa	<> 55 - 90 kPa	<> 55 - 90 kPa	<> 55 - 90 kPa	<> 55 - 90 kPa				
RVP 37.8°C (Transition)			<> 45 - 80 kPa	<> 45 - 80 kPa						
MTBE	< 15 %v/v	< 15 %v/v	< 15 %v/v	< 15 %v/v	< 15 %v/v	< 15 %v/v	-	-	-	-
Ethers 5 or more C Atoms	< 22 %v/v	< 22 %v/v	< 22 %v/v	< 22 %v/v	< 22 %v/v	< 22 %v/v	Based on oxygen content	<22 %v/v	Based on oxygen content	<22 %v/v

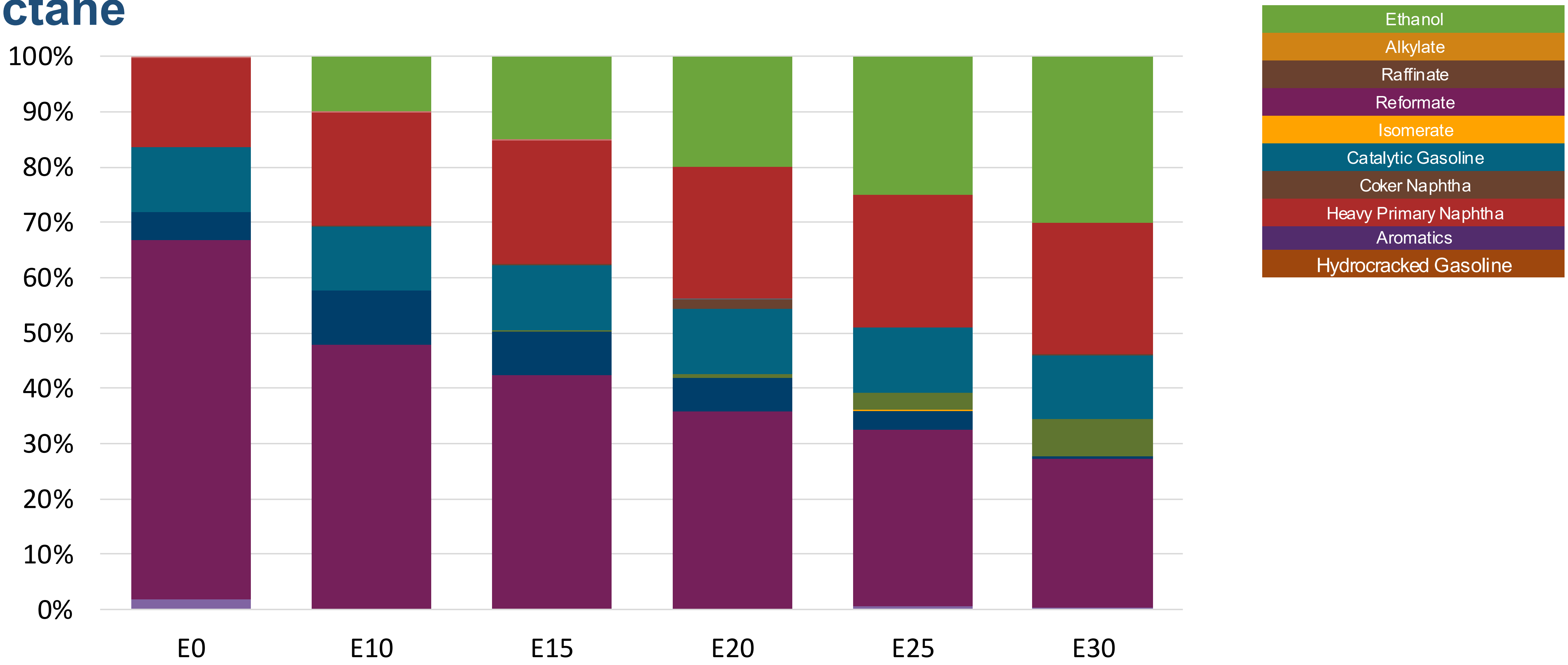
Source: Ministerio de Hacienda, 2019 and 2023

Available Blending Components



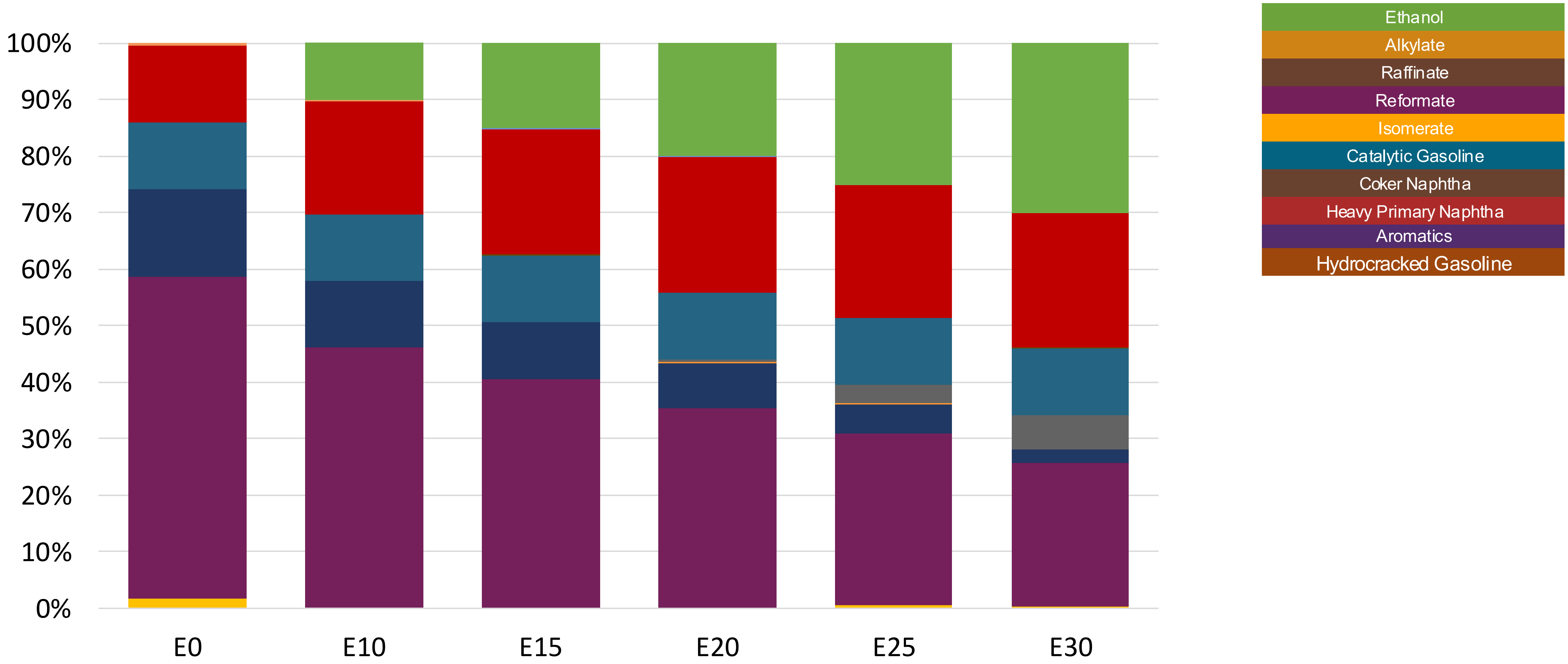
Source: Faro90

Argentina – Nafta Grado 2 – North & Central Zones– Constant Octane



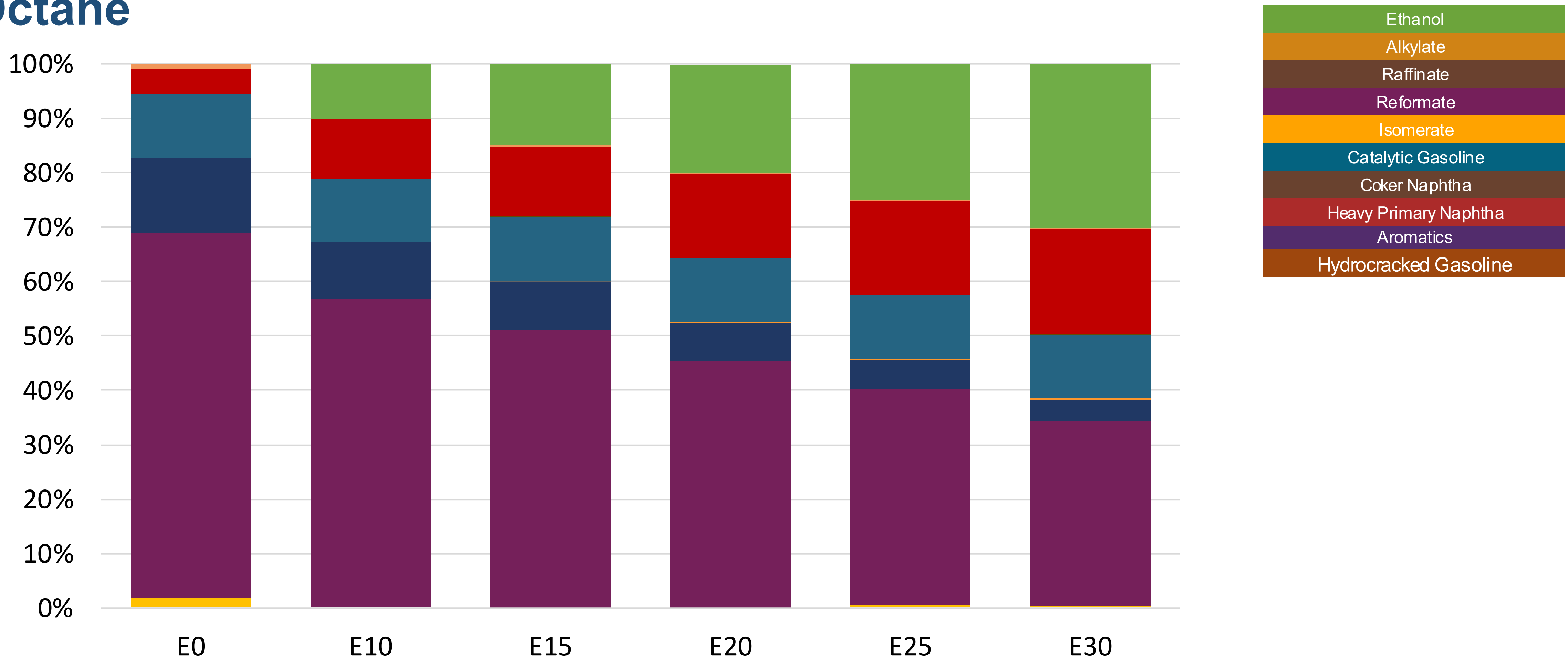
Octane (RON)	93.0	93.0	93.0	93.0	93.0	93.0
Price (US$/gal)	$ 2.28	$ 2.04	$ 1.98	$ 1.92	$ 1.86	1.79

Argentina – Nafta Grado 2 – South Zone – Constant Octane



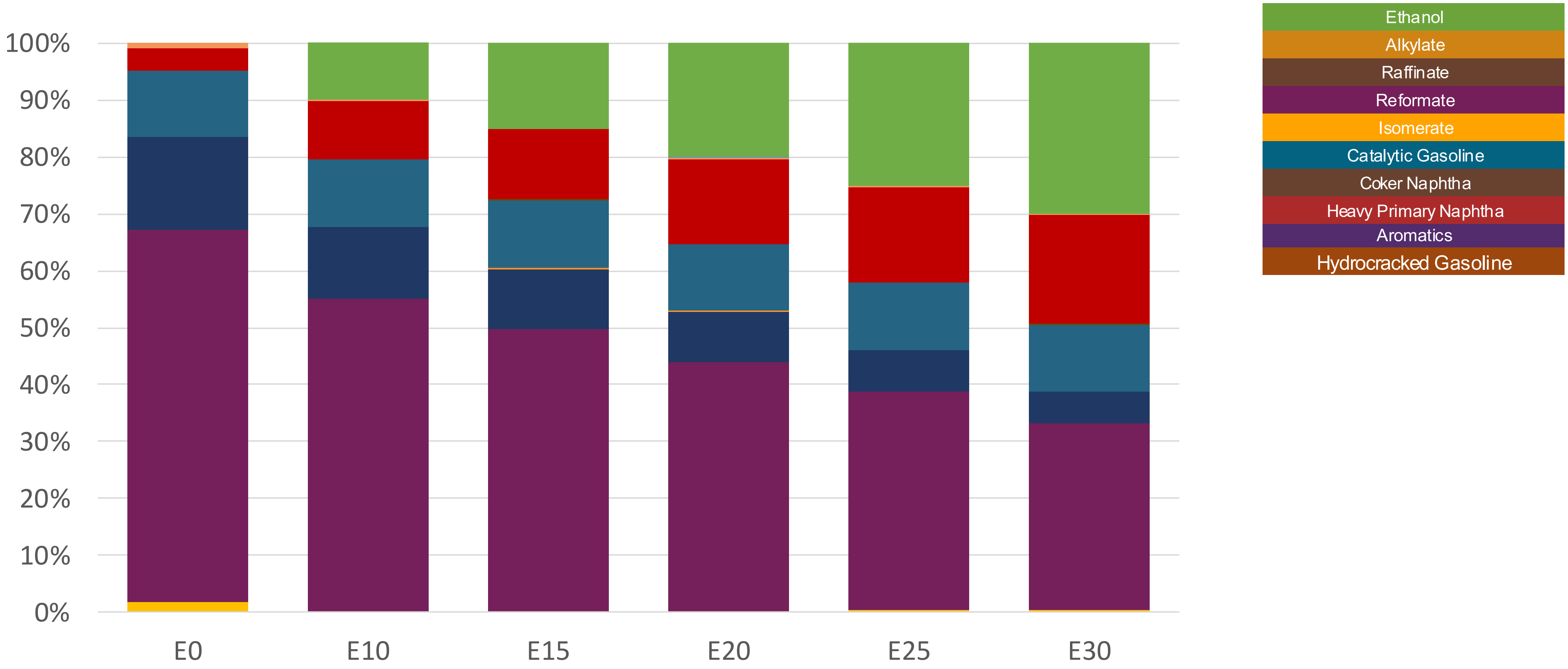
Octane (RON)	93.0	93.0	93.0	93.0	93.0	93.0
Price (US$/gal)	$ 2.13	$ 2.01	$ 1.95	$ 1.89	$ 1.83	1.77

Argentina – Nafta Grado 3 – North & Central Zones– Constant Octane



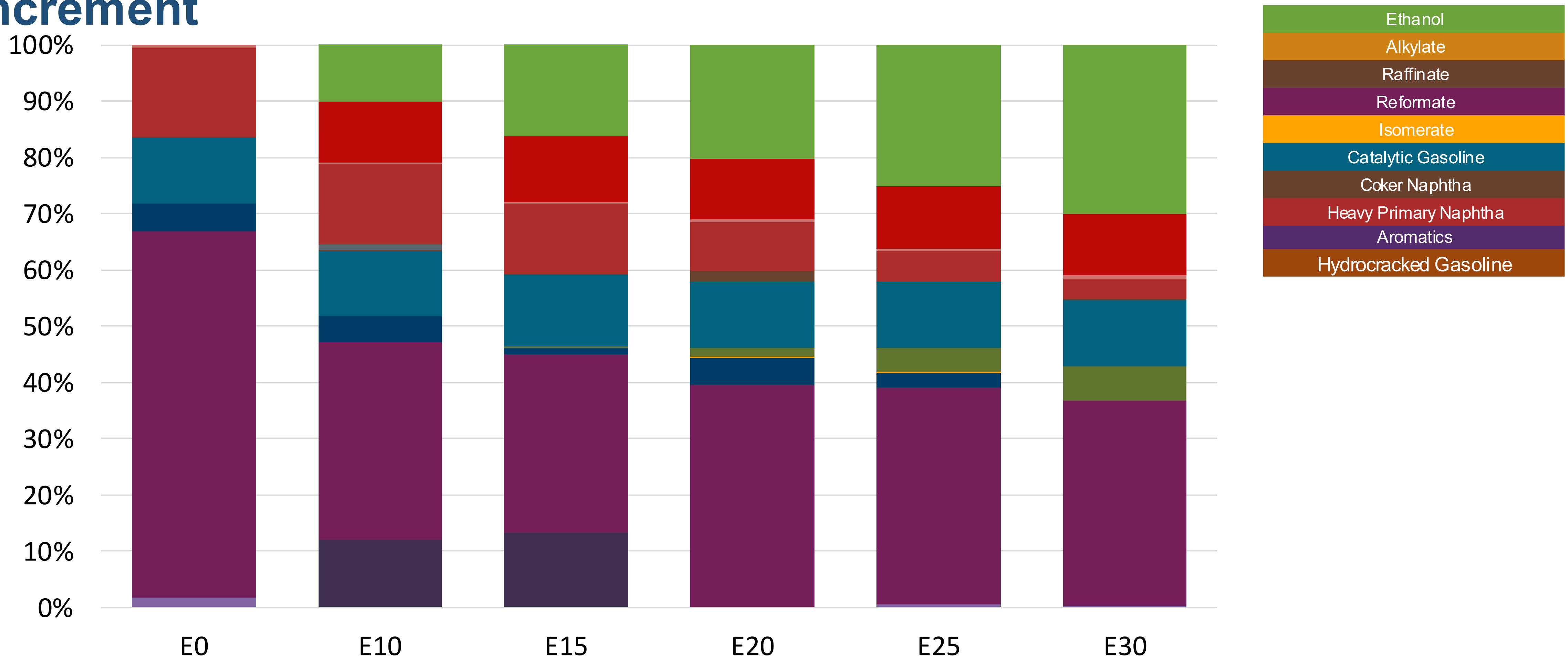
Octane (RON)	97.0	97.0	97.0	97.0	97.0	97.0
Price (US$/gal)	$ 2.35	$ 2.22	$ 2.16	$ 2.10	2.03	$ 1.97

Argentina – Nafta Grado 3 – South Zone – Constant Octane



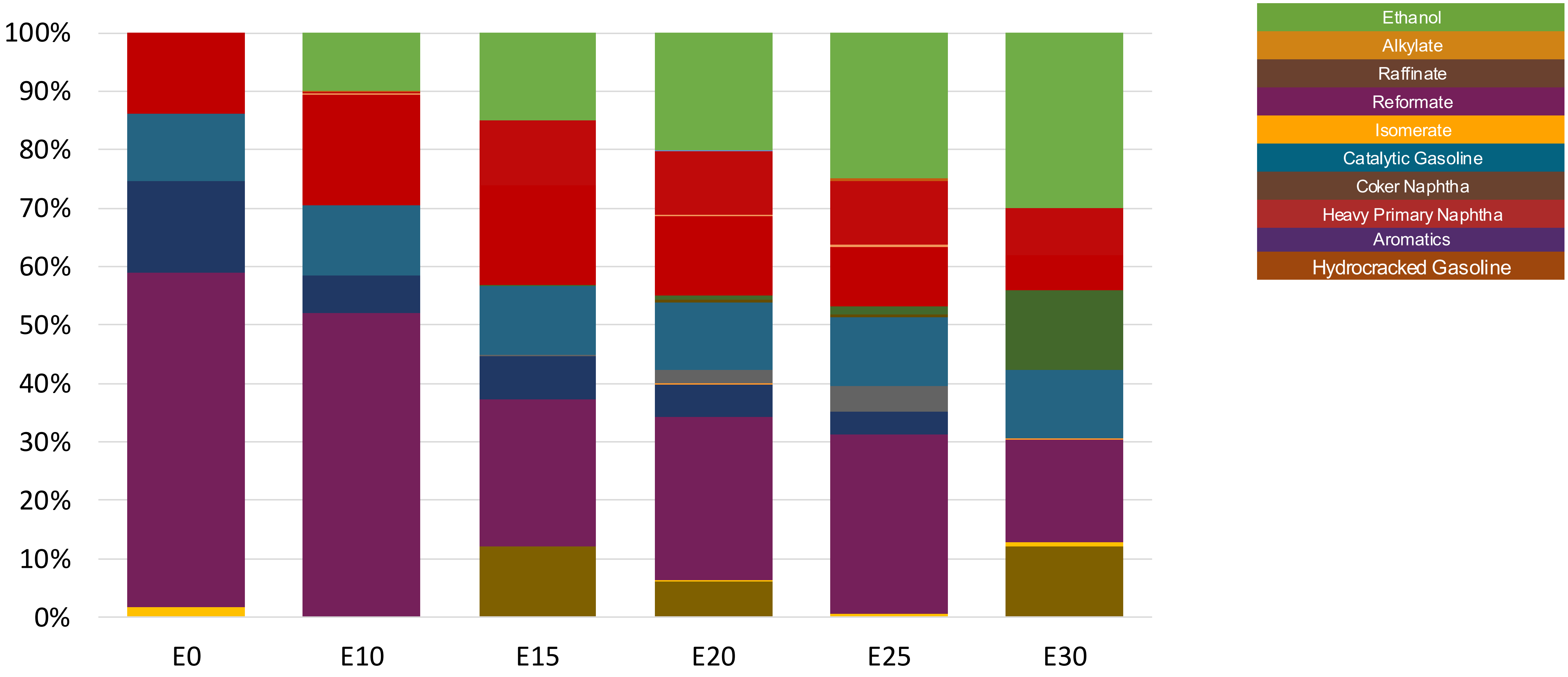
Octane (RON)	97.0	97.0	97.0	97.0	97.0	97.0
Price (US$/gal)	$ 2.32	$ 2.19	$ 2.13	$ 2.07	2.01	$ 1.95

Argentina – Nafta Grado 2 – North & Central Zones– Octane Increment



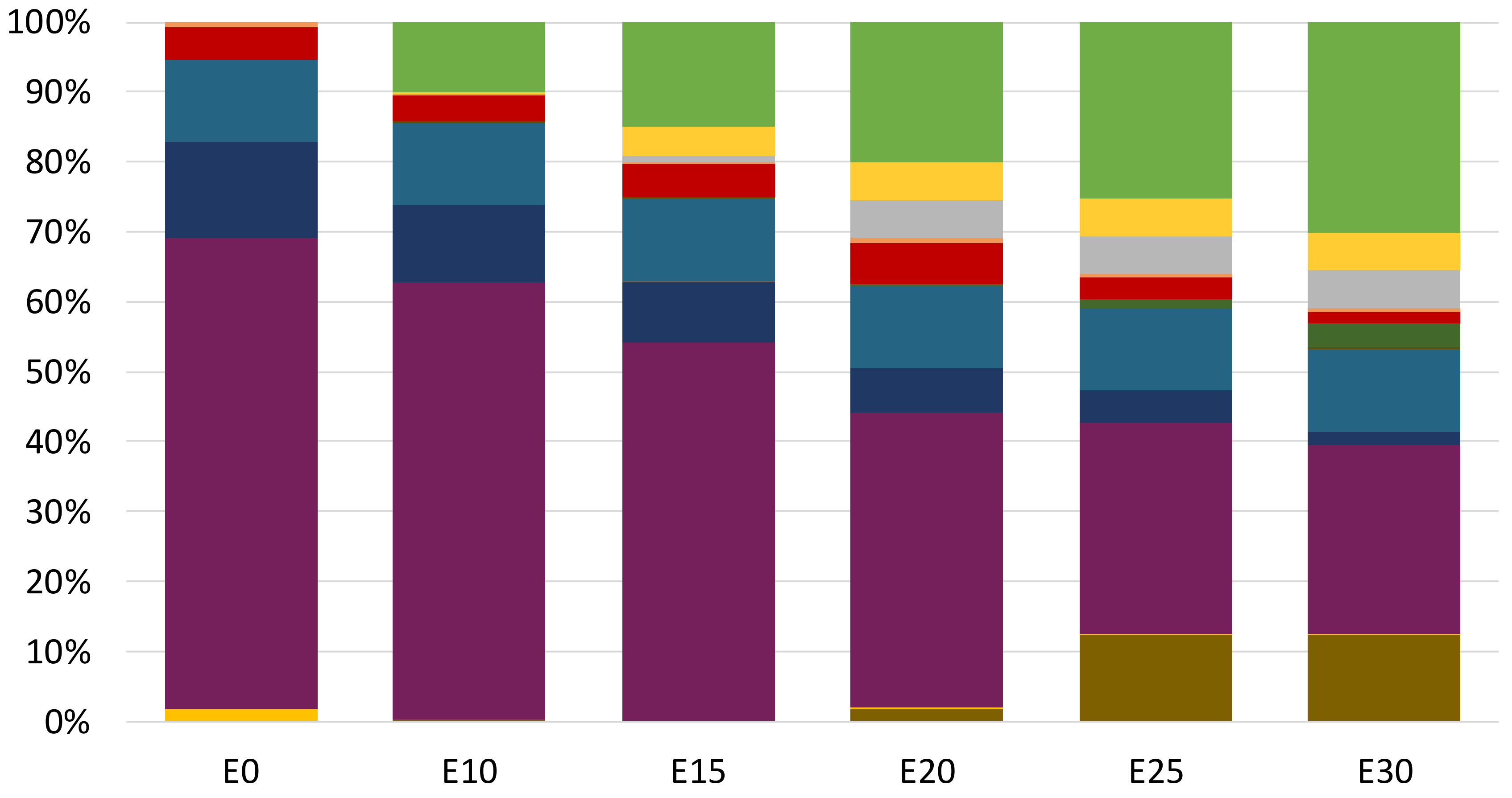
Octane (RON)	93.0	95.4	96.0	99.9	101.3	102.6
Price (US$/gal)	$ 2.28	$ 2.28	$ 2.28	$ 2.28	$ 2.28	$ 2.28

Argentina – Nafta Grado 2 – South Zone – Octane Increment



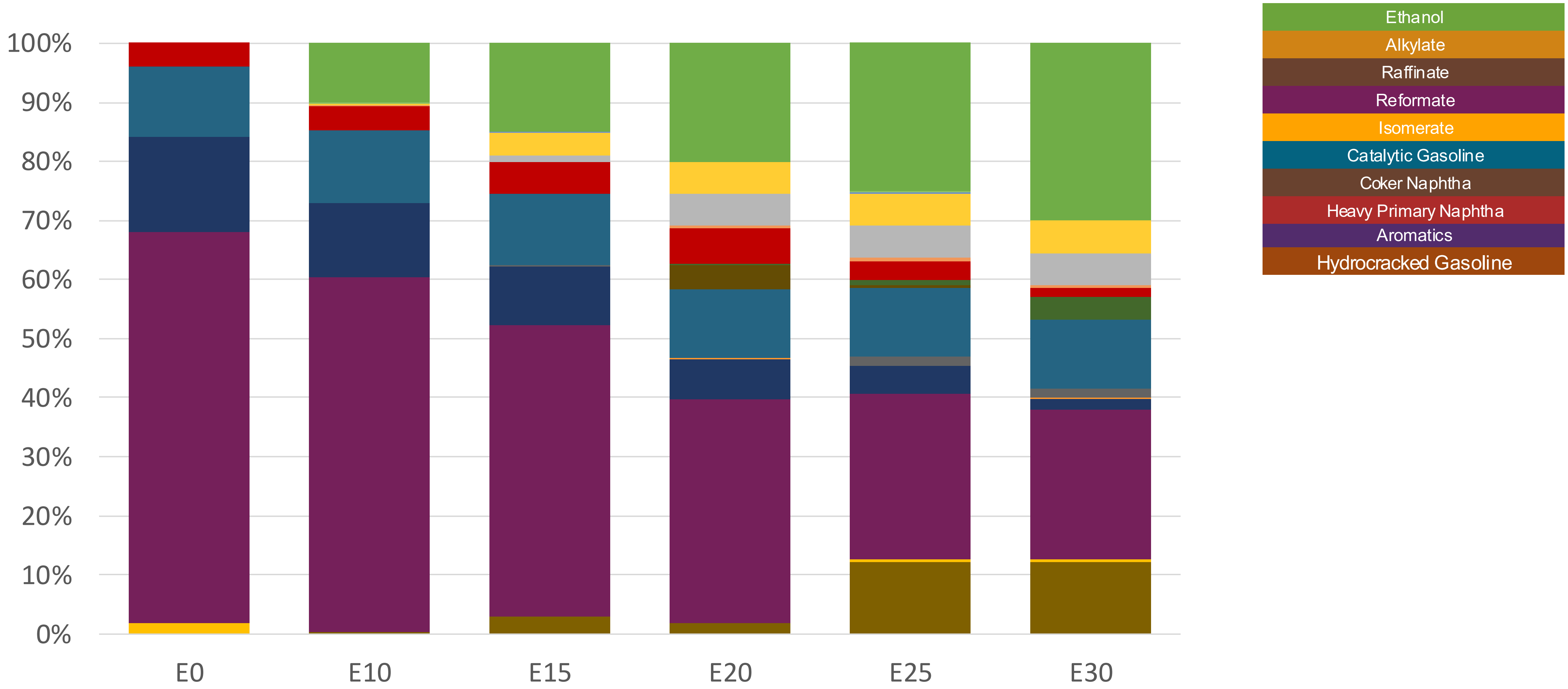
Octane (RON)	93.0	93.9	94.9	96.8	98.6	98.1
Price (US$/gal)	$ 2.13	$ 2.13	$ 2.13	$ 2.13	$ 2.13	$ 2.13

Argentina – Nafta Grado 3 – North & Central Zones– Octane Increment



Octane (RON)	97.0	99.8	100.8	101.6	102.2	103.2
Price (US$/gal)	$ 2.35	$ 2.35	$ 2.35	$ 2.35	$ 2.35	$ 2.35

Argentina – Nafta Grado 3 – South Zone – Octane Increment



Octane (RON)	97.0	99.5	100.2	100.9	101.6	102.6
Price (US$/gal)	$ 2.32	$ 2.32	$ 2.32	$ 2.32	$ 2.32	$ 2.32

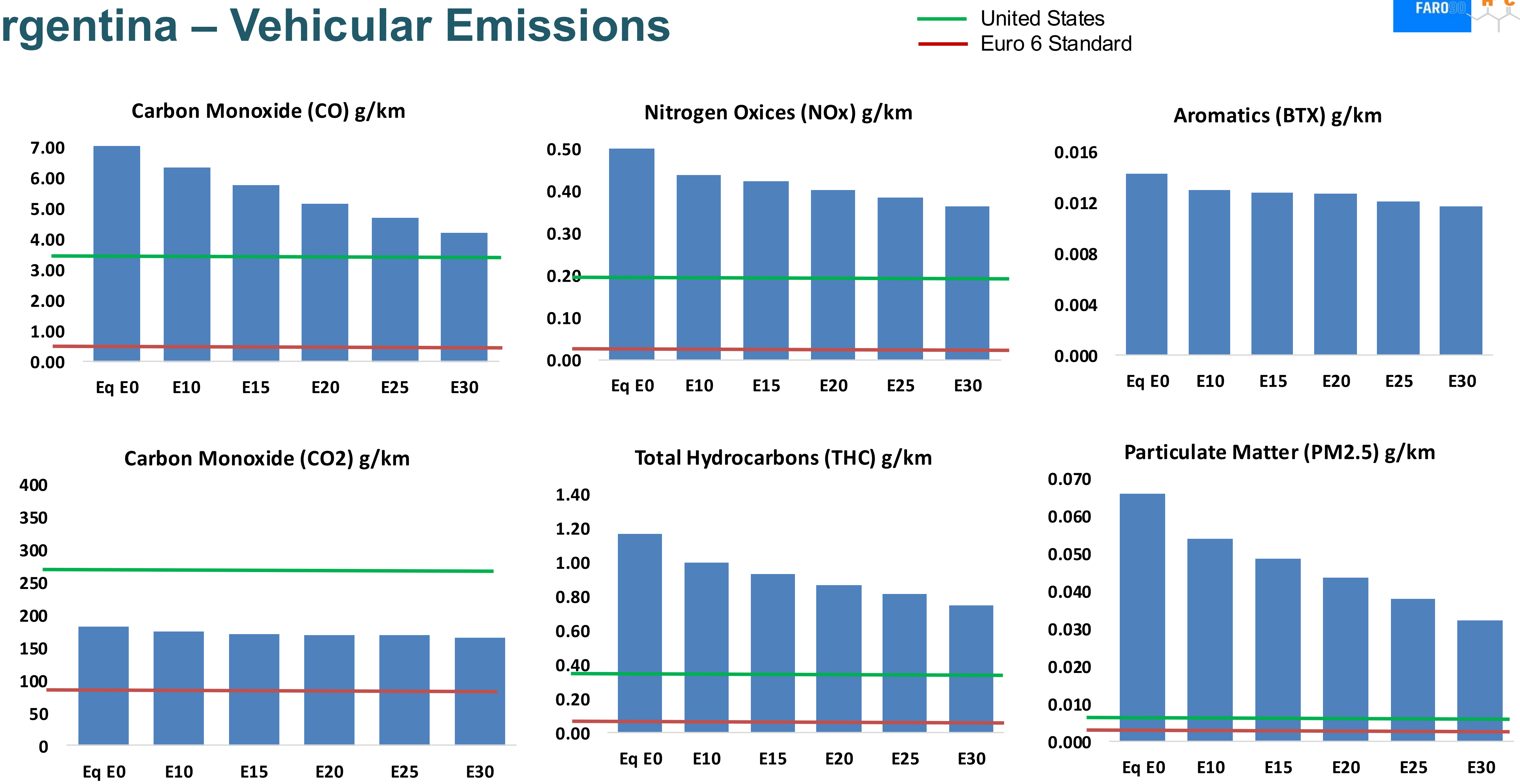
Gasoline Vehicle Fleet - Argentina

Vehicle Type	Motor	Tailpipe	Recuperator	Age	Number of vehicles
Auto/Sml Truck	Multi FI-Pt	Euro V	PCV Tank	>4years, >80 mkm	1,797,207
				4-8 years, 80-160 mkm	2,398,818
				<8 years, < 160 mkm	2,004,100
		Euro IV		<8 years, < 160 mkm	235,115
		Euro III		<8 years, < 160 mkm	548,601
		3-Ways		<8 years, < 160 mkm	2,550,413
Sml Engine	FI 4-cycle	Euro III	PCV	>4years, >80 mkm	2,019,500
				4-8 years, 80-160 mkm	1,232,840
		Euro II		4-8 years, 80-160 mkm	528,360
				<8 years, < 160 mkm	3,219,300
Truck	FI	Euro V	PCV	>4years, >80 mkm	55,584
				4-8 years, 80-160 mkm	74,190
				<8 years, < 160 mkm	61,982
		Euro IV		<8 years, < 160 mkm	16,967
		Euro III		<8 years, < 160 mkm	7,272
		3-Ways		<8 years, < 160 mkm	78,879

Vehicular Fleet: **16,829,128**
Average age: **12,5 years**
Motorcycles: **41.6%**

Source: Ministerio de Transporte, CAFAM, AFAC, 2025

Argentina – Vehicular Emissions

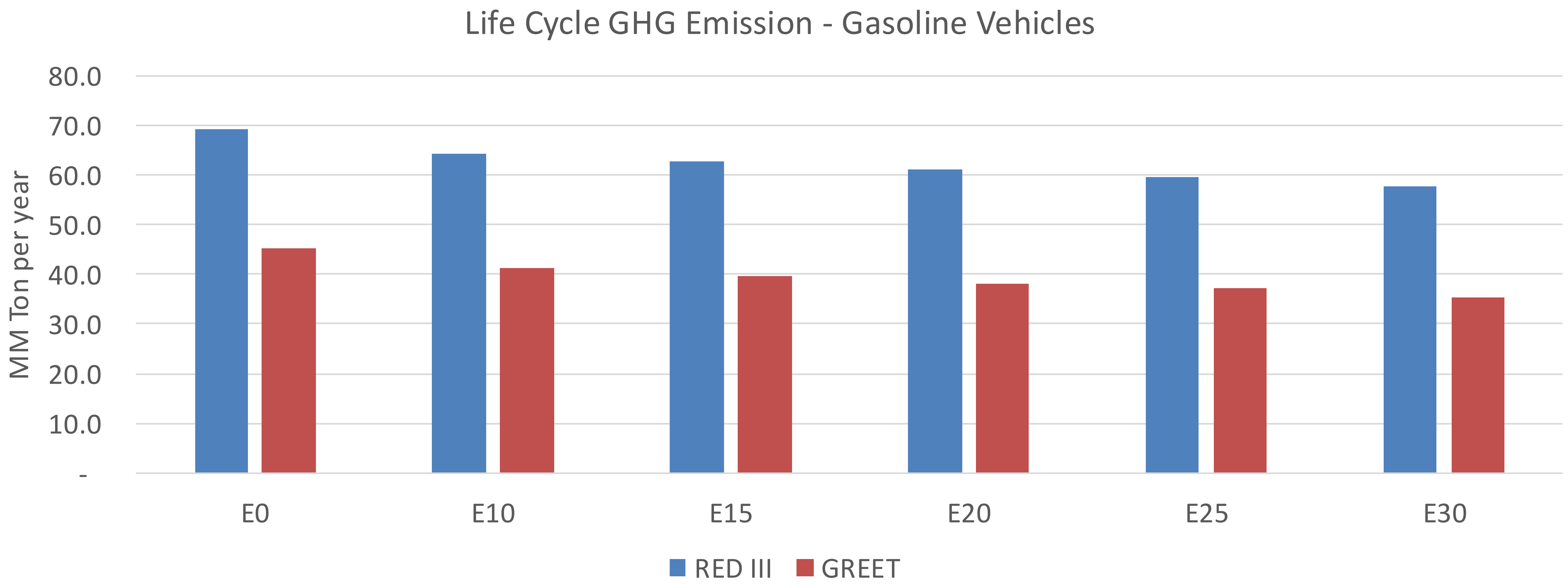


Source: Faro90

Argentina – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	2,490,584	2,269,882	2,049,179	1,862,618	1,672,251	1,525,078	1,368,593	-18%	-33%
CO2	59,094,716	57,575,137	56,139,980	55,011,271	54,452,741	54,228,972	53,229,376	-5%	-8%
VOC	376,201	349,287	322,374	300,711	279,091	262,521	240,217	-14%	-26%
NOx	162,969	150,474	141,783	137,293	130,129	124,482	117,852	-13%	-20%
SOx	670	619	568	526	485	440	393	-15%	-28%
NH3	25,823	25,567	25,312	25,432	25,718	25,919	26,086	-2%	0%
N2O	1,088	1,082	1,078	1,083	1,106	1,118	1,131	-1%	2%
THC	82,636	82,636	82,636	83,875	85,693	87,263	88,751	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	2,739	2,575	2,410	2,285	2,162	2,070	1,942	-12%	-21%
Acetaldehyde	7,765	8,911	13,076	19,913	27,129	31,000	36,630	68%	249%
Formaldehyde	30,810	29,955	34,919	41,002	42,956	47,520	51,731	13%	39%
Benzene	4,625	4,445	4,212	4,145	4,111	3,931	3,804	-9%	-11%
PM2.5	3,932	3,574	3,217	2,902	2,594	2,265	1,919	-18%	-34%
PM10	17,365	15,786	14,208	12,818	11,459	10,002	8,477	-18%	-34%

Argentina – Gasoline Vehicle Fleet GEI Emissions



Country Emissions 2024 (MMT/año)	359.7
Target @ 2035 (MMT/year)	197.5
Delta	162.3

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	9.5%	12.5%	15.8%	18.4%	22.0%
RED II/III	0.0%	7.0%	9.4%	11.9%	14.0%	16.6%

% 2035 Target	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	2.7%	3.5%	4.4%	5.2%	6.2%
RED II/III	0.0%	3.0%	4.0%	5.1%	6.0%	7.1%

Source: Greet, RED II/III, climateactiontracker.org, F90

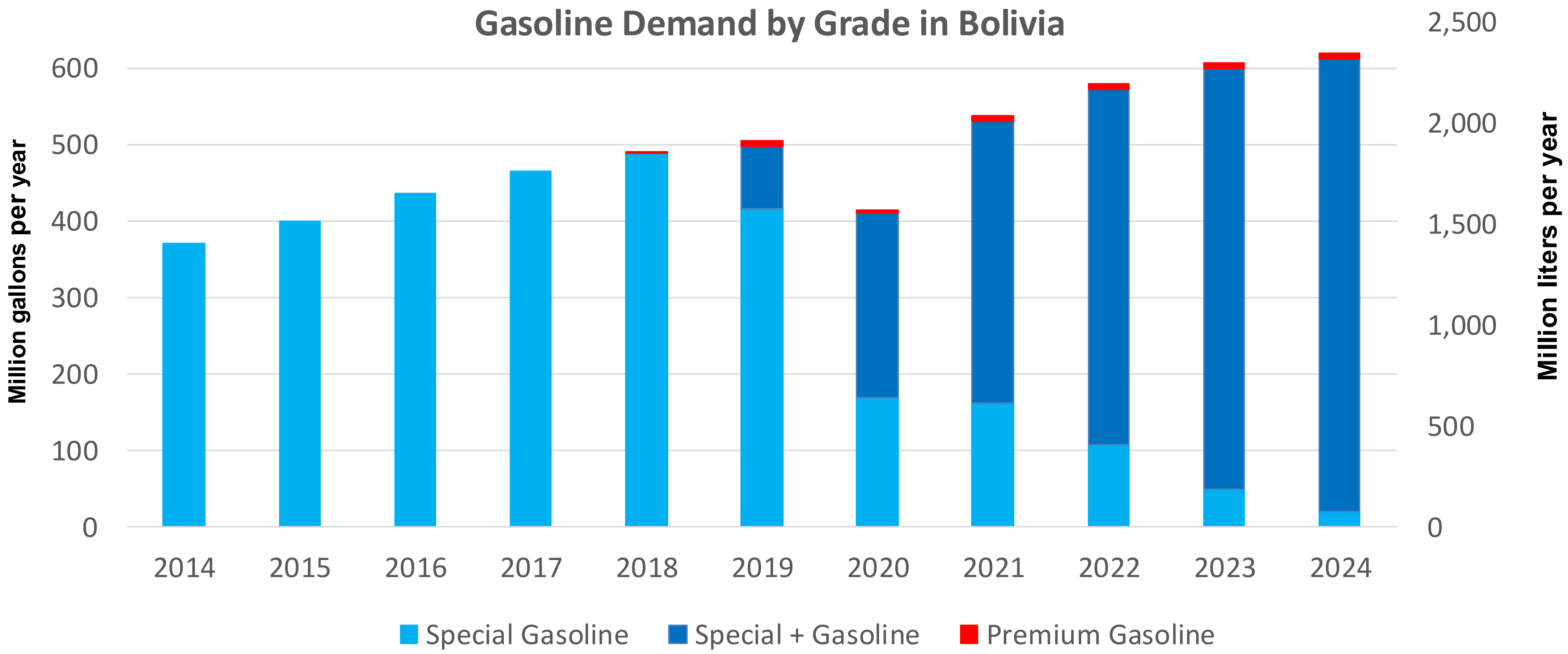
Ethanol Blending in Gasoline - Bolivia

In 2024, gasoline demand was 620 million gallons (2,348 million liters). Gasoline Imports represented 51.4% of demand, and are supplied from Brazil, Chile, Argentina, and the United States. Regular Gasoline (Especial) with 85 RON Octane represented 96% of demand and Premium Gasoline with a 95 RON octane the rest. Gasoline production has decreased in the last years.

Gasoline E8 and E12 are currently marketed (Decreto Supremo No. 4718:ANH). About 80% of gasoline demand contains ethanol. All ethanol consumed in Bolivia is produced by five local companies: Azúcar Aguaí, Unagro, Poplar Capital, Granosol, and Guabirá.

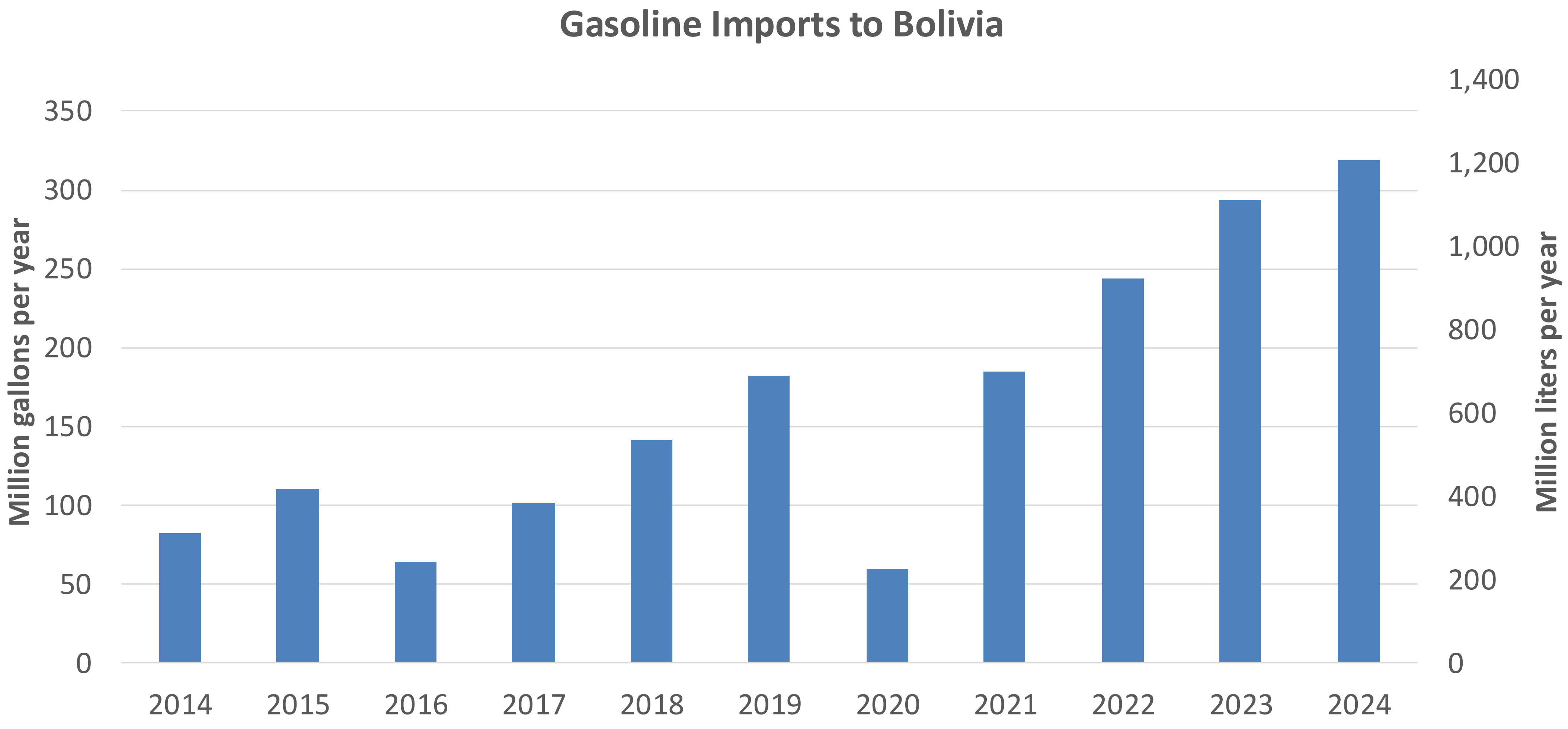
Source: ANH- Bolivia

Gasoline Demand in Bolivia



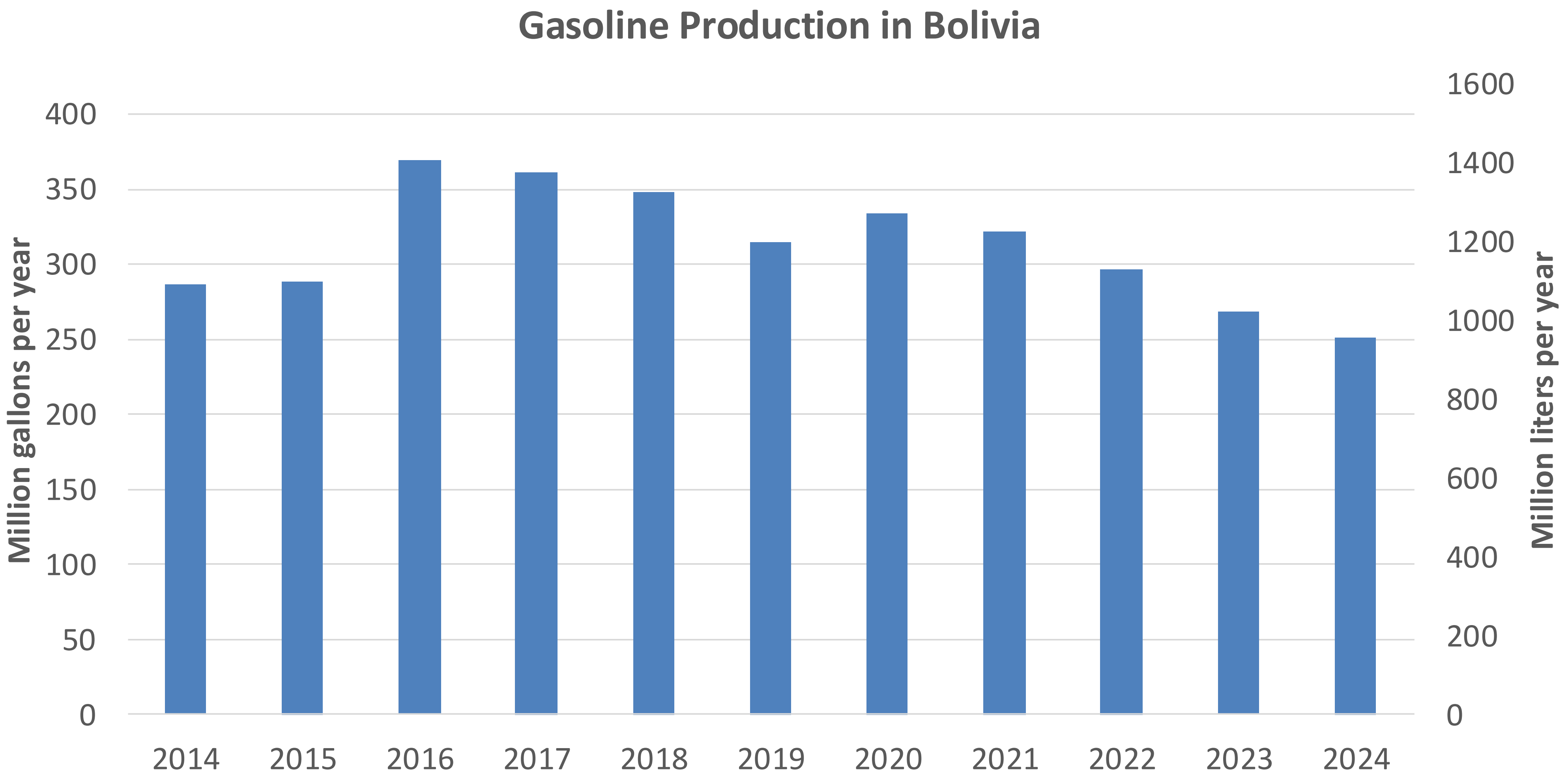
Source: ANH,INE - Bolivia

Gasoline Imports in Bolivia



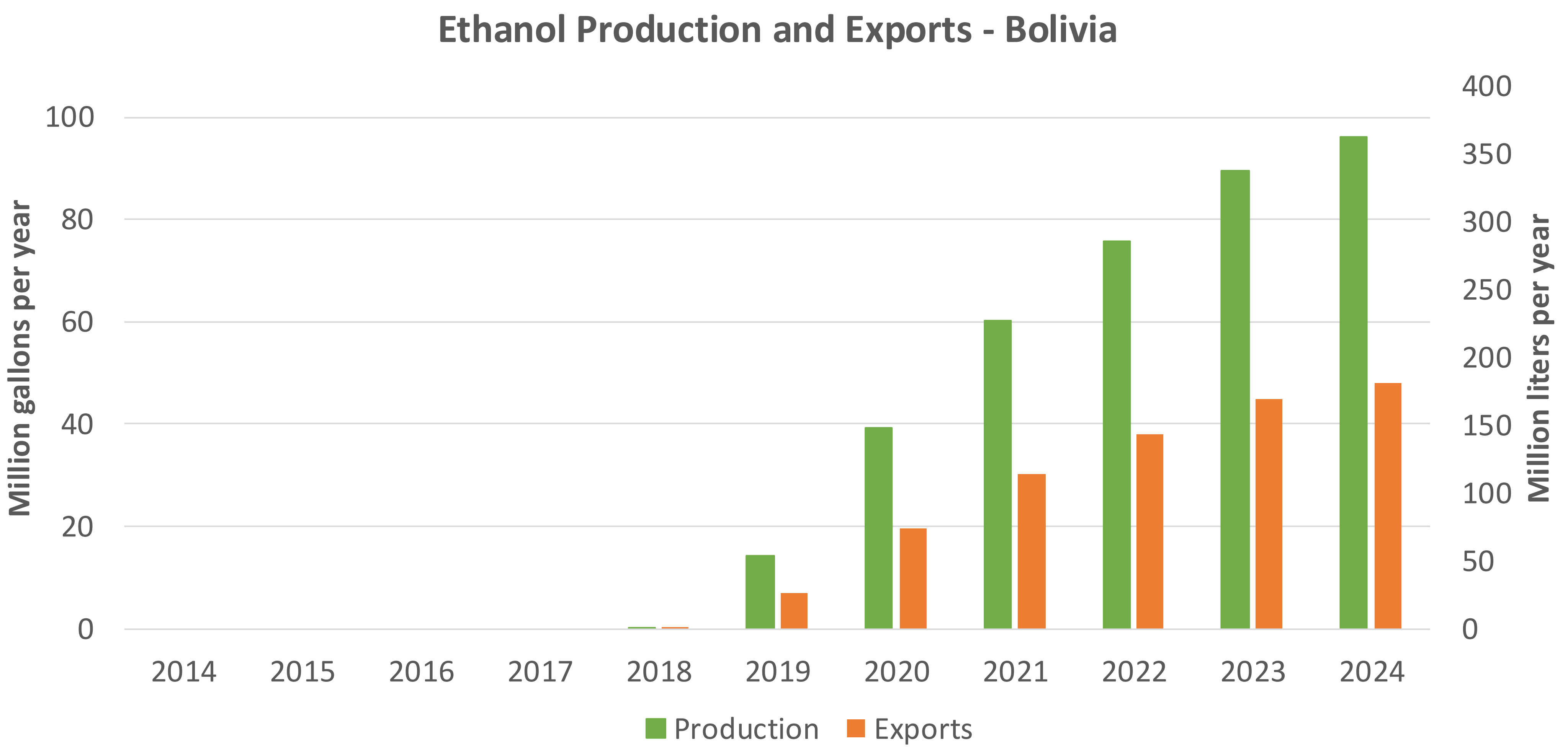
Source: ANH,INE - Bolivia

Gasoline Production in Bolivia



Source: ANH,INE - Bolivia

Ethanol Production in Bolivia

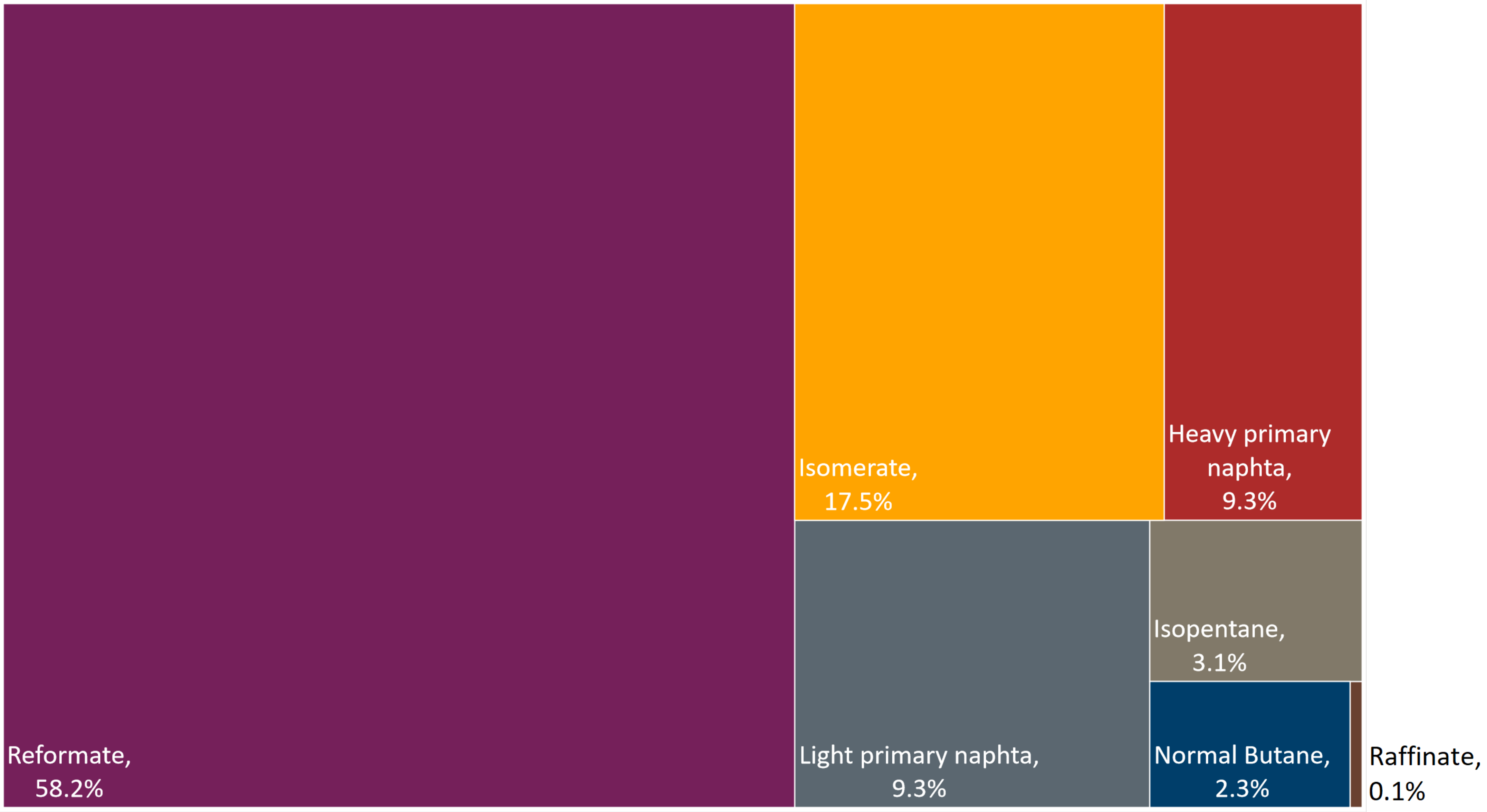


Source: ANH,INE - Bolivia

Gasoline Quality in Bolivia

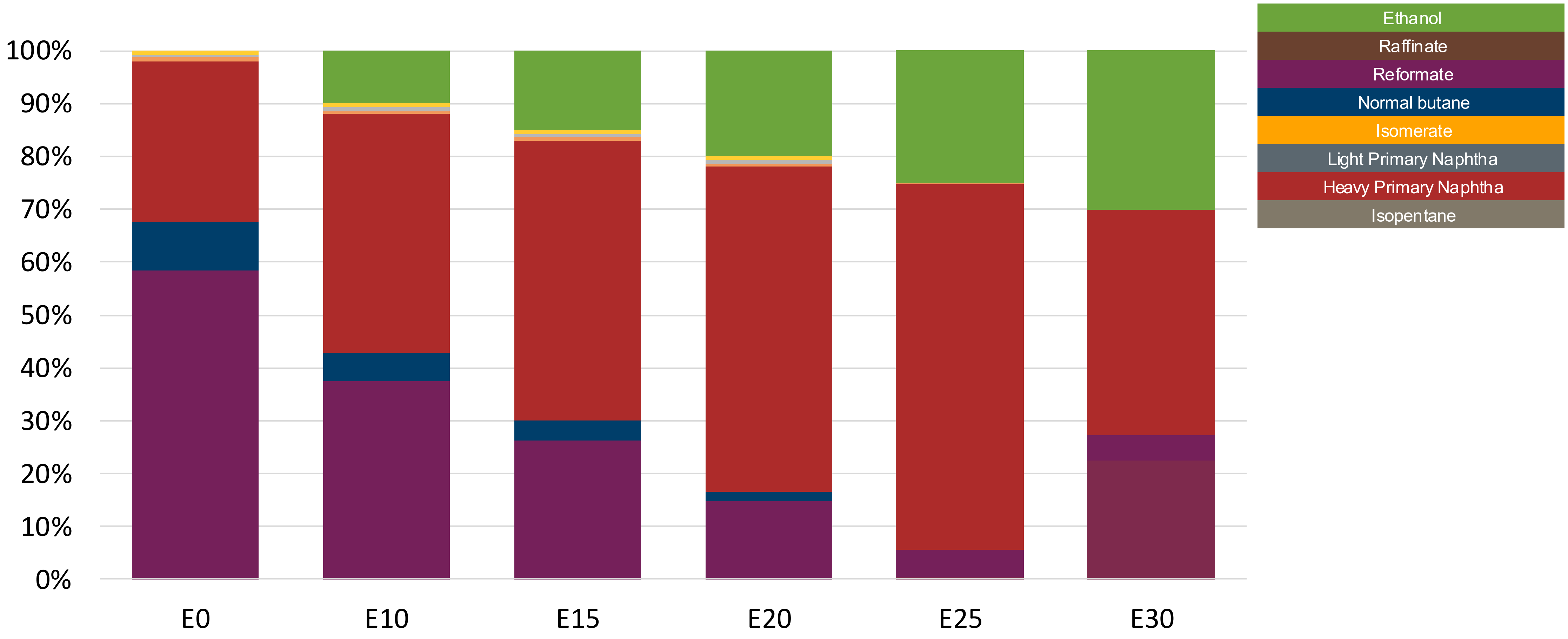
Name	Supreme Decree 4718		EN 228:2012 + A1:2017 (Euro 6 enabling)			
Implementation Date	2022		2017			
Applicability	Whole country	Whole country	All countries			
Selected Grade	Gasoline Regular	Gasoline Premium	RON 95 E5	RON 95 E10	RON 98 E5	RON 98 E10
Benzene Content	< 3%v/v (2,7%v/v for E12 in summer)	< 3%v/v (2,7%v/v for E12 in summer)	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v
Aromatics	< 42 % v/v	< 48 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v
Olefins	< 18 %v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v
Lead Content	< 0,013 g/l	< 0,013 g/l	< 5 mg/l	< 5 mg/l	< 5 mg/l	< 5 mg/l
Manganese	< 18 mg/l	< 18 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l
RON	> 85	> 95	> 95	> 95	> 98	> 98
MON	-	-	> 85	> 88	> 85	> 88
AKI						
Sulfur Content	< 500 mg/kg	< 500 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg
Oxygen Content	< 2,7 %m/m	< 2,7 %m/m	<2,7 % m/m	<3,7 % m/m	<2,7 % m/m	<3,7 % m/m
Ethanol (EtOH)	< 12 %v/v	< 12 %v/v	<5 %v/v	<10 %v/v	<5 %v/v	<10 %v/v
RVP 37.8°C (Summer)	<>7-9 psi	<> 7-9 psi	<> 60 - 70 kPa *Depends on the country, RVP is regulated in the EU Fuel Quality Directive			
RVP 37.8 °C(Winter)	<>7-9.5 psi	<> 7-9.5 psi				
RVP 37.8°C (Transition)	<>7-9 psi	<> 7-9 psi				
MTBE			-	-	-	-
Ethers 5 or more C Atoms	-	-	Based on oxygen content	<22 %v/v	Based on oxygen content	<22 %v/v

Available Blending Components



Source: Faro90

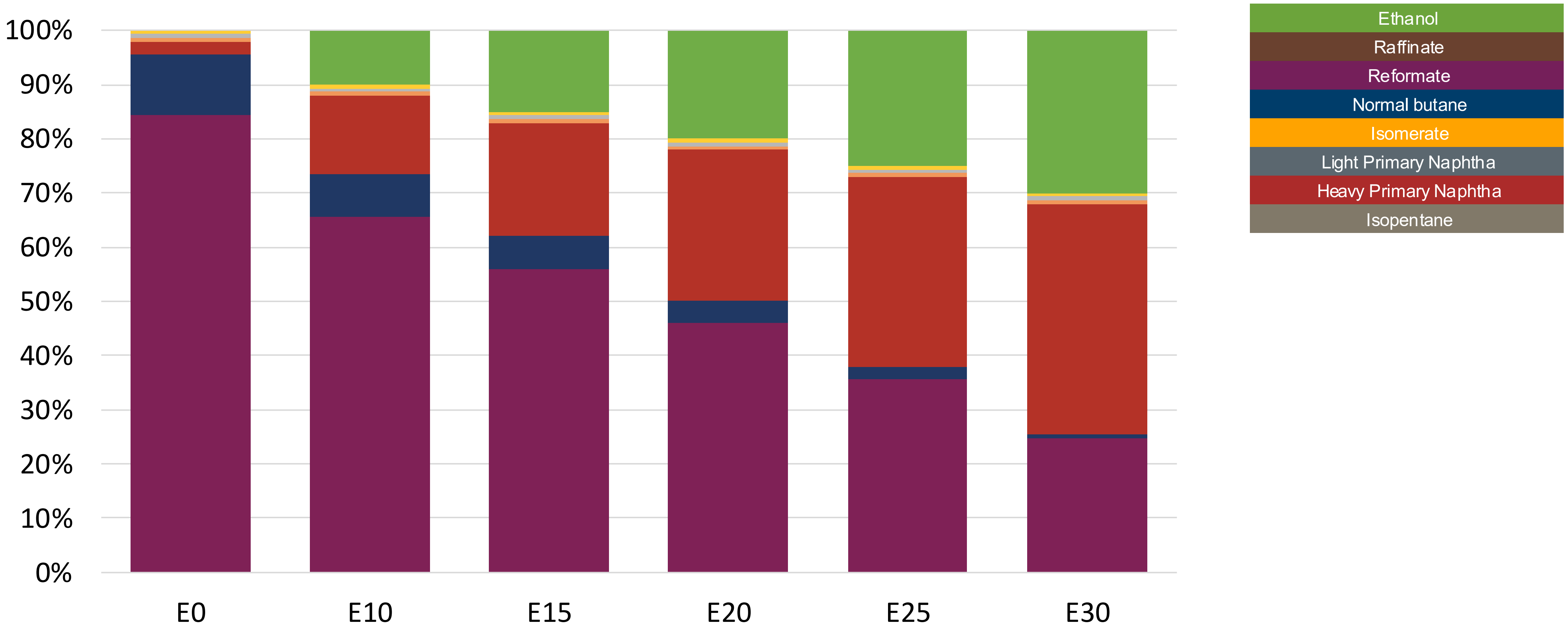
Ethanol Blending - Gasoline Especial – Constant Octane



Octane (RON)	85.0	85.0	85.0	85.0	85.0	89.2
Price (US$/gal)	$2.26	$1.96	$1.79	$1.62	$1.44	$1.54

Source: Faro90

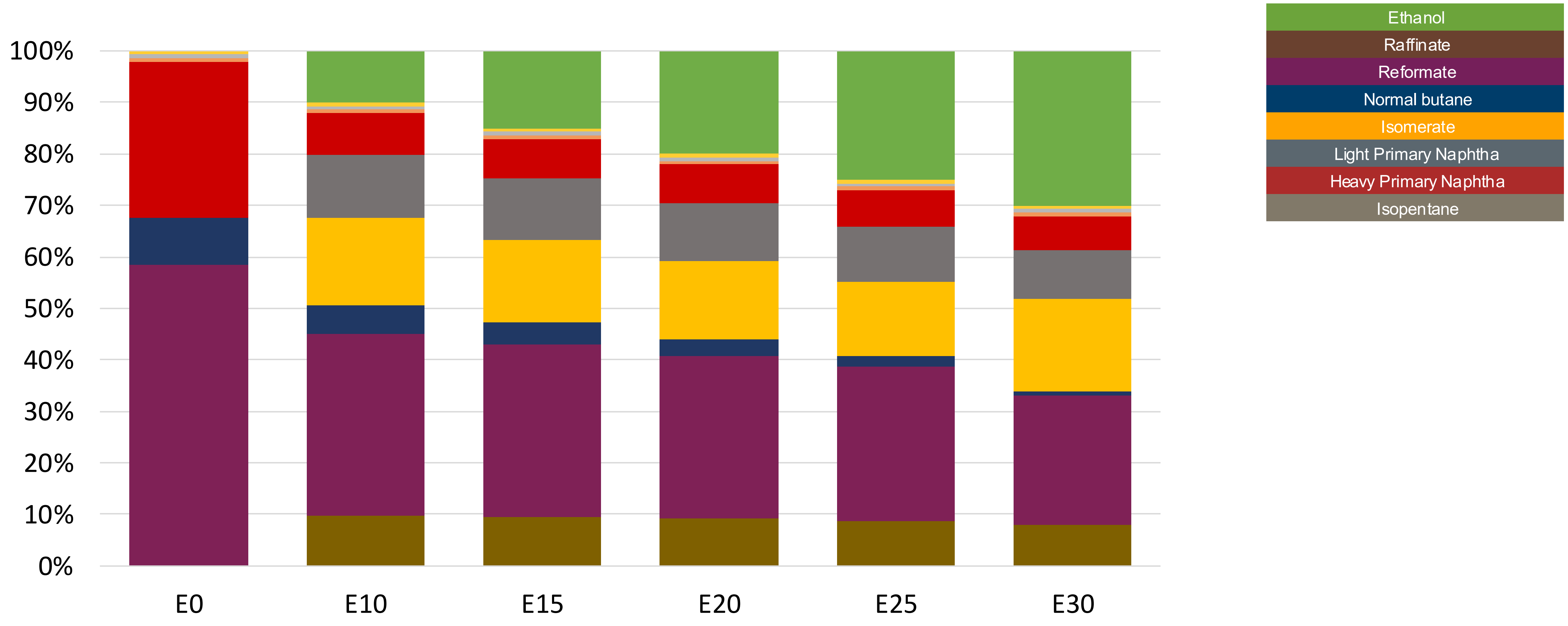
Ethanol Blending – Gasoline Premium – Constant Octane



Octane (RON)	95.0	95.0	95.0	95.0	95.0	95.0
Price (US$/gal)	$2.76	$2.50	$2.37	$2.22	$2.08	$1.91

Source: Faro90

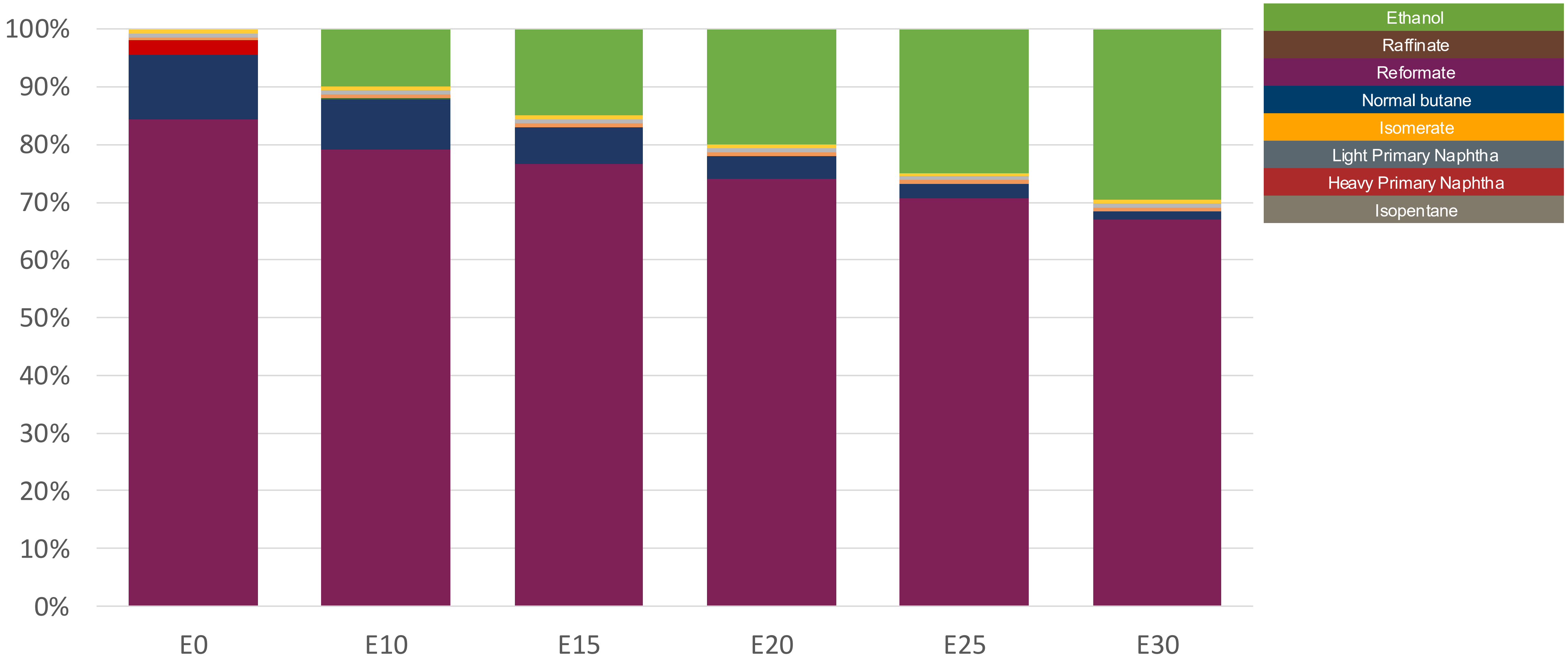
Ethanol Blending - Gasoline Especial – Octane Increment



Octane (RON)	85.0	90.2	92.8	95.3	97.7	99.9
Price (US$/gal)	$2.26	$2.26	$2.26	$2.26	$2.26	$2.26

Source: Faro90

Ethanol Blending - Gasoline Premium – Octane Increment



	E0	E10	E15	E20	E25	E30
Octane (RON)	95.0	99.7	101.6	103.4	105.4	107.7
Price (US$/gal)	$2.76	$2.76	$2.76	$2.76	$2.76	$2.76

Source: Faro90

Gasoline Vehicle Fleet Bolivia

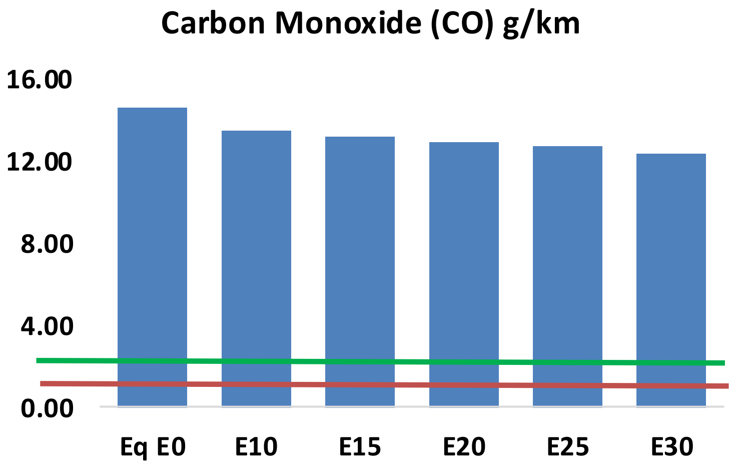
Vehicle Type	Motor	Tailpipe	Recuperator	Age	Number of vehicles
Auto/Sml Truck	Multi FI-Pt	Euro V	PCV Tank	>4years, >80 mkm	39,474
				4-8 years, 80-160 mkm	56,957
				<8 years, < 160 mkm	39,689
		Euro IV		<8 years, < 160 mkm	43,498
		Euro III		<8 years, < 160 mkm	24,308
		3-Ways		<8 years, < 160 mkm	249,368
Sml Engine	FI 4-cycle	Euro III	PCV	>4years, >80 mkm	258,749
				4-8 years, 80-160 mkm	89,752
		Euro II		4-8 years, 80-160 mkm	89,752
				<8 years, < 160 mkm	440,804
Truck	FI	Euro V	PCV	>4years, >80 mkm	178,544
				4-8 years, 80-160 mkm	203,281
				<8 years, < 160 mkm	135,905
		Euro IV		<8 years, < 160 mkm	132,960
		Euro III		<8 years, < 160 mkm	126,191
		3-Ways		<8 years, < 160 mkm	280,182

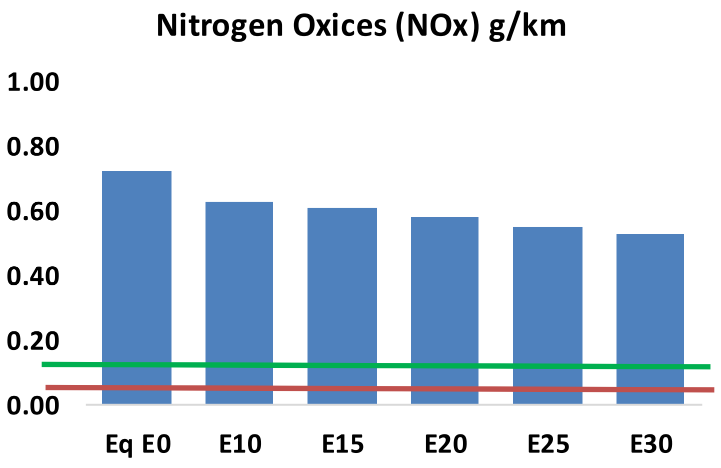
Vehicular Fleet: **2,389,413** Average Age:
12 years Motorcycles**: 36.8%**

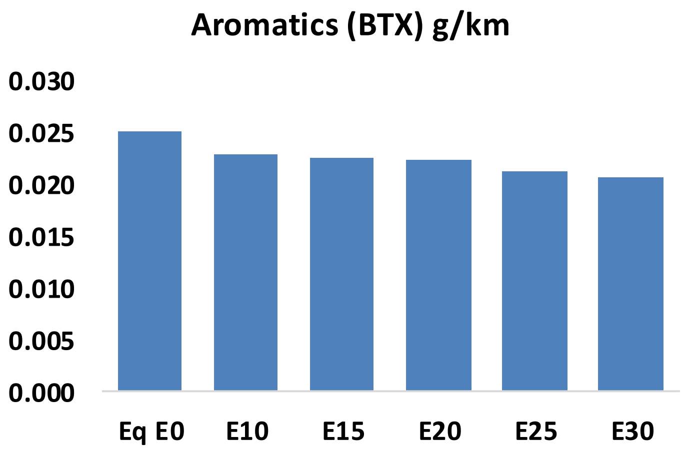
Source: Instituto Nacional de Estadística – Bolivia (INE), analysis Faro 90

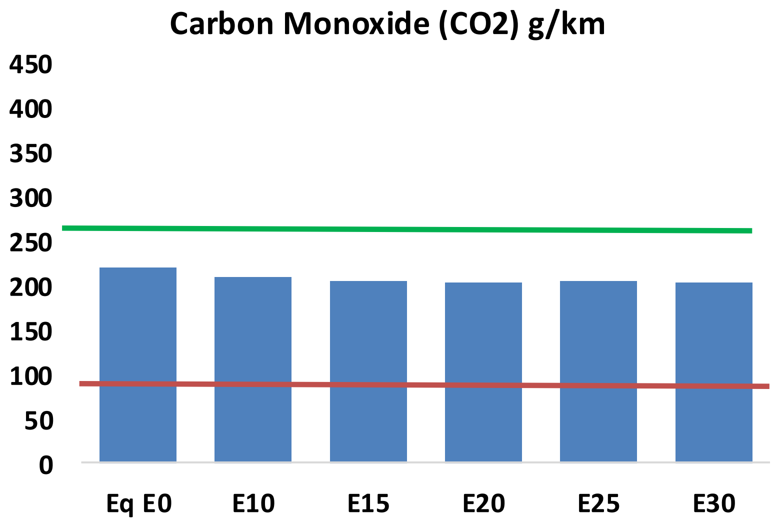
Bolivia – Vehicular Emissions

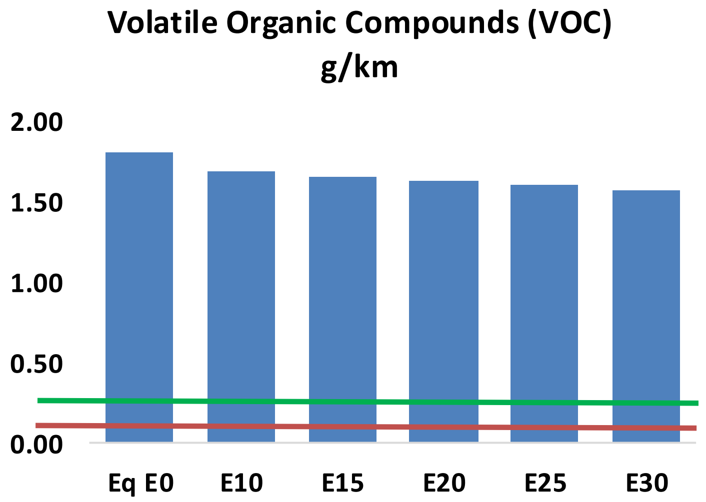
United States
Euro 6

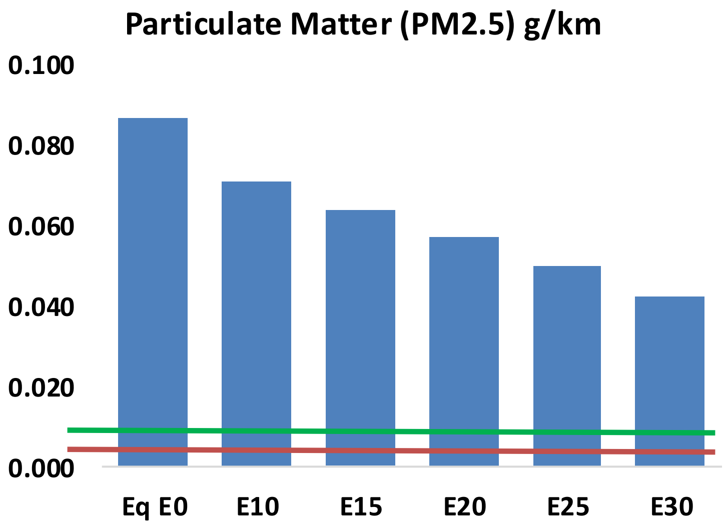












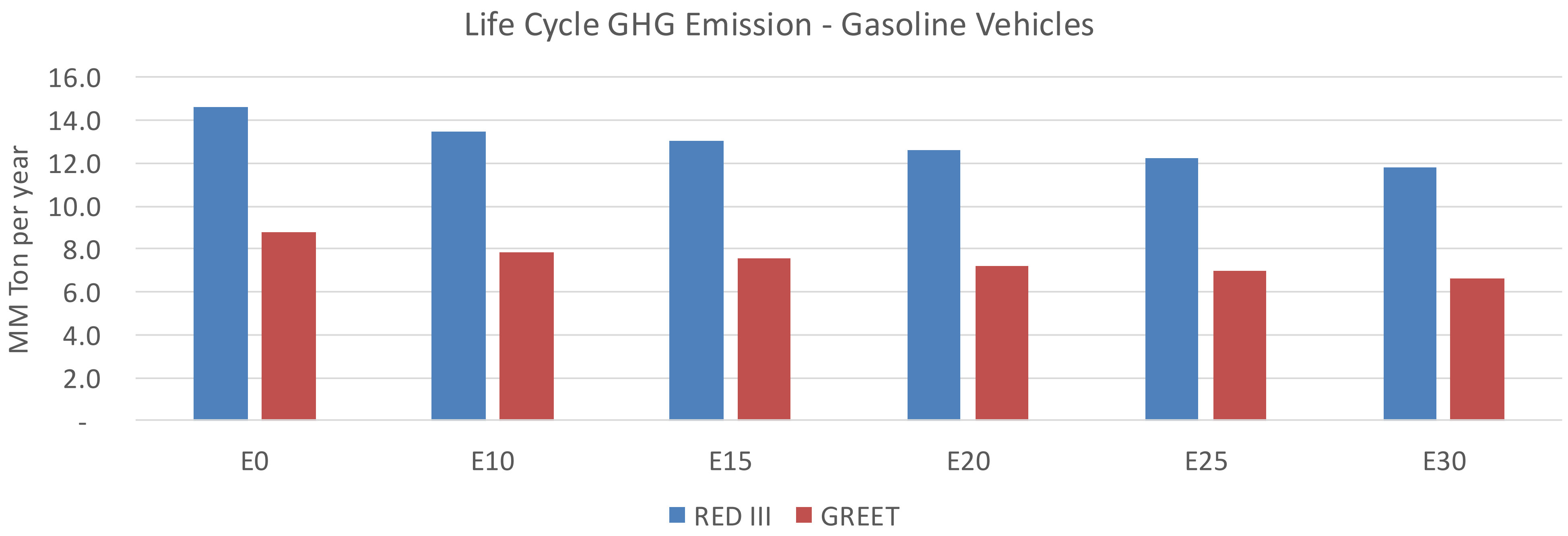
Source: Faro90

Bolivia – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	675,843	650,569	625,296	609,792	596,587	587,156	572,017	-7%	-12%
CO2	10,167,733	9,906,277	9,659,346	9,465,142	9,369,043	9,514,488	9,339,109	-5%	-8%
VOC	83,716	80,888	78,059	76,445	75,145	74,243	72,694	-7%	-10%
NOx	33,422	30,859	29,077	28,156	26,687	25,529	24,169	-13%	-20%
SOx	126	116	107	99	91	83	74	-15%	-28%
NH3	2,920	2,891	2,862	2,875	2,908	2,930	2,949	-2%	0%
N2O	993	988	984	988	1,009	1,020	1,032	-1%	2%
CH4	18,986	18,986	18,986	19,271	19,689	20,049	20,391	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	353	337	322	312	302	295	285	-9%	-14%
Acetaldehyde	1,118	1,283	1,883	2,868	3,907	4,464	5,275	68%	249%
Formaldehyde	4,767	4,635	5,403	6,344	6,646	7,352	8,004	13%	39%
Benzene	1,159	1,114	1,056	1,039	1,030	985	954	-9%	-11%
PM2.5	817	743	669	603	539	471	399	-18%	-34%
PM10	3,181	2,891	2,602	2,348	2,099	1,832	1,553	-18%	-34%

Source: Faro90

Bolivia – Gasoline Vehicle Fleet GEI Emissions



Country Emissions 2024 (MMT/year)	63.8
2035 Target (MMT/year)	44.6
Delta	19.1

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	10.7%	14.3%	18.1%	20.7%	24.9%
RED II/III	0.0%	7.9%	10.7%	13.7%	15.9%	19.0%

% 2035 Target	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	4.9%	6.6%	8.3%	9.5%	11.4%
RED II/III	0.0%	6.0%	8.2%	10.5%	12.1%	14.5%

Source: Greet, RED II/III, climateactiontracker.org, F90

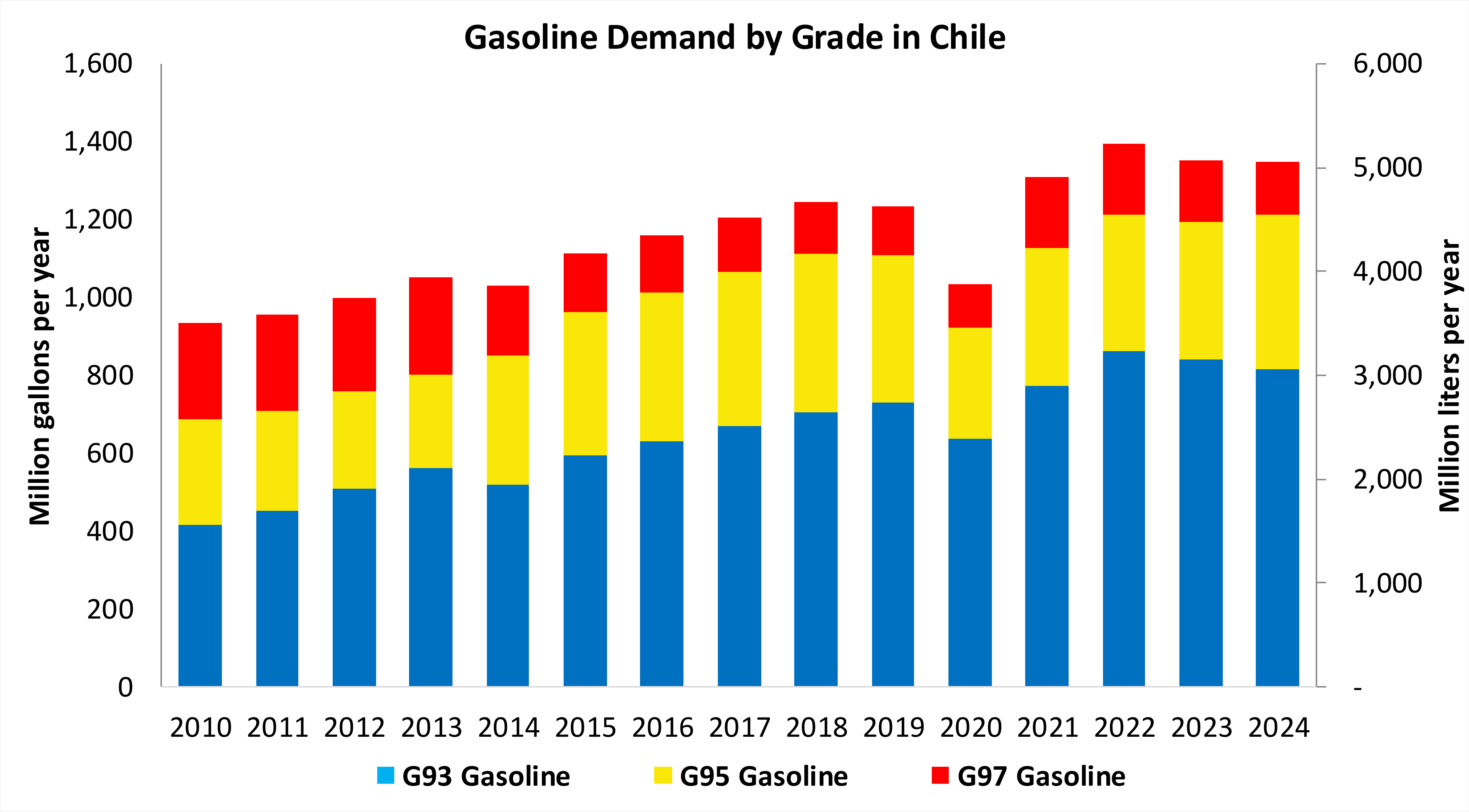
Ethanol Blending in Ethanol - Chile

Chile's gasoline consumption is currently over 1,300 million gallons per year (5,000 million liters) in three grades: RON 93 (AKI 87), RON 95 (AKI 88) and RON 97 (AKI 92). There is a more stringent quality specification for Santiago Metropolitan Area. In 2024, the market share of RON 93 gasoline was 28%. Government company ENAP produces 80% of the total demand (CNE). Gasoline imports made in the country come mainly from the United States. Refineries only produce RON 93 and RON 97, gasoline RON 95 is blended from these grades in gas stations.

Ethanol can be blended up to 5% v/v (Decreto Supremo 56) but it is not used because of regulatory barriers. Chile does not produce or imports ethanol. The use of ethanol is being analyzed to achieve short term goals in the national economy decarbonization program.

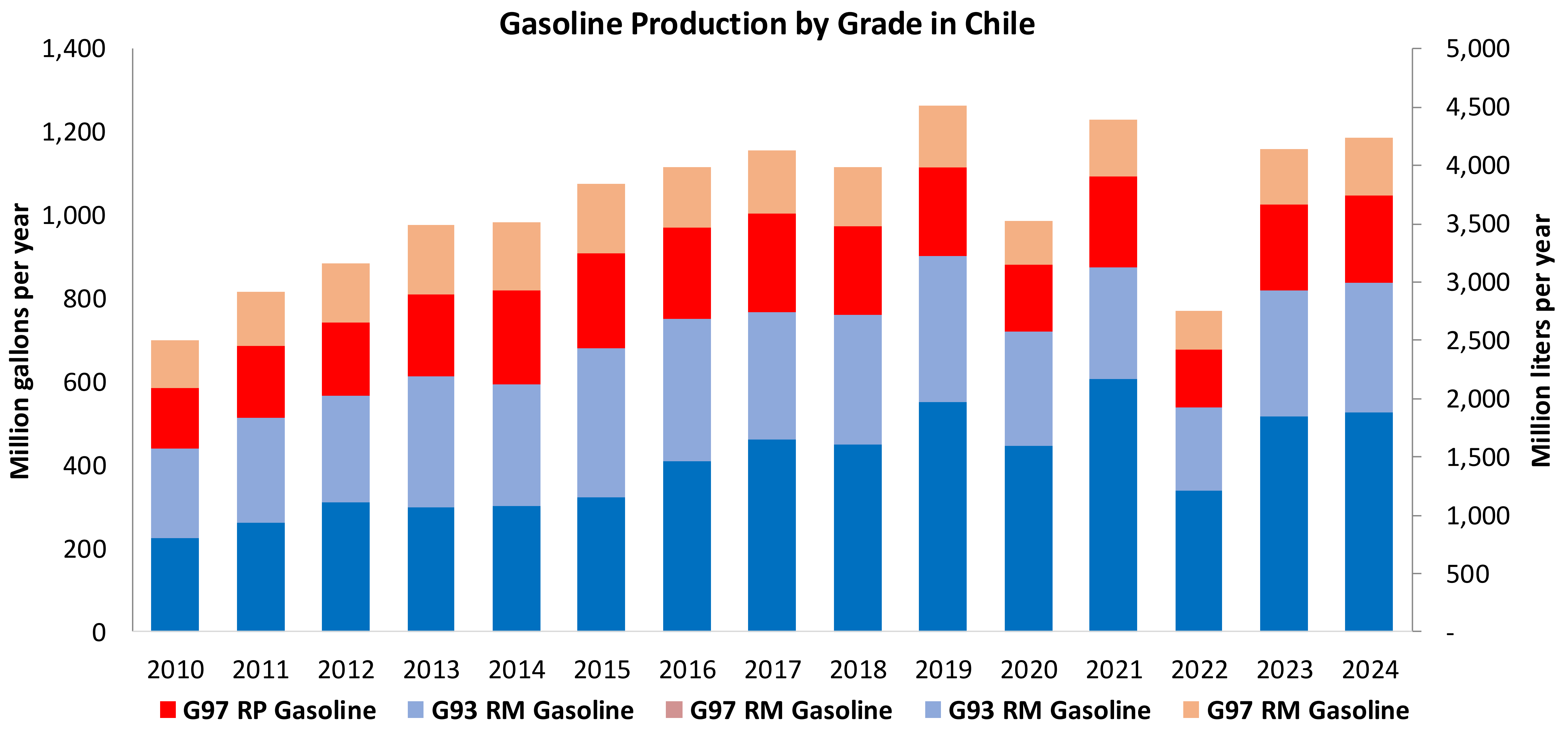
Source: CNE, SEC, 2025

Gasoline Demand in Chile



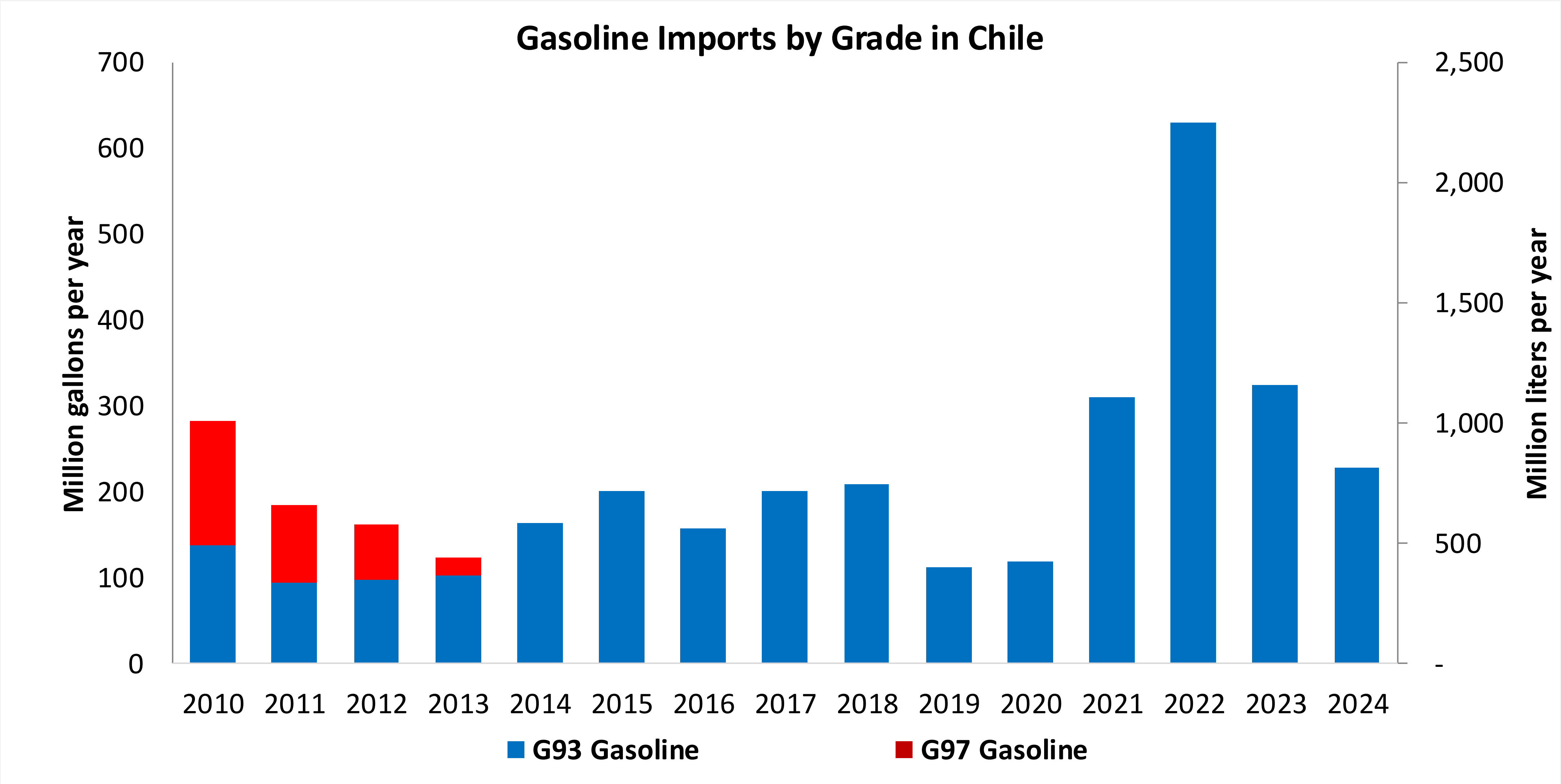
Fuente: CNE, 2025

Gasoline Production in Chile



Fuente: CNE, 2025

Gasoline Imports in Chile



Fuente: CNE, 2025

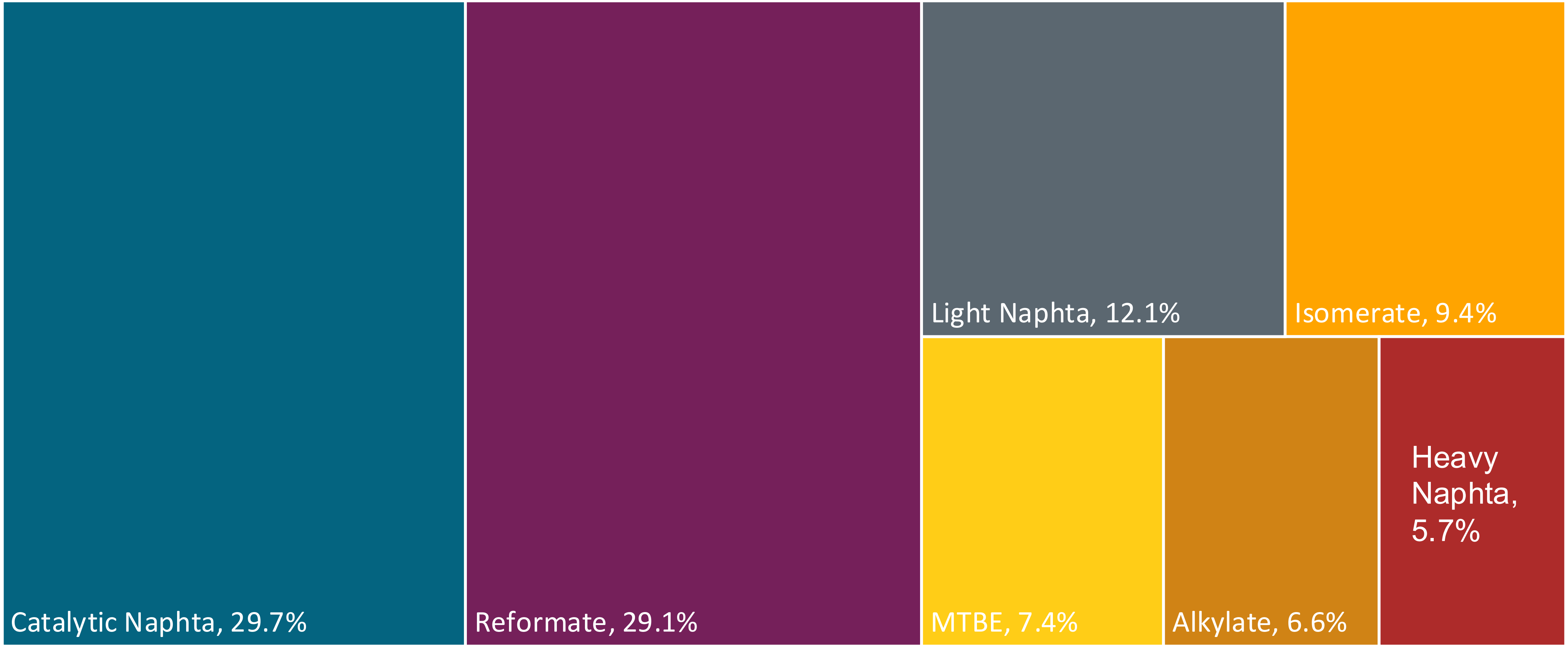
45

Gasoline Quality in Chile

Name	PPDA DS31	PPDA DS31	DS 60	DS 60	EN 228:2012 + A1:2017 (Euro 6 enabling)			
Implementation Date	2017	2017	2012	2012	2017			
Applicability	Metropolitan Region	Metropolitan Region	Rest of the country	Rest of the country	All countries			
Selected Grade	G93	G97	G93	G97	RON 95 E5	RON 95 E10	RON 98 E5	RON 98 E10
Benzene Content	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v
Aromatics	< 38 %v/v	< 38 %v/v	< 38 %v/v	< 38 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v
Olefins	12 %v/v	12 %v/v	20 %v/v	20 %v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v
Lead Content	< 0,013 g/l	< 0,013 g/l	< 0,013 g/l	< 0,013 g/l	< 5 mg/l	< 5 mg/l	< 5 mg/l	< 5 mg/l
Manganese	Report	Report	Report	Report	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l
RON	93	97	93	97	> 95	> 95	> 98	> 98
MON	Report	Report	Report	Report	> 85	> 88	> 85	> 88
AKI								
Sulfur Content	< 15 mg/kg	< 15 mg/kg	< 15 mg/kg	< 15 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg
Oxygen Content	< 2 %m/m	< 2 %m/m	< 2 %m/m	< 2 %m/m	<2,7 % m/m	<3,7 % m/m	<2,7 % m/m	<3,7 % m/m
Ethanol (EtOH)	<5 %v/v	<5 %v/v	<5 %v/v	<5 %v/v	<5 %v/v	<10 %v/v	<5 %v/v	<10 %v/v
RVP 37.8°C (Summer)	<> 55 kPa	<> 55 kPa	<> 69 kPa, <> 83 kPa (Magallanes y región Antártica)	<> 69 kPa, <> 83 kPa (Magallanes y región Antártica)	<> 60 - 70 kPa *Depends on the country, RVP is regulated in the EU Fuel Quality Directive			
RVP 37.8 °C(Winter)	<> 69 kPa	<> 69 kPa	<> 45 - 80 kPa	<> 45 - 80 kPa				
RVP 37.8°C (Transition)								
MTBE					-	-	-	-
Ethers 5 or more C Atoms	-	-	-	-	Based on oxygen content	<22 %v/v	Based on oxygen content	<22 %v/v

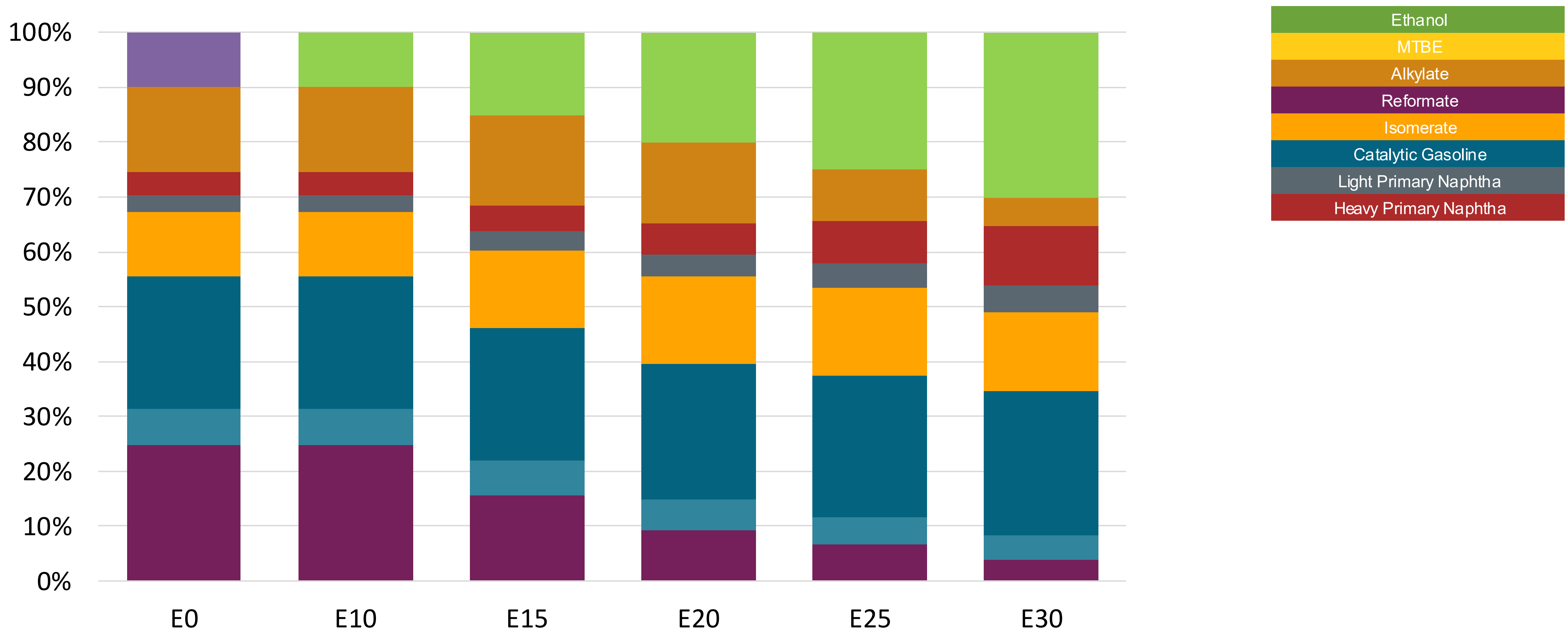
Source: ENAP, 2022

Available Blending Components



Fuente: ENAP, 2025

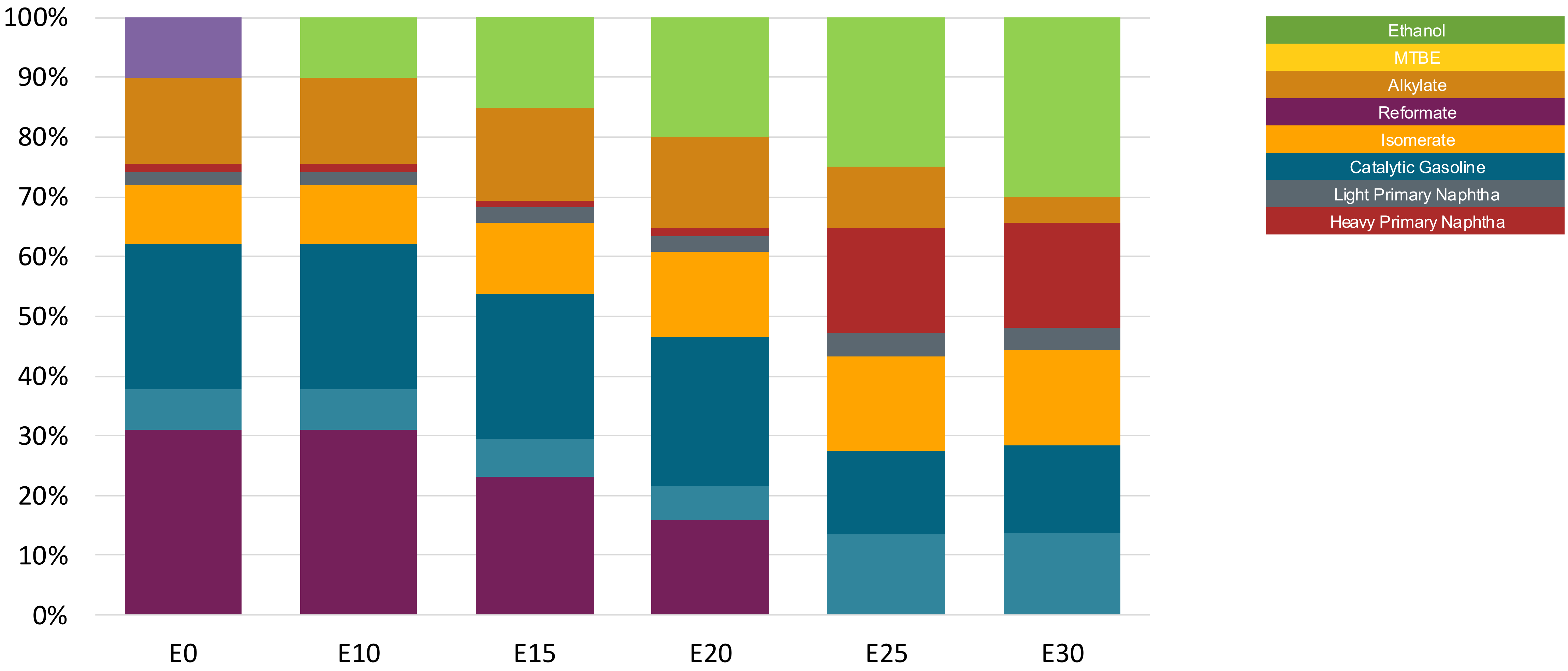
Chile – G93 RP Aconcagua – Constant Octane



Octane (RON)	93	93	93	93	93	93
Price (US$/gal)	$2.601	$2.519	$2.430	$2.341	$2.249	$2.160

Source: Faro90, 2025

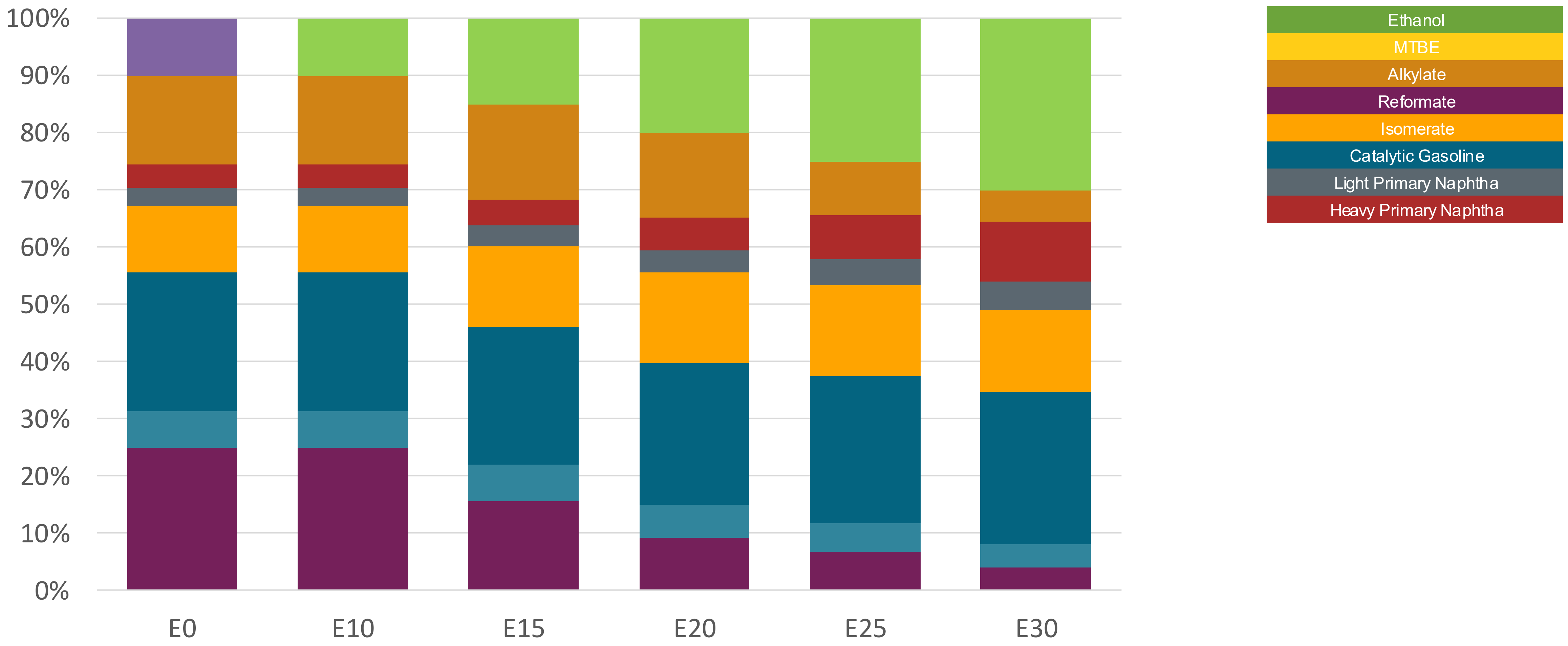
Chile – G97 RP Aconcagua – Constant Octane



Octane (RON)	97	97	97	97	97	97
Price (US$/gal)	$2.727	$2.645	$2.567	$2.487	$2.133	2.077

Source: Faro90, 2025

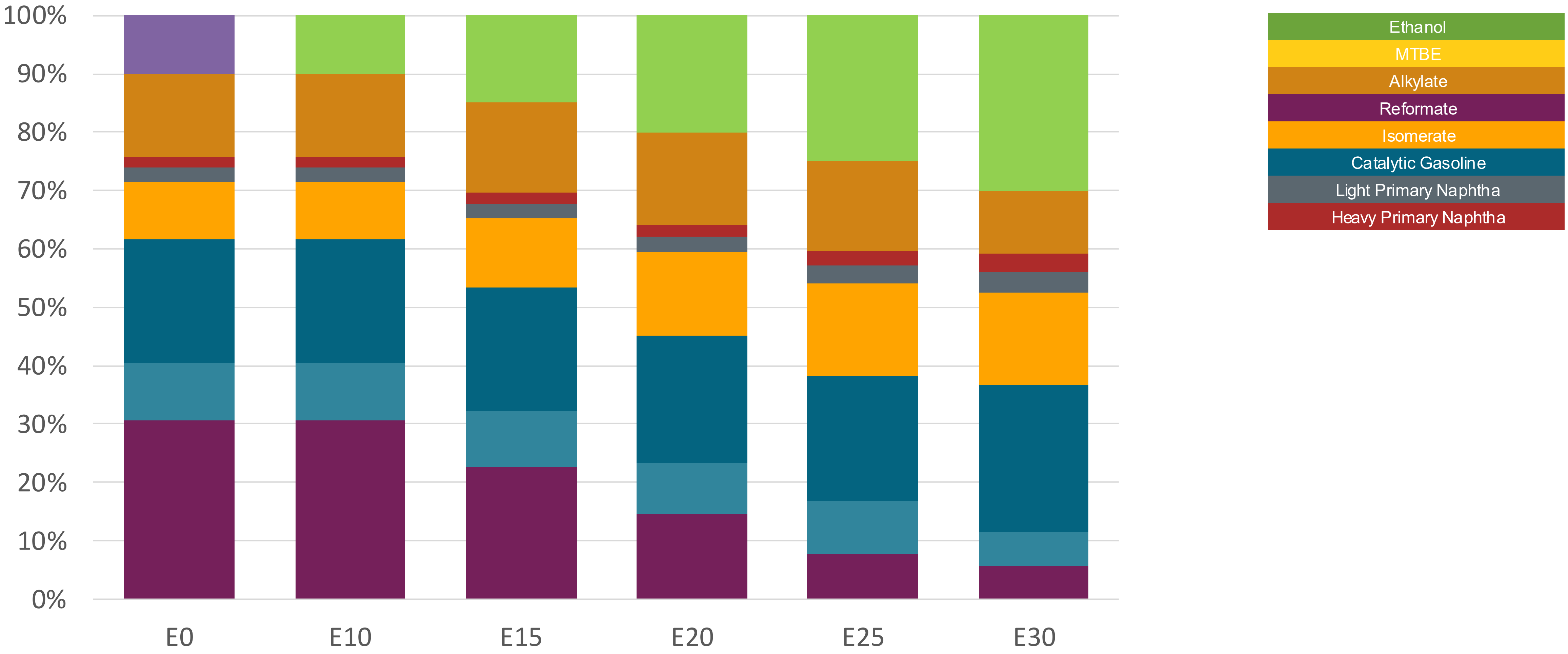
Chile – G93 RM Aconcagua – Constant Octane



Octane (RON)	93	93	93	93	93	93
Price (US$/gal)	$2.601	$2.519	$2.430	$2.341	$2.249	$2.160

Source: Faro90, 2025

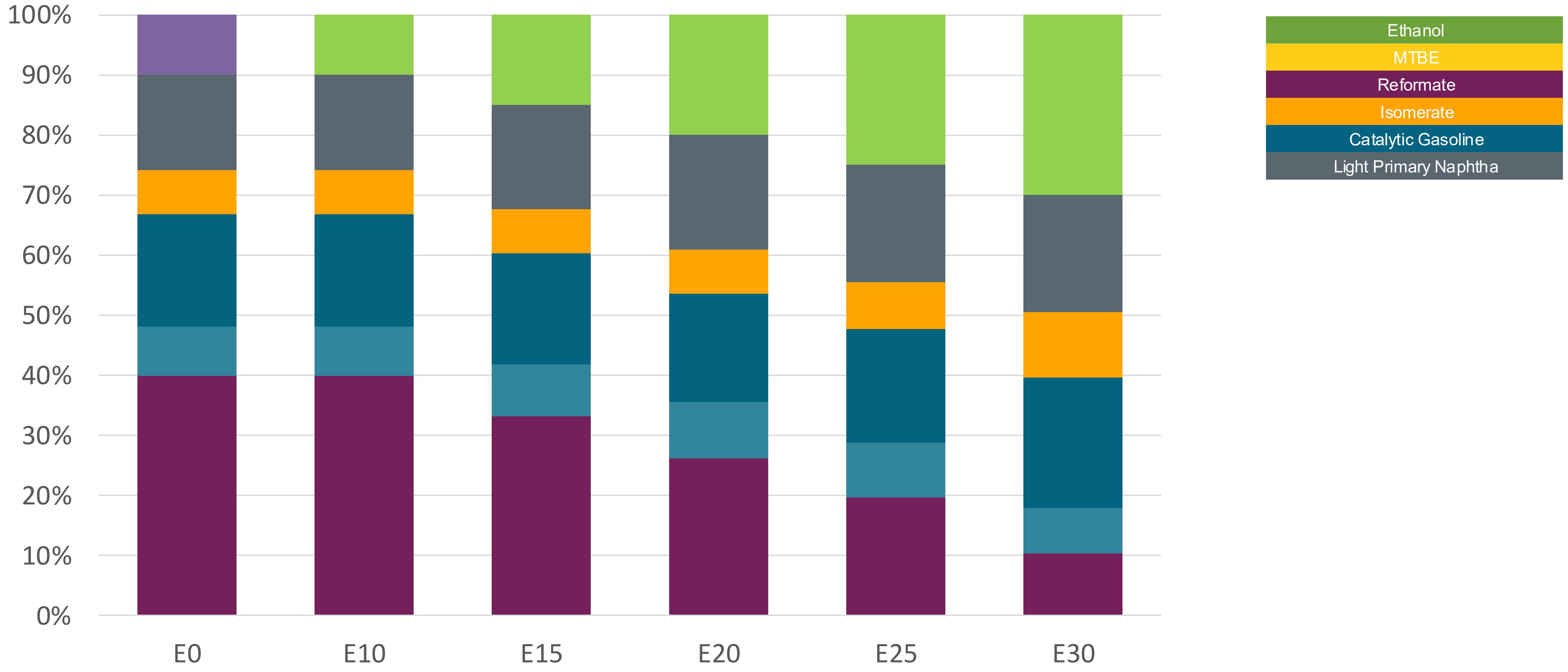
Chile – G97 RM Aconcagua – Constant Octane



Octane (RON)	97	97	97	97	97	97
Price (US$/gal)	$2.71	$2.63	$2.55	$2.47	$2.39	$2.32

Source: Faro90, 2025

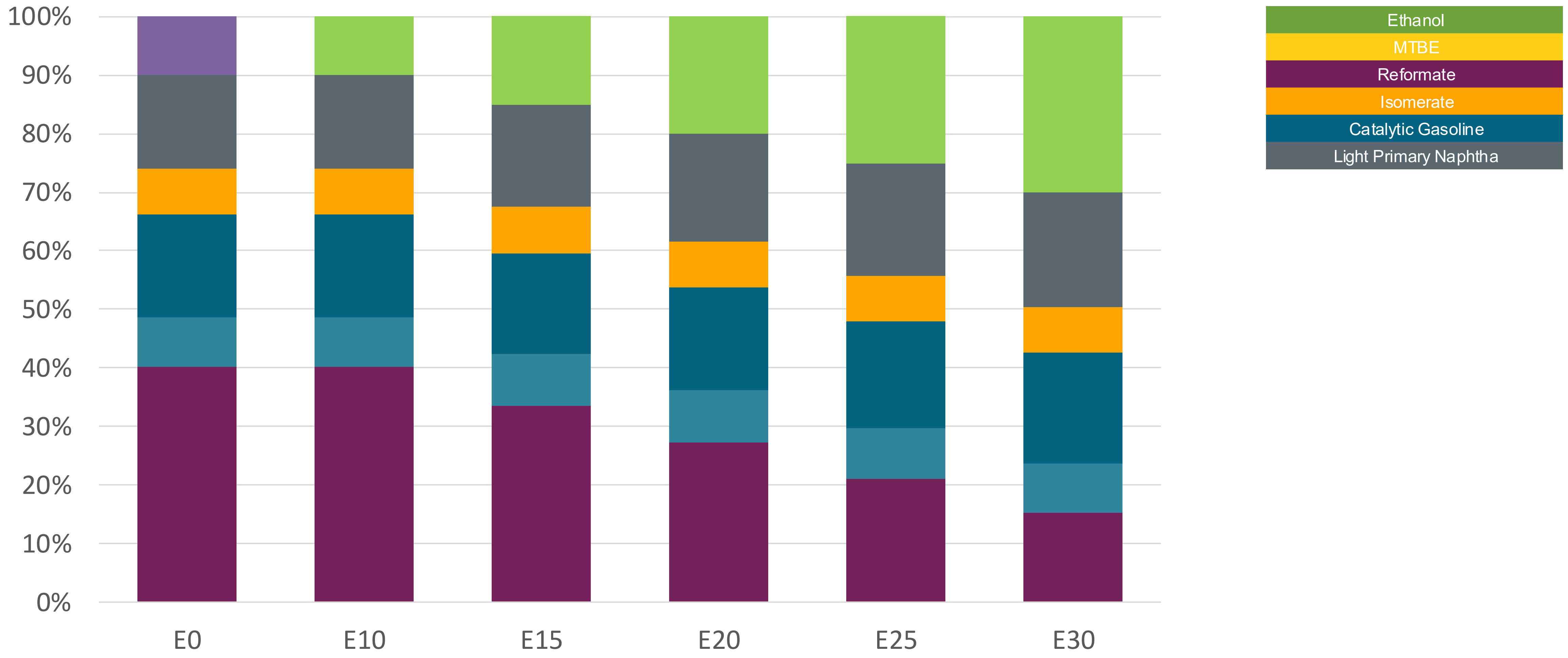
Chile – G93 RP Biobío – Constant Octane



Octane (RON)	93.0	93.0	93.0	93.0	93.0	93.0
Price (US$/gal)	$ 2.46	$ 2.38	$ 2.29	$ 2.21	$ 2.14	$ 2.06

Source: Faro90, 2025

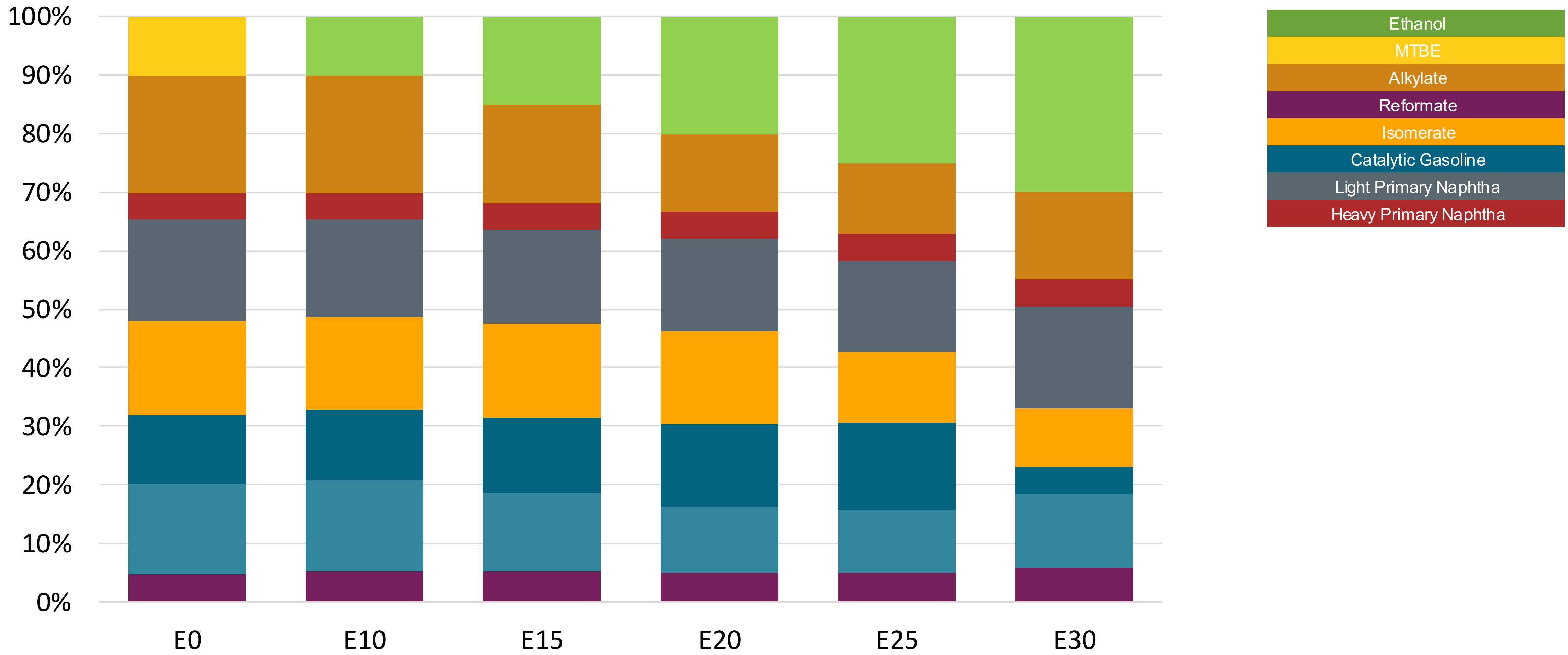
Chile – G97 RP Biobío – Constant Octane



Octane (RON)	97.0	97.0	97.0	97.0	97.0	97.0
Price (US$/gal)	$ 2.56	$ 2.47	$ 2.39	$ 2.31	$ 2.24	$ 2.17

Source: Faro90, 2025

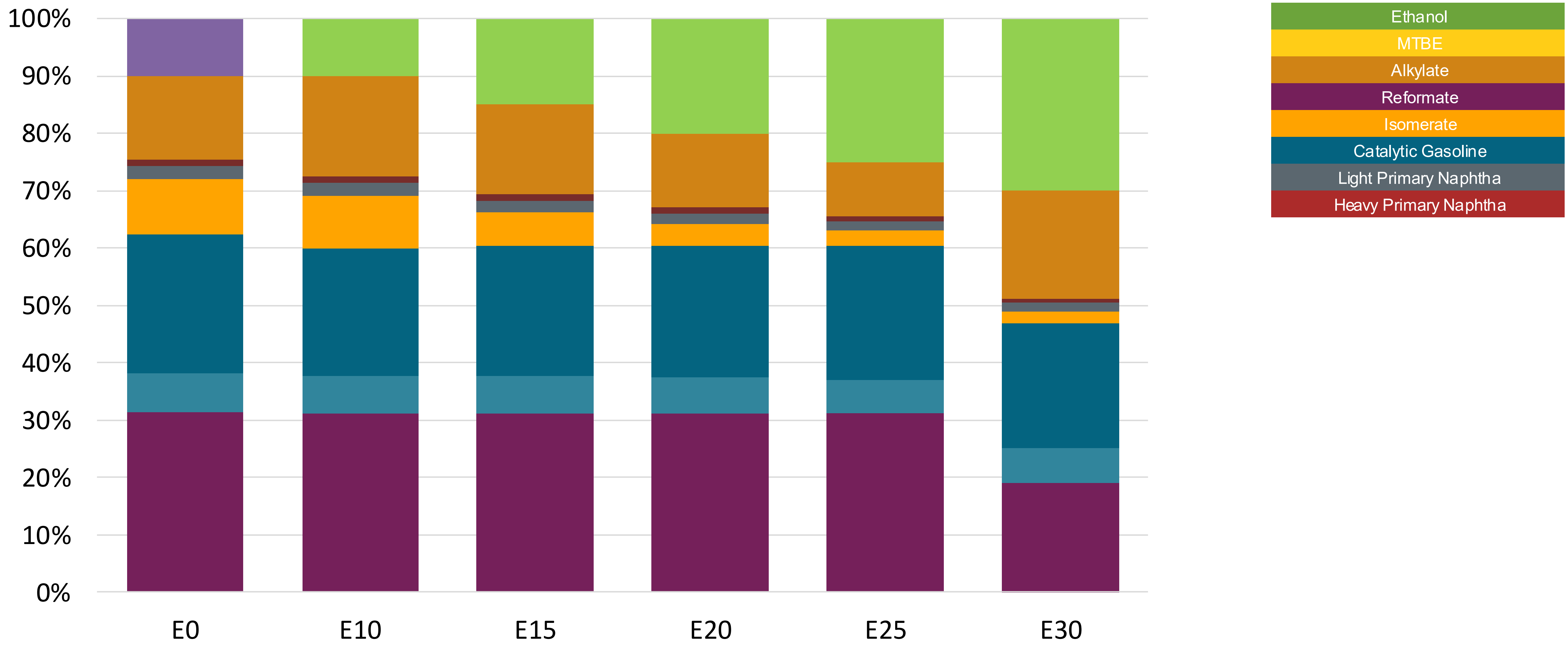
Chile – G93 RP Aconcagua – Octane Increment



Octane (RON)	93.0	95.1	97.5	100.1	102.0	104.3
Price (US$/gal)	$ 2.40	$ 2.40	$ 2.40	$ 2.40	$ 2.40	$ 2.40

Source: Faro90, 2025

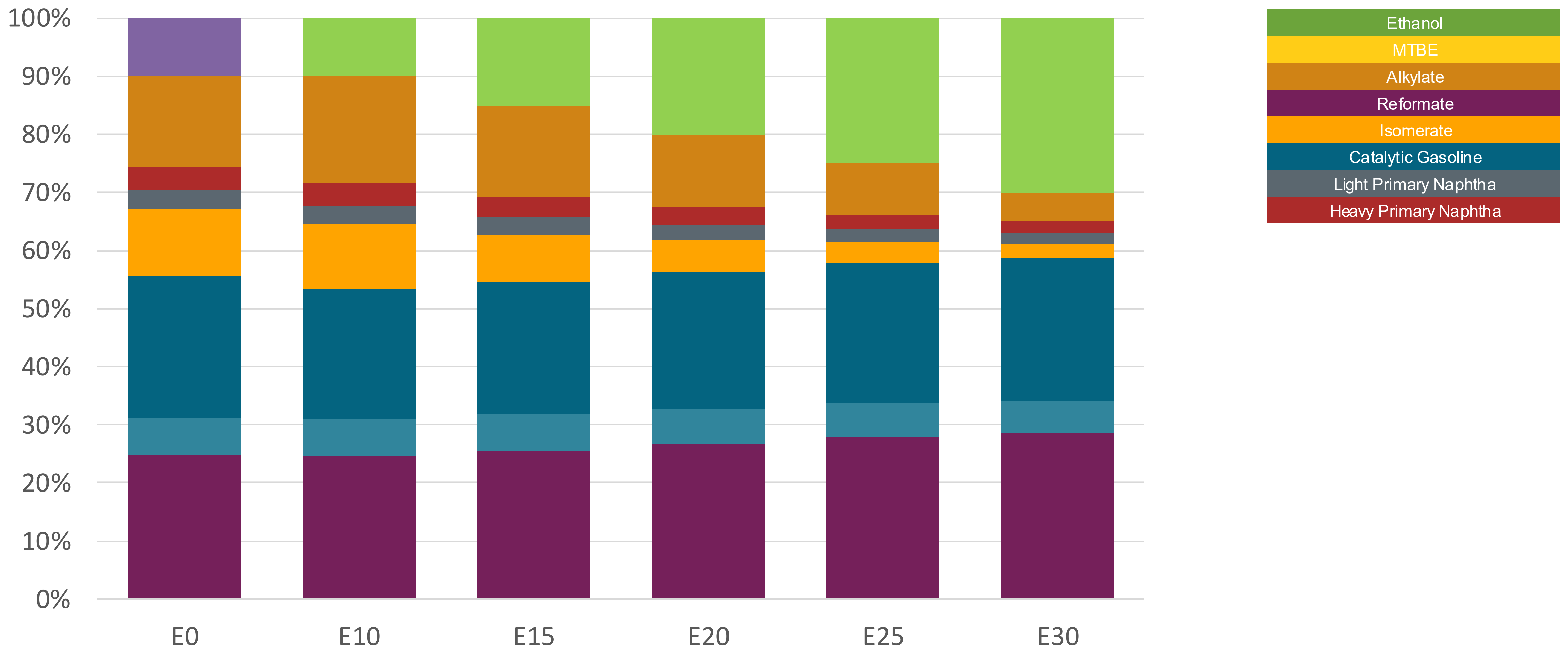
Chile – G97 RP Aconcagua – Octane Increment



Octane (RON)	97.0	100.1	102.1	104.3	106.9	109.4
Price (US$/gal)	$2.73	$2.73	$2.73	$2.73	$2.73	$2.73

Source: Faro90, 2025

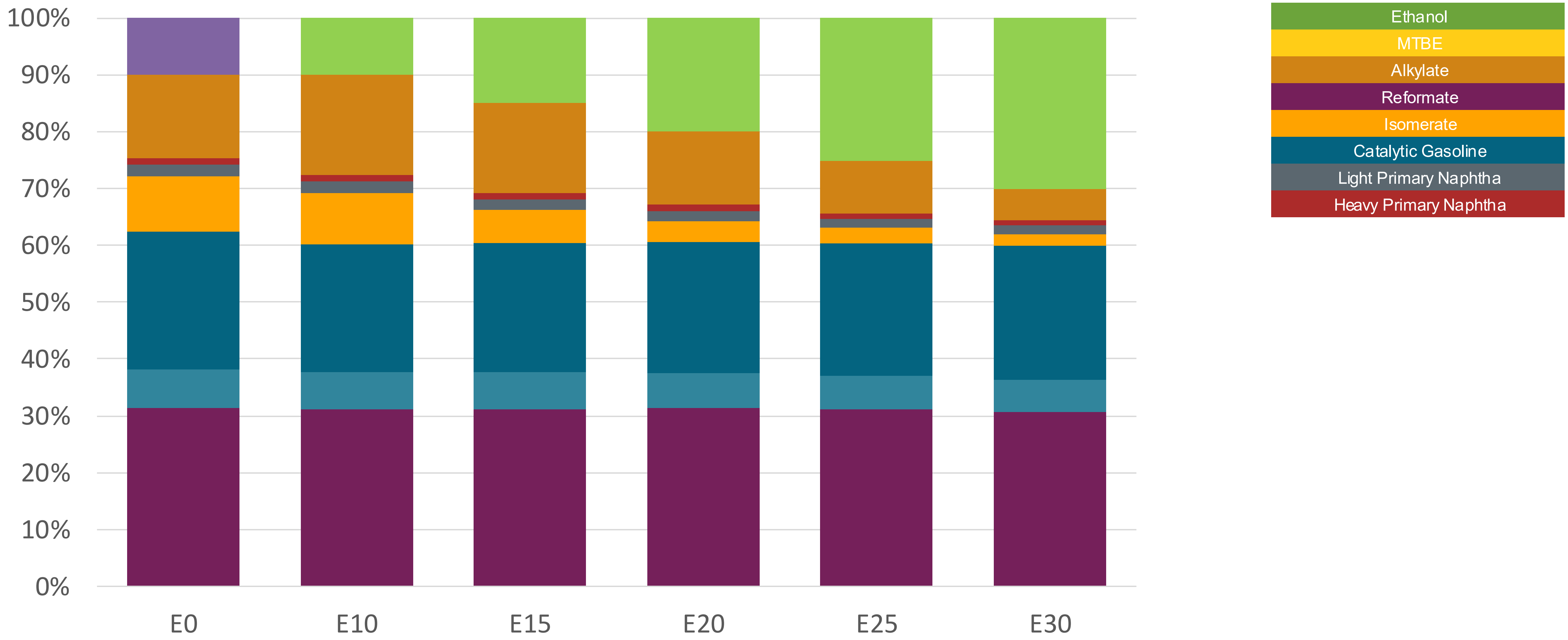
Chile – G93 RM Aconcagua – Octane Increment



Octane (RON)	93.0	96.2	98.1	100.0	102.1	104.3
Price (US$/gal)	$ 2.601	$ 2.601	$ 2.601	$ 2.601	$ 2.601	$ 2.601

Source: Faro90, 2025

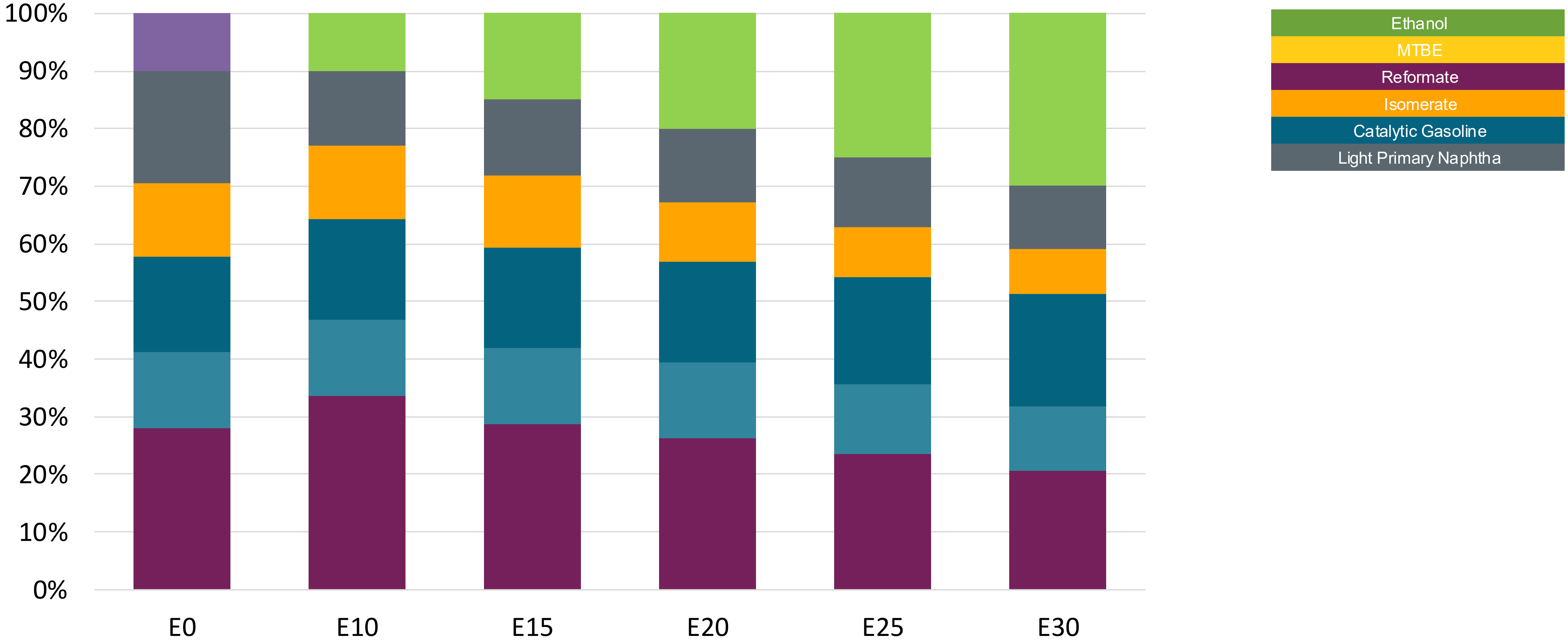
Chile – G97 RM Aconcagua – Octane Increment



	E0	E10	E15	E20	E25	E30
Octane (RON)	97.0	100.1	102.1	104.3	106.9	109.5
Price (US$/gal)	$ 2.73	$ 2.73	$ 2.73	$ 2.73	$ 2.73	$ 2.73

Source: Faro90, 2025

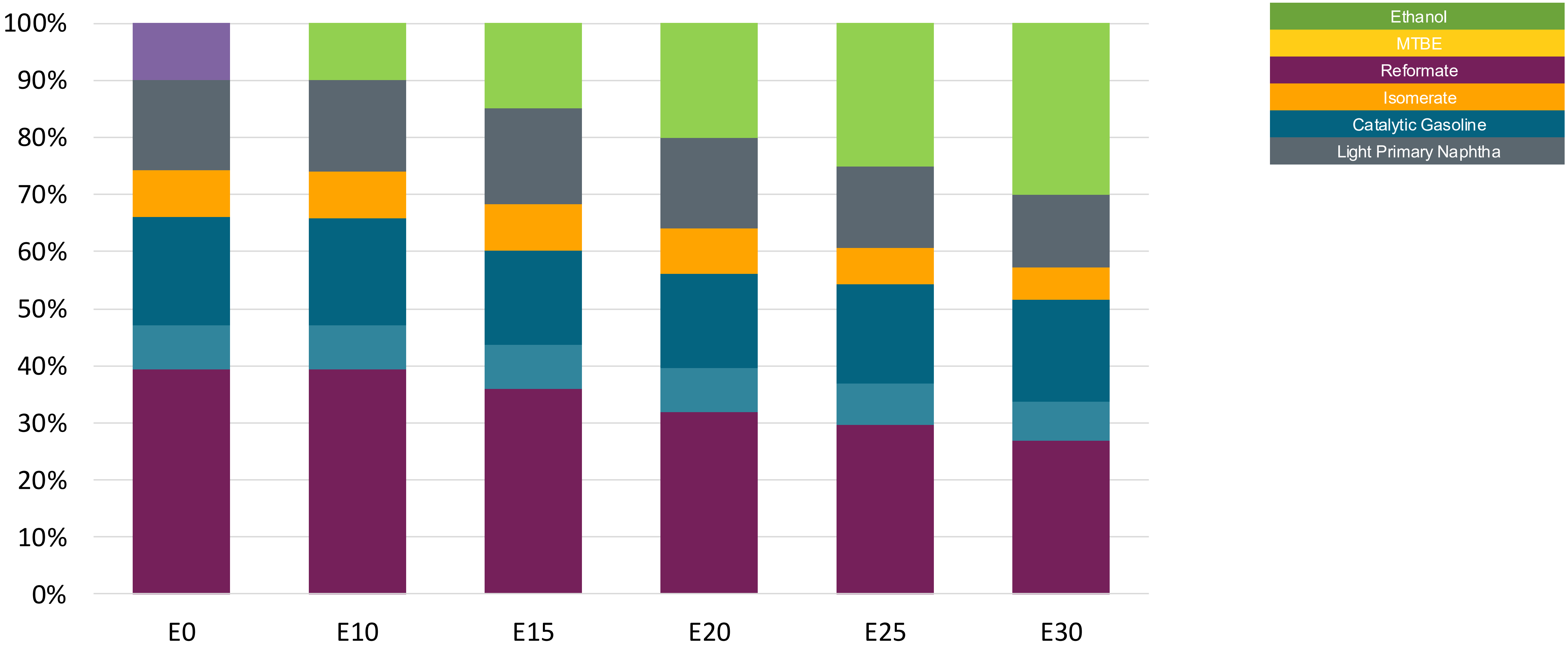
Chile – G93 RP Biobio – Octane Increment



	E0	E10	E15	E20	E25	E30
Octane (RON)	93.0	93.0	95.9	98.2	100.4	102.7
Price (US$/gal)	$ 2.33	$ 2.33	$ 2.33	$ 2.33	$ 2.33	$ 2.33

Source: Faro90, 2025

Chile – G97 RP Biobío – Octane Increment



Octane (RON)	97.0	101.0	103.7	107.1	108.9	111.1
Price (US$/gal)	$ 2.558	$ 2.558	$ 2.558	$ 2.558	$ 2.558	$ 2.558

Source: Faro90, 2025

Chile – Vehicular Fleet

Tipo	Peso	Motor	Escape	Recuperator	Age	Index	Number of Vehicles
Auto/Sml Truck	Light	Carburator	3-Ways	PCV Tank	+8años	29	42,608
		Single FI	3-Ways		+8años	83	298,823
		Multi FI	3-Ways		+8años	119	1,120,144
			Euro III		+8años	191	690,653
			Euro IV		+8años	200	787,336
			Euro V		+8años	1253	464,935
					4-8años	1252	803,943
			Euro VI		-4años	1251	415,697
		Hybrid	Hybrid		4-8años	208	898
					-4años	207	3,650
		Electric					1,340
	Medium	Carburator	3-Ways	PCV Tank	+8años	32	246,934
		Single FI	3-Ways		+8años	86	337,941
		Multi FI	3-Ways		+8años	122	393,950
			Euro III		+8años	194	167,702
			Euro IV		+8años	203	291,291
			Euro V		+8años	1256	89,733
					4-8años	1255	263,841
			Euro VI		-4años	1254	236,646
		Hybrid	Hybrid		4-8años	211	0
					-4años	210	495
		Electric					415
Sml Engines	Ligeros	4-Cycle	Euro II	PCV	+8años	1235	33,031
					+8 años	1234	999
			Euro III		+8 años	1243	145,315
					-4 años	1242	
			Euro IV		4-8años		21,990
			Euro V		4-8años		85,681
		Multi FI	Hibrido				0
		Electric	-		-8años		258
Truck	Pesados	Carburator	3-Ways	PCV	+8años	863	105,504
		Fuel-Injection	3-Ways		+8años	908	177,934
			Euro III		+8años	944	35,057
			Euro IV		+8años	953	45,369
			Euro V		+8años	962	25,786
					4-8años	961	12,630
			Euro VI		-4años	960	31,201

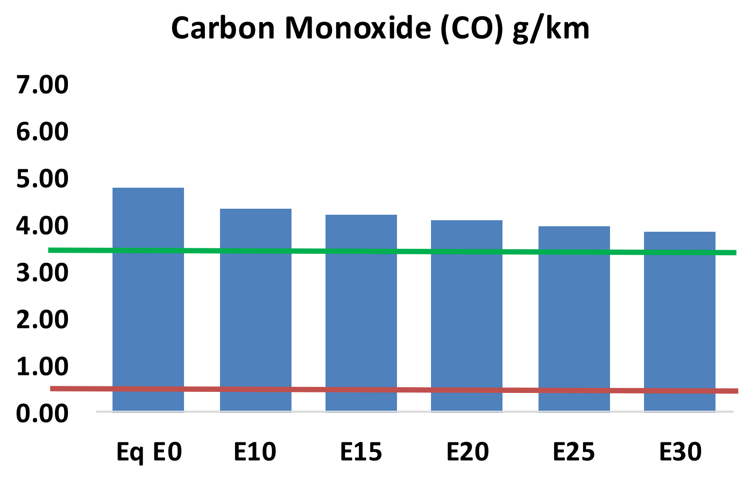
Vehicular Fleet: **7,379,730 vehicles that use gasoline**
Average Age: **10 year**

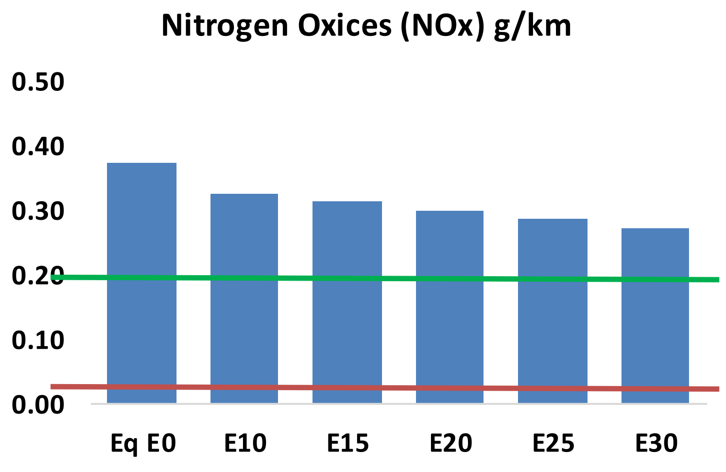
Main Technology: **Motor Multi-Fuel-Injection, Euro V, PCV Tank**

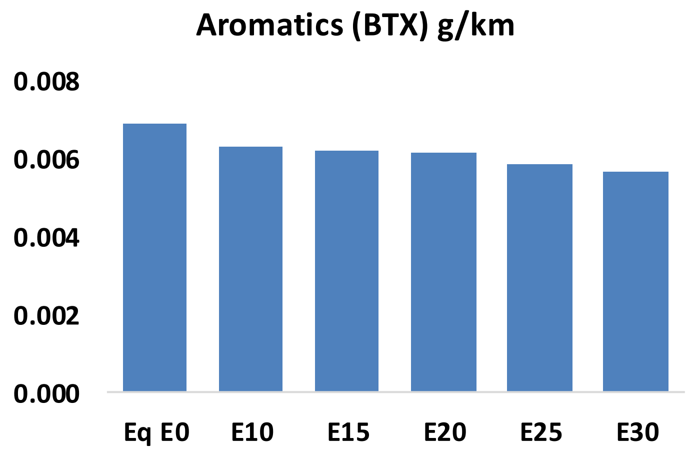
Source: INE, 2024

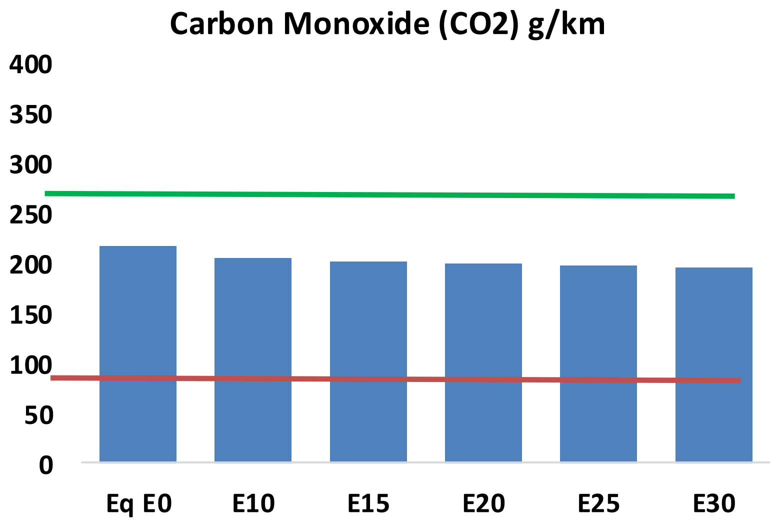
Chile – Gasoline Vehicular Fleet Emissions

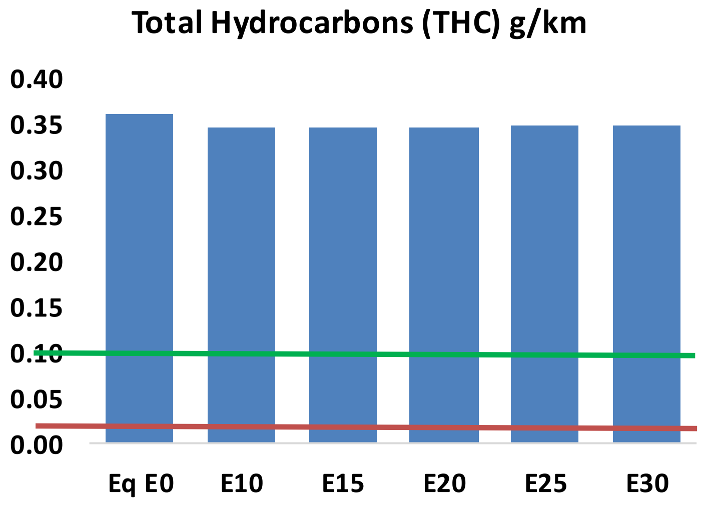
United States
Euro 6

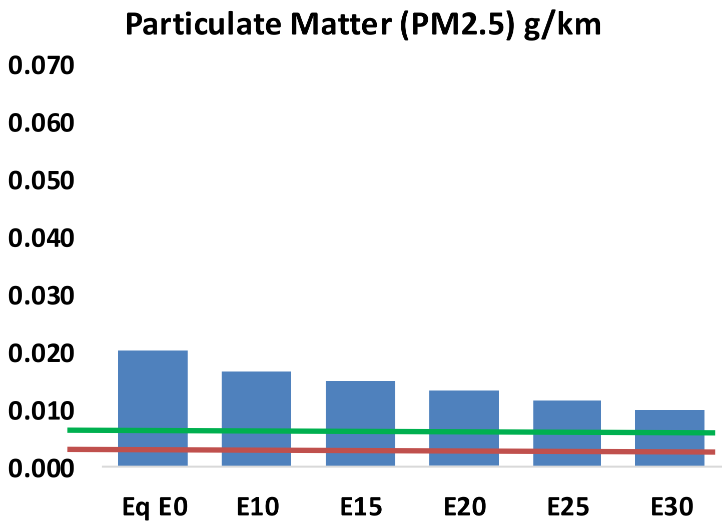












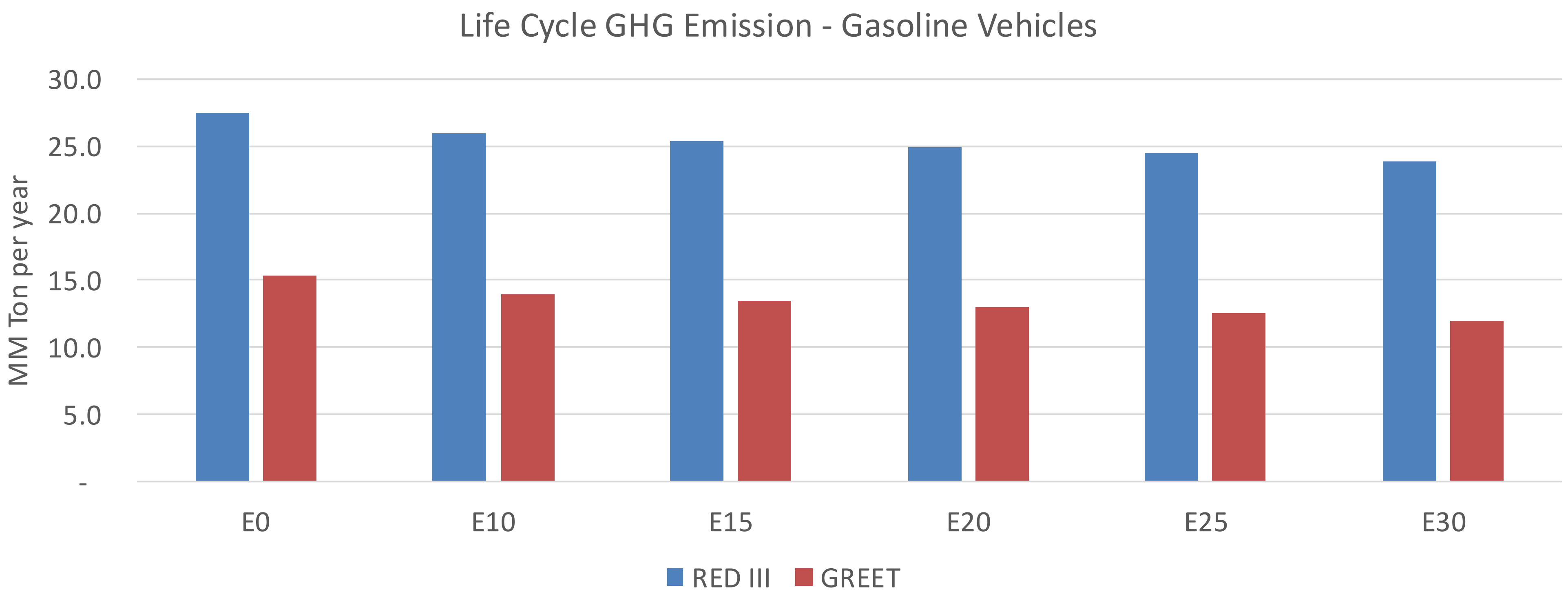
Source: Faro 90

Chile – Gasoline Vehicular Fleet Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	487,387	465,588	443,789	428,983	415,490	405,540	391,301	-9%	-15%
CO2	22,058,884	21,491,656	20,955,940	20,534,616	20,326,128	20,205,060	19,832,623	-5%	-8%
VOC	36,645	35,990	35,335	35,221	35,299	35,418	35,371	-4%	-4%
NOx	37,886	34,981	32,987	31,971	30,335	29,052	27,542	-13%	-20%
SOx	256	236	217	201	185	168	150	-15%	-28%
NH3	7,199	7,128	7,057	7,090	7,170	7,226	7,273	-2%	0%
N2O	390	388	386	388	396	400	405	-1%	2%
CH4	8,281	8,281	8,281	8,405	8,587	8,745	8,894	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	447	440	433	433	435	437	438	-3%	-3%
Acetaldehyde	800	918	1,346	2,050	2,794	3,192	3,772	68%	249%
Formaldehyde	2,296	2,232	2,602	3,056	3,201	3,541	3,855	13%	39%
Benzene	703	676	640	630	625	598	578	-9%	-11%
PM2.5	1,272	1,156	1,041	939	839	733	621	-18%	-34%
PM10	770	700	630	568	508	443	376	-18%	-34%

Source: Faro 90

Chile – Gasoline Vehicle Fleet GEI Emissions



Country Emissions 2024 (MMT/year)	118.1
2035 Target (MMT/year)	90.1
Delta	28.0

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	8.9%	12.2%	15.3%	18.1%	21.8%
RED II/III	0.0%	5.5%	7.5%	9.4%	11.1%	13.3%

% 2035 Target	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	4.9%	6.6%	8.4%	9.9%	11.9%
RED II/III	0.0%	5.4%	7.3%	9.2%	10.9%	13.0%

Source: Greet, RED II/III, climateactiontracker.org, F90

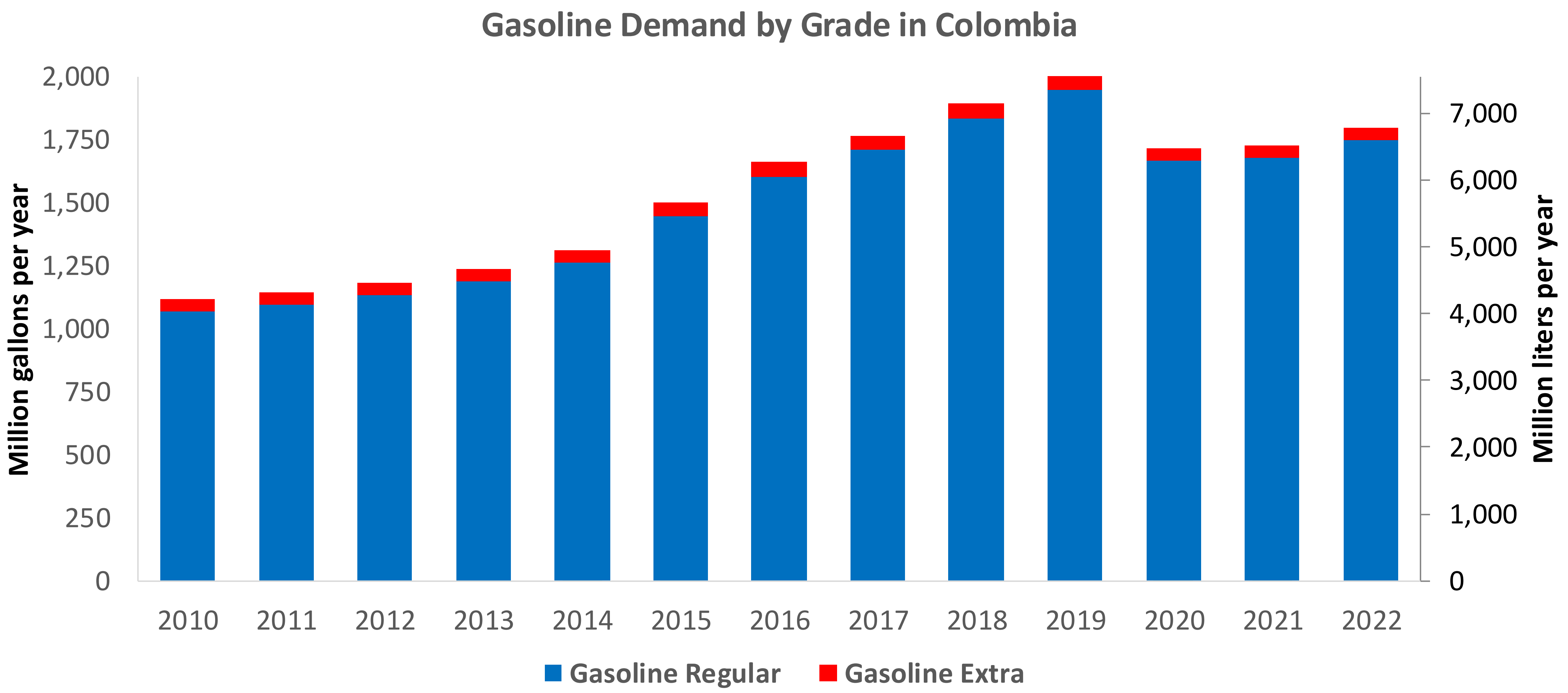
Ethanol Blending in Ethanol - Colombia

In 2024, gasoline consumption in Colombia was 2,023 million gallons (7,658 million liters). Market share was 3% for Extra gasoline E10 (RON 97 – AKI 94) and 97% for Regular gasoline (regular) E10 (RON 89 – AKI 84). Local production supplies 65% of total demand. United States is the main source of gasoline imports.

Ethanol blends are obligatory in cities with more than 500,000 habitants. Resolution 40444 of 2023 authorizes the blending of up to 10% ethanol in the country and allows for lower percentages in particular cases. E10 is the predominant grade in the country except on the border with Venezuela. The ethanol supply comes from imports from the United States and from local production.

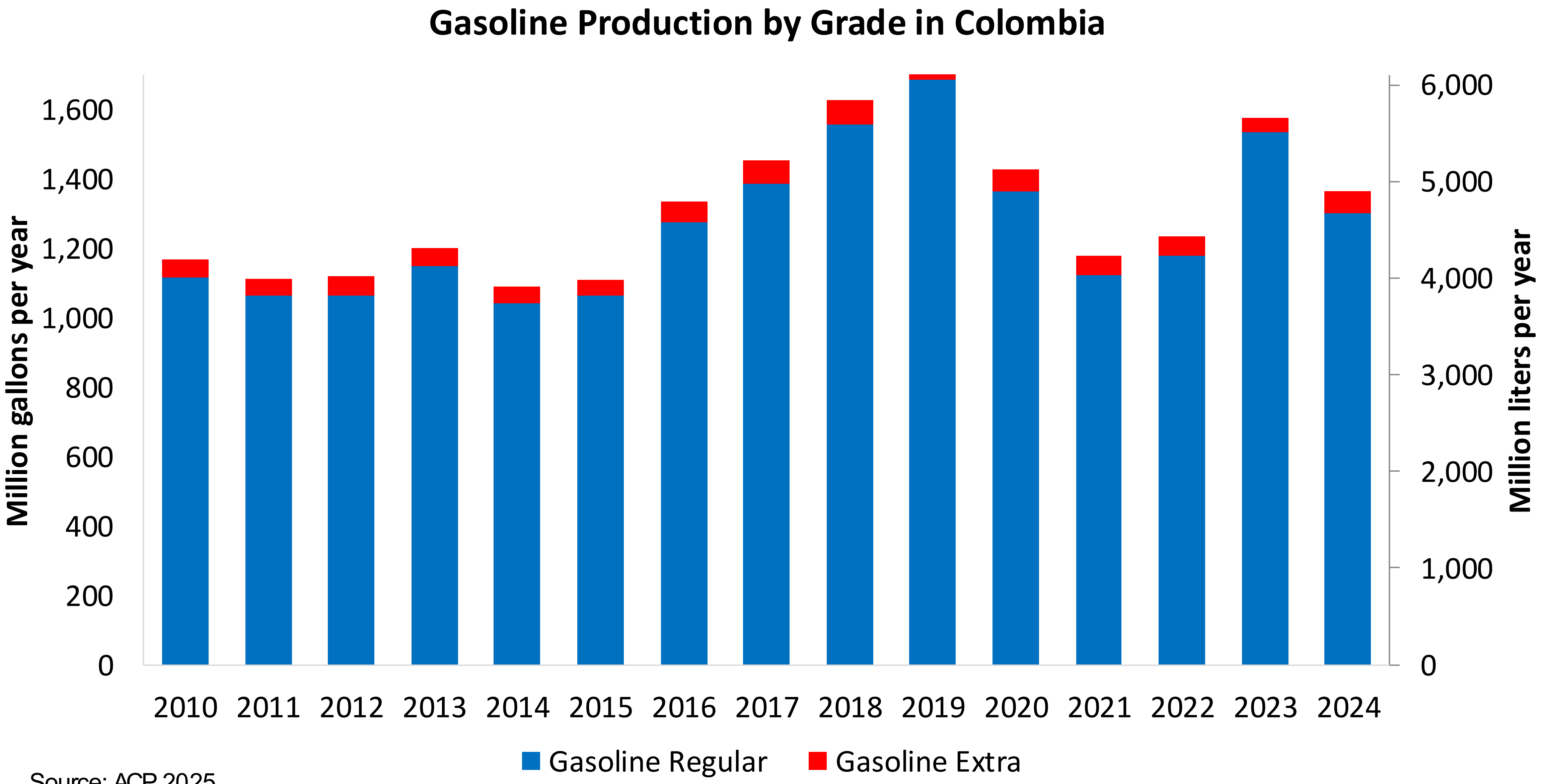
Source: ACP, 2024

Gasoline Demand in Colombia



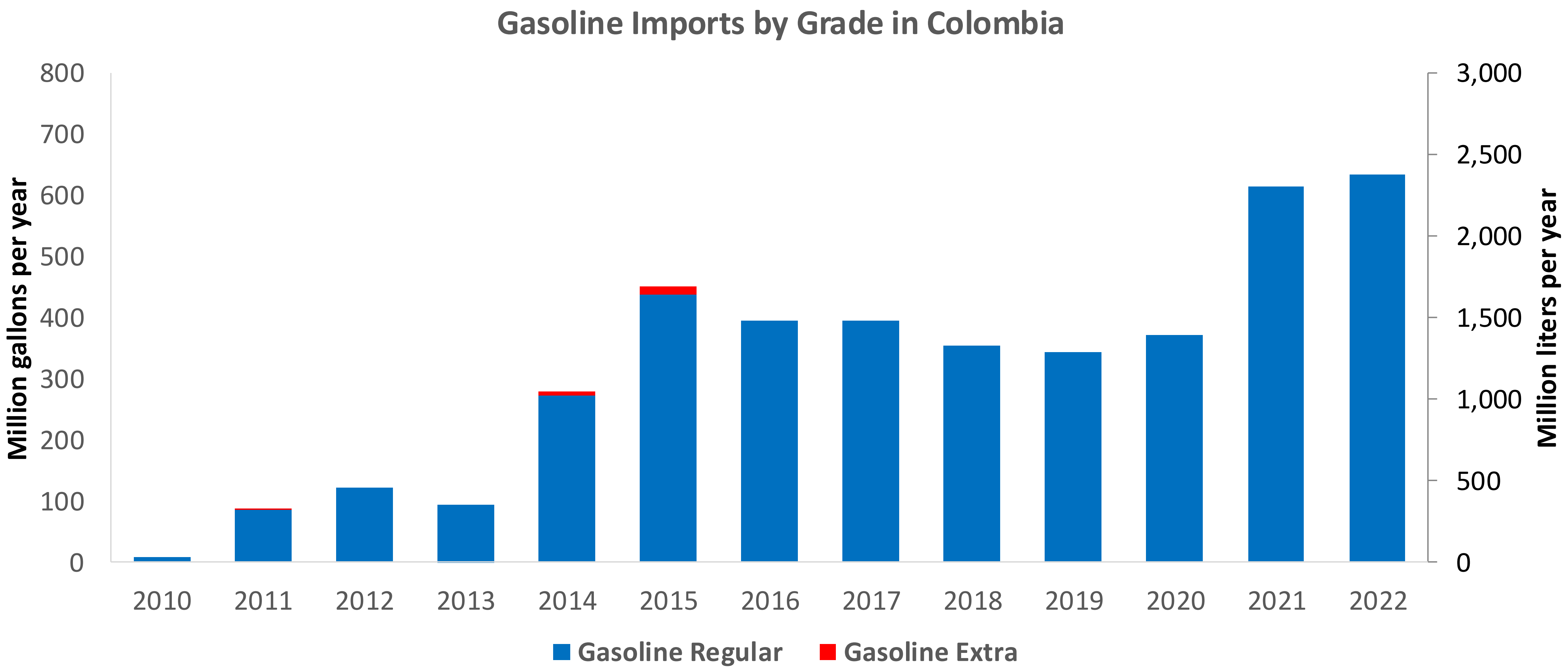
Source: ACP, 2025

Gasoline Production in Colombia



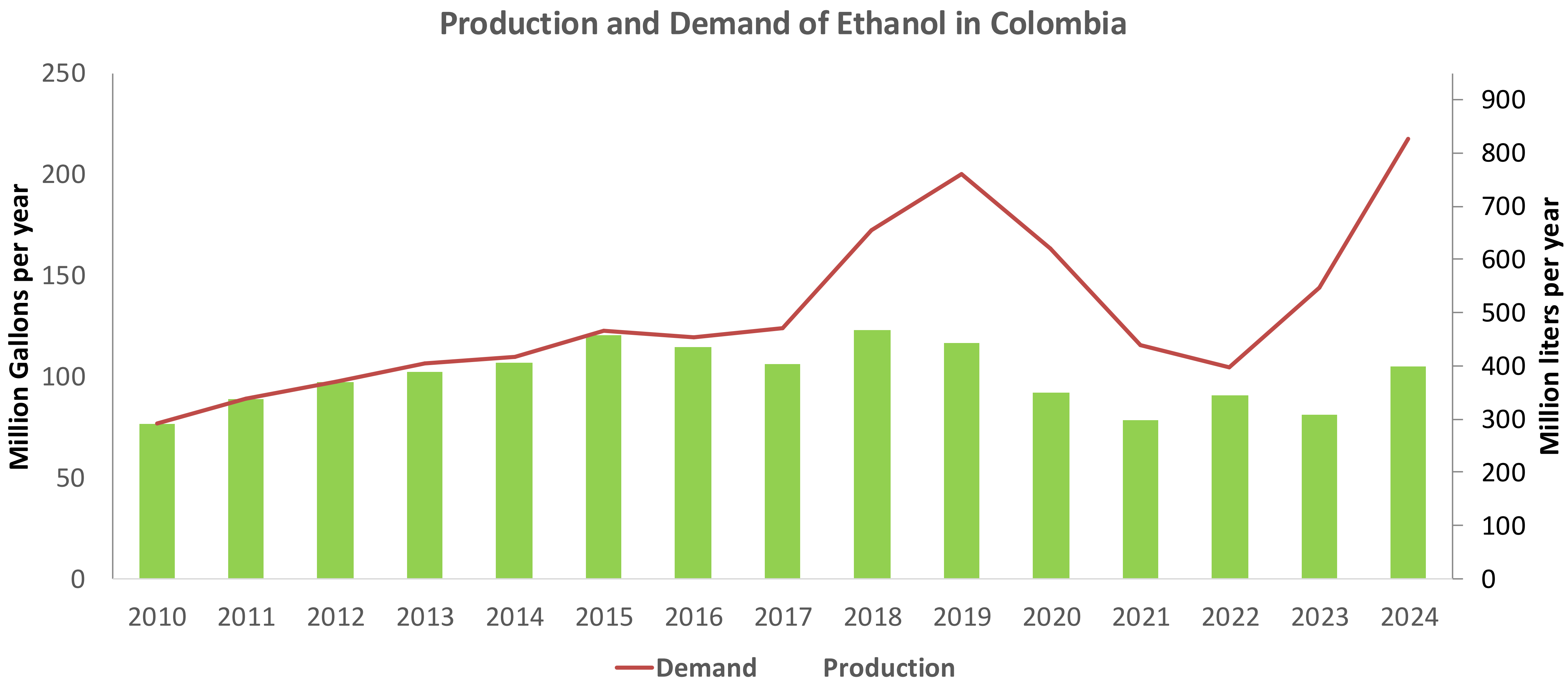
Source: ACP, 2025

Gasoline Imports in Colombia



Source: ACP, 2025

Ethanol Balance in Colombia



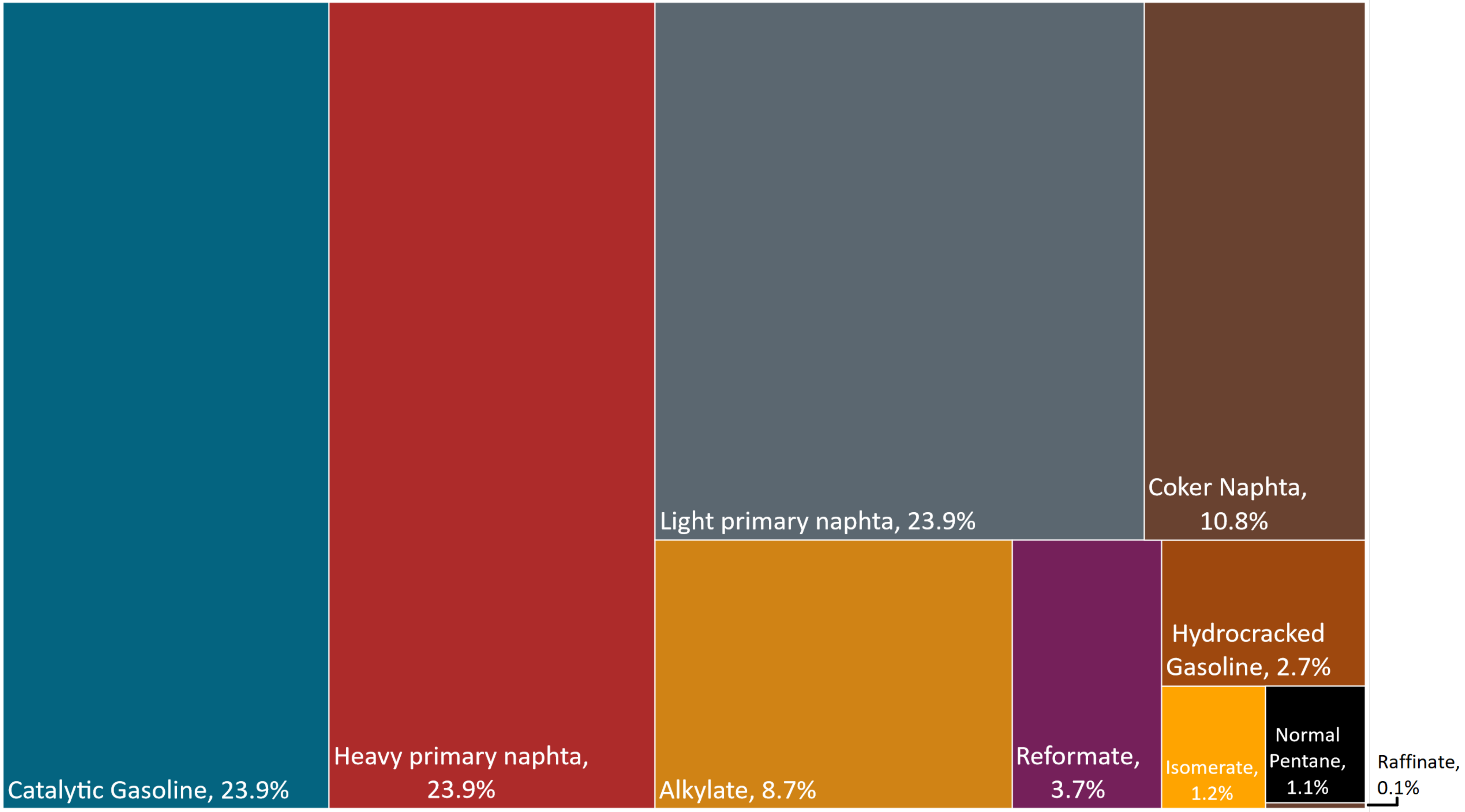
Source: Asocaña - Balance Azucarero Colombiano, 2024; EIA, 2024

Gasoline Quality in Colombia

Name	Resolution 40103 de 2021				EN 228:2012 + A1:2017 (Euro 6 enabling)			
Implementation Date	07/01/2023				2017			
Applicability	Whole country	Whole country	Whole country	Whole country	All countries			
Selected Grade	Gasoline Regular	Gasoline Extra	Gasoline Regular E10	Gasoline Extra E10	RON 95 E5	RON 95 E10	RON 98 E5	RON 98 E10
Benzene Content	< 1,0 %v/v	< 2,0 %v/v	< 0,9 %v/v	< 1,8 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v
Aromatics	< 28 %v/v	< 35 %v/v	< 25 %v/v	< 31,5 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v
Olefins	-	-	-	-	< 18 %v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v
Lead Content	< 0,013 g/l	< 0,013 g/l	< 0,013 g/l	< 0,013 g/l	< 5 mg/l	< 5 mg/l	< 5 mg/l	< 5 mg/l
Manganese	-	-	-	-	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l
RON	> 84	> 93	> 89	> 97	> 95	> 95	> 98	> 98
MON	-	-	-	-	> 85	> 88	> 85	> 88
AKI	> 81	> 91	> 84	> 94				
Sulfur Content	< 50 mg/kg	< 50 mg/kg	< 50 mg/kg	< 50 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg
Oxygen Content			< 3,7 %m/m	< 3,7 %m/m	<2,7 % m/m	<3,7 % m/m	<2,7 % m/m	<3,7 % m/m
Ethanol (EtOH)	<> 9,5 - 10,5 %v/v	<> 9,5 - 10,5 %v/v	<> 9,5 - 10,5 %v/v	<> 9,5 - 10,5 %v/v	<5 %v/v	<10 %v/v	<5 %v/v	<10 %v/v
RVP 37.8°C (Summer)	<> 65 kPa	<> 65 kPa	<> 65 kPa	<> 65 kPa	<> 60 - 70 kPa *Depends on the country, RVP is regulated in the EU Fuel Quality Directive			
RVP 37.8 °C(Winter)								
RVP 37.8°C (Transition)								
MTBE					-	-	-	-
Ethers 5 or more C Atoms	-	-	-	-	Based on oxygen content	<22 %v/v	Based on oxygen content	<22 %v/v

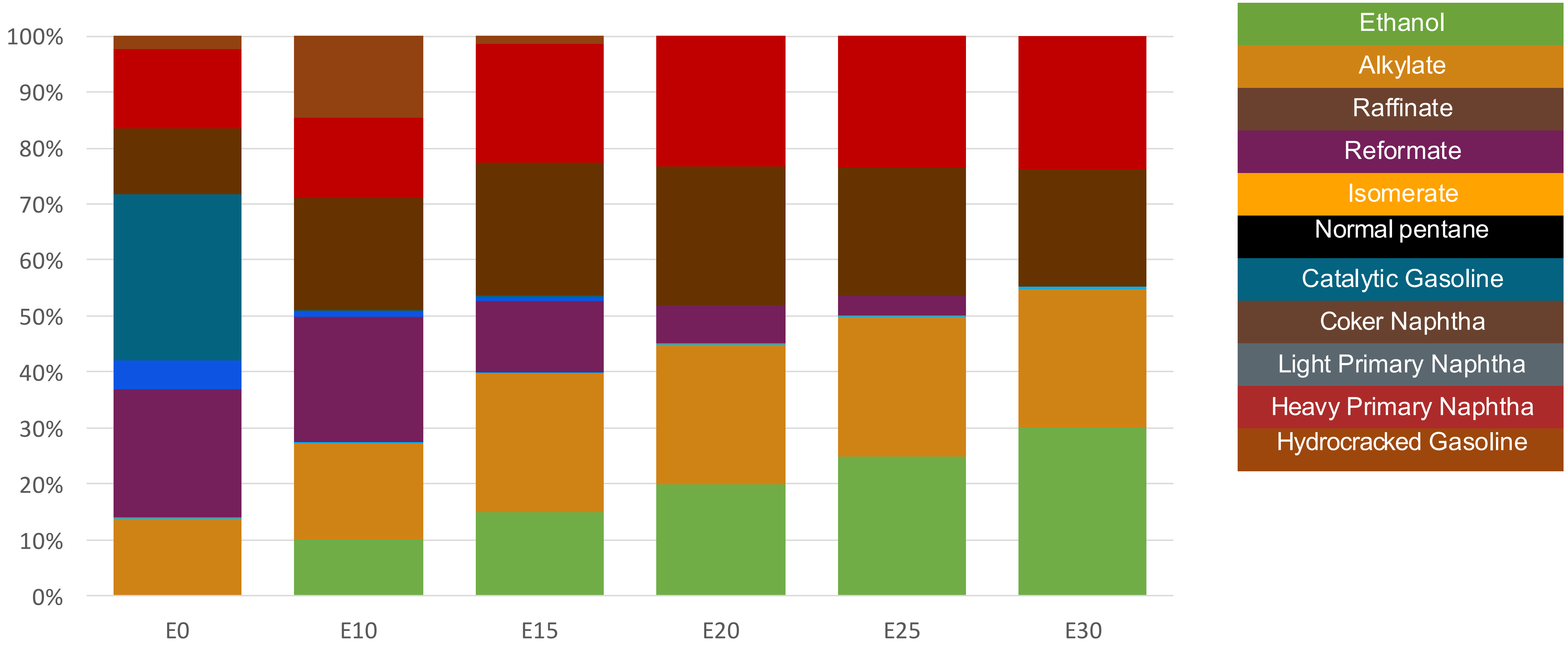
Source: Ministerio de Energía y Minas, 2021

Available Blending Components



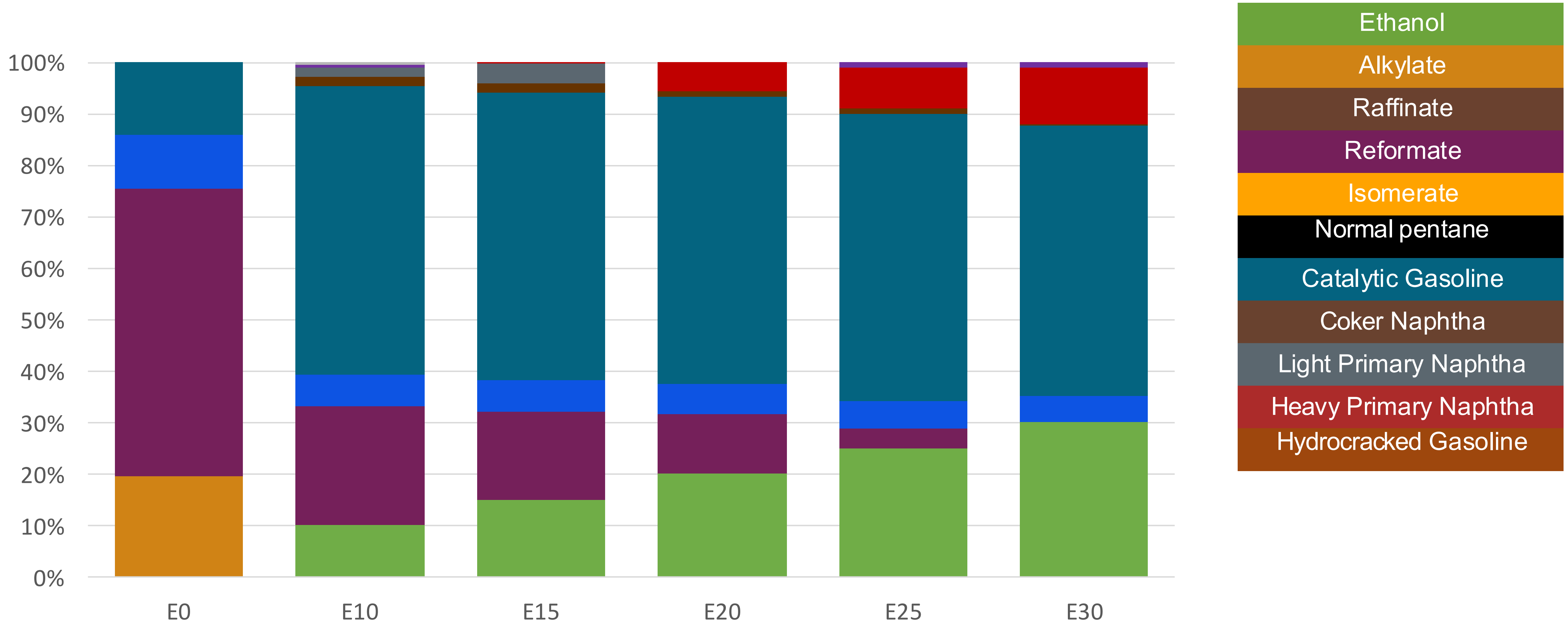
Source: Faro90

Colombia – Regular Gasoline – Constant Octane



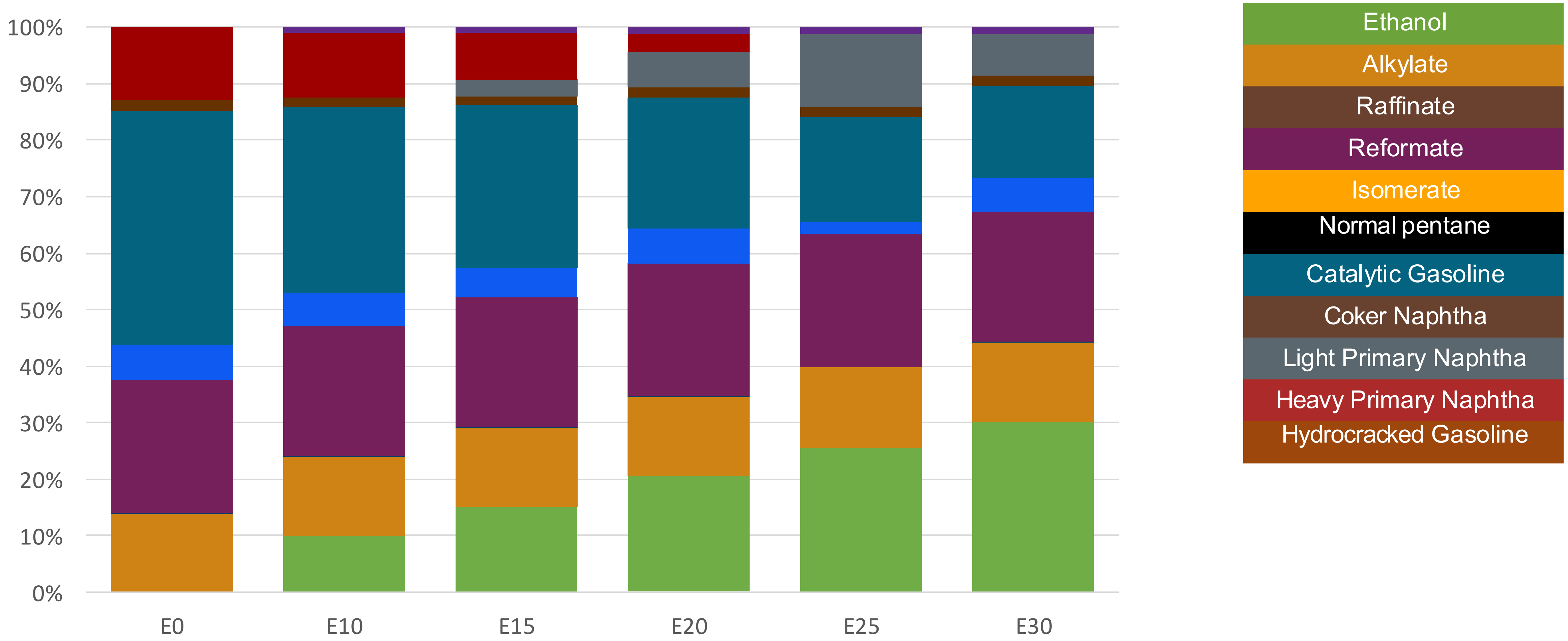
Octane (RON)	89.0	89.0	89.0	89.3	90.3	91.3
Price (US$/gal)	2.260	2.133	2.044	1.971	1.935	1.900

Colombia – Extra Gasoline – Constant Octane



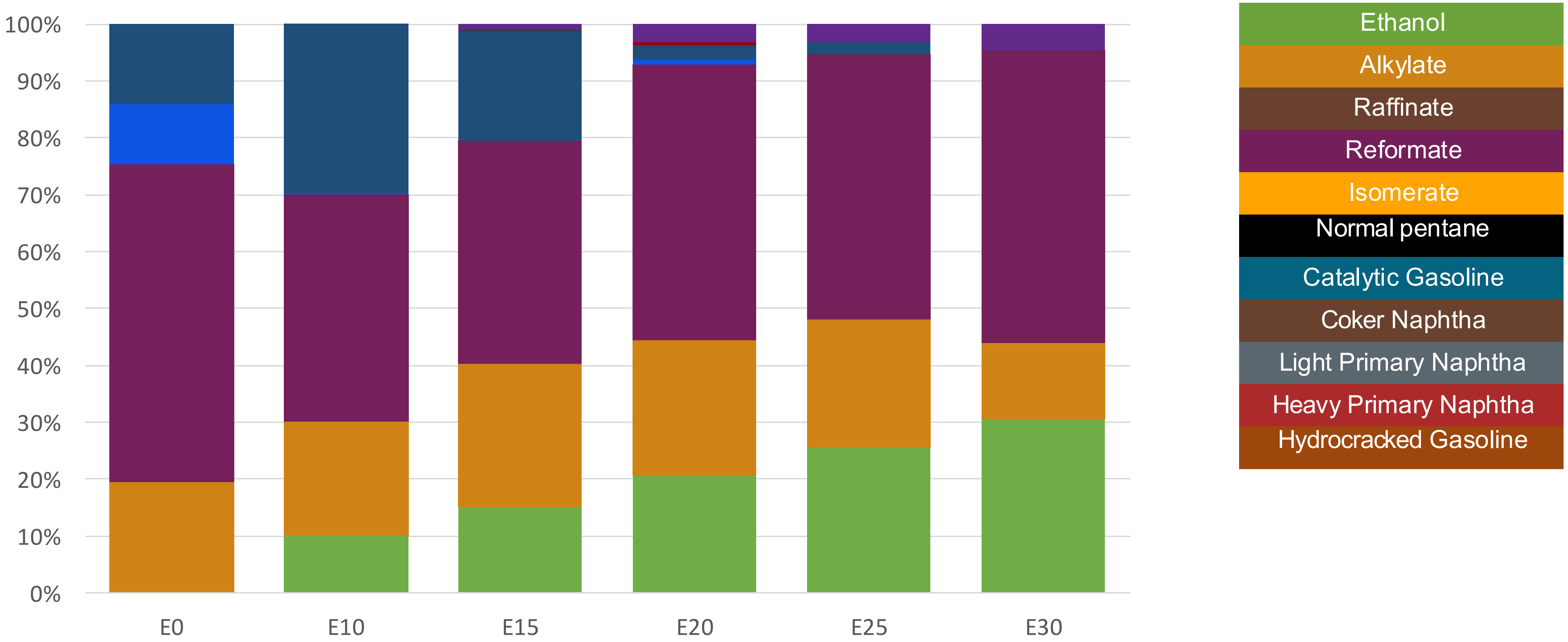
Octane (RON)	97.0	97.0	97.0	97.0	97.0	97.0
Price (US$/gal)	2.660	2.451	2.352	2.264	2.185	2.100

Colombia – Regular Gasoline – Increased Octane



Octane (RON)	90.1	93.5	95.2	97.2	97.9	100.7
Price (US$/gal)	2.328	2.328	2.328	2.328	2.328	2.328

Colombia – Extra Gasoline – Increased Octane



Octane (RON)	97.0	98.2	99.6	101.4	103.0	105.1
Price (US$/gal)	$ 2.660	$ 2.660	$ 2.660	$ 2.660	$ 2.660	$ 2.660

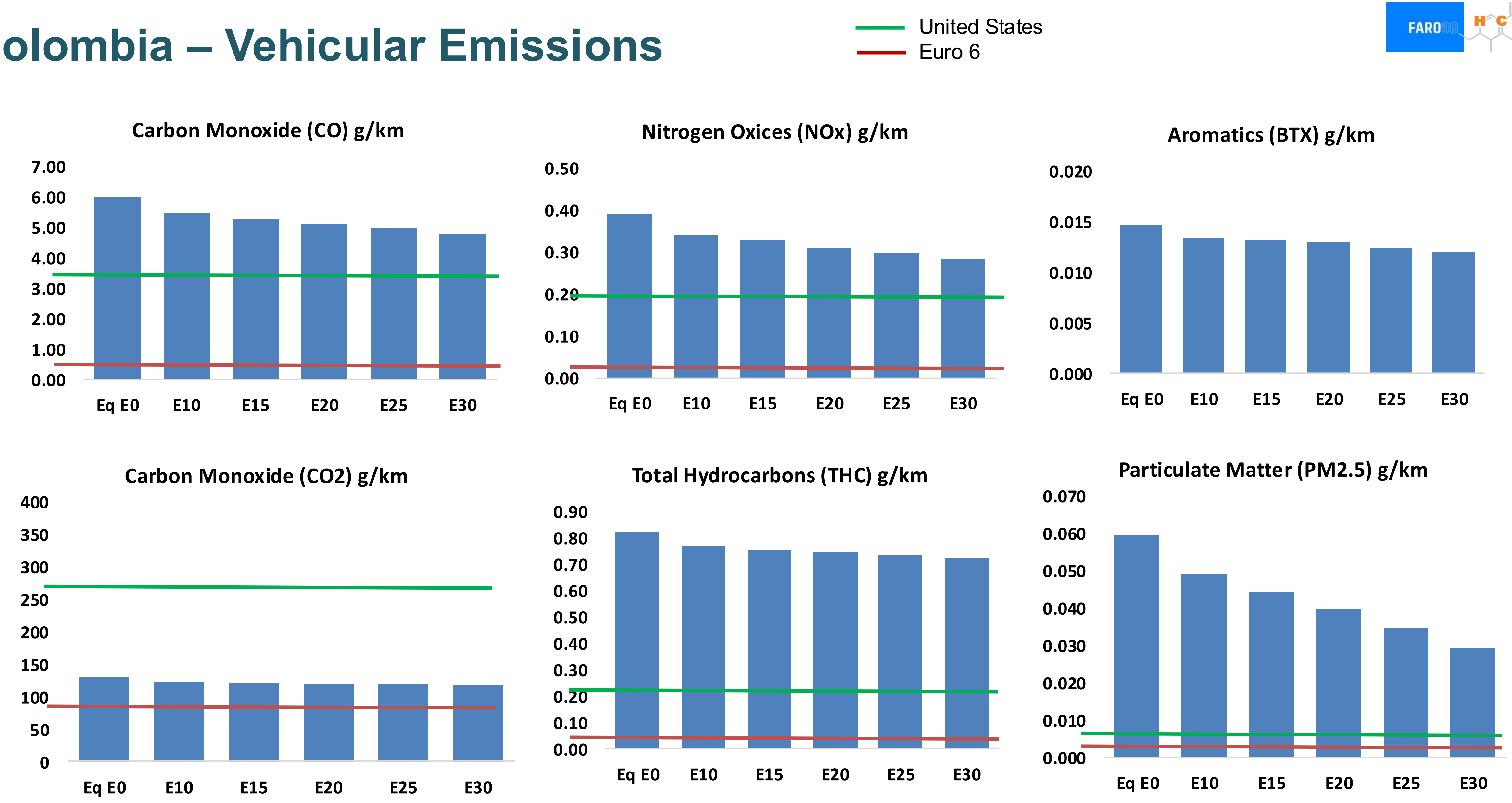
Gasoline Vehicle Fleet - Colombia

Tipo	Motor	Escape	Recuperator	Age	Number of vehicles
Light	Multi FI-Pt	Euro VI	PCV Tank	>4years, >80 mkm	386,511
		Euro V		4-8 years, 80-160 mkm	999,675
				<8 years, < 160 mkm	469,176
		Euro IV		<8 years, < 160 mkm	853,423
		Euro III		<8 years, < 160 mkm	1,011,893
		3-Ways		<8 years, < 160 mkm	2,487,481
Medium	Multi FI-Pt	Euro VI	PCV Tank	>4years, >80 mkm	300,891
		Euro V		4-8 years, 80-160 mkm	305,827
				<8 years, < 160 mkm	200,914
		Euro IV		<8 years, < 160 mkm	223,590
		Euro III		<8 years, < 160 mkm	201,232
		3-Ways		<8 years, < 160 mkm	60,439
Small Engines	FI 4-cycles	Euro V	PCV	>4years, >80 mkm	2,122,342
				4-8 years, 80-160 mkm	1,268,237
		Euro IV		4-8 years, 80-160 mkm	1,665,139
				<8 years, < 160 mkm	574,481
		Euro III		<8 years, < 160 mkm	5,852,593
		Euro II		<8 years, < 160 mkm	543,536
		none		<8 years, < 160 mkm	326,463
Heavy	FI	Euro VI	PCV	>4years, >80 mkm	54,048
		Euro V		4-8 years, 80-160 mkm	79,046
				<8 years, < 160 mkm	45,268
		Euro IV		<8 years, < 160 mkm	44,747
		Euro III		<8 years, < 160 mkm	28,167
		3-Ways		<8 years, < 160 mkm	14,108
			Total	**19,453,333**	

Vehicular Fleet: **19,453,333**
Average Age: **12.8 years**
Motorcycles Fleet: **62%**

Source: RUNT, Andemos, 2024, Faro90

Colombia – Vehicular Emissions



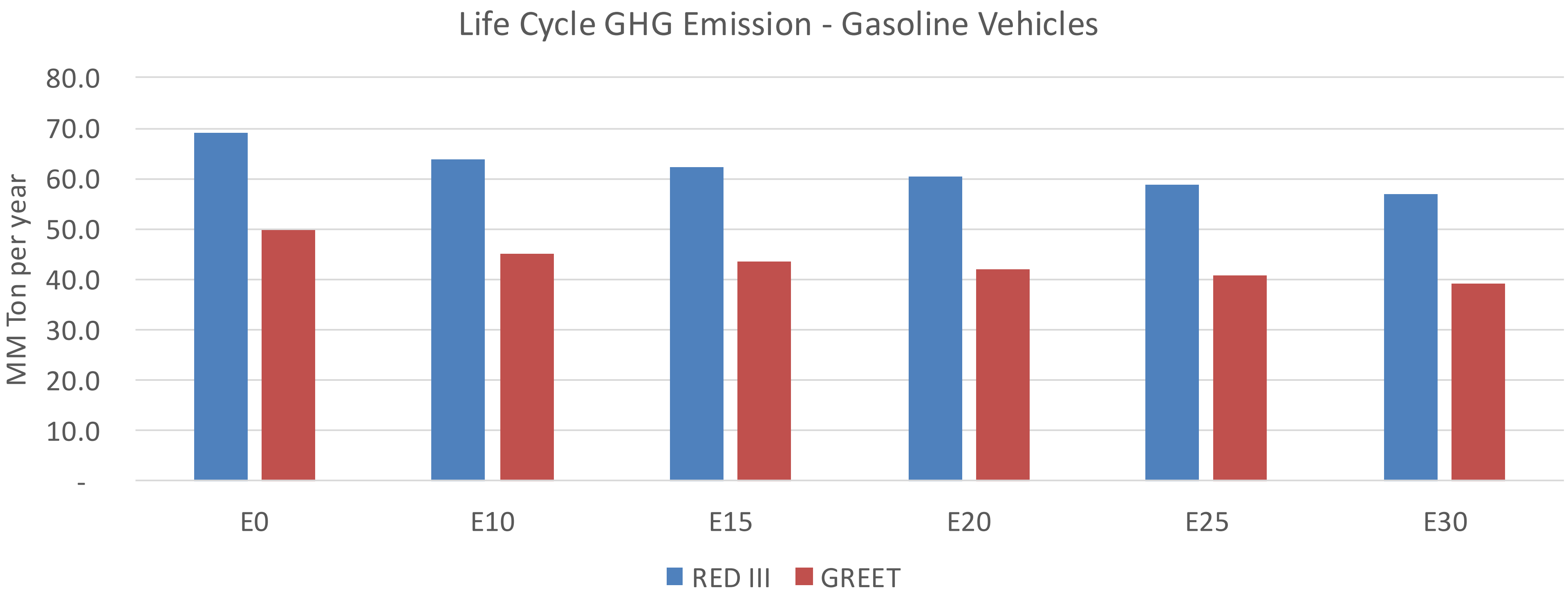
Source: Faro90

Colombia – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	2,961,579	2,828,085	2,694,592	2,603,576	2,520,444	2,459,073	2,368,164	-9%	-15%
CO2	63,944,216	62,299,937	60,747,006	59,525,671	58,921,307	58,658,193	57,576,953	-5%	-8%
VOC	403,824	391,100	378,376	371,525	366,275	362,735	356,258	-6%	-9%
NOx	192,505	177,746	167,480	162,176	153,714	147,043	139,212	-13%	-20%
SOx	732	677	622	575	530	481	430	-15%	-28%
NH3	26,039	25,781	25,523	25,644	25,933	26,136	26,304	-2%	0%
N2O	1,510	1,501	1,495	1,503	1,534	1,551	1,568	-1%	2%
CH4	103,662	103,662	103,662	105,217	107,497	109,467	111,333	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	3,460	3,330	3,200	3,120	3,052	3,004	2,926	-8%	-12%
Acetaldehyde	9,203	10,562	15,498	23,601	32,154	36,742	43,415	68%	249%
Formaldehyde	36,989	35,961	41,921	49,224	51,570	57,050	62,105	13%	39%
Benzene	7,238	6,956	6,591	6,487	6,433	6,152	5,953	-9%	-11%
PM2.5	4,924	4,477	4,029	3,635	3,250	2,836	2,404	-18%	-34%
PM10	24,495	22,268	20,041	18,082	16,164	14,109	11,958	-18%	-34%

Source: Faro90

Colombia – Gasoline Vehicle Fleet GEI Emissions



Country Emissions 2024 (MMT/year)	217.6
2035 Target (MMT/year)	125.5
Delta	92.0

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	9.4%	12.3%	15.4%	18.0%	21.4%
RED II/III	0.0%	7.5%	9.9%	12.4%	14.5%	17.2%

% 2035 Target	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	5.1%	6.6%	8.3%	9.7%	11.5%
RED II/III	0.0%	5.6%	7.4%	9.3%	10.9%	12.9%

Source: Greet, RED II/III, climateactiontracker.org, F90

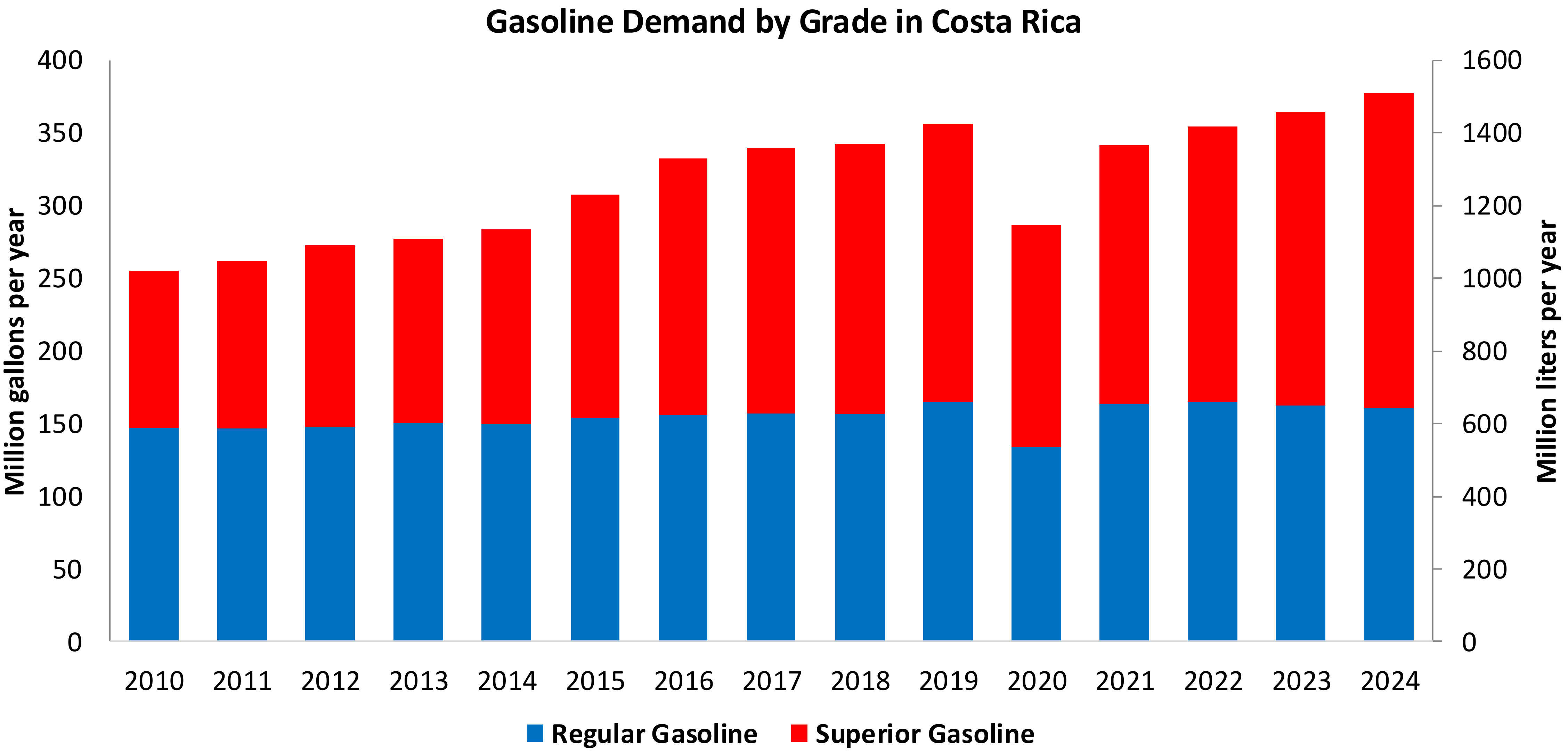
Ethanol Blending in Gasoline - Costa Rica

In 2024, gasoline demand was 378 million gallons (1429 million liters). Costa Rica has two gasoline RON 95 and RON 91, with a market share of 45% for RON 91 and 55% for RON 95. The Limon refinery stopped production in 2012. Gasoline supply is done with imports mainly from the United States and some from Europe.

Gasoline blends with up to 10% v/v of ethanol are allowed (INTE E1:2019). However, this mandate is not yet reflected on the gasoline market. Current ethanol consumption is less than 0.2% because there is no stakeholder's consensus for implementation. In 2024, ethanol production was 8.9 million gallons (33 million liters) and ethanol exports mainly to Europe was 7.2 million gallons (27 million liters). Introduction of E10 is expected by the Government in 2026.

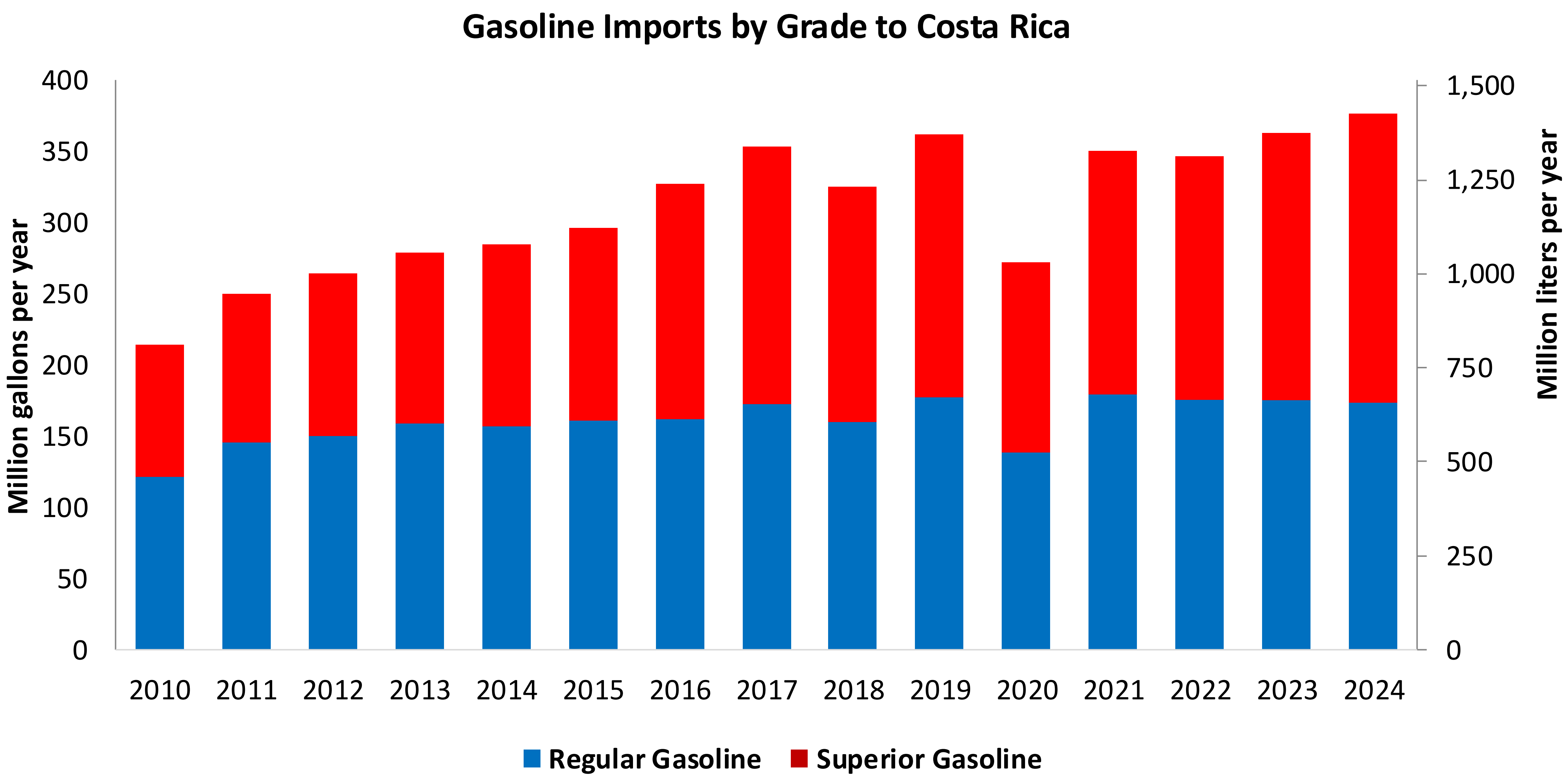
Source: Sepse

Gasoline Demand in Costa Rica



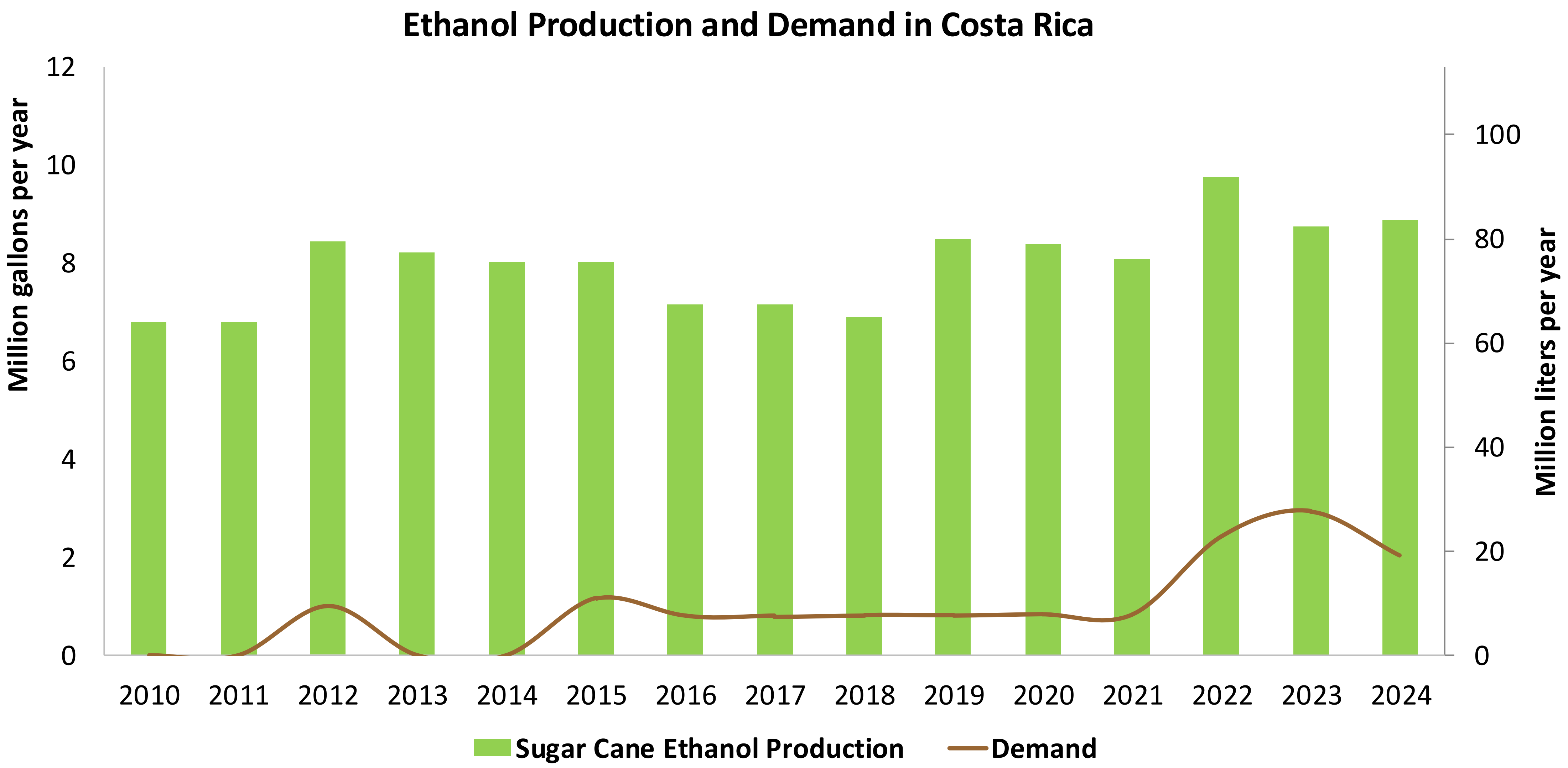
Source: Sepse

Gasoline Imports in Costa Rica



Source: Sepse

Ethanol Demand in Costa Rica



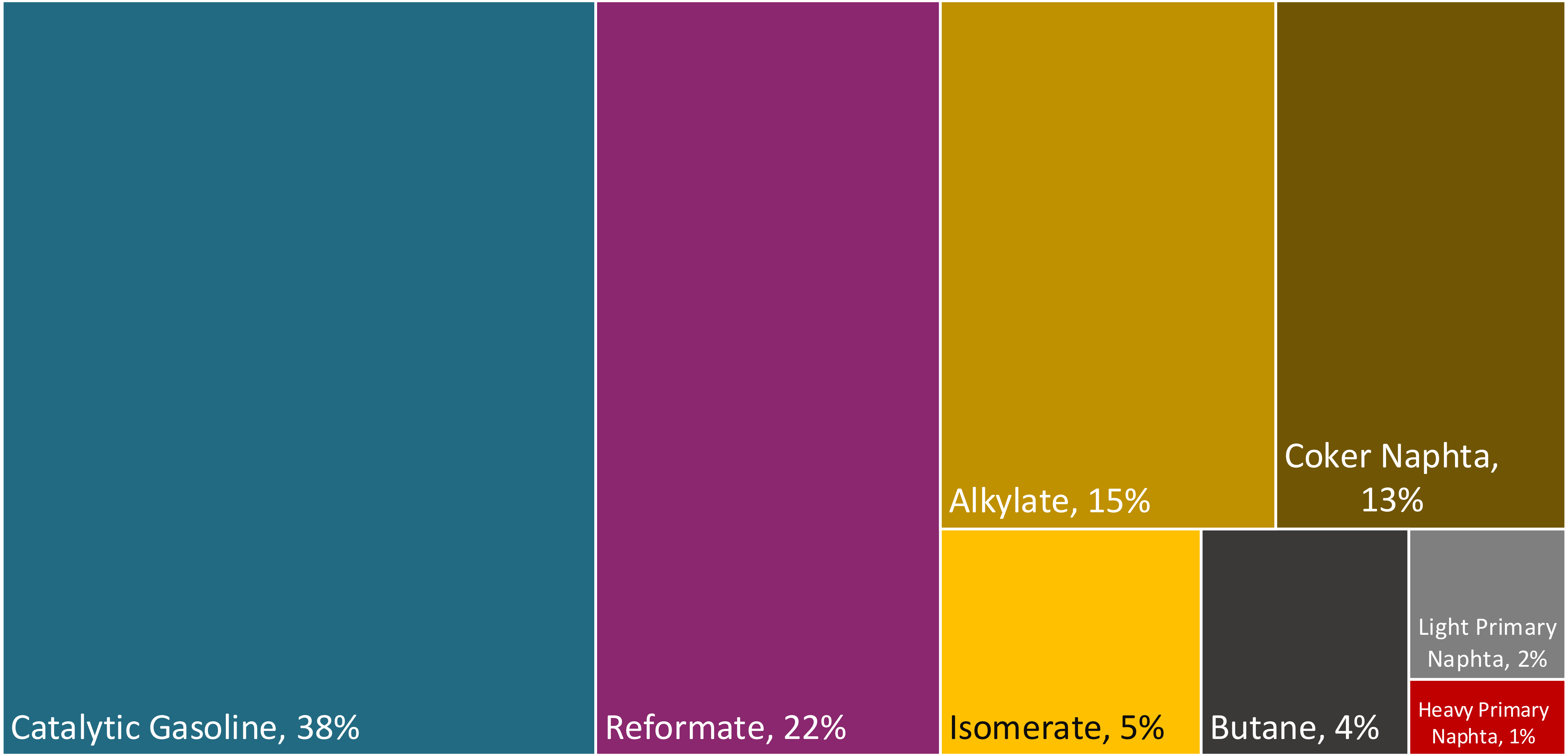
Source: Sepse

Gasoline Quality in Costa Rica

Name	INTE E1:2019		EN 228:2012 + A1:2017 (Euro 6 enabling)			
Implementation Date	2019		2017			
Applicability	Whole country	Whole country	All countries			
Selected Grade	RON 91	RON 95	RON 95 E5	RON 95 E10	RON 98 E5	RON 98 E10
Benzene Content	< 1,5% v/v	< 1,5% v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v
Aromatics	< 35% v/v	< 35% v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v
Olefins	< 18% v/v	< 18% v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v
Lead Content	< 0,013 g/l	< 0,013 g/l	< 5 mg/l	< 5 mg/l	< 5 mg/l	< 5 mg/l
Manganese	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l
RON	> 91	> 95	> 95	> 95	> 98	> 98
MON	> 79	> 83	> 85	> 88	> 85	> 88
AKI						
Sulfur Content	< 50 mg/kg	< 50 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg
Oxygen Content	2,7% m/m (3,7% m/m if ethanol is added)	2,7% m/m (3,7% m/m if ethanol is added)	<2,7 % m/m	<3,7 % m/m	<2,7 % m/m	<3,7 % m/m
Ethanol (EtOH)	< 10% v/v	< 10% v/v	<5 %v/v	<10 %v/v	<5 %v/v	<10 %v/v
RVP 37.8°C (Summer)	< 69 kPa (< 76 kPa if ethanol is added)	< 69 kPa (< 76 kPa if ethanol is added)	<> 60 - 70 kPa *Depends on the country, RVP is regulated in the EU Fuel Quality Directive			
RVP 37.8 °C(Winter)						
RVP 37.8°C (Transition)						
MTBE	-		-	-	-	-
Ethers 5 or more C Atoms	-		Based on oxygen content	<22 %v/v	Based on oxygen content	<22 %v/v

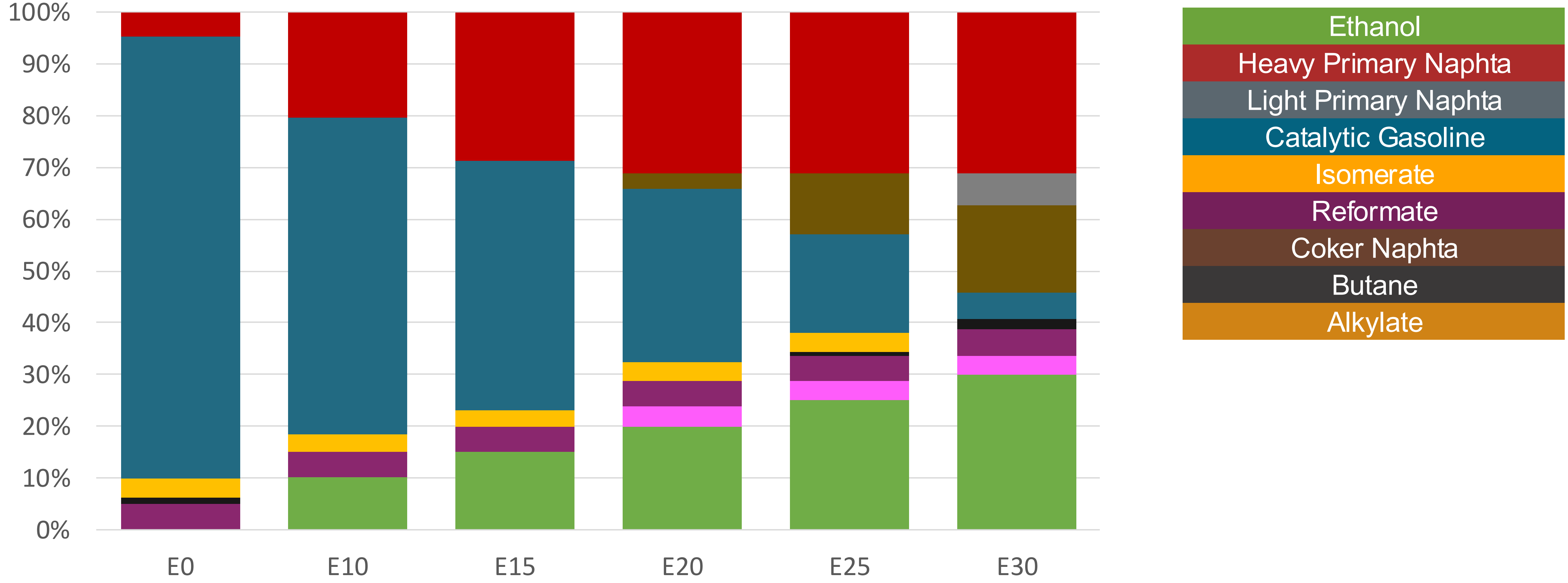
Source: INTECO

Blending Components - Costa Rica



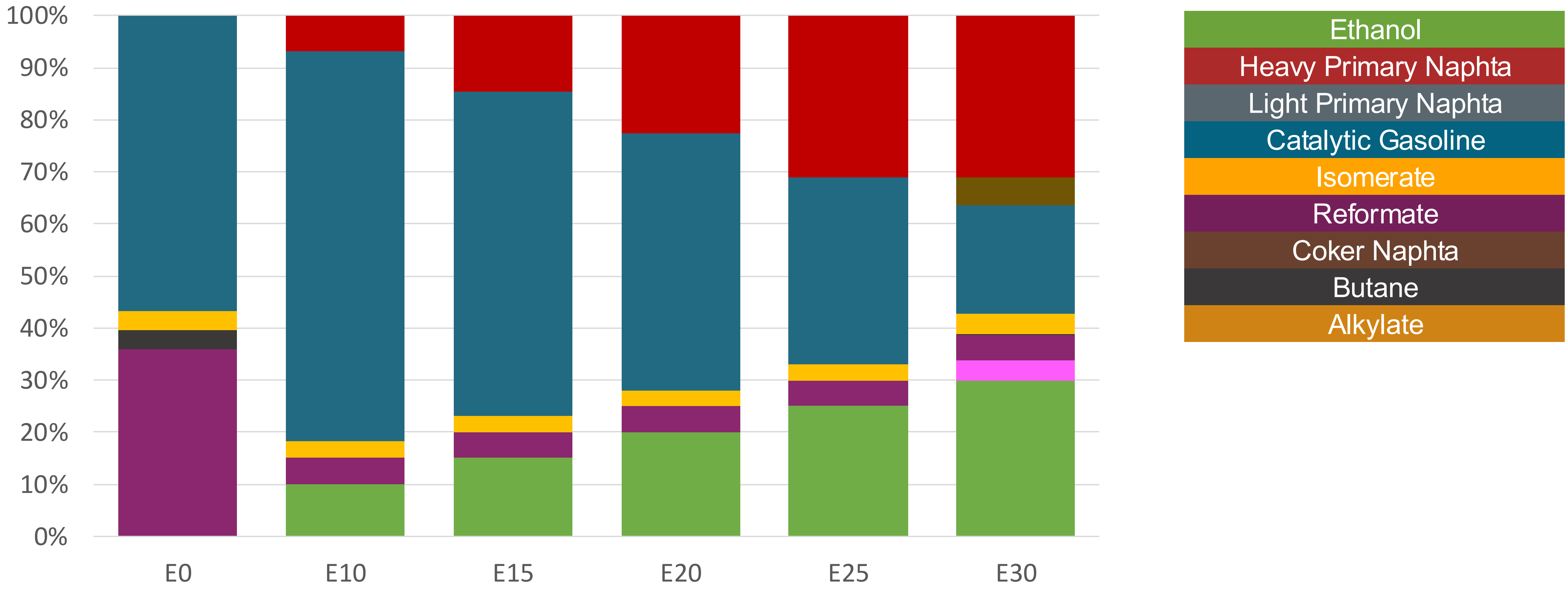
Source: Faro90

Costa Rica – Regular – Constant Octane



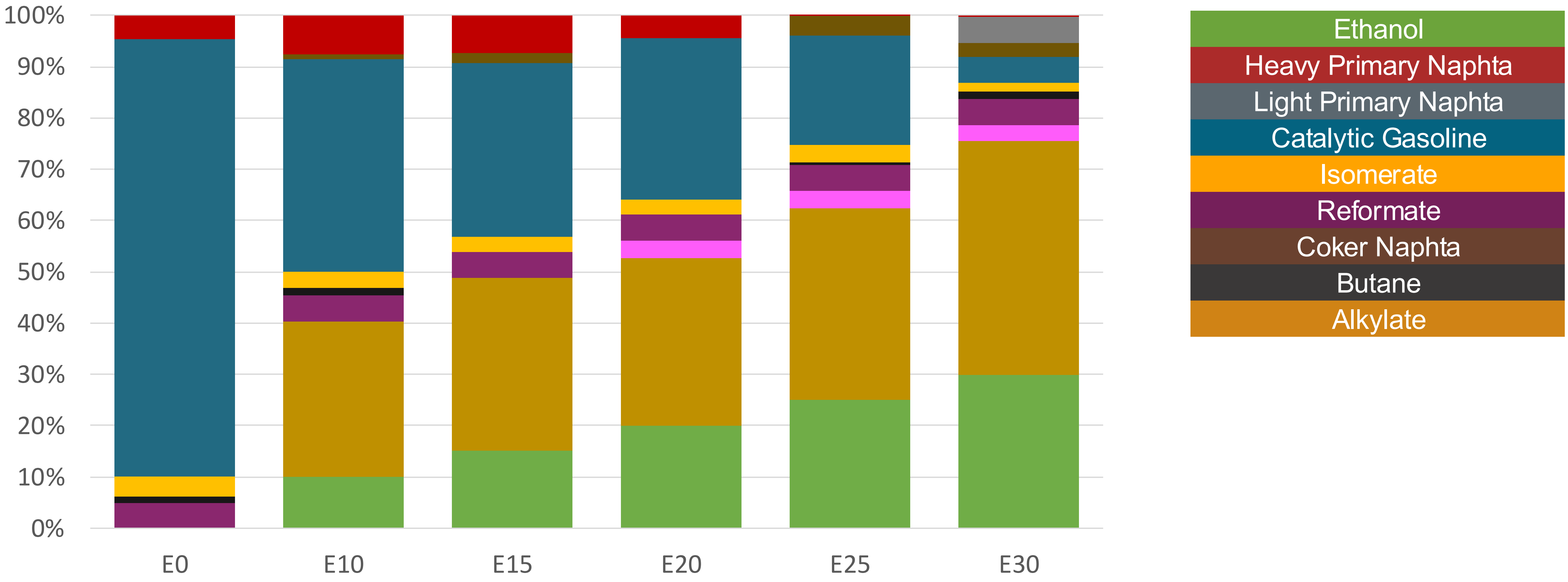
Octane (RON)	91.0	91.0	91.0	91.0	91.0	91.0
Price (USD/gal)	$ 2.286	$ 2.028	$ 1.884	$ 1.737	$ 1.586	$ 1.468

Costa Rica – Premium – Constant Octane



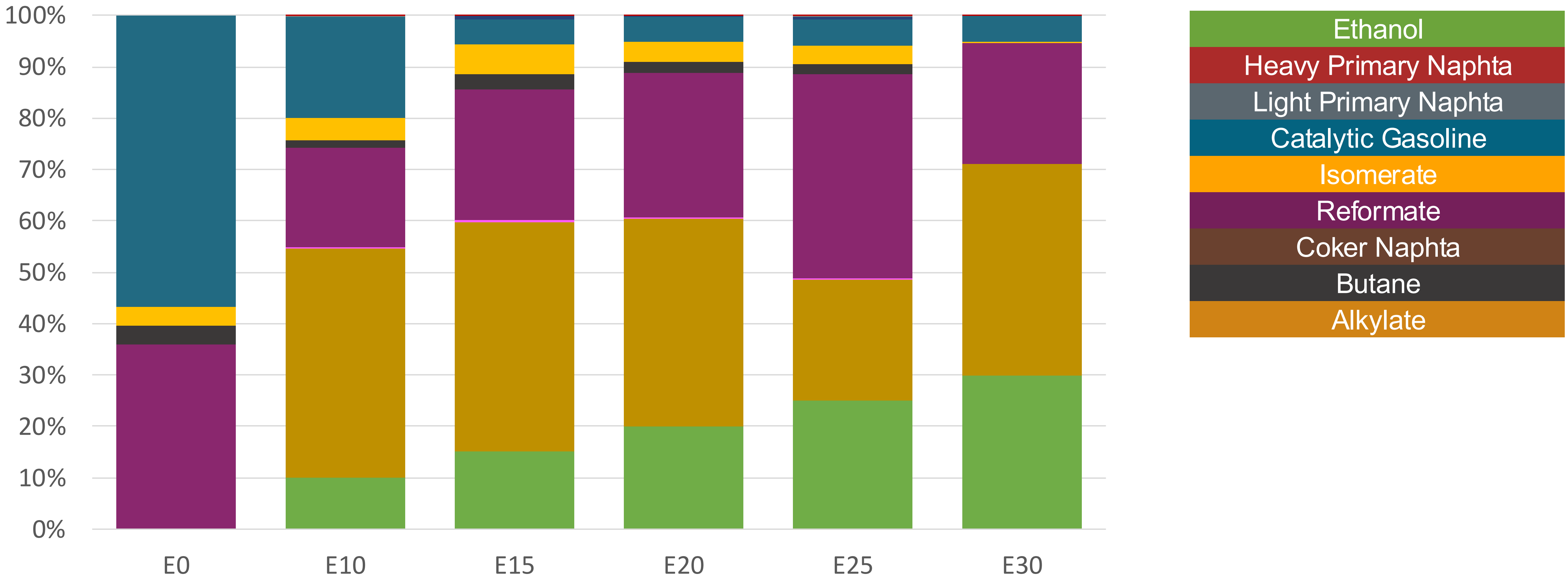
Octane (RON)	95.0	95.0	95.0	95.0	95.0	95.0
Price (USD/gal)	$ 2.536	$ 2.233	$ 2.099	$ 1.957	$ 1.807	$ 1.659

Costa Rica – Regular – Increased Octane



Octane (RON)	91.0	94.6	96.6	99.1	101.2	102.4
Price (USD/gal)	$ 2.286	$ 2.286	$ 2.286	$ 2.286	$ 2.286	$ 2.286

Costa Rica – Premium – Increased Octane



Octane (RON)	95.0	98.0	100.3	102.6	104.9	105.8
Price (USD/gal)	$ 2.536	$ 2.536	$ 2.536	$ 2.536	$ 2.536	$ 2.536

Gasoline Vehicular Fleet in Costa Rica

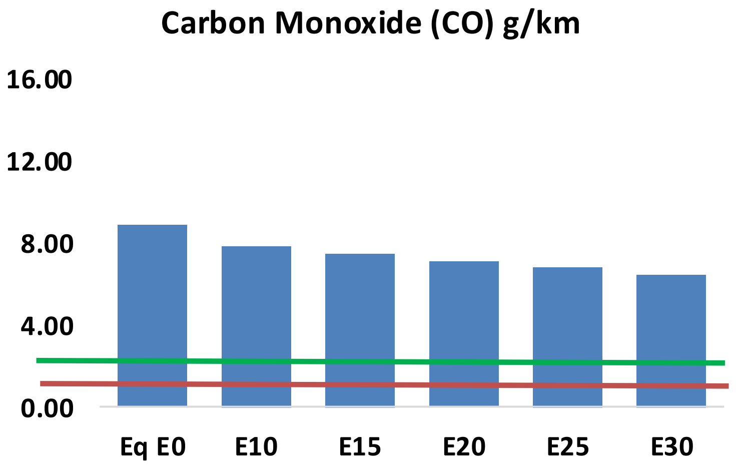
Type	Motor	Exhaust	Recuperator	Age	Number of Vehicles
Auto/Light-Medium	Multi FI-Pt	Euro V	PCV Tank	>4years, >80 mkm	177,395
				4-8 years, 80-160 mkm	201,973
				<8 years, < 160 mkm	135,031
		Euro IV		<8 years, < 160 mkm	132,105
		Euro III		<8 years, < 160 mkm	125,379
		3-Ways		<8 years, < 160 mkm	278,379
Small Engines	FI 4-cycles	Euro III	PCV	>4years, >80 mkm	52,143
				4-8 years, 80-160 mkm	29,684
		Euro II		4-8 years, 80-160 mkm	29,684
				<8 years, < 160 mkm	197,199
Trucks	FI	Euro V	PCV	>4years, >80 mkm	36,587
				4-8 years, 80-160 mkm	41,656
				<8 years, < 160 mkm	27,850
		Euro IV		<8 years, < 160 mkm	27,246
		Euro III		<8 years, < 160 mkm	25,859
		3-Ways		<8 years, < 160 mkm	57,415

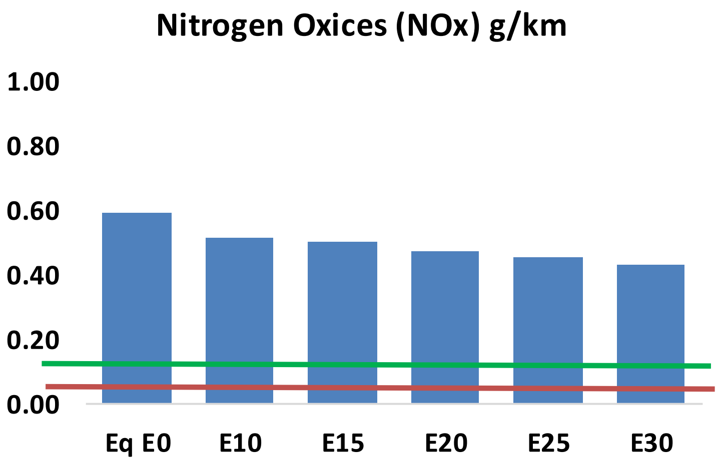
Vehicular Fleet: **1,575,585**
Average Age: **11 years**
Motorcycles: **19.6%**

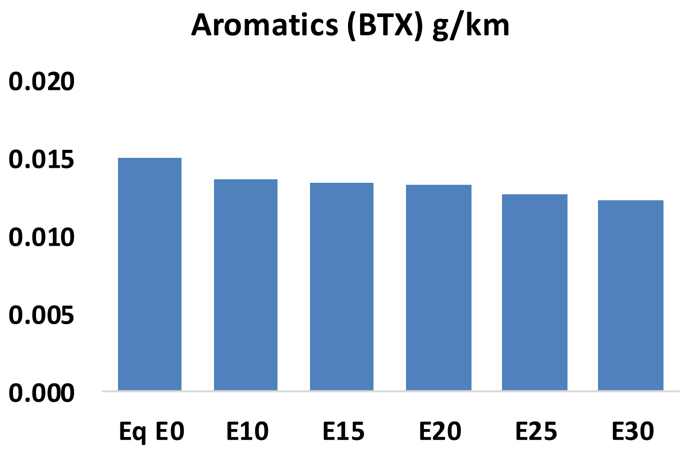
Source: INEC , Faro 90

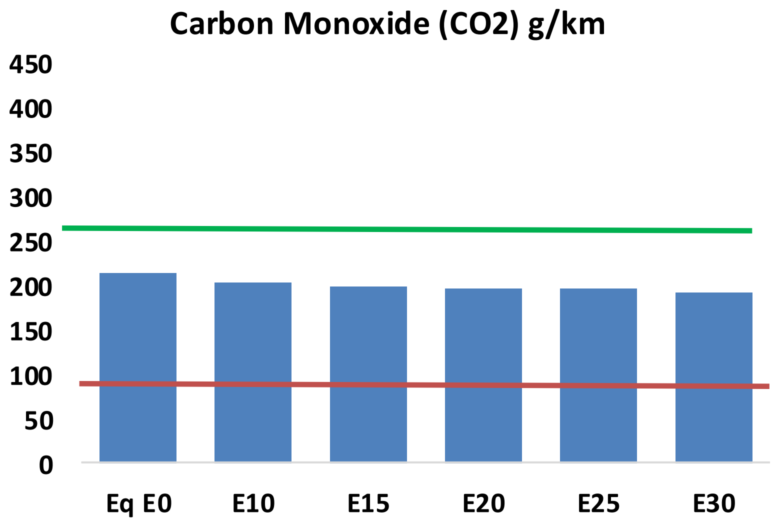
Costa Rica – Vehicular Emissions

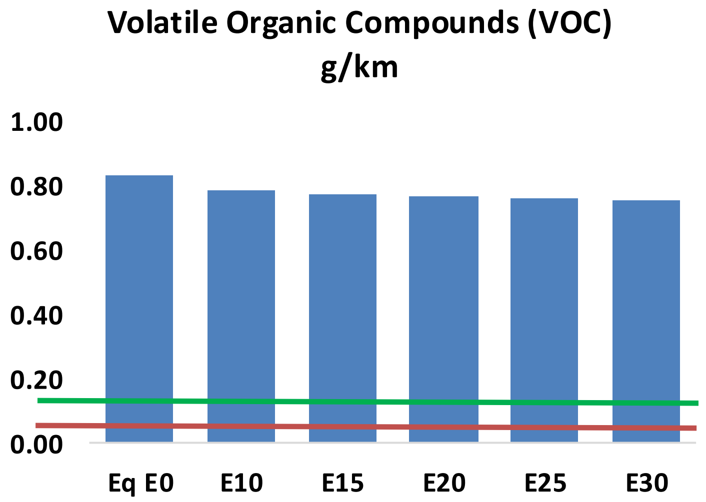
United States
Euro 6

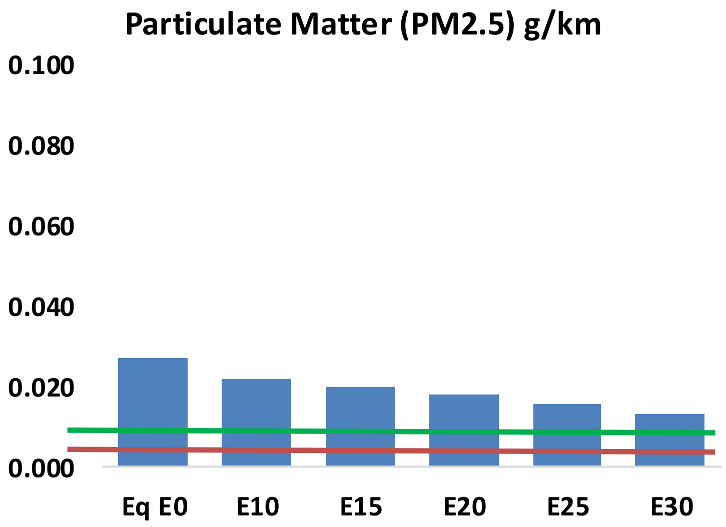












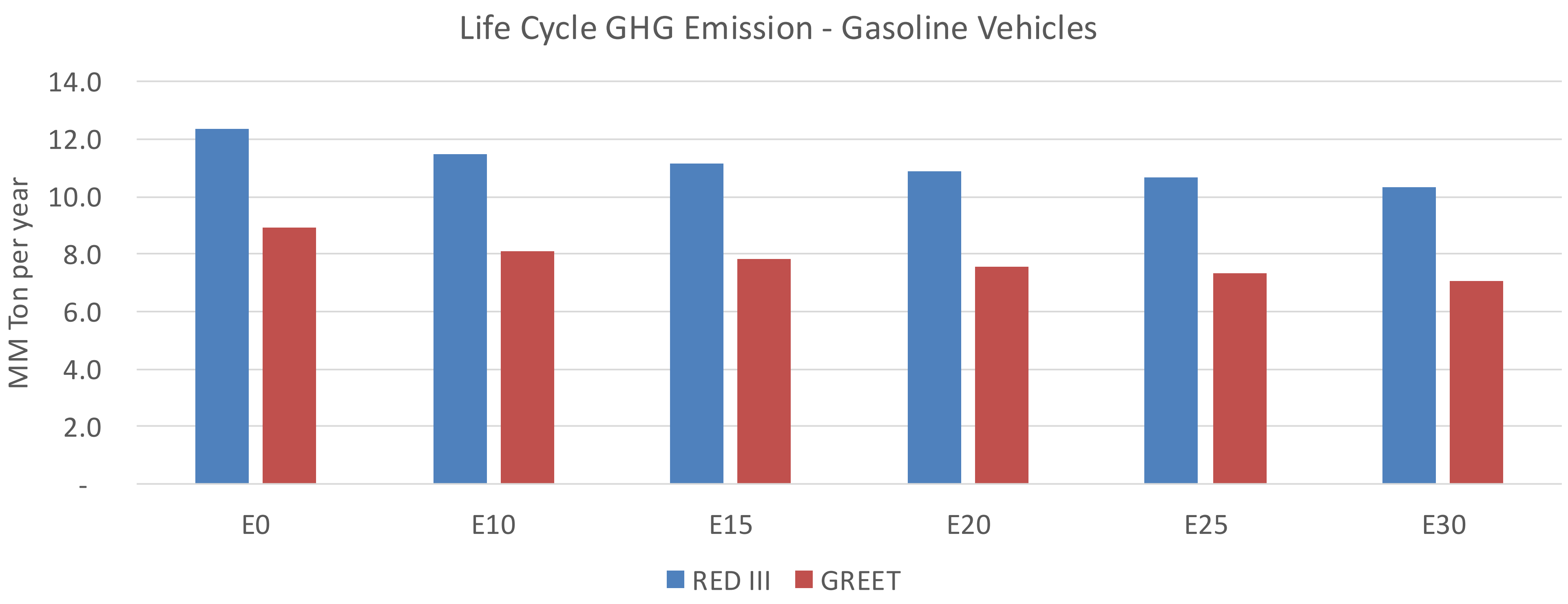
Source: Faro90

Costa Rica – Vehicular Emissions

Vehicular Emissions (kg/day)	E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	497,036	468,122	439,208	417,328	396,144	380,107	359,765	-12%	-20%
CO2	11,924,378	11,617,751	11,328,159	11,100,403	10,987,701	10,923,615	10,722,262	-5%	-8%
VOC	46,603	45,283	43,962	43,321	42,880	42,602	42,014	-6%	-8%
NOx	32,987	30,458	28,763	27,921	26,541	25,470	24,202	-13%	-20%
SOx	137	126	116	107	99	90	80	-15%	-28%
NH3	4,183	4,142	4,100	4,120	4,166	4,199	4,226	-2%	0%
N2O	166	165	164	165	169	170	172	-1%	2%
THC	9,847	9,847	9,847	9,995	10,211	10,398	10,576	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	378	368	357	352	348	346	341	-6%	-8%
Acetaldehyde	784	899	1,319	2,009	2,738	3,128	3,696	68%	249%
Formaldehyde	2,382	2,316	2,700	3,170	3,322	3,674	4,000	13%	39%
Benzene	838	805	763	751	745	712	689	-9%	-11%
PM2.5	701	637	573	517	462	404	342	-18%	-34%
PM10	790	718	646	583	521	455	386	-18%	-34%

Source: Faro90

Costa Rica – Gasoline Vehicle Fleet GEI Emissions



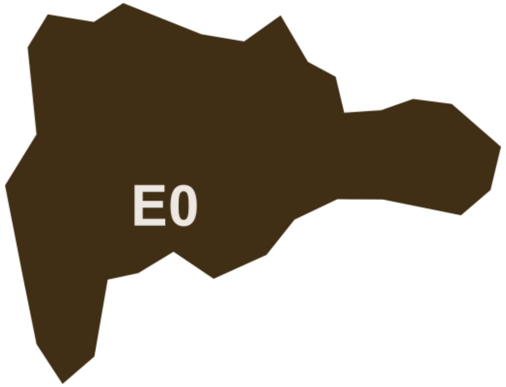
Country Emissions 2024 (MMT/year)	17.1
2035 Target (MMT/year)	8.5
Delta	8.6

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	9.2%	11.9%	14.9%	17.3%	20.6%
RED II/III	0.0%	7.2%	9.4%	11.8%	13.7%	16.3%

% 2035 Target	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	9.5%	12.3%	15.4%	17.9%	21.3%
RED II/III	0.0%	10.3%	13.5%	16.9%	19.7%	23.3%

Source: Greet, RED II/III, climateactiontracker.org, F90

Ethanol Blending in Gasoline – Dominican Republic

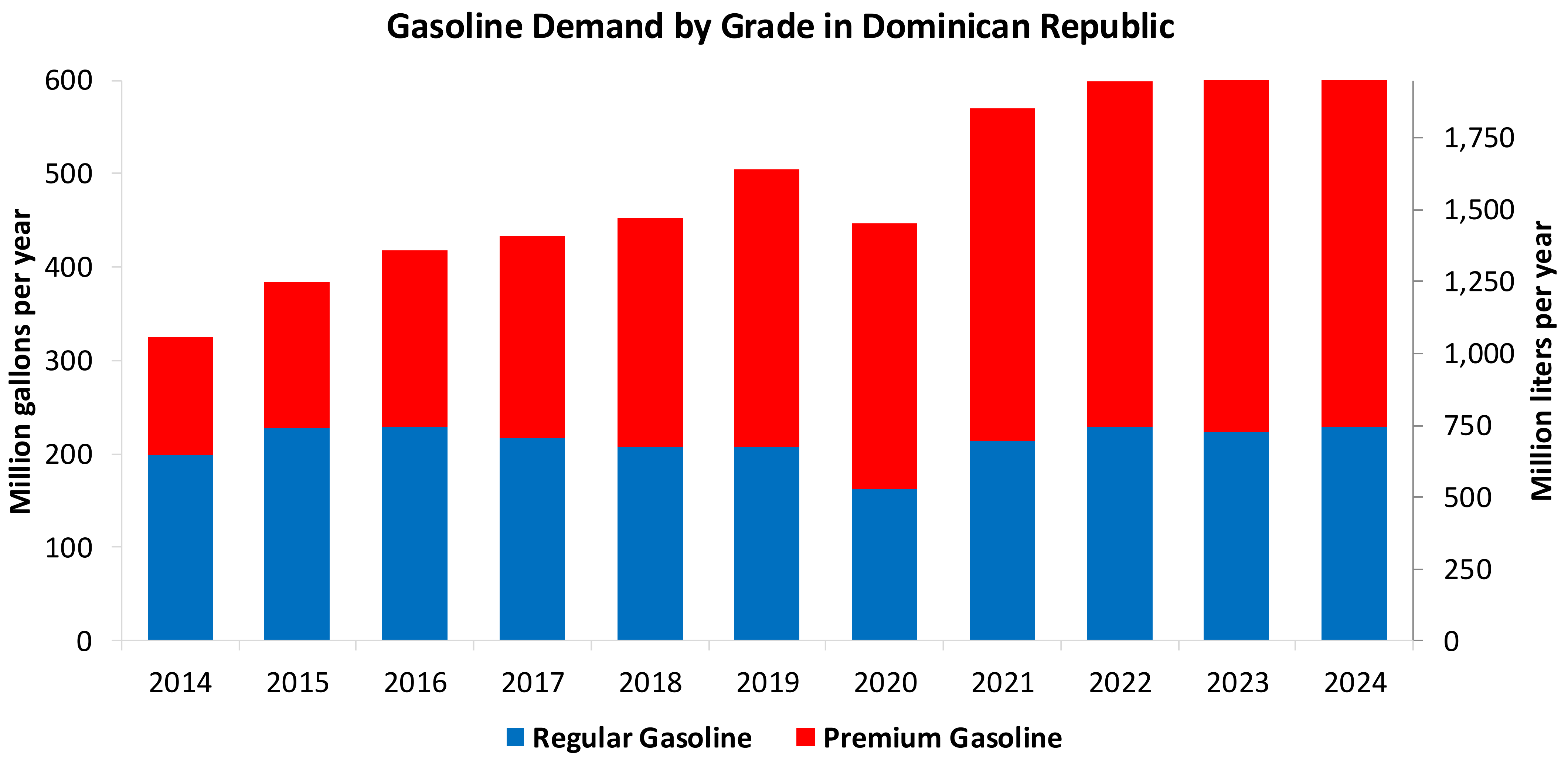


National demand is covered up by local production representing 15%, and imports from United States, Trinidad and Tobago, Caribbean, among other countries. In 2024, gasoline consumption was 562 million gallons (2,126 million liters). Regular gasoline consumption was 35% of total demand and premium 65%. The Dominican Government applies subsidies to gasoline prices.

Dominican Republic does not currently have an ethanol mandate.

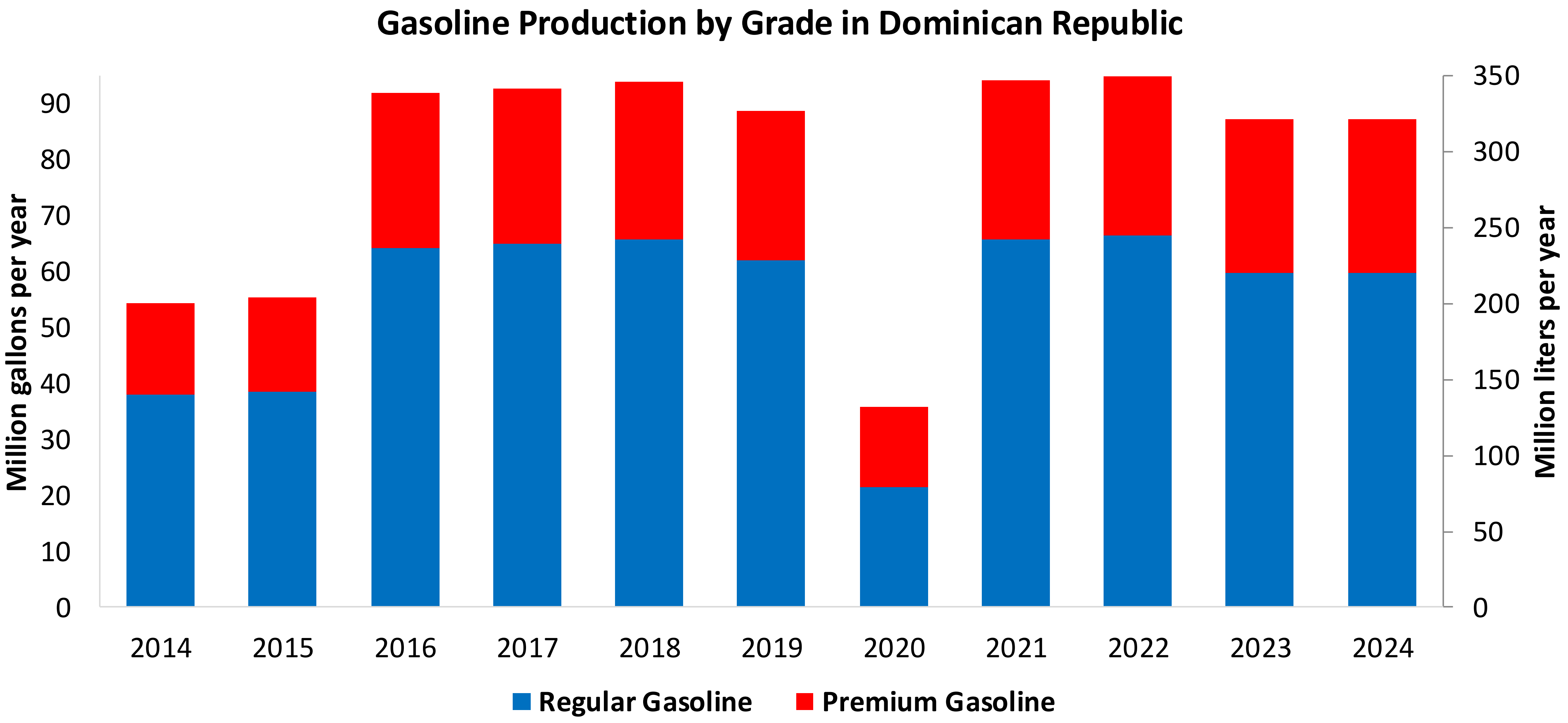
Source: DGII

Gasoline Demand in Dominican Republic



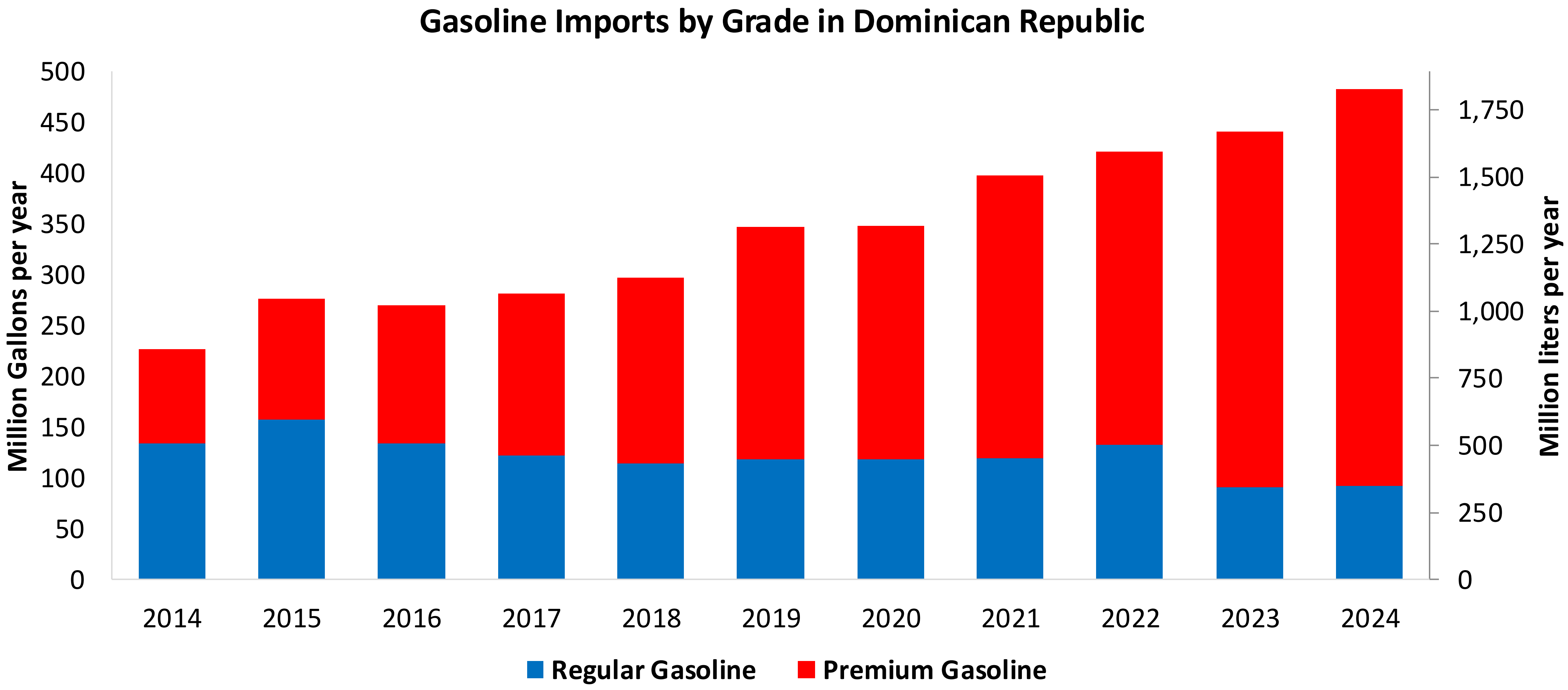
Source:
DGII

Gasoline Production in Dominican Republic



Source:
DGII

Gasoline Imports in Dominican Republic



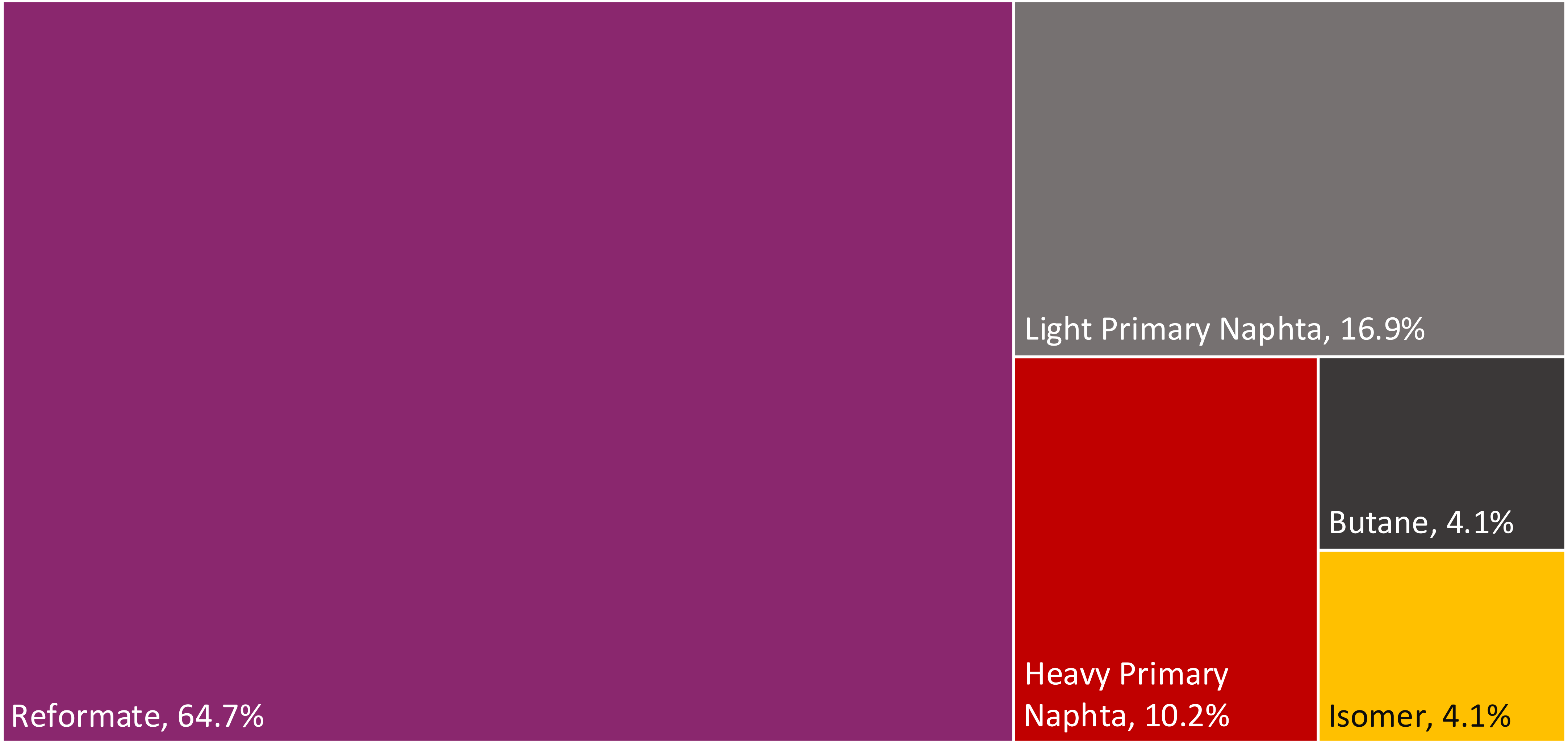
Source:
DGII

Gasoline Quality in Dominican Republic

Name	NORDOM 476 3rd Revision				EN 228:2012 + A1:2017 (Euro 6 enabling)			
Implementation Date	2015				2017			
Applicability	Whole country	Whole country	Whole country	Whole country	All countries			
Selected Grade	Regular Gasoline	Premium Gasoline	Oxygenated Regular Gasoline	Oxygenated Premium Gasoline	RON 95 E5	RON 95 E10	RON 98 E5	RON 98 E10
Benzene Content	-	-	-	-	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v
Aromatics	-	-	-	-	< 35 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v
Olefins	-	-	-	-	< 18 %v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v
Lead Content	< 0,013 g/l	< 0,013 g/l	< 0,013 g/l	< 0,013 g/l	< 5 mg/l	< 5 mg/l	< 5 mg/l	< 5 mg/l
Manganese	< 8,3 mg/l	< 8,3 mg/l	< 8,3 mg/l	< 8,3 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l
RON	> 89	> 95	> 90	> 96	> 95	> 95	> 98	> 98
MON	> 76	> 82	> 77	> 83	> 85	> 88	> 85	> 88
AKI								
Sulfur Content	< 1.500 mg/kg	< 1.500 mg/kg	< 1.500 mg/kg	< 1.500 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg
Oxygen Content			< 3.5 %m/m	< 3.5 %m/m	<2,7 % m/m	<3,7 % m/m	<2,7 % m/m	<3,7 % m/m
Ethanol (EtOH)			< 10 %v/v	< 10 %v/v	<5 %v/v	<10 %v/v	<5 %v/v	<10 %v/v
RVP 37.8°C (Summer)	< 61 kPa	< 61 kPa	< 69 kPa	< 69 kPa	<> 60 - 70 kPa *Depends on the country, RVP is regulated in the EU Fuel Quality Directive			
RVP 37.8 °C(Winter)								
RVP 37.8°C (Transition)								
MTBE					-	-	-	-
Ethers 5 or more C Atoms	-	-	-	-	Based on oxygen content	<22 %v/v	Based on oxygen content	<22 %v/v

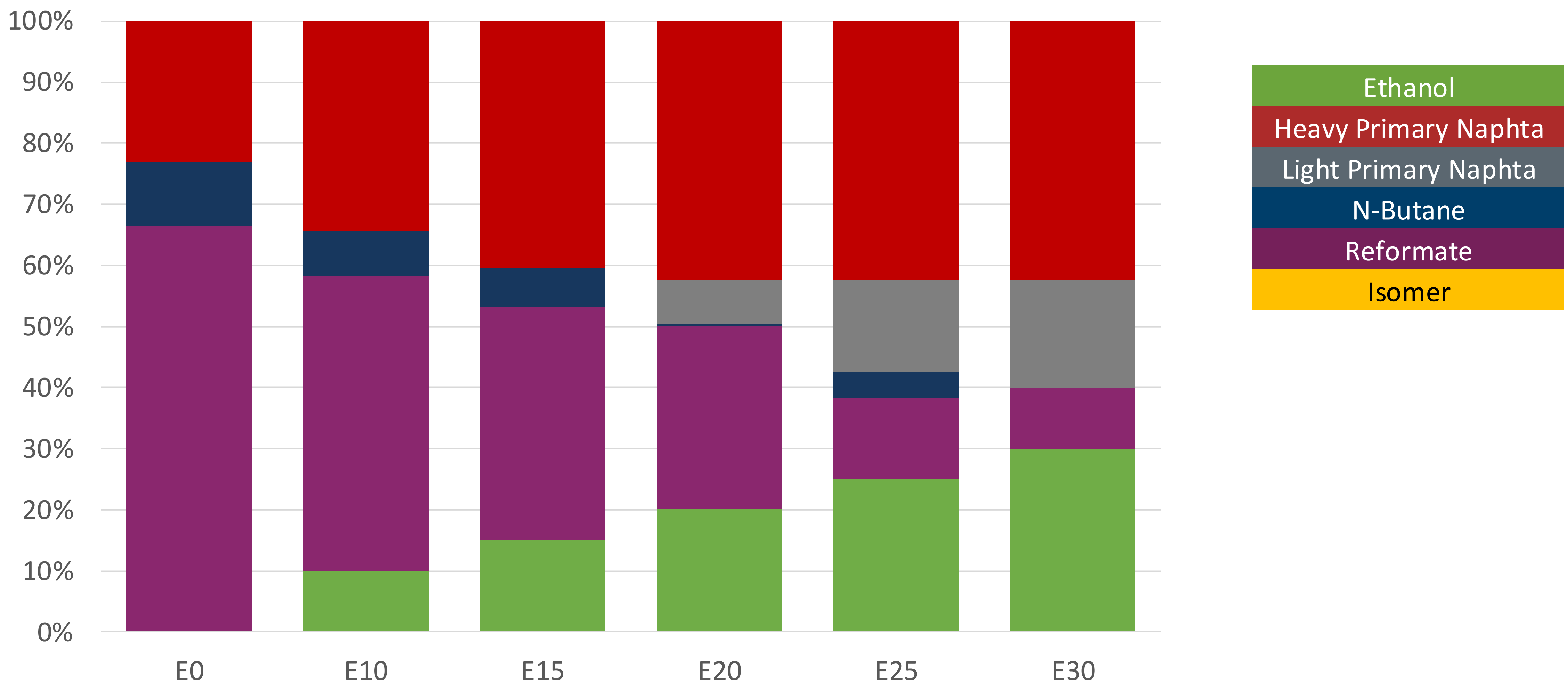
Source: NORDOM

Available Blending Components



Source: Faro90

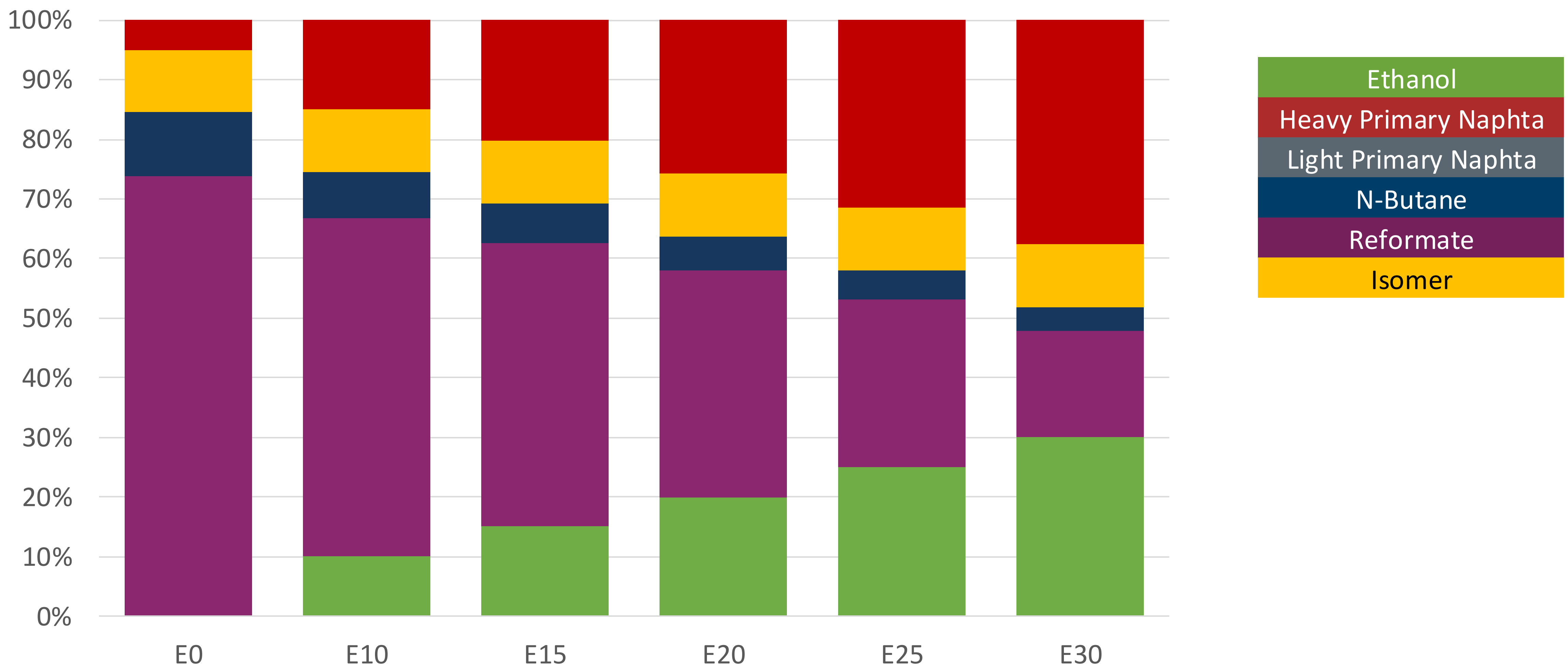
Dominican Republic – Regular – Constante Octane



Octane (RON)	90.0	90.0	90.0	90.0	90.0	91.8
Price (US$/gal)	$ 2.357	$ 2.060	$ 1.894	$ 1.800	$ 1.549	$ 1.539

Source: Faro90

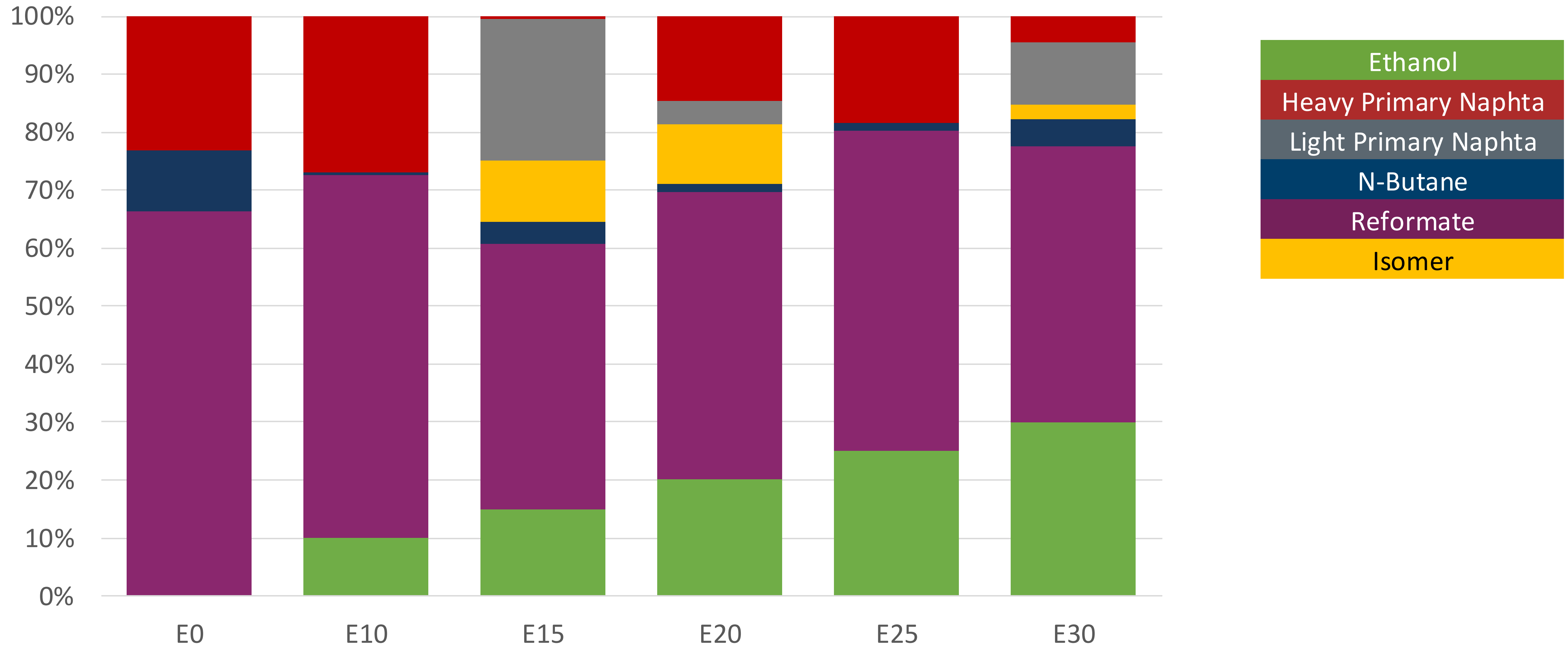
Dominican Republic– Premium – Constant Octane



Octane (RON)	96.0	96.0	96.0	96.0	96.0	96.0
Price (US$/gal)	$ 2.665	$ 2.395	$ 2.246	$ 2.088	$ 1.922	$ 1.749

Source: Faro90

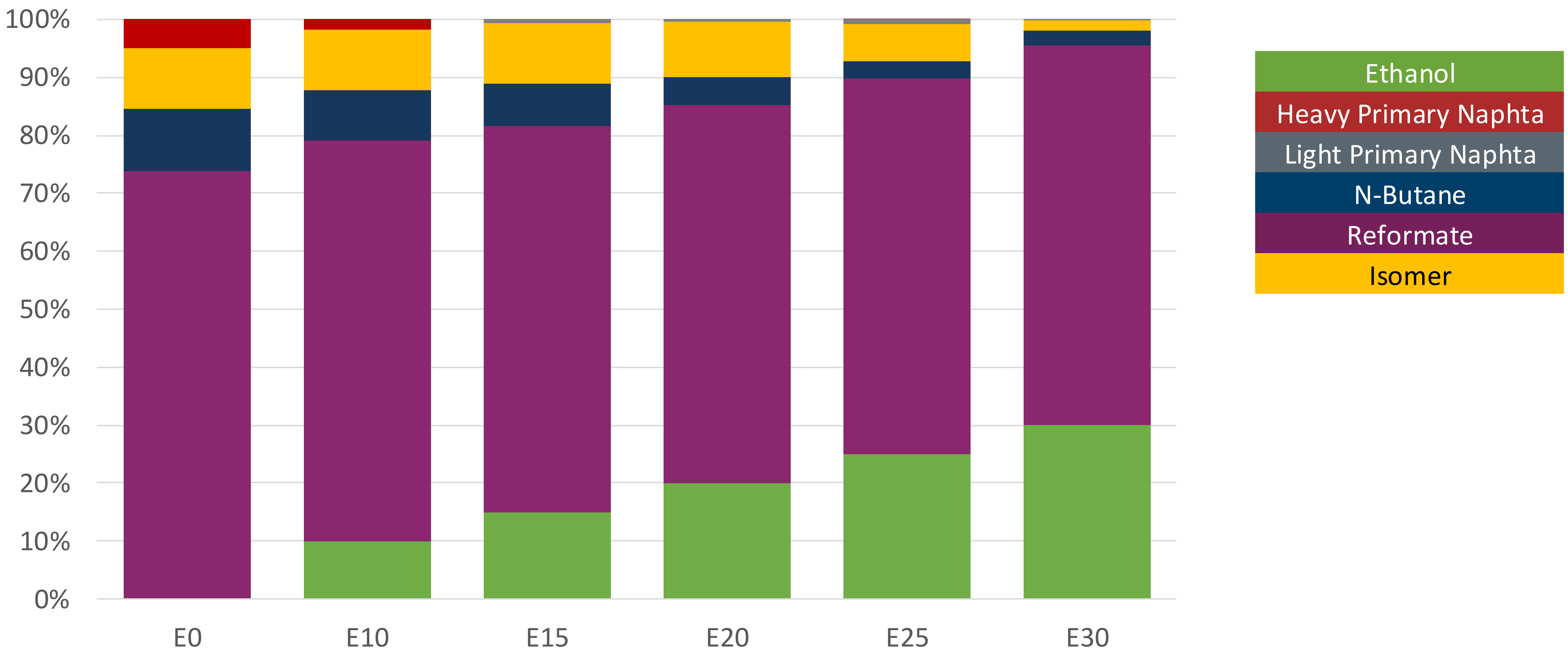
Dominican Republic – Regular – Increased Octane



	E0	E10	E15	E20	E25	E30
Octane (RON)	90.0	93.2	96.8	99.0	101.3	104.3
Price (US$/gal)	$ 2.357	$ 2.357	$ 2.357	$ 2.357	$ 2.357	$ 2.357

Source: Faro90

Dominican Republic – Premium – Increased Octane

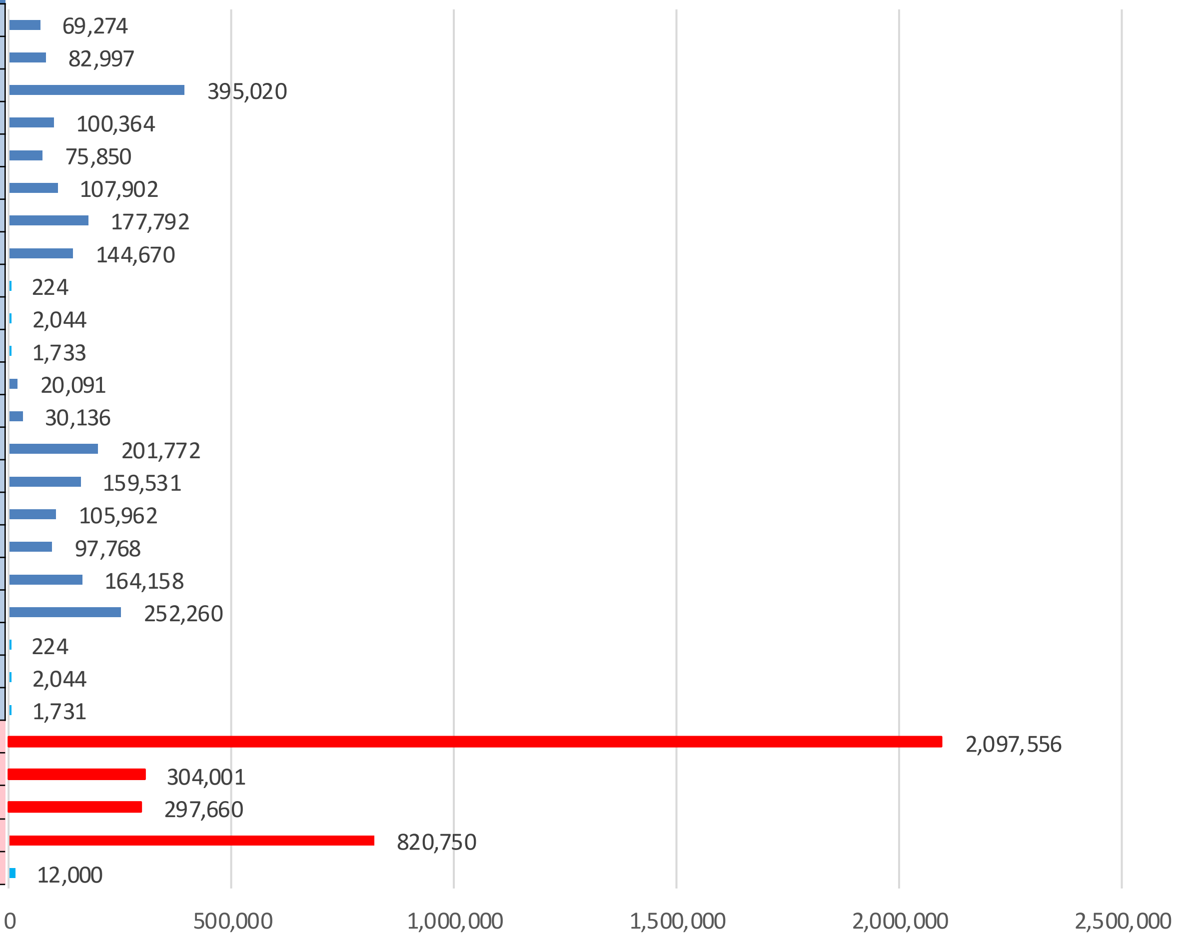


Octane (RON)	96.0	100.8	102.9	104.6	106.3	108.2
Price (US$/gal)	$ 2.665	$ 2.665	$ 2.665	$ 2.665	$ 2.665	$ 2.665

Source: Faro90

Gasoline Vehicular Fleet in Dominican Republic

Type	Wheight	Motor	Exhaust	Recuperator	Age
Auto/Sml Truck	Light	Carburator	3-Ways	PCV Tank	+8years
		Single FI	3-Ways		+8years
		Multi FI	3-Ways		+8years
			Euro III		+8years
			Euro IV		+8years
			Euro V/VI		+8years
					4-8years
					-4years
		Hybrid	Hybrid		4-8years
					-4years
		Electric			
	Medium	Carburator	3-Ways	PCV Tank	+8years
		Single FI	3-Ways		+8years
		Multi FI	3-Ways		+8years
			Euro III		+8years
			Euro IV		+8years
			Euro V/VI		+8years
					4-8years
					-4years
		Hybrid	Hybrid		4-8years
					-4years
		Electric			
Sml Engines	Light	4-Cycle	Euro II	PCV	+8years
					4-8years
			Euro III		4-8years
					-4years
		Electric			-8years

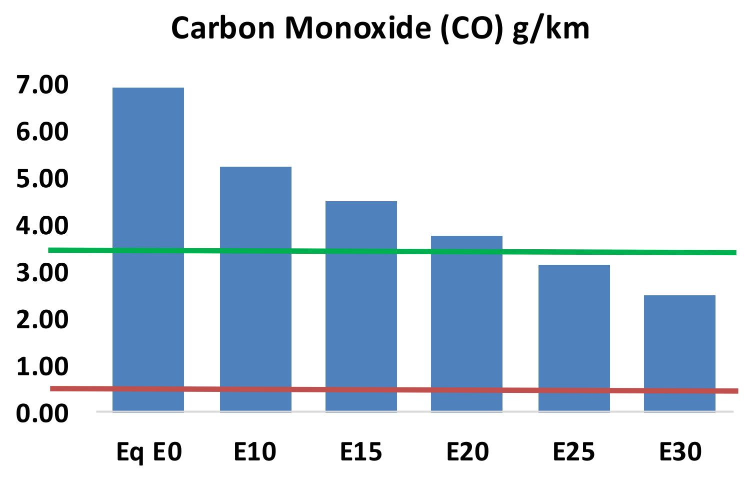


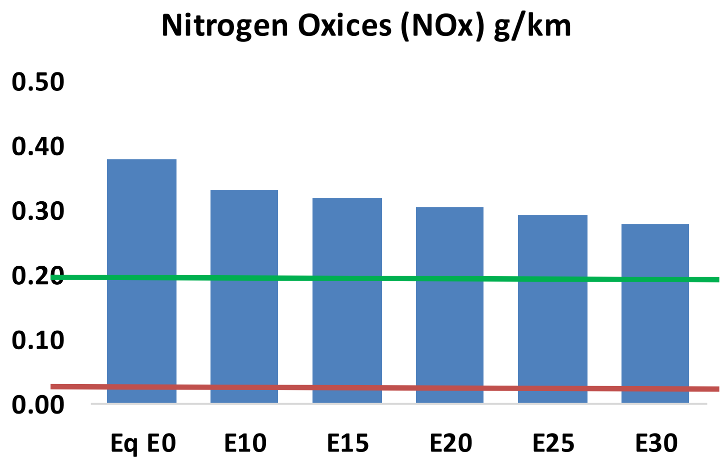
Gasoline Vehicular Fleet: **5,710,050**
Average Age: **10 years**
Motorcycles: **62%**

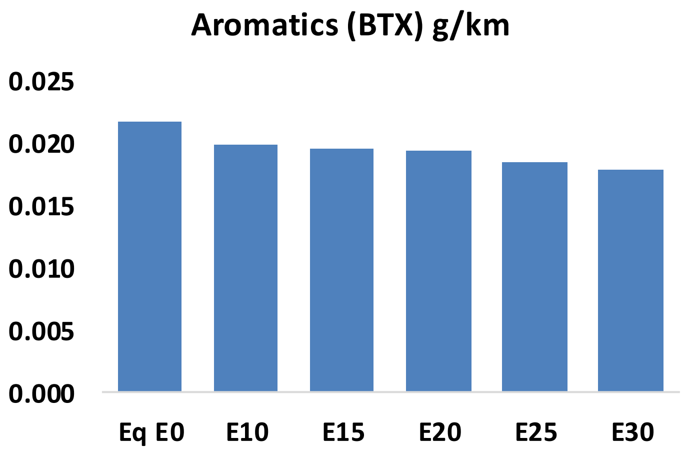
Source: Dirección General de Impuestos Internos (DGII), 2024

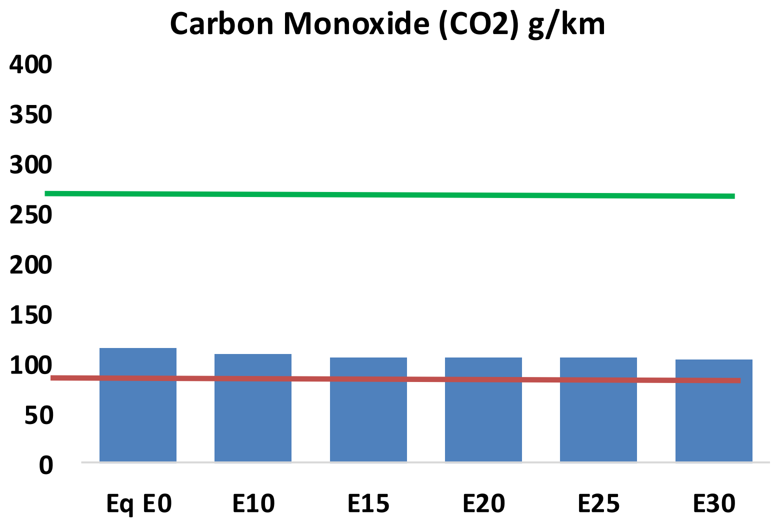
Vehicular Emissions – Dominican Republic

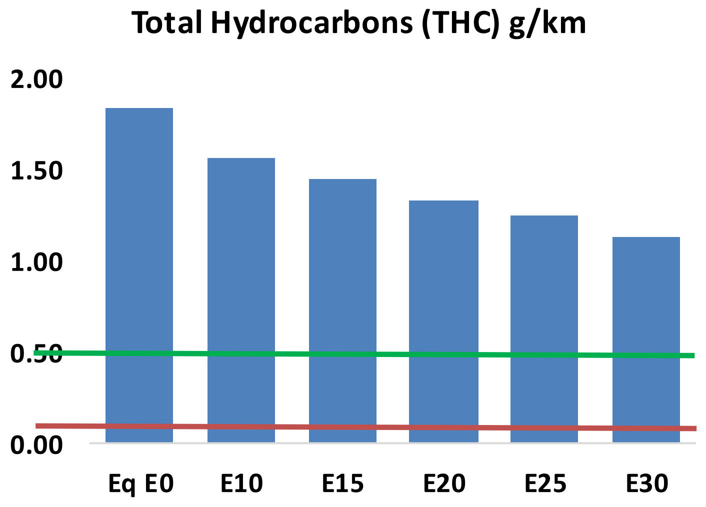
United States
Euro 6 Standard

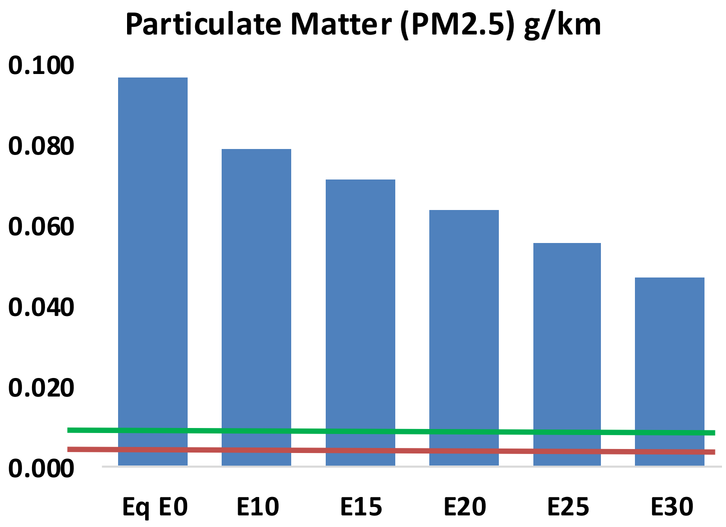












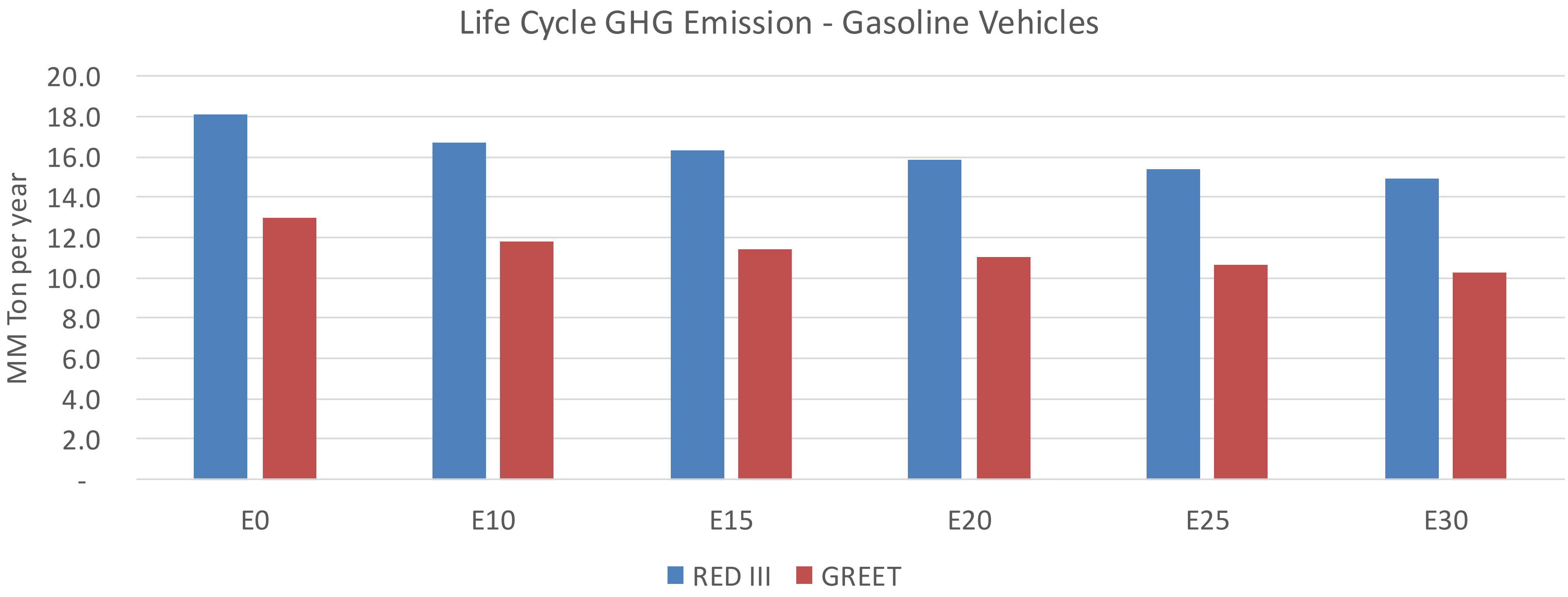
Source: Faro90

Dominican Republic – Vehicular Fleet

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	941,919	828,510	715,101	614,167	508,913	426,867	336,644	-24%	-46%
CO2	15,484,691	15,086,513	14,710,456	14,414,699	14,268,346	14,209,005	13,947,092	-5%	-8%
VOC	248,823	230,217	211,611	196,467	181,273	169,604	153,982	-15%	-27%
NOx	51,415	47,473	44,844	43,545	41,407	39,752	37,790	-13%	-19%
SOx	171	158	145	134	124	112	100	-15%	-28%
NH3	8,851	8,763	8,676	8,717	8,815	8,884	8,941	-2%	0%
N2O	195	194	193	194	198	200	202	-1%	2%
CH4	55,203	55,203	55,203	56,031	57,245	58,294	59,288	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	1,204	1,102	1,000	915	829	762	674	-17%	-31%
Acetaldehyde	4,414	5,066	7,433	11,320	15,422	17,623	20,823	68%	249%
Formaldehyde	18,492	17,978	20,957	24,608	25,781	28,520	31,048	13%	39%
Benzene	2,959	2,843	2,694	2,651	2,629	2,515	2,433	-9%	-11%
PM2.5	1,101	1,000	900	812	726	634	537	-18%	-34%
PM10	12,004	10,913	9,822	8,861	7,922	6,914	5,860	-18%	-34%

Source: Faro90

Dominican Republic – Gasoline Vehicle Fleet GEI Emissions



Country Emissions 2024 (MMT/year)	51.6
2035 Target (MMT/year)	36.1
Delta	15.5

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	9.5%	12.3%	15.4%	18.0%	21.3%
RED II/III	0.0%	7.7%	10.2%	12.8%	15.0%	17.8%

% 2035 Target	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	8.0%	10.4%	13.0%	15.1%	17.9%
RED II/III	0.0%	9.1%	11.9%	15.0%	17.6%	20.8%

Source: Greet, RED II/III, climateactiontracker.org, F90

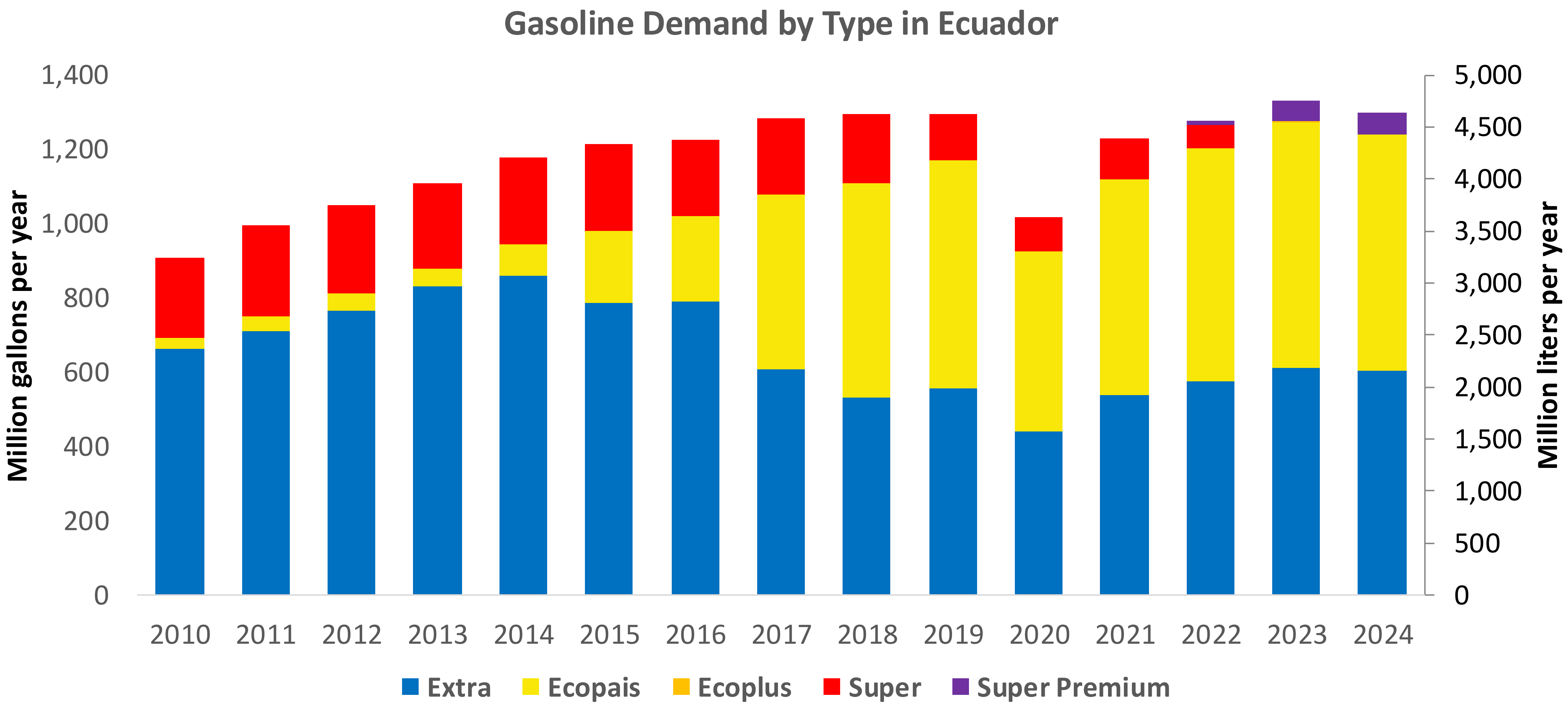
Ethanol Blending in Gasoline - Ecuador

In 2024, gasoline consumption in Ecuador was 1,188 million gallons (4,498 million liters). A new plan was stablished to implement new grades of gasoline: four grades with 5 different types according to the current quality norm (NTE INEN 935:2021): Extra (RON 85), Ecopais E10 (RON 85), Ecoplus (RON 89), Súper (RON 92) and Super Premium (RON 95). Super Premium (RON 95) will gradually substitute Super (RON 92). Gasoline Ecoplus (RON 89) was recently introduced in the market. Ecuador has local production and imports naphtha (unfinished gasolines) to supply national demand.

There is no specific mandate for ethanol blending, however there is a permit to blend it with gasoline. Ethanol has been blending in gasoline since 2009. Today the usual blend is E10 for Ecoplus and Super. Ecopais gasoline is supplied in some provinces with an ethanol content between 3-4%. Gasoline Ethanol is produced ,consumed and exported to neighboring countries for fuel and industrial use.

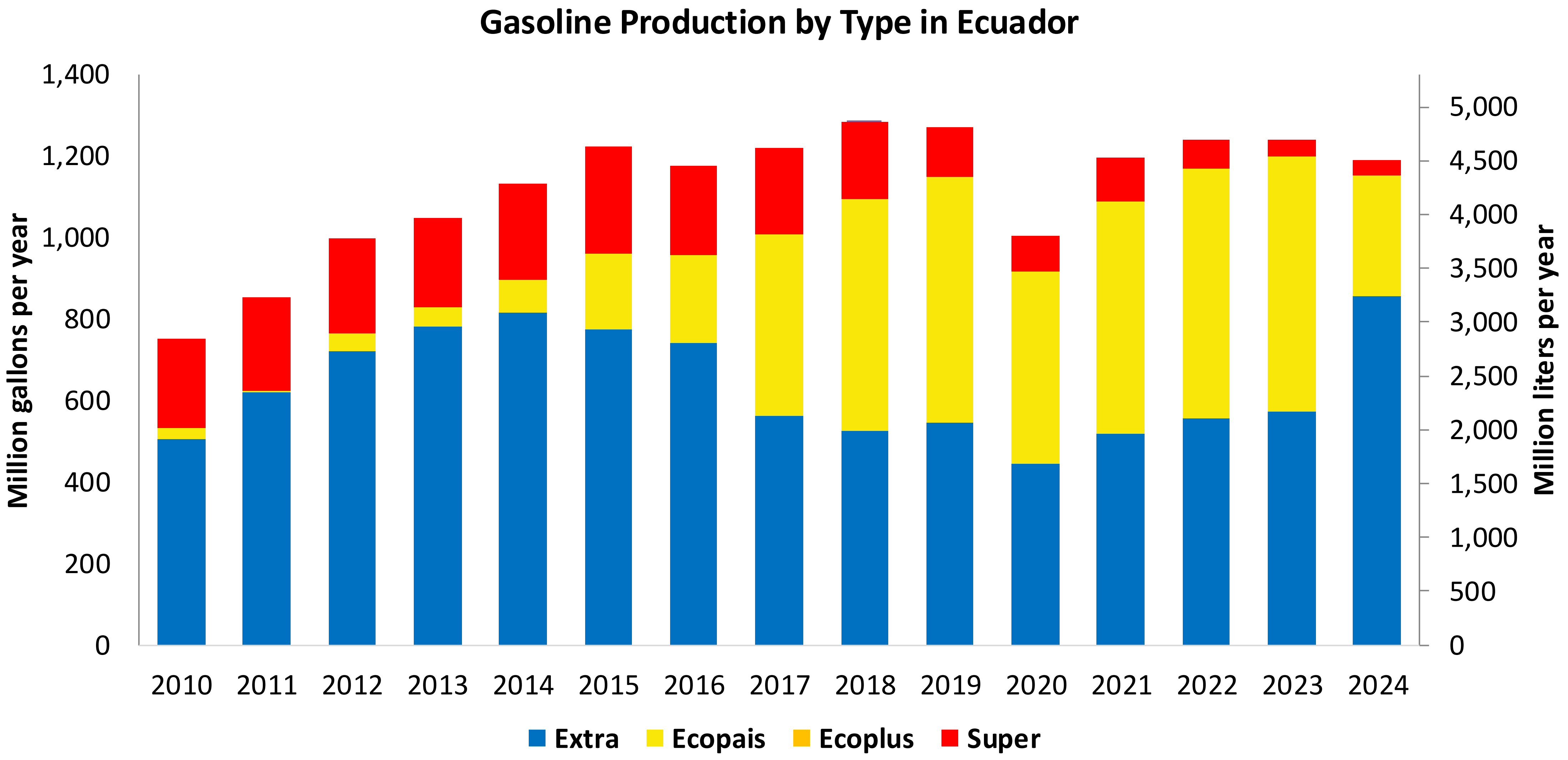
Source: Ministrio de Energía y Minas, EP PETROECUADOR, 2024

Gasoline Demand in Ecuador



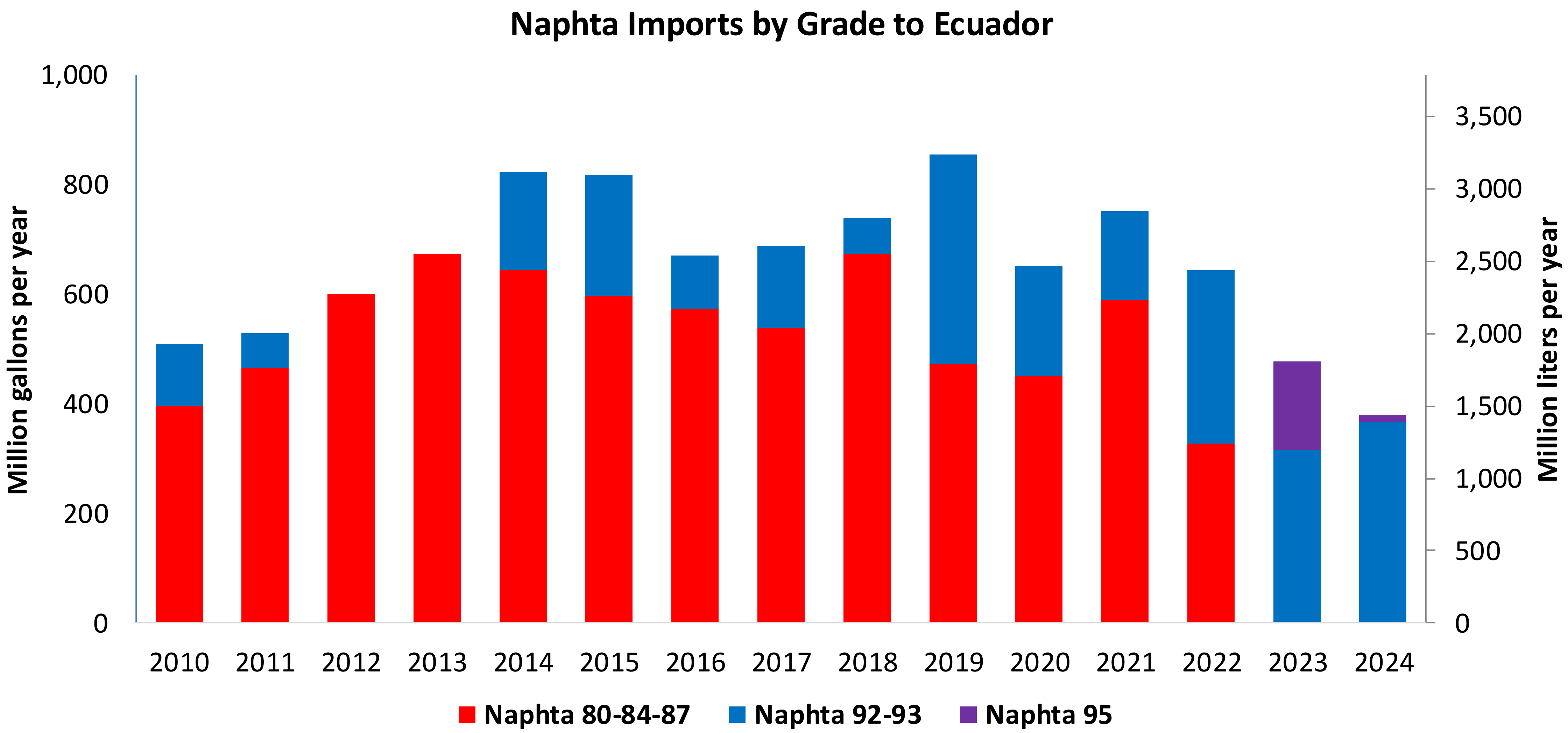
Source: EP PETROECUADOR, 2024

Gasoline Production in Ecuador



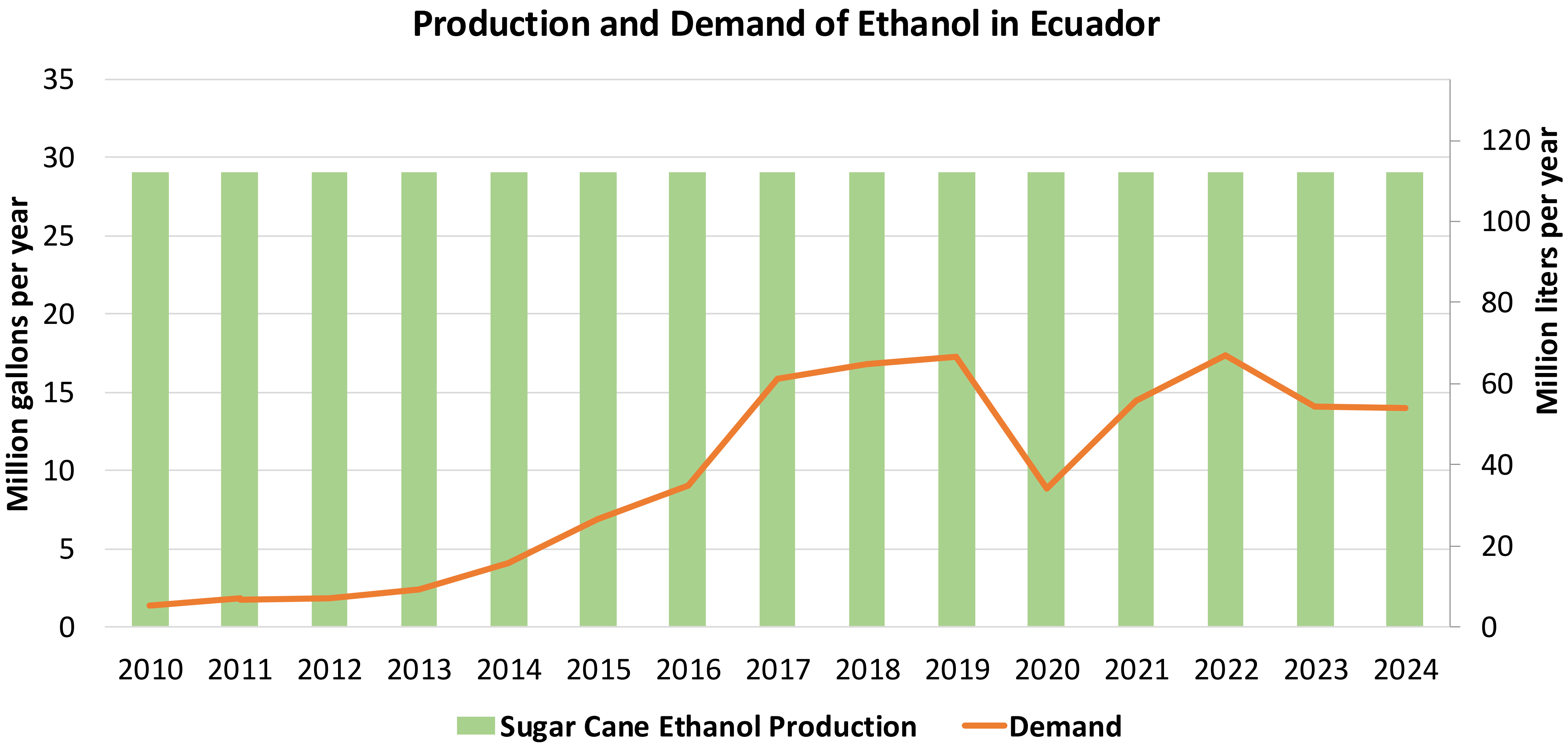
Source: EP PETROECUADOR, 2024

Gasoline Imports in Ecuador



Source: EP PETROECUADOR, 2024

Ethanol Balance in Ecuador



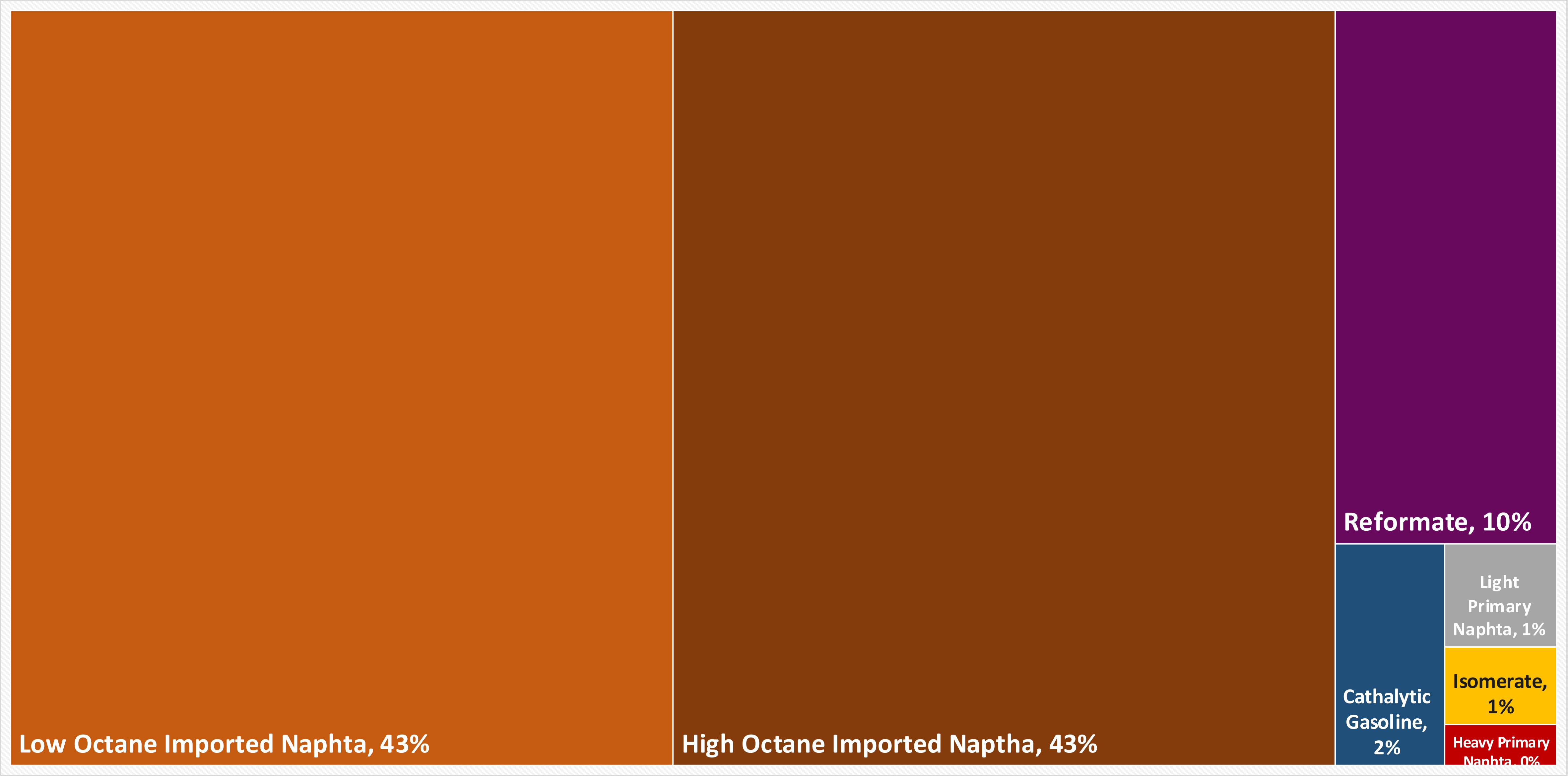
Source: EP PETROECUADOR, 2024; CEDRSSA, 2020

Gasoline Quality in Ecuador

Name	NTE INEN 935:2021				EN 228:2012 + A1:2017 (Euro 6 enabling)			
Implementation Date	2021				2017			
Applicability	Whole country	Whole country	Whole country	Whole country	All countries			
Selected Grade	RON 85	RON 89	RON 92	RON 93	RON 95 E5	RON 95 E10	RON 98 E5	RON 98 E10
Benzene Content	< 1 %v/v	< 2 %v/v	<2 %v/v	<1,3 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v
Aromatics	< 30 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v
Olefins	< 18 %v/v	< 25 %v/v	< 25 %v/v	< 25 %v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v
Lead Content	< 0 mg/l	< 0 mg/l	< 0 mg/l	< 0 mg/l	< 5 mg/l	< 5 mg/l	< 5 mg/l	< 5 mg/l
Manganese	0 mg/l	0 mg/l	0 mg/l	0 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l
RON	85	89	92	95	> 95	> 95	> 98	> 98
MON					> 85	> 88	> 85	> 88
AKI								
Sulfur Content	650 mg/kg	650 mg/kg	450 mg/kg	<300 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg
Oxygen Content	< 2,7 %m/m	< 2,7 %m/m	< 2,7 %m/m	< 2,7 %m/m	<2,7 % m/m	<3,7 % m/m	<2,7 % m/m	<3,7 % m/m
Ethanol (EtOH)					<5 %v/v	<10 %v/v	<5 %v/v	<10 %v/v
RVP 37.8°C (Summer)	<60 kPa	<60 kPa	<60 kPa	<62 kPa	<> 60 - 70 kPa *Depends on the country, RVP is regulated in the EU Fuel Quality Directive			
RVP 37.8 °C(Winter)								
RVP 37.8°C (Transition)								
MTBE					-	-	-	-
Ethers 5 or more C Atoms	-	-	-	-	Based on oxygen content	<22 %v/v	Based on oxygen content	<22 %v/v

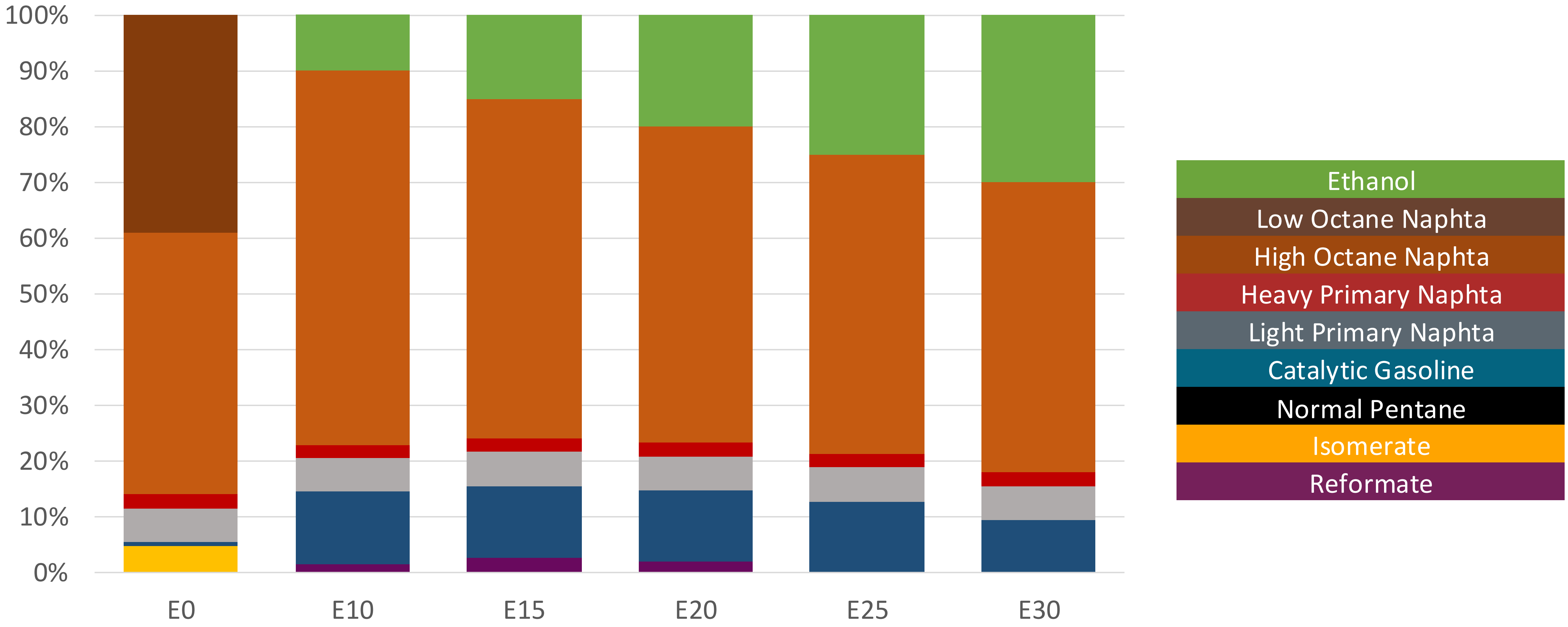
Source: Servicio Ecuatoriano de Normalización , 2024

Available Blending Components



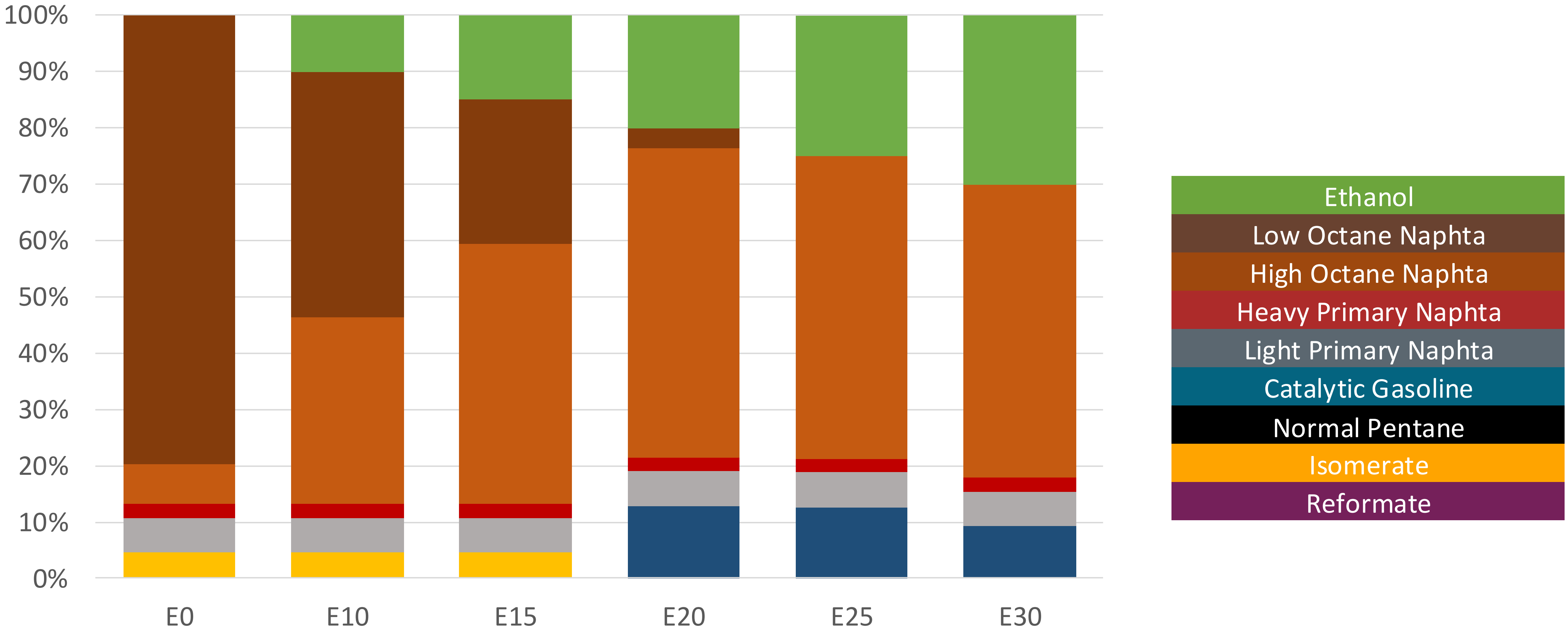
Source: Faro90

Ecuador –RON 85 Gasoline – Constant Octane



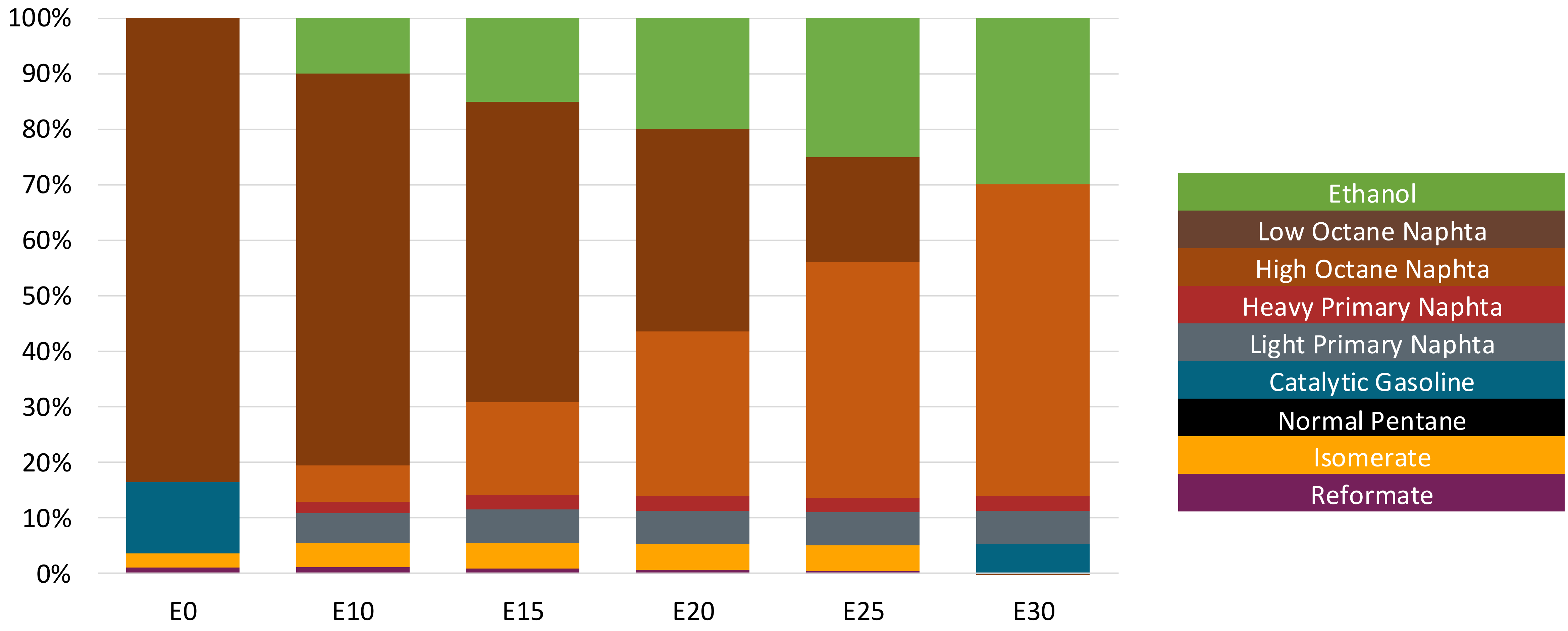
Octane (RON)	85.0	85.9	87.9	89.6	91.0	92.5
Price (US$/gal)	$ 2.386	$ 2.321	$ 2.310	$ 2.288	$ 2.258	$ 2.234

Ecuador – RON 89 Gasoline– Constant Octane



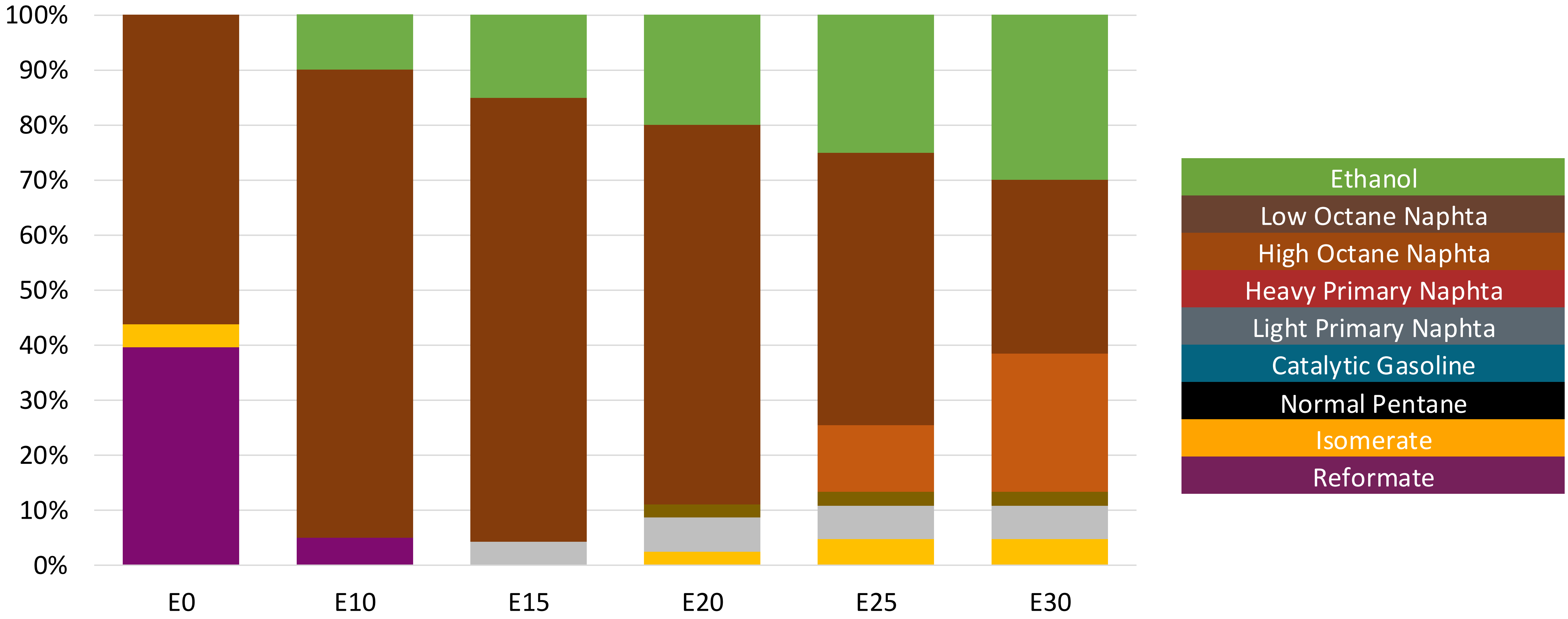
Octane (RON)	89.0	89.0	89.0	89.6	91.0	92.5
Price (US$/gal)	$ 2.453	$ 2.357	$ 2.310	$ 2.282	$ 2.258	$ 2.234

Ecuador – RON 92 Gasoline – Constant Octane



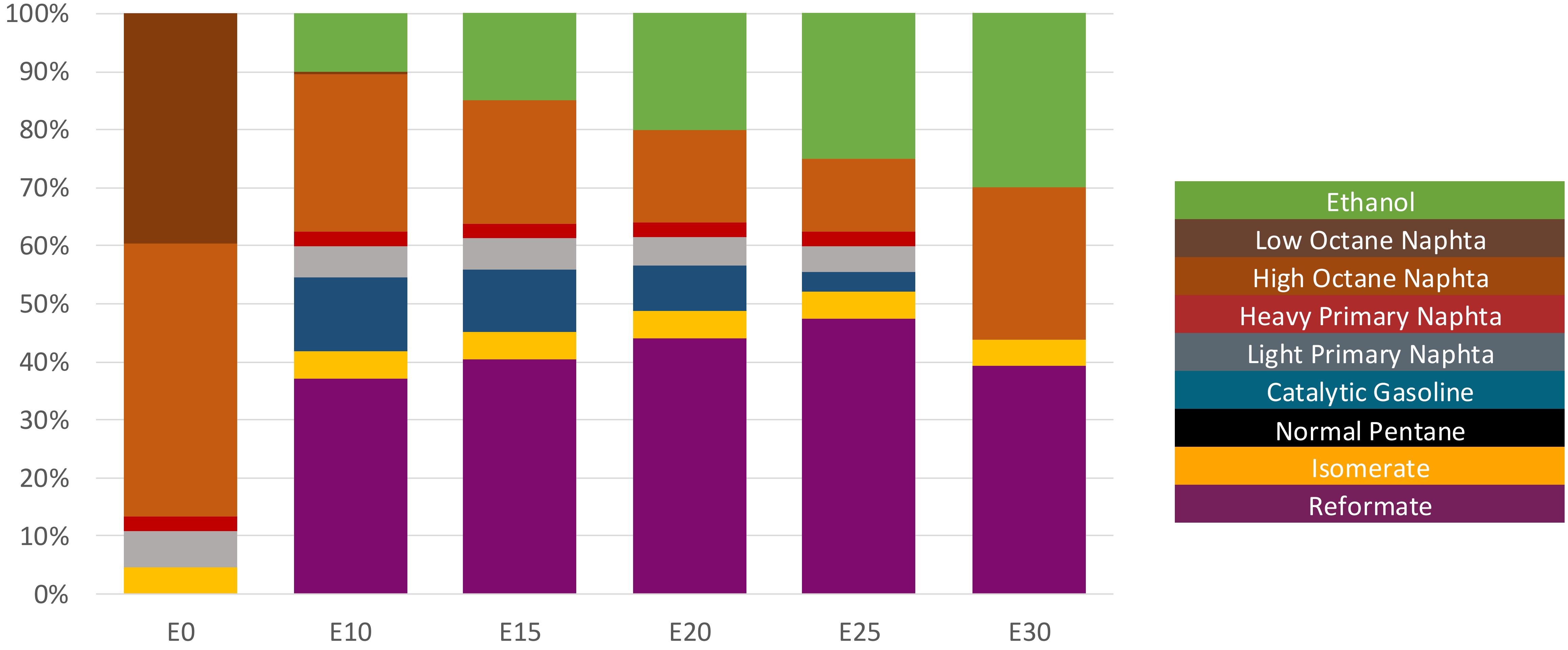
Octane (RON)	92.0	92.0	92.0	92.0	92.0	92.1
Price (US$/gal)	$ 2.584	$ 2.419	$ 2.362	$ 2.314	$ 2.266	$ 2.225

Ecuador – RON 95 Gasoline – Constant Octane



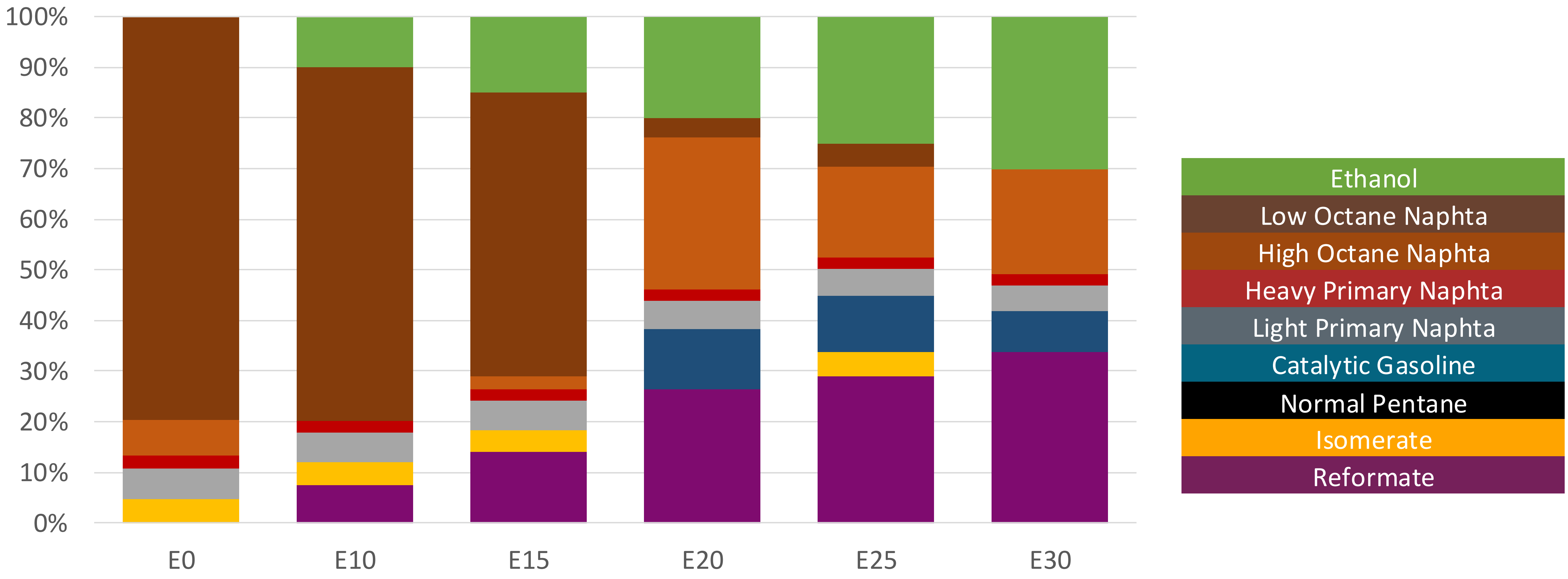
	E0	E10	E15	E20	E25	E30
Octane (RON)	95.0	95.0	95.0	95.0	95.0	95.0
Price (US$/gal)	$ 2.748	$ 2.548	$ 2.454	$ 2.364	$ 2.315	$ 2.267

Ecuador – RON 85 Gasoline – Increased Octane



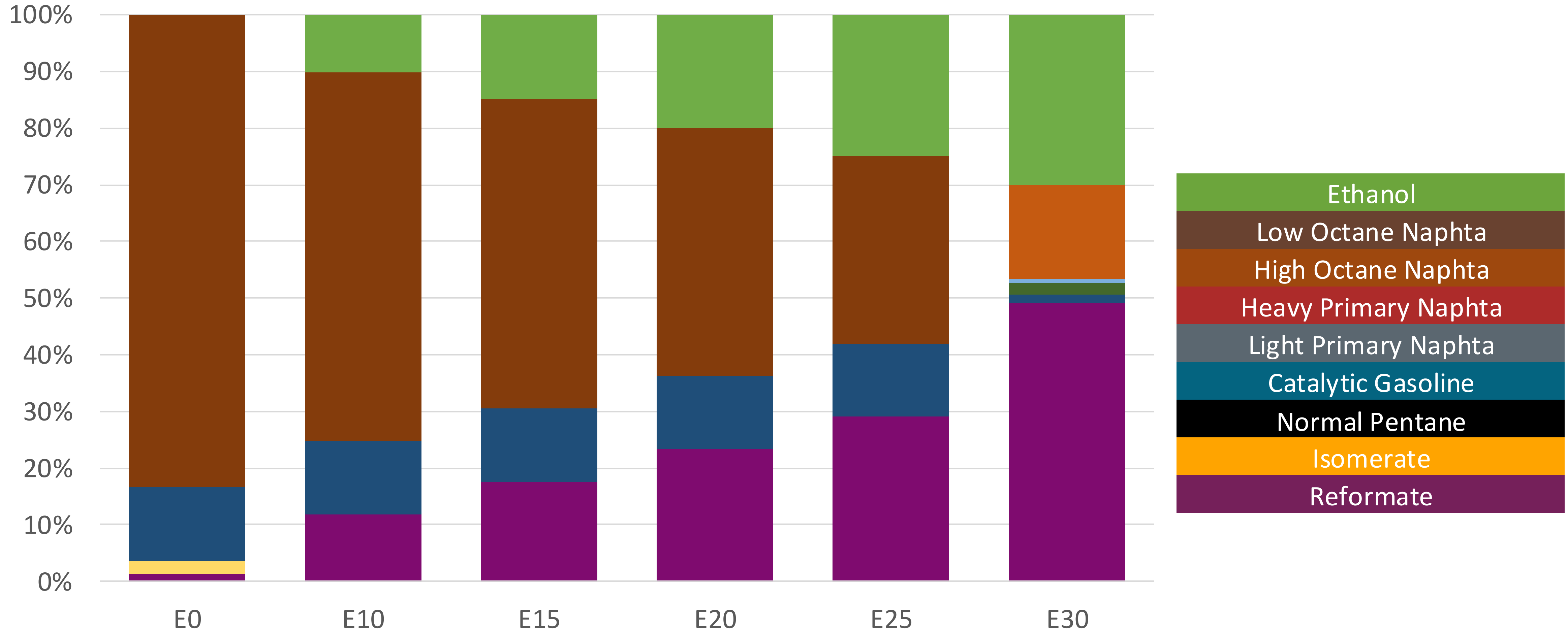
Octane (RON)	85.0	92.7	94.9	97.1	99.1	100.1
Price (US$/gal)	$ 2.386	$ 2.386	$ 2.386	$ 2.386	$ 2.386	$ 2.386

Ecuador – Gasoline RON 89 – Increased Octane



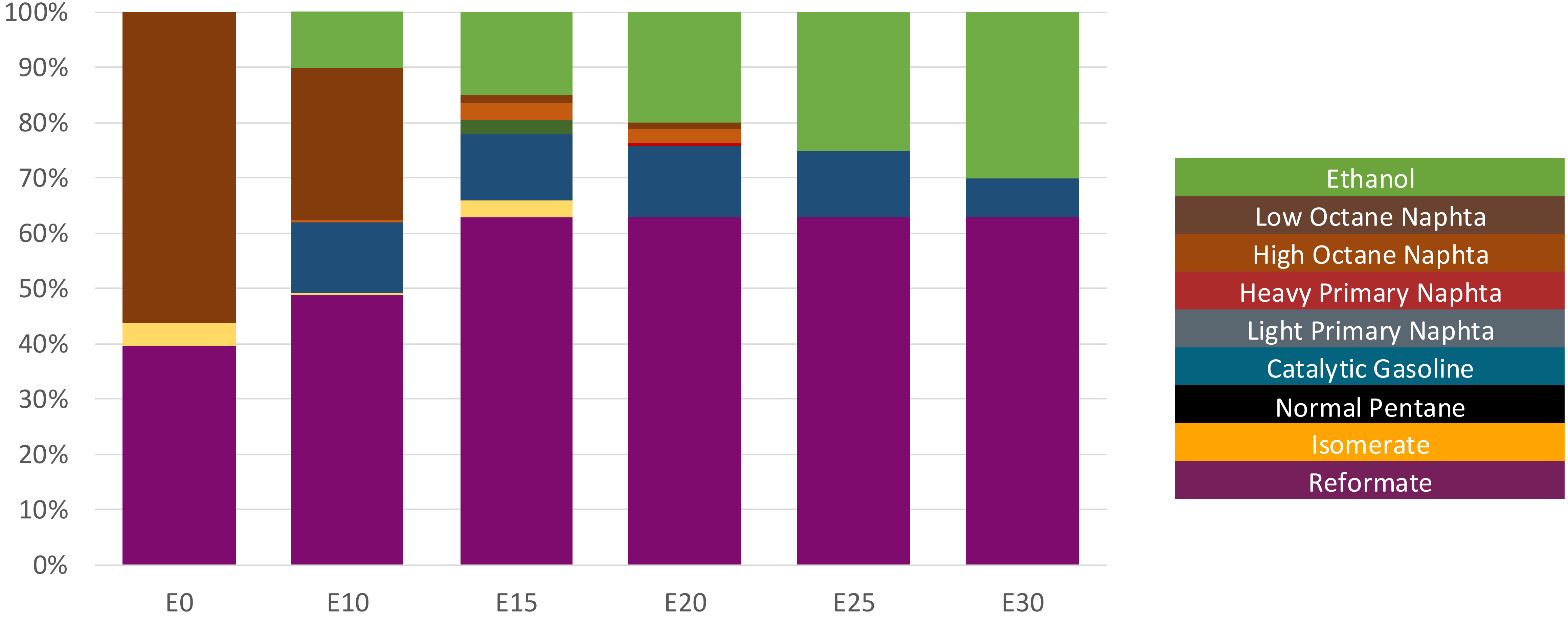
Octane (RON)	89.0	93.0	94.6	94.4	97.0	98.6
Price (US$/gal)	$ 2.453	$ 2.453	$ 2.453	$ 2.453	$ 2.453	$ 2.453

Ecuador – RON 92 Gasoline – Increased Octane



	E0	E10	E15	E20	E25	E30
Octane (RON)	92.0	95.5	97.3	99.1	100.8	101.4
Price (US$/gal)	$ 2.584	$ 2.584	$ 2.584	$ 2.584	$ 2.584	$ 2.584

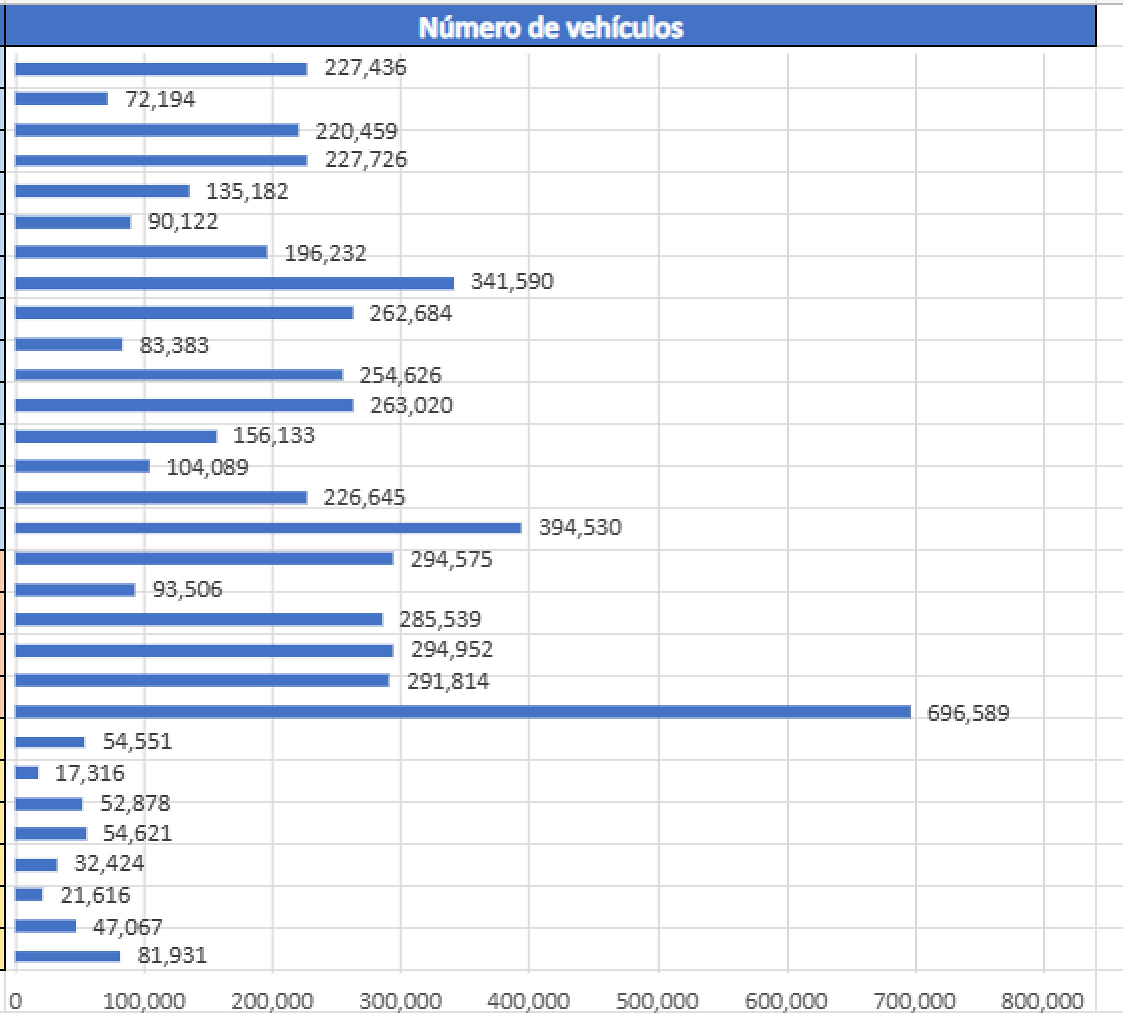
Ecuador – RON 95 Gasoline – increased Octane



	E0	E10	E15	E20	E25	E30
Octane (RON)	95.0	98.4	99.9	101.8	103.5	104.8
Price (US$/gal)	$ 2.748	$ 2.748	$ 2.748	$ 2.748	$ 2.748	$ 2.748

Gasoline Vehicle Fleet - Ecuador

Tipo	Motor	Escape	Recuperador	Edad
Automóviles ligeros	Multi FI-Pt	Euro V	PCV Tank	>4años, >80 mkm
				4-8 años, 80-160 mkm
				<8 años, < 160 mkm
		Euro IV		<8 años, < 160 mkm
		Euro III		<8 años, < 160 mkm
		3-Ways		<8 años, < 160 mkm
Automóviles medianos	Multi FI-Pt	Euro V	PCV Tank	>4años, >80 mkm
				4-8 años, 80-160 mkm
				<8 años, < 160 mkm
		Euro IV		<8 años, < 160 mkm
		Euro III		<8 años, < 160 mkm
		3-Ways		<8 años, < 160 mkm
Pequeños motores	FI 4-ciclos	Euro III	PCV	>4años, >80 mkm
				4-8 años, 80-160 mkm
		Euro II		4-8 años, 80-160 mkm
				<8 años, < 160 mkm
Camiones y pesados	FI	Euro V	PCV	>4años, >80 mkm
				4-8 años, 80-160 mkm
				<8 años, < 160 mkm
		Euro IV		<8 años, < 160 mkm
		Euro III		<8 años, < 160 mkm
		3-Ways		<8 años, < 160 mkm
			Total	**3,411,980**

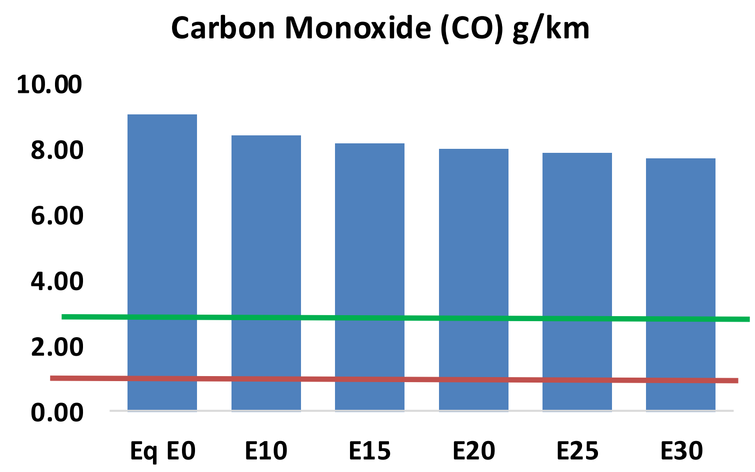


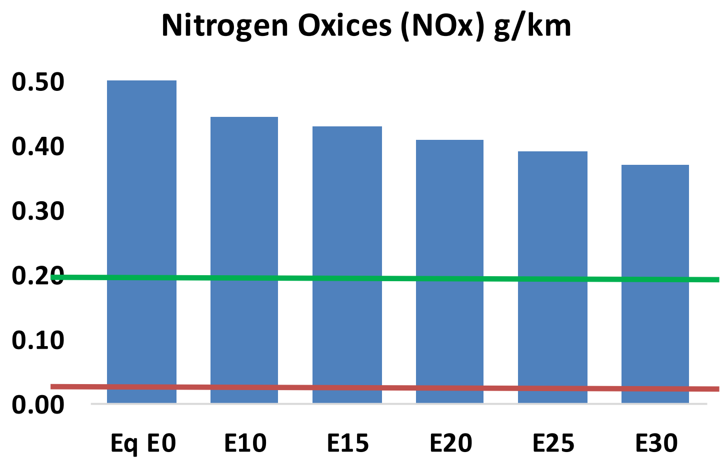
Vehicle Fleet: **3,411,980**
Age: 13 years

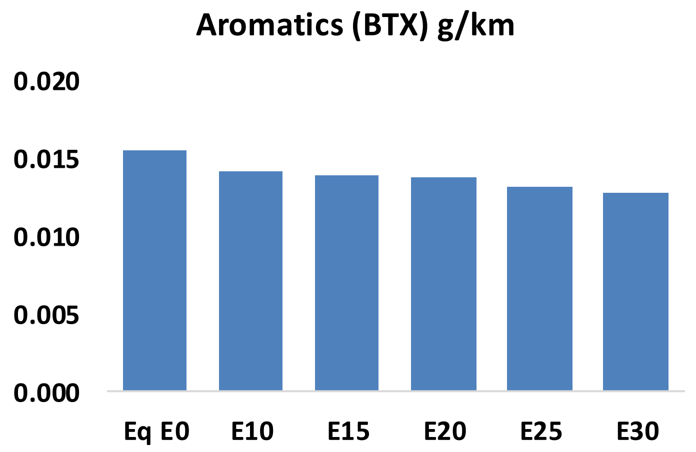
Source: CINAE, 2023; analysis de Faro90

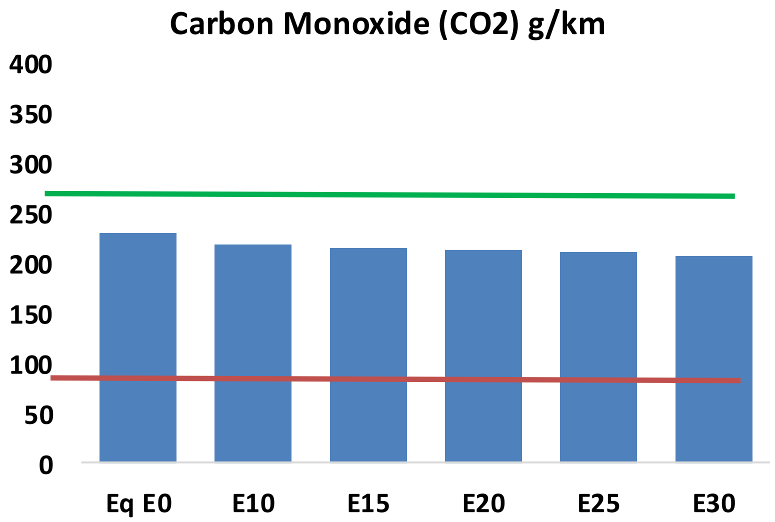
Ecuador – Vehicular Emissions

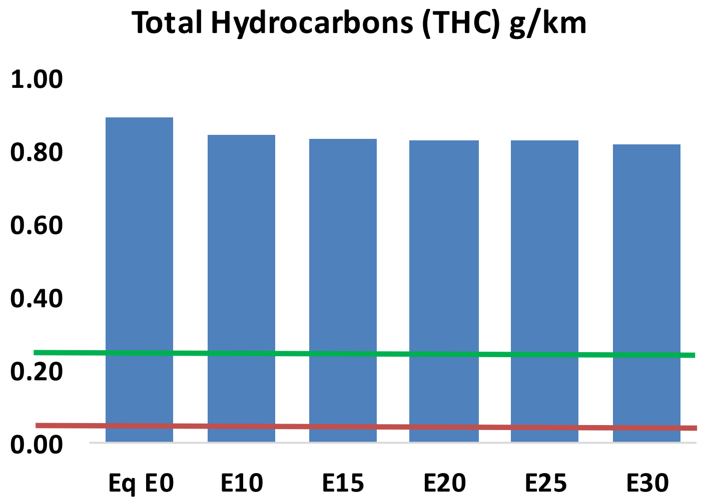
United States
Euro 6

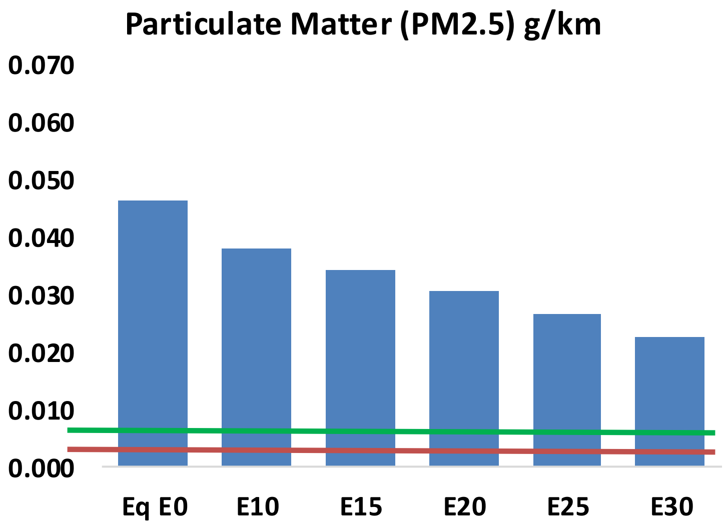












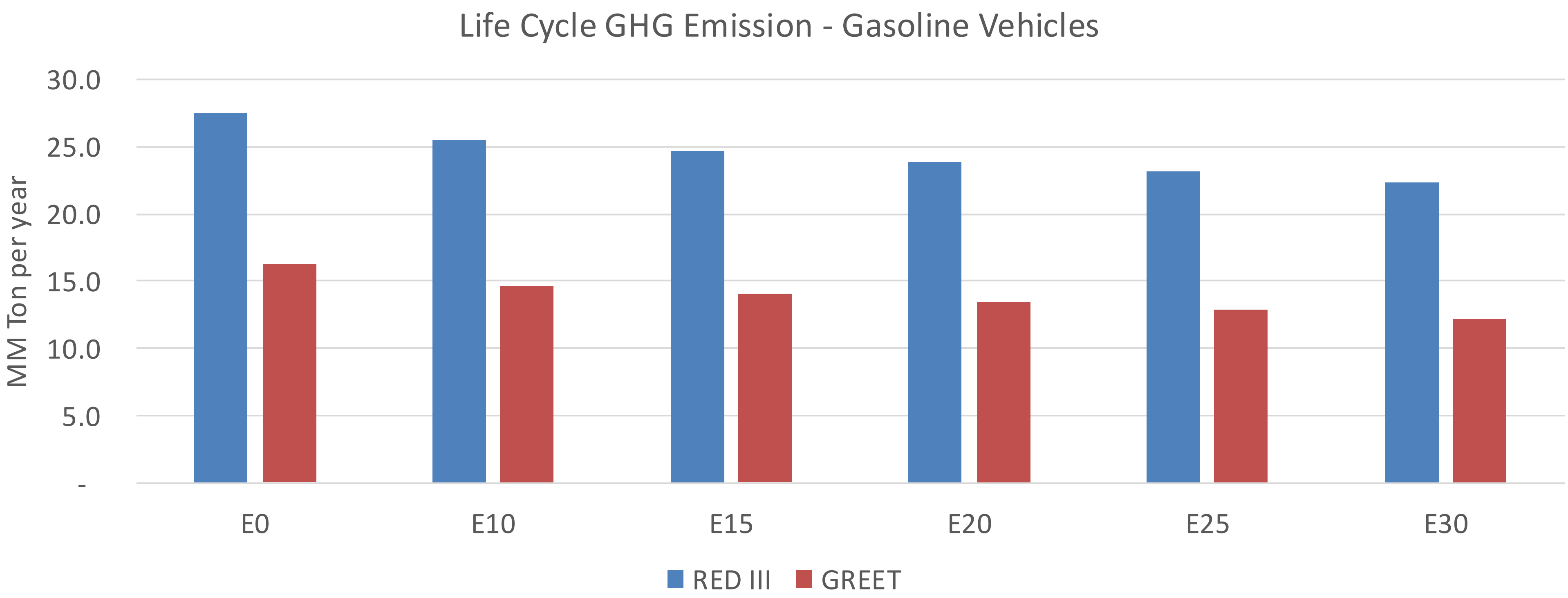
Source: Faro 90

Ecuador – Gasoline Vehicle Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	870,841	838,578	806,315	786,644	769,963	758,077	738,897	-7%	-12%
CO2	22,073,764	21,506,153	20,970,076	20,548,467	20,339,838	20,243,798	19,870,646	-5%	-8%
VOC	85,477	83,277	81,077	80,128	79,565	79,250	78,413	-5%	-7%
NOx	48,947	45,194	42,584	41,235	39,084	37,388	35,397	-13%	-20%
SOx	265	245	225	208	192	174	155	-15%	-28%
NH3	6,263	6,201	6,139	6,168	6,238	6,287	6,327	-2%	0%
N2O	824	819	816	820	837	846	856	-1%	2%
CH4	19,548	19,548	19,548	19,841	20,271	20,643	20,995	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	695	677	659	652	647	645	638	-5%	-7%
Acetaldehyde	1,415	1,624	2,383	3,628	4,943	5,649	6,674	68%	249%
Formaldehyde	5,271	5,124	5,974	7,014	7,348	8,129	8,850	13%	39%
Benzene	1,491	1,433	1,357	1,336	1,325	1,267	1,226	-9%	-11%
PM2.5	1,483	1,348	1,213	1,094	978	854	724	-18%	-34%
PM10	2,943	2,675	2,408	2,172	1,942	1,695	1,437	-18%	-34%

Source: Faro90

Ecuador – Gasoline Vehicle Fleet GEI Emissions



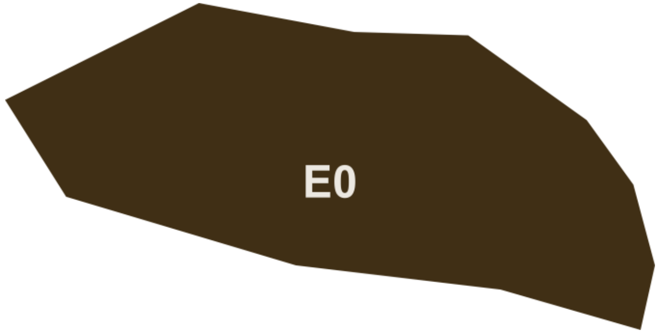
Country Emissions 2024 (MMT/year)	101.4
2035 Target (MMT/year)	71.0
Delta	30.4

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	10.1%	13.7%	17.5%	20.7%	24.8%
RED II/III	0.0%	7.5%	10.4%	13.2%	15.8%	18.8%

% 2035 Target	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	5.4%	7.3%	9.3%	11.1%	13.3%
RED II/III	0.0%	6.8%	9.4%	12.0%	14.3%	17.0%

Source: Greet, RED II/III, climateactiontracker.org, F90

Ethanol Blending in Gasoline - El Salvador

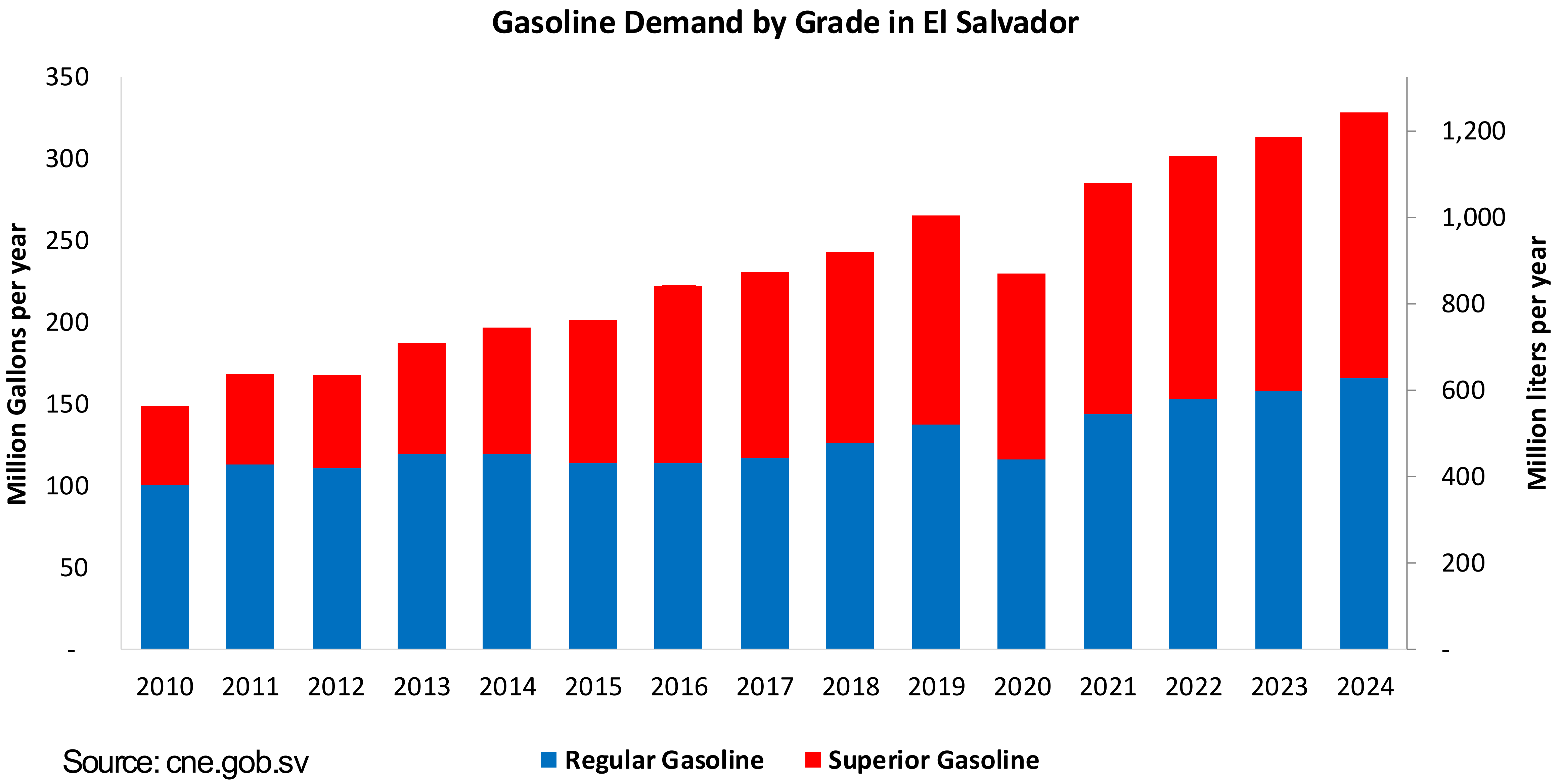


In 2024, gasoline demand was 328 million gallons (1,243 million liters). Regular gasoline RON 91 represented 51% of the volume consumed while Superior gasoline RON 95 reached 49%. Gasoline is supplied only by imports, distributed mainly by Chevron, Puma and Uno companies and imported mainly from the United States.

El Salvador has no national mandate for ethanol. It has the potential to produce ethanol from maize and sugar cane. In the last year there has been an interest for its use.

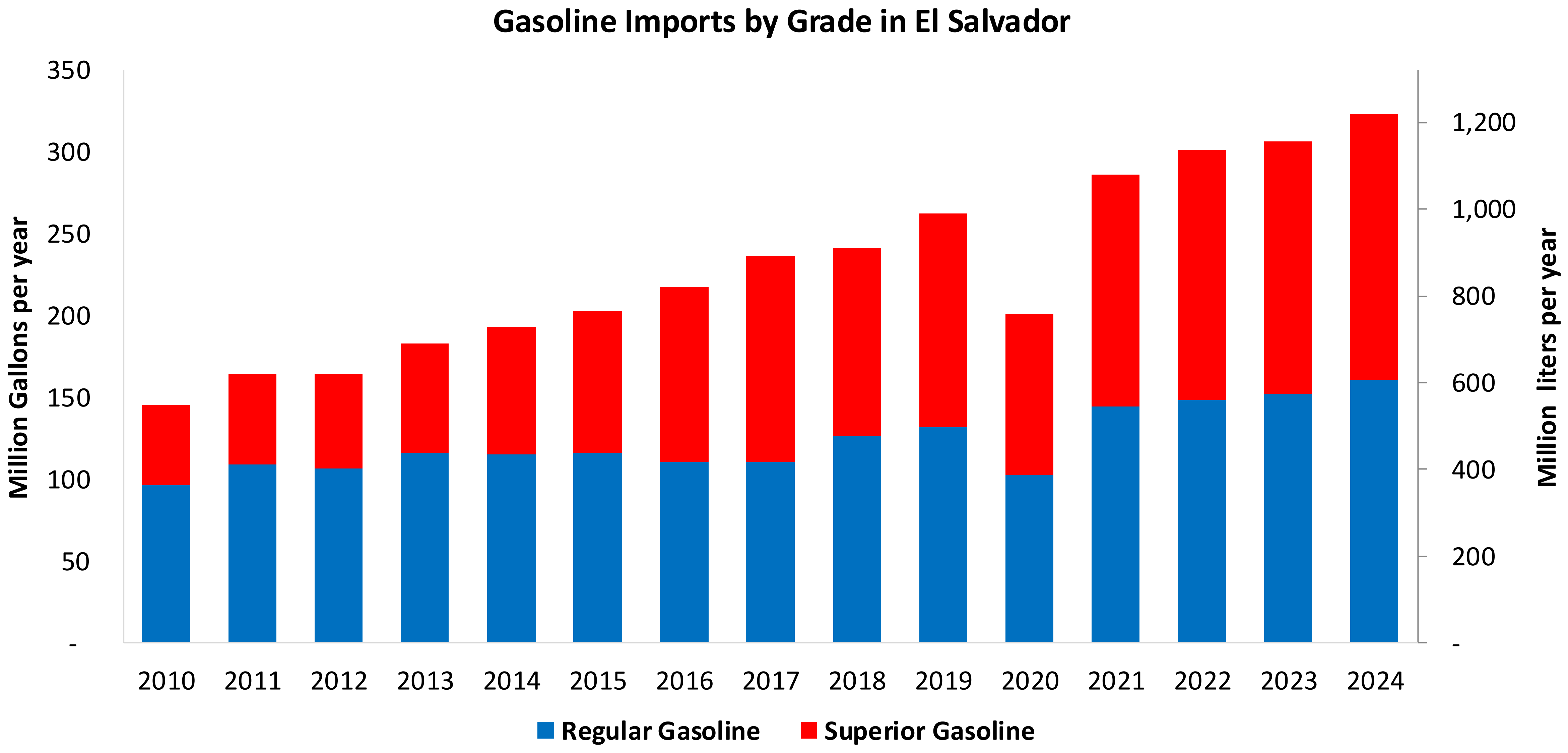
Sources: cne.gob.sv

Gasoline Demand in El Salvador



Source: cne.gob.sv

Gasoline Imports in El Salvador



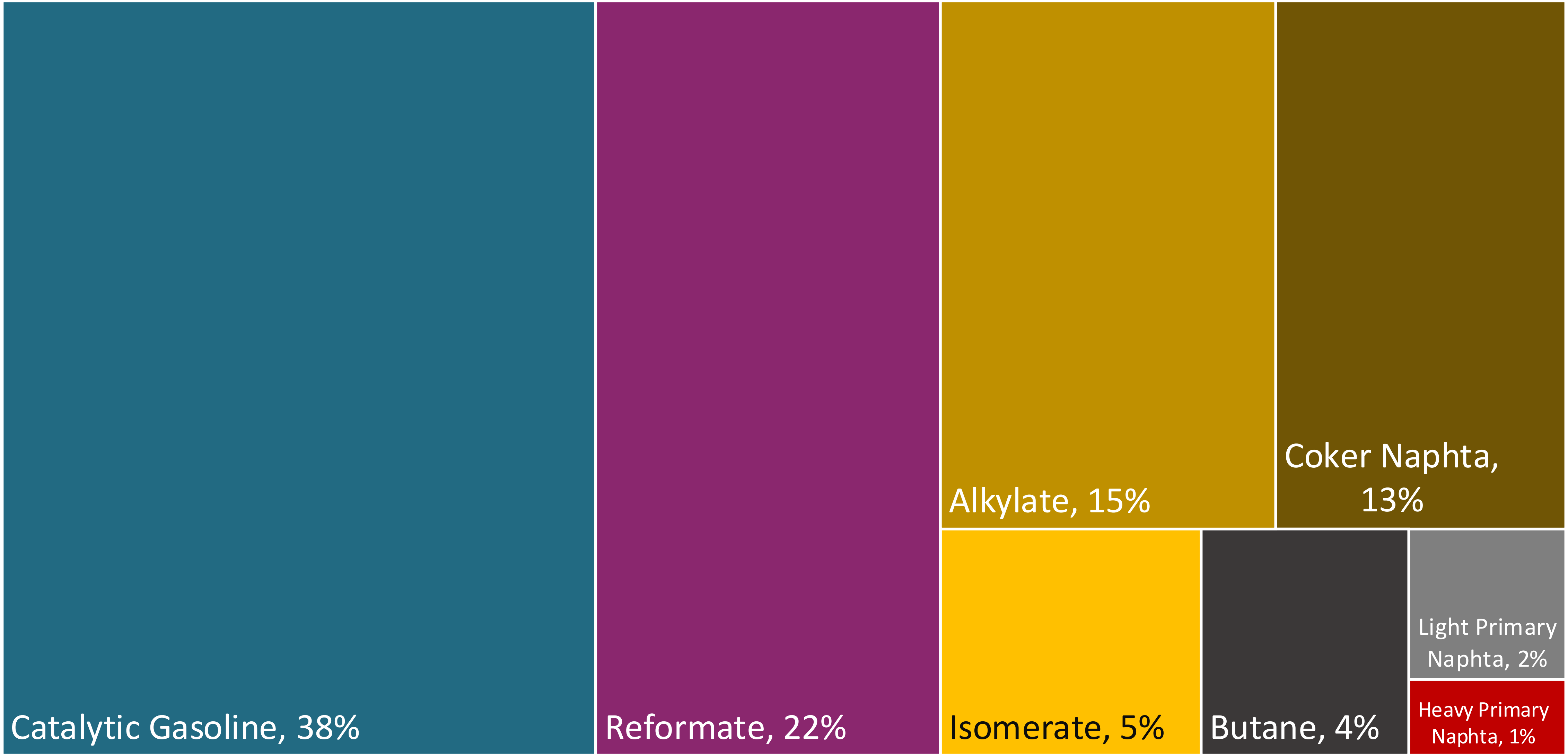
Source: cne.gob.sv

Gasoline Quality in El Salvador

Name	RTCA 75.01.19:19	RTCA 75.01.20:19	EN 228:2012 + A1:2017 (Euro 6 enabling)			
Implementation Date	2021	2021	2017			
Applicability	Whole country	Whole country	All countries			
Selected Grade	Gasoline Regular	Gasoline Premium	RON 95 E5	RON 95 E10	RON 98 E5	RON 98 E10
Benzene Content	5% v/v	5% v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v
Aromatics	50% v/v	50% v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v
Olefins	30% v/v	30% v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v
Lead Content	< 0,013 g/l	< 0,013 g/l	< 5 mg/l	< 5 mg/l	< 5 mg/l	< 5 mg/l
Manganese	2,0% v/v	2,0% v/v	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l
RON	> 91	> 95	> 95	> 95	> 98	> 98
MON	-	-	> 85	> 88	> 85	> 88
AKI						
Sulfur Content	< 500 mg/kg	< 500 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg
Oxygen Content	0,7% v/v	0,7% v/v	<2,7 % m/m	<3,7 % m/m	<2,7 % m/m	<3,7 % m/m
Ethanol (EtOH)	-	-	<5 %v/v	<10 %v/v	<5 %v/v	<10 %v/v
RVP 37.8°C (Summer)	< 69 kPa	< 69 kPa	<> 60 - 70 kPa *Depends on the country, RVP is regulated in the EU Fuel Quality Directive			
RVP 37.8 °C(Winter)						
RVP 37.8°C (Transition)						
MTBE	-	-	-	-	-	-
Ethers 5 or more C Atoms	-	-	Based on oxygen content	<22 %v/v	Based on oxygen content	<22 %v/v

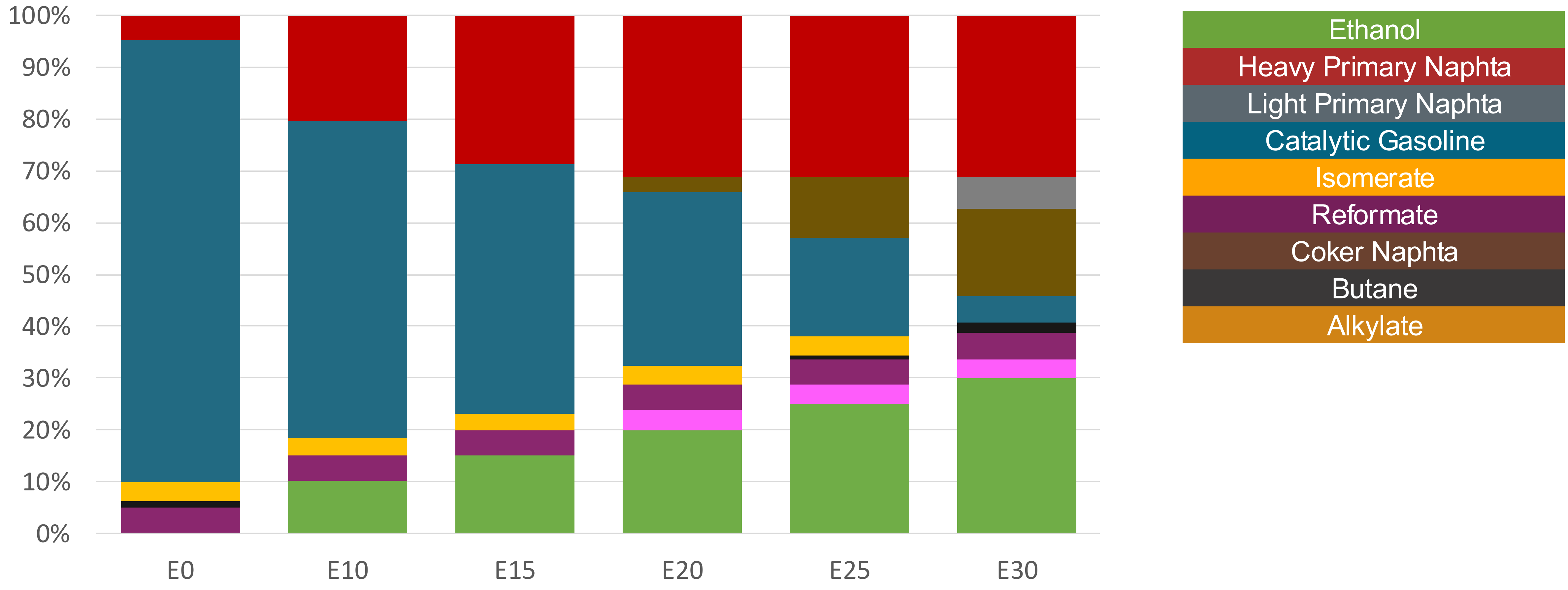
Source: RTCA

Blending Components- El Salvador



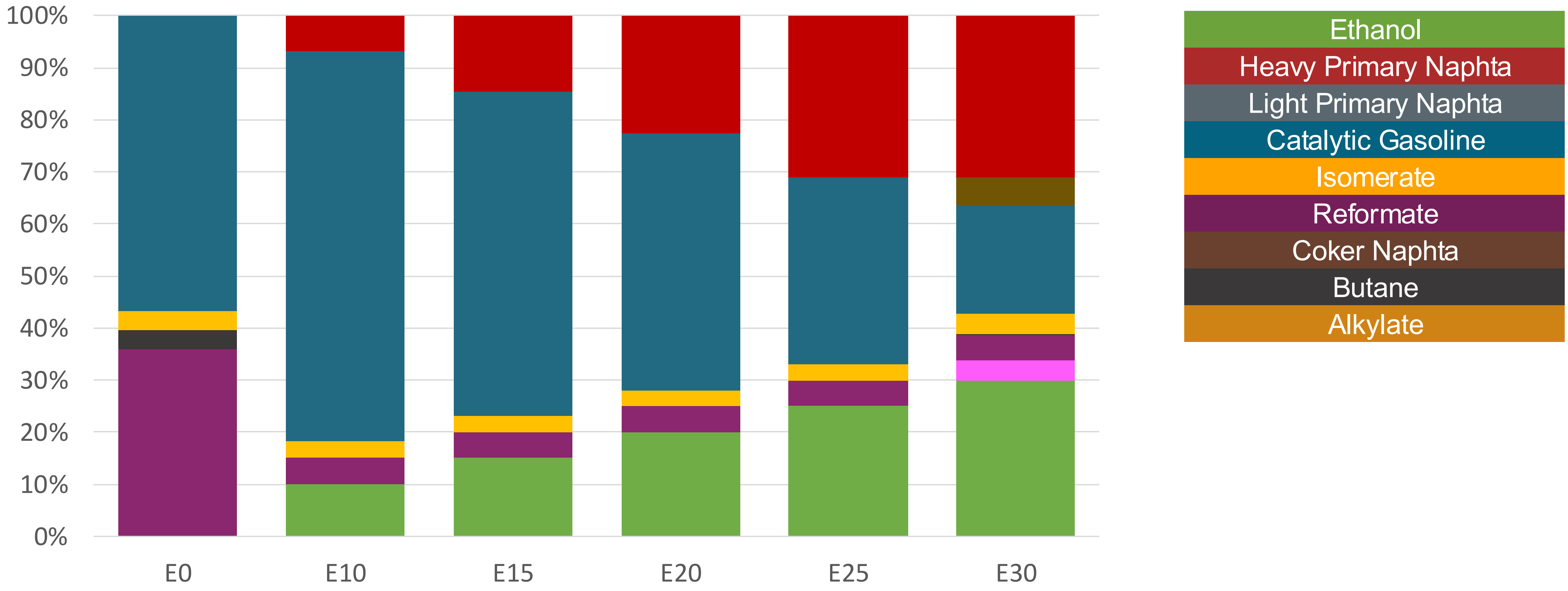
Source: Faro90

El Salvador – Regular – Constant Octane



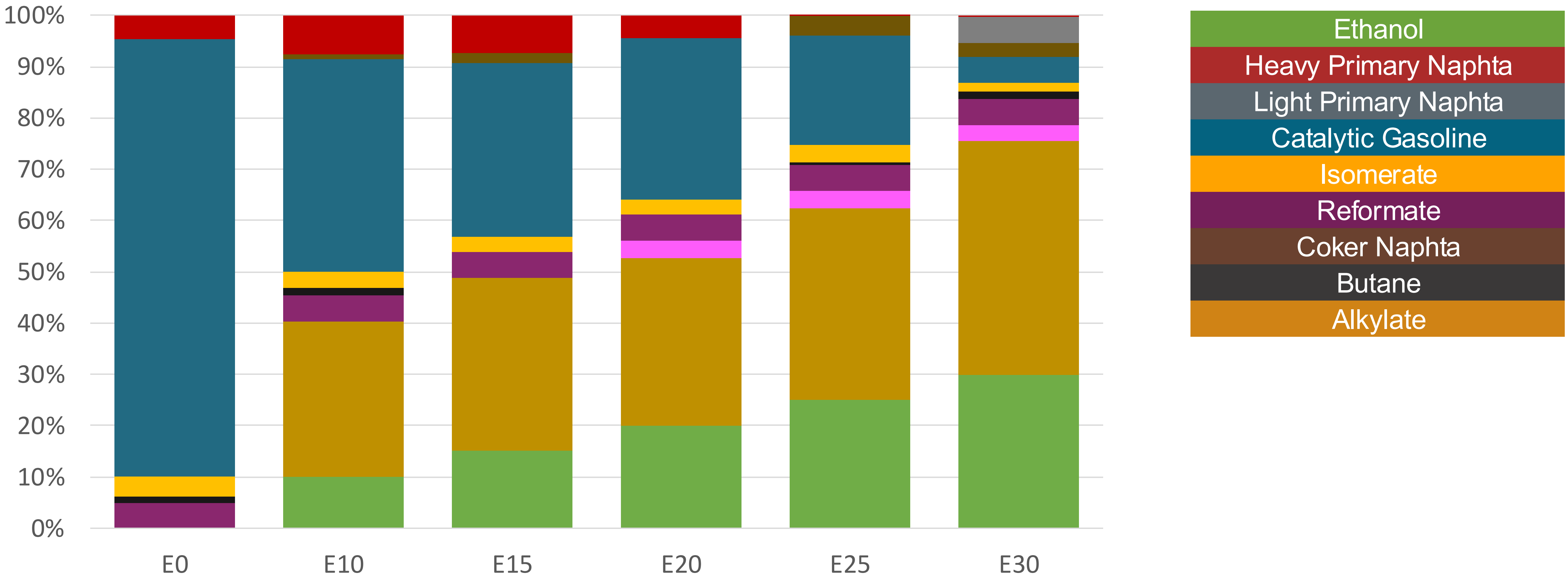
Octane (RON)	91.0	91.0	91.0	91.0	91.0	91.0
Price (USD/gal)	$ 2.336	$. 2.078	$. 1.934	$. 1.787	$. 1.636	$. 1.518

El Salvador – Premium – Constant Octane



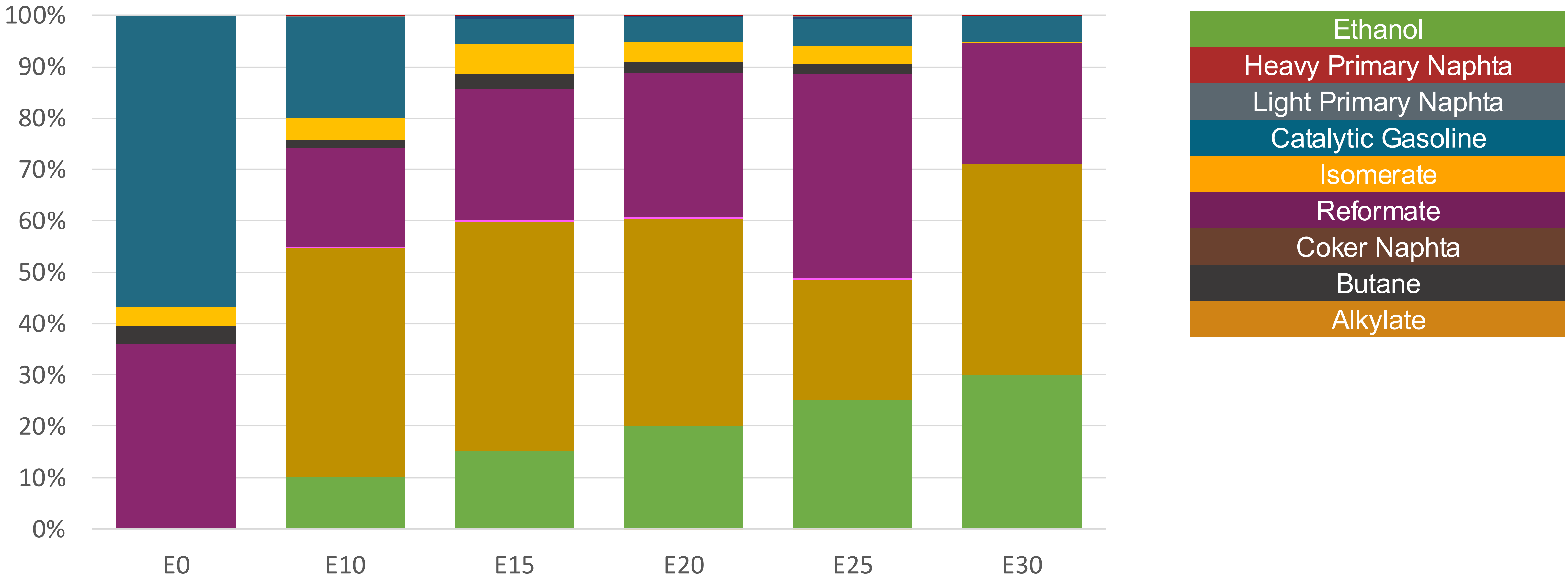
Octane (RON)	95.0	95.0	95.0	95.0	95.0	95.0
Price (USD/gal)	$ 2.586	$ 2.283	$ 2.149	$ 2.007	$ 1.857	$ 1.709

El Salvador – Regular – Increased Octane



	E0	E10	E15	E20	E25	E30
Octane (RON)	91.0	94.6	96.6	99.1	101.2	102.4
Price (USD/gal)	$ 2.336	$ 2.336	$ 2.336	$ 2.336	$ 2.336	$ 2.336

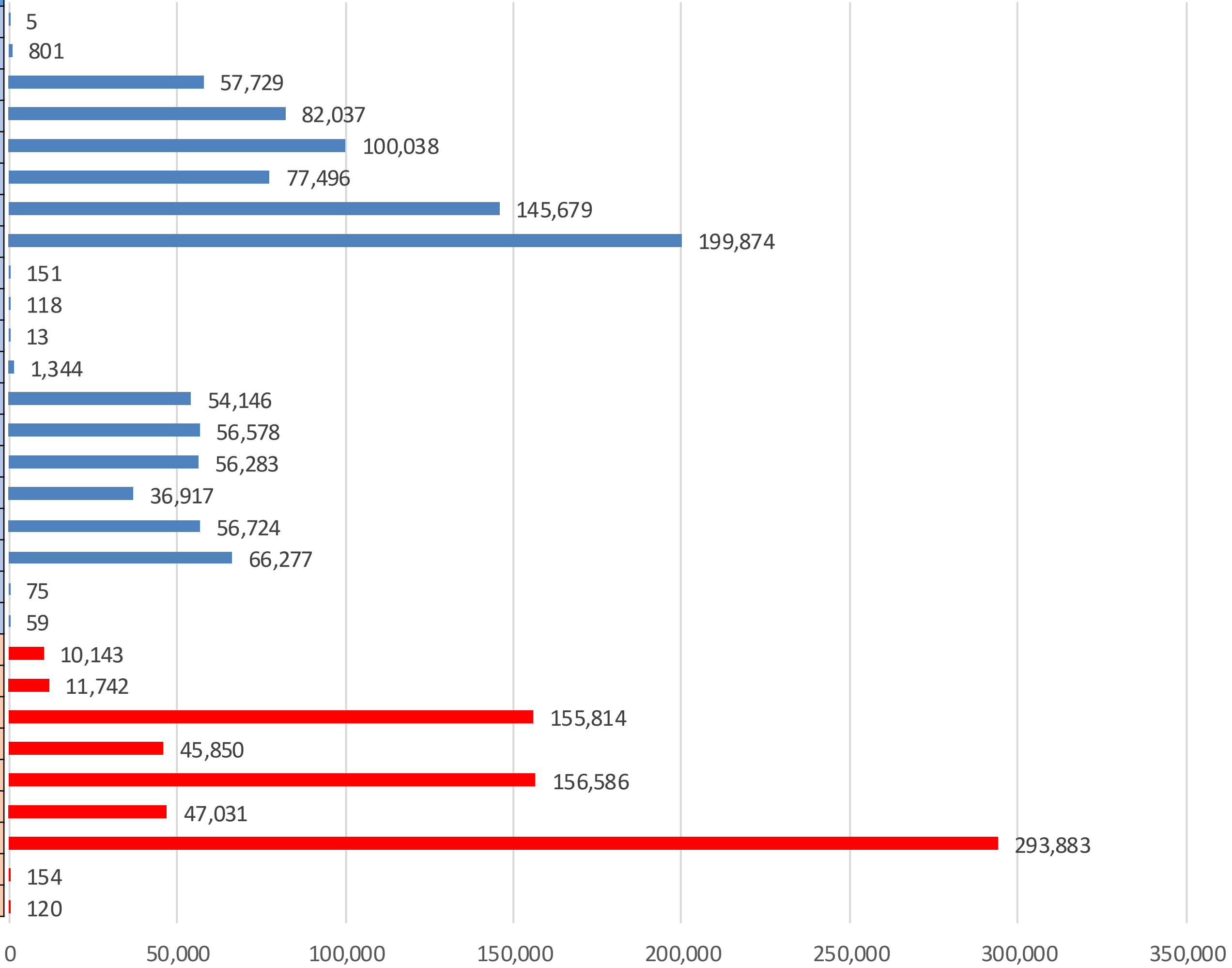
El Salvador – Premium – Increased Octane



Octane (RON)	95.0	98.0	100.3	102.6	104.9	105.8
Price (USD/gal)	$ 2.586	$ 2.586	$ 2.586	$ 2.586	$ 2.586	$ 2.586

Gasoline Vehicular Fleet in El Salvador

Type	Motor	Exhaust	Recuperator	Age
Auto/Light	Carburator	3-Ways	PCV Tank	<8 years, < 160 mkm
	Single FI-Pt	3-Ways		<8 years, < 160 mkm
	Multi FI-Pt	3-Ways		<8 years, < 160 mkm
		Euro III		<8 years, < 160 mkm
		Euro IV		<8 years, < 160 mkm
		Euro V / Euro 6		<8 years, < 160 mkm
				4-8 years, 80-160 mkm
				>4years, >80 mkm
	Hybrid			All
	Electric			All
Auto/Medium	Carburator	3-Ways	PCV Tank	<8 years, < 160 mkm
	Single FI-Pt	3-Ways		<8 years, < 160 mkm
	Multi FI-Pt	3-Ways		<8 years, < 160 mkm
		Euro III		<8 years, < 160 mkm
		Euro IV		<8 years, < 160 mkm
		Euro V / Euro 6		<8 years, < 160 mkm
				4-8 years, 80-160 mkm
				>4years, >80 mkm
	Hybrid			All
	Electric			All
Small Engines	FI 4-cycle	None	PCV	<8 years, < 160 mkm
		Euro II		<8 years, < 160 mkm
		Euro III		<8 years, < 160 mkm
		Euro IV		<8 years, < 160 mkm
				4-8 years, 80-160 mkm
		Euro V		4-8 years, 80-160 mkm
				>4years, >80 mkm
	Hybrid			All
	Electric			All

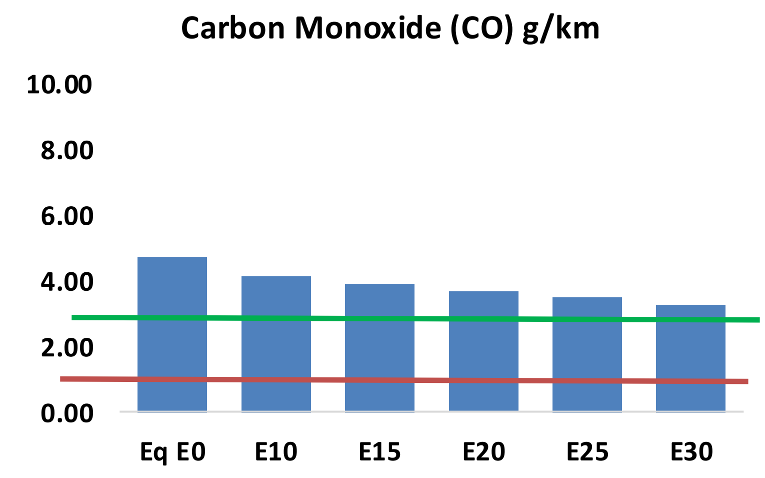


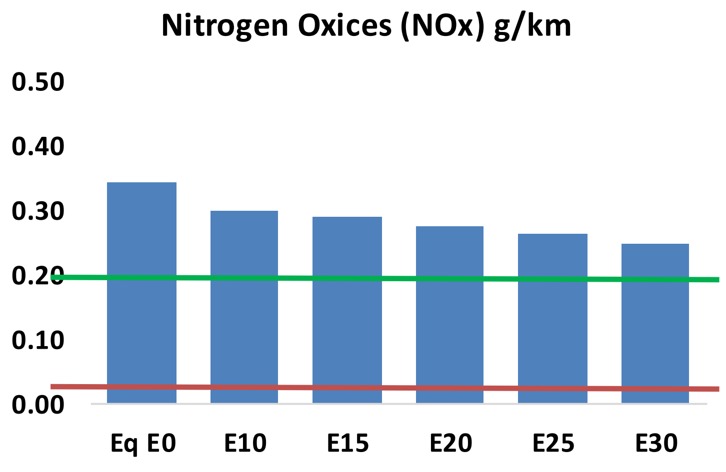
Vehicular Fleet: **1,712,991**
Average Age: **9 years**
Motorcycles: **33%**

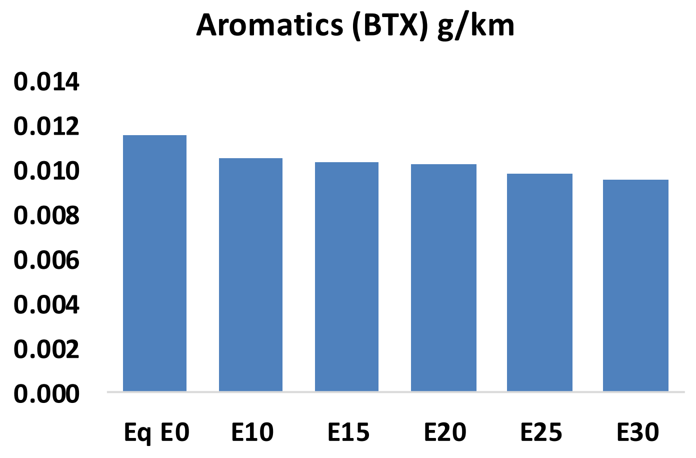
Source: Registro Publico, Faro 90

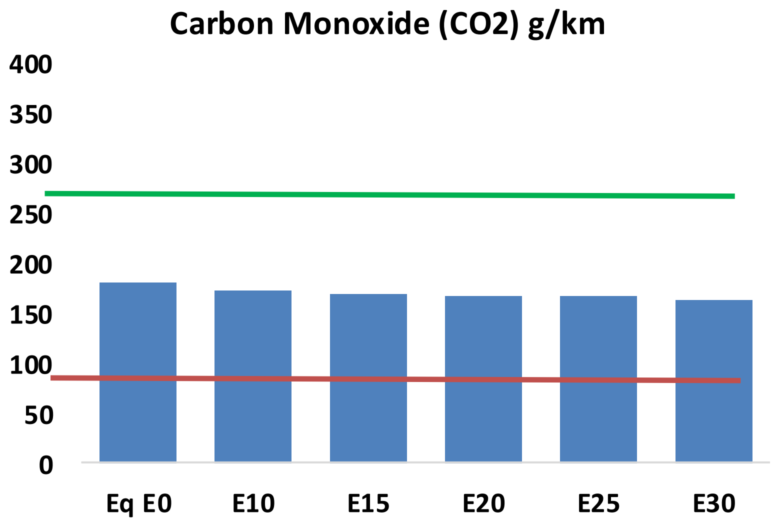
El Salvador - Vehicular Emissions

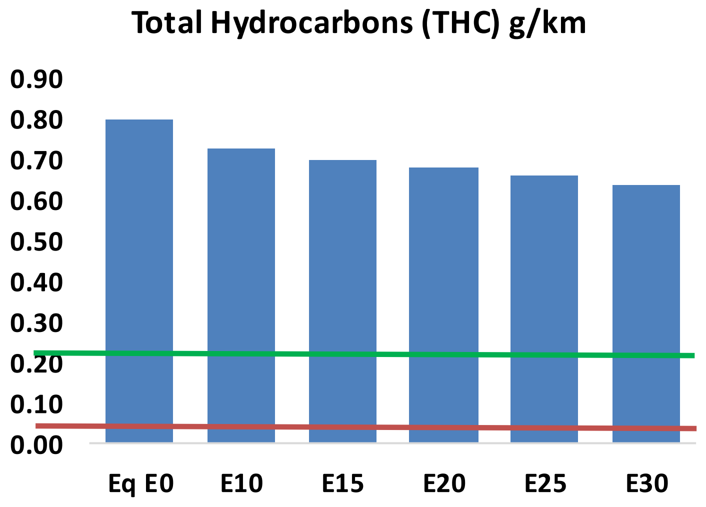
United States
Euro 6

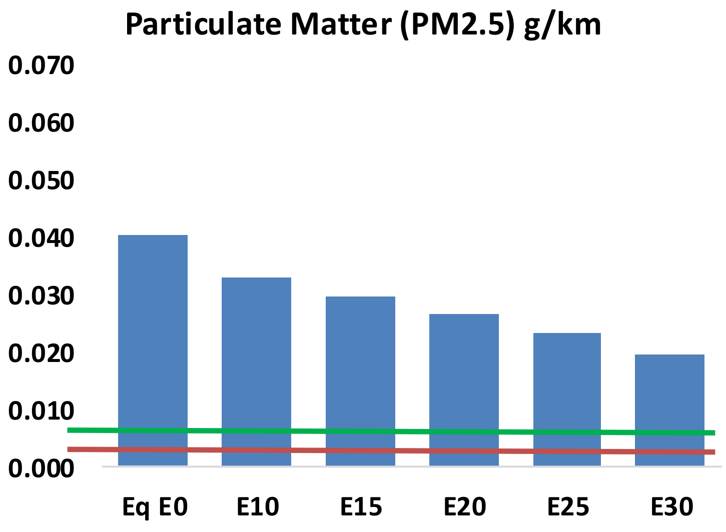












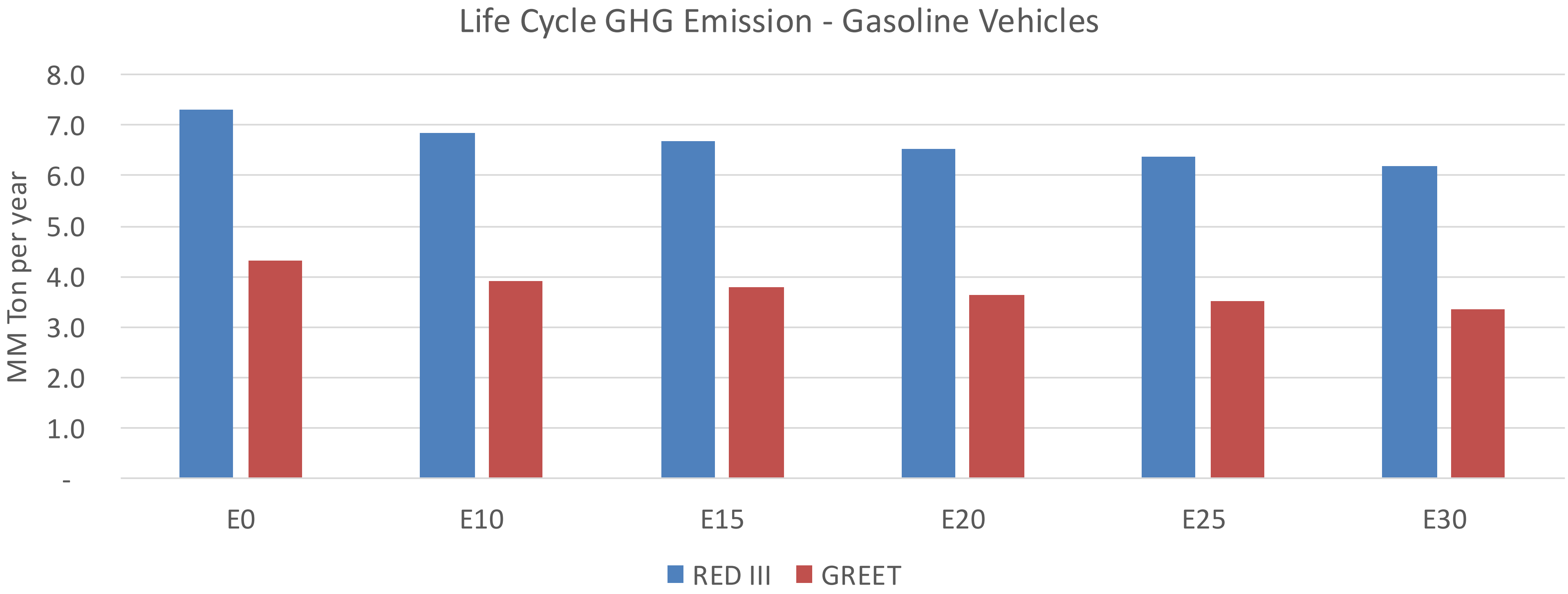
Source: Faro90

El Salvador - Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	159,777	149,748	139,718	131,939	124,313	118,510	111,443	-13%	-22%
CO2	6,127,680	5,970,111	5,821,296	5,704,257	5,646,342	5,621,675	5,518,052	-5%	-8%
VOC	26,993	25,768	24,542	23,704	22,937	22,370	21,532	-9%	-15%
NOx	11,643	10,750	10,130	9,811	9,300	8,897	8,425	-13%	-20%
SOx	70	64	59	55	50	46	41	-15%	-28%
NH3	2,338	2,315	2,292	2,303	2,329	2,347	2,362	-2%	0%
N2O	74	74	73	74	75	76	77	-1%	2%
CH4	6,335	6,335	6,335	6,430	6,569	6,689	6,803	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	132	124	116	110	104	100	93	-12%	-21%
Acetaldehyde	369	423	621	946	1,289	1,473	1,740	68%	249%
Formaldehyde	1,456	1,415	1,650	1,937	2,029	2,245	2,444	13%	39%
Benzene	392	377	357	352	349	333	323	-9%	-11%
PM2.5	376	342	307	277	248	216	183	-18%	-34%
PM10	980	891	802	723	647	564	478	-18%	-34%

Source: Faro90

El Salvador – Gasoline Vehicle Fleet GEI Emissions



Country Emissions 2024 (MMT/year)	14.8
2035 Target (MMT/year)	10.3
Delta	4.4

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	9.3%	12.6%	15.9%	18.7%	22.4%
RED II/III	0.0%	6.2%	8.5%	10.7%	12.7%	15.1%

% 2035 Target	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	9.0%	12.2%	15.5%	18.2%	21.8%
RED II/III	0.0%	10.3%	14.0%	17.7%	20.9%	24.9%

Source: Greet, RED II/III, climateactiontracker.org, F90

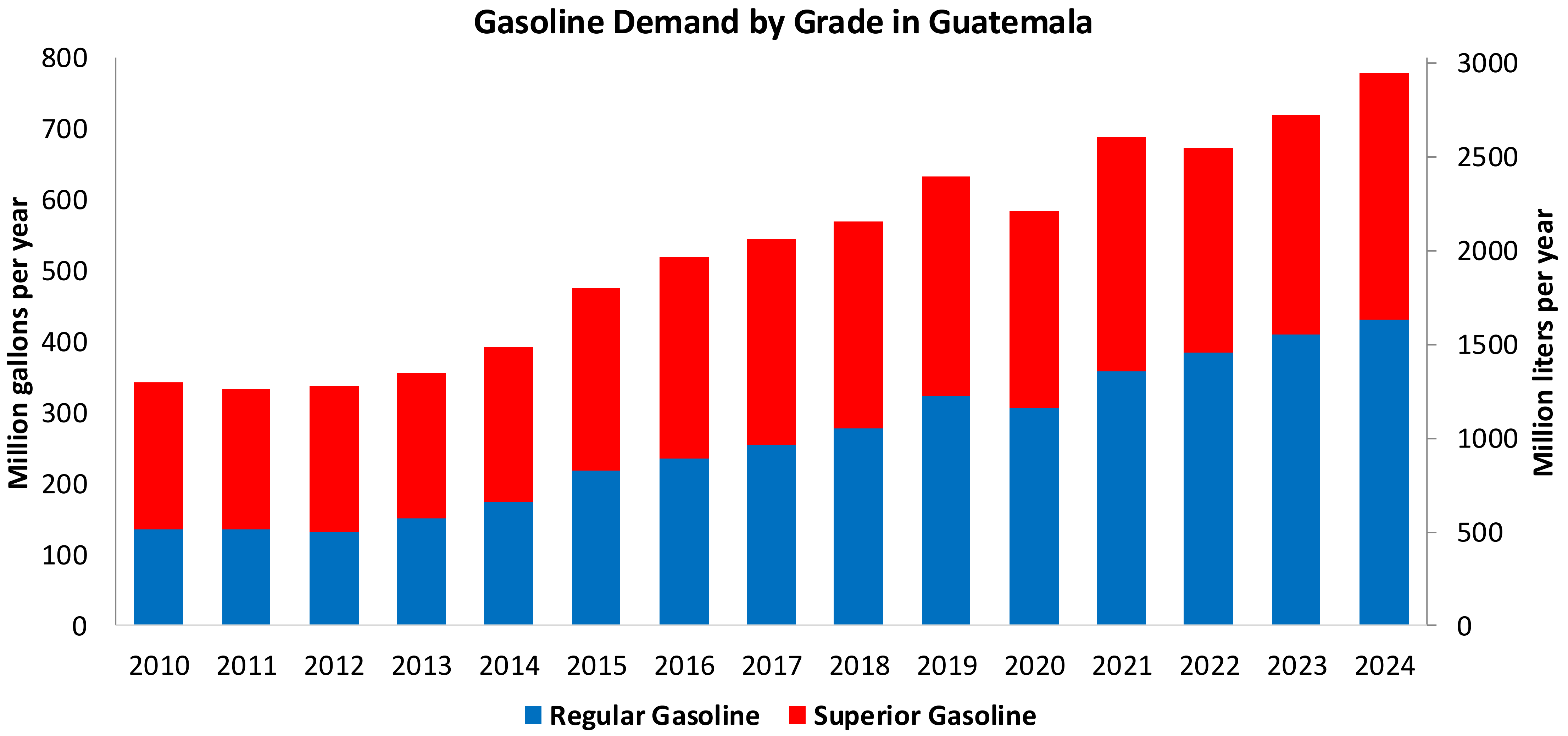
Ethanol Blending in Gasoline - Guatemala

In 2024, gasoline consumption was 778 million gallons (2,946 million liters). Gasoline is supplied only by imports, mainly from United States. There are two grades of gasoline: (RON 91 - AKI 85) and Superior (RON 95 - AKI 89). 55% of gasoline is gasoline Regular.

Guatemala allows for blends up to 10% v/v (Acuerdo Gubernativo 159-2023, MEM) but no ethanol is blended in the country because there is no stakeholder's consensus regarding its implementation. It has local ethanol advanced ethanol production that is exported to Europe. Guatemala is the main ethanol productor in Central America. The E10 mandate is currently under analsys for implementation for 2026-2027.

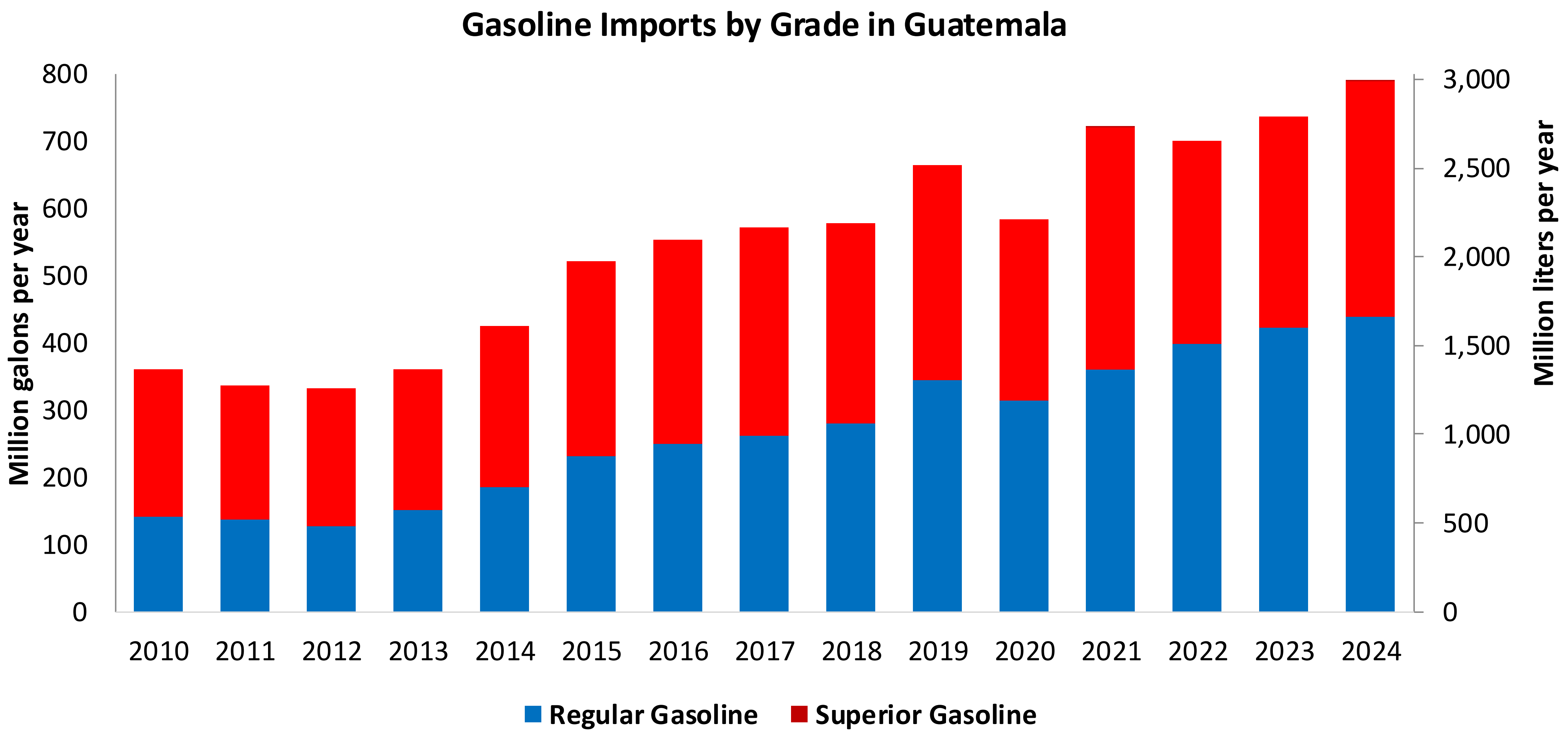
Source: MEM, 2025

Gasoline Demand in Guatemala



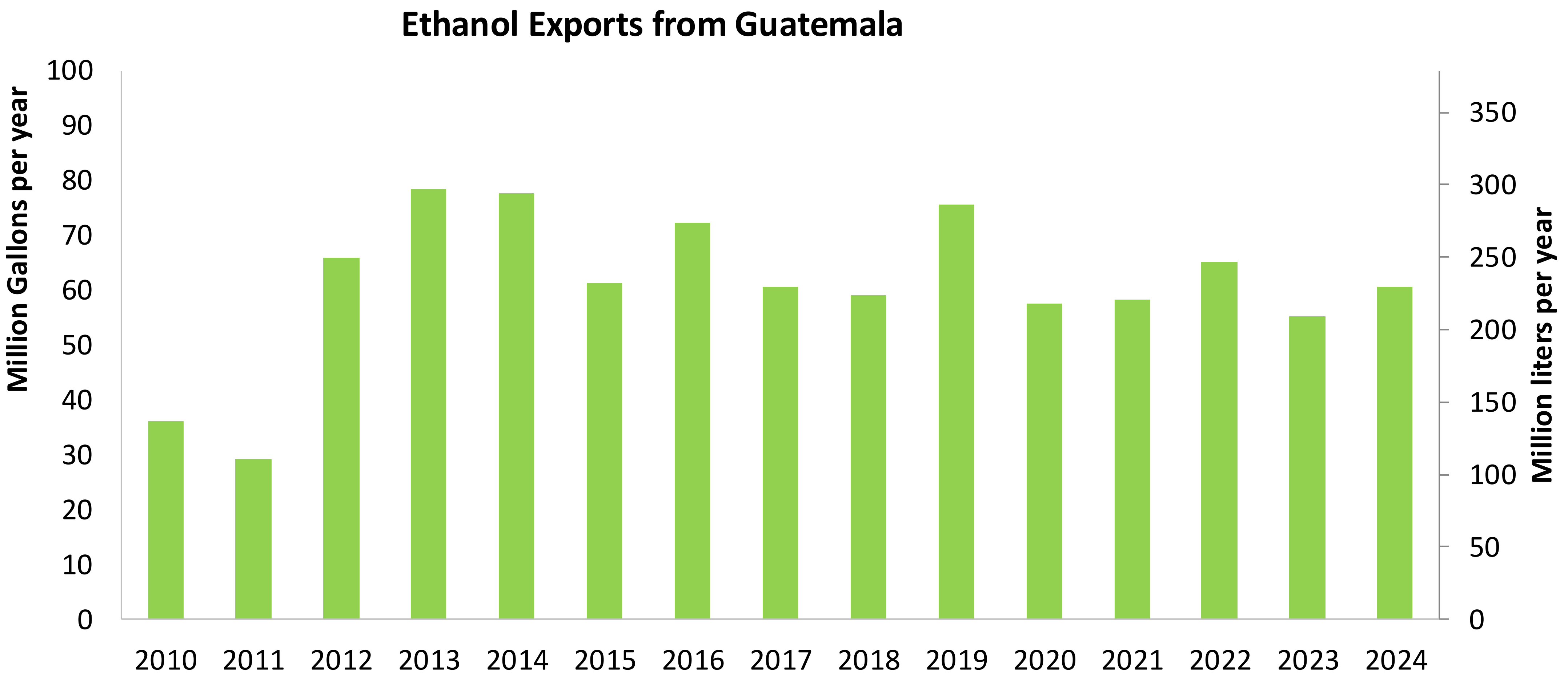
Source: MEM, 2025

Gasoline Imports in Guatemala



Source: MEM, 2025

Ethanol Exports from Guatemala



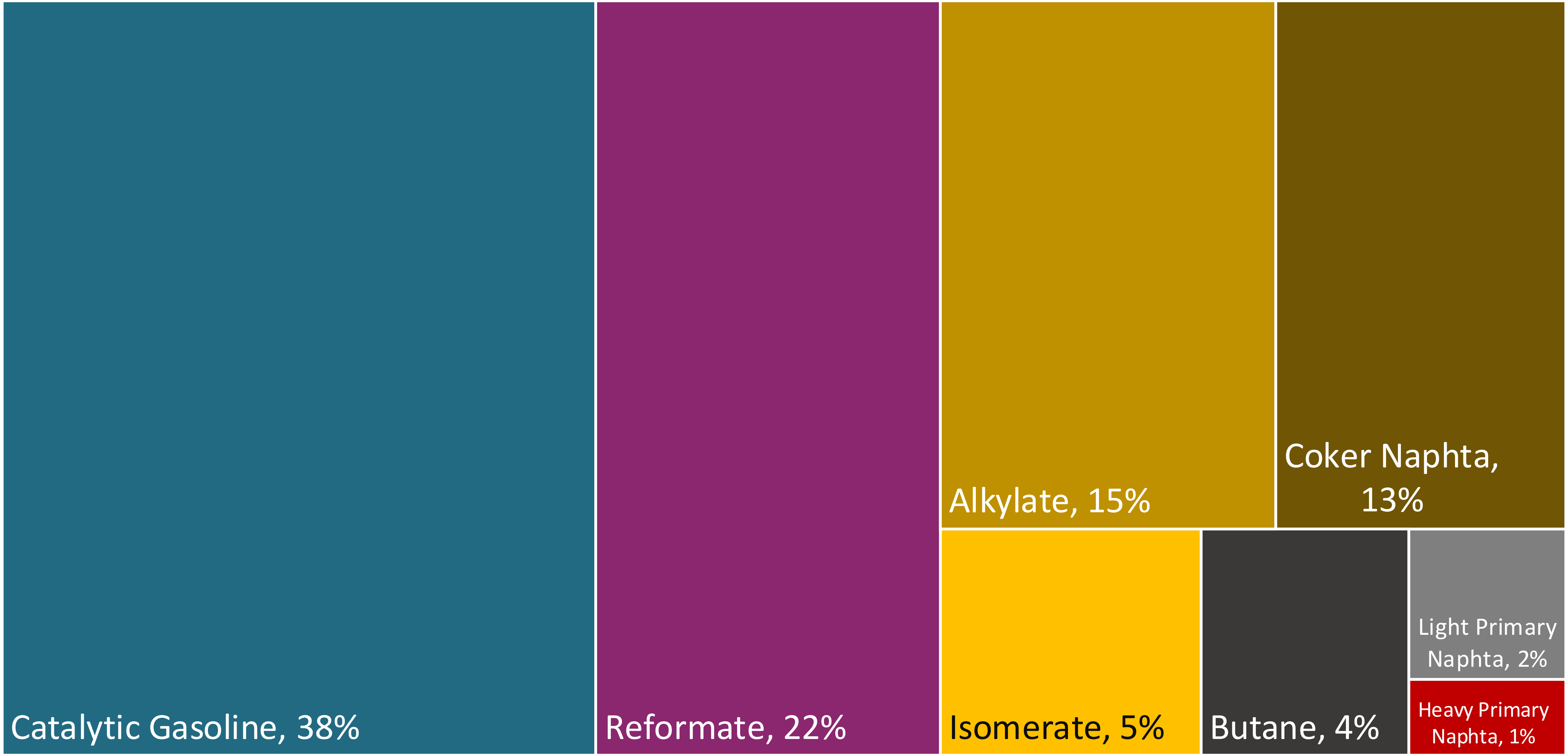
Source: Asociación de productores de etanol de Guatemala

Gasoline Quality in Guatemala

Name	Ministerial Accord Number 320-2022		EN 228:2012 + A1:2017 (Euro 6 enabling)			
Implementation Date	2022		2017			
Applicability	Whole country	Whole country	All countries			
Selected Grade	Gasoline Regular	Gasoline Superior	RON 95 E5	RON 95 E10	RON 98 E5	RON 98 E10
Benzene Content	< 2,5% v/v	< 2,5% v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v
Aromatics	- / < 50% v/v	- / < 50% v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v
Olefins	- / < 30% v/v	- / < 30% v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v
Lead Content	< 0,013 g/l	< 0,013 g/l	< 5 mg/l	< 5 mg/l	< 5 mg/l	< 5 mg/l
Manganese	< 2,5 mg/l	< 2,5 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l
RON	> 91	> 95	> 95	> 95	> 98	> 98
MON	-	-	> 85	> 88	> 85	> 88
AKI						
Sulfur Content	< 500 mg/kg	< 500 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg
Oxygen Content	-	-	<2,7 % m/m	<3,7 % m/m	<2,7 % m/m	<3,7 % m/m
Ethanol (EtOH)	< 10 %v/v	< 10 %v/v	<5 %v/v	<10 %v/v	<5 %v/v	<10 %v/v
RVP 37.8°C (Summer)	< 69 kPa	< 69 kPa	<> 60 - 70 kPa *Depends on the country, RVP is regulated in the EU Fuel Quality Directive			
RVP 37.8 °C(Winter)						
RVP 37.8°C (Transition)						
MTBE	10% v/v	10% v/v	-	-	-	-
Ethers 5 or more C Atoms	-	-	Based on oxygen content	<22 %v/v	Based on oxygen content	<22 %v/v

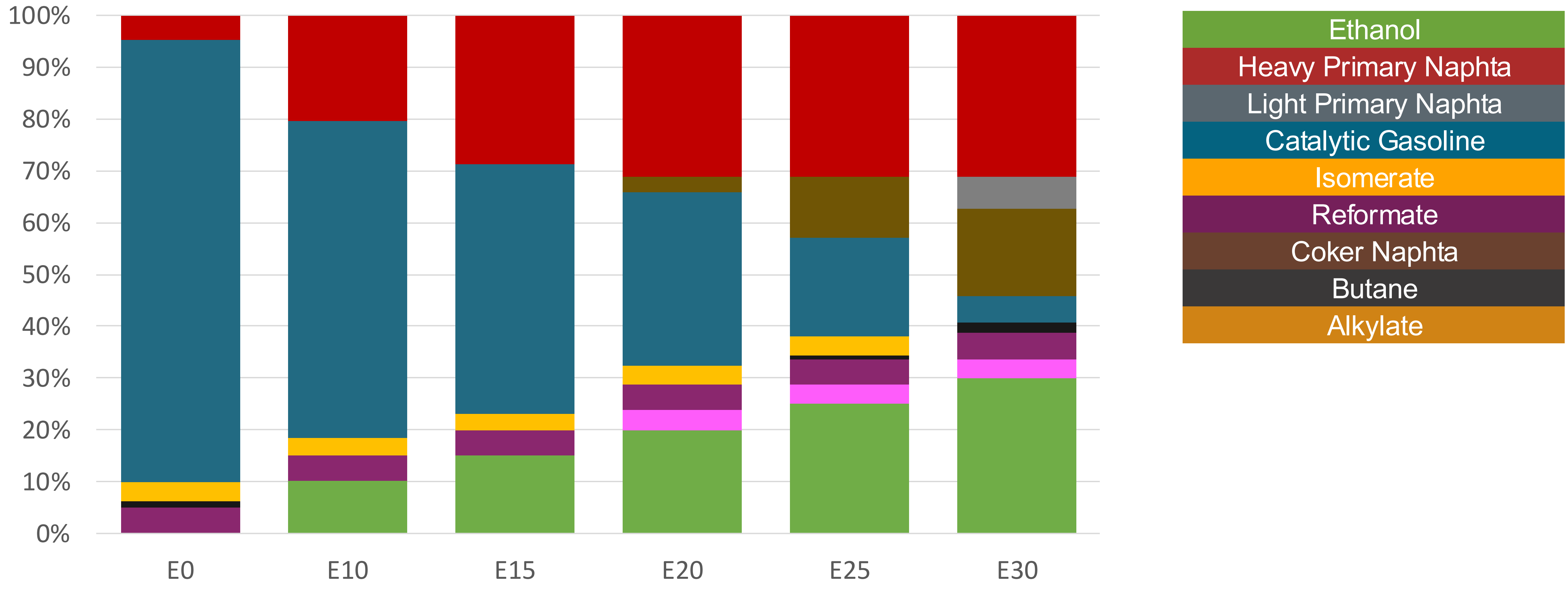
Source: MEM, 2022

Blending Components - Guatemala



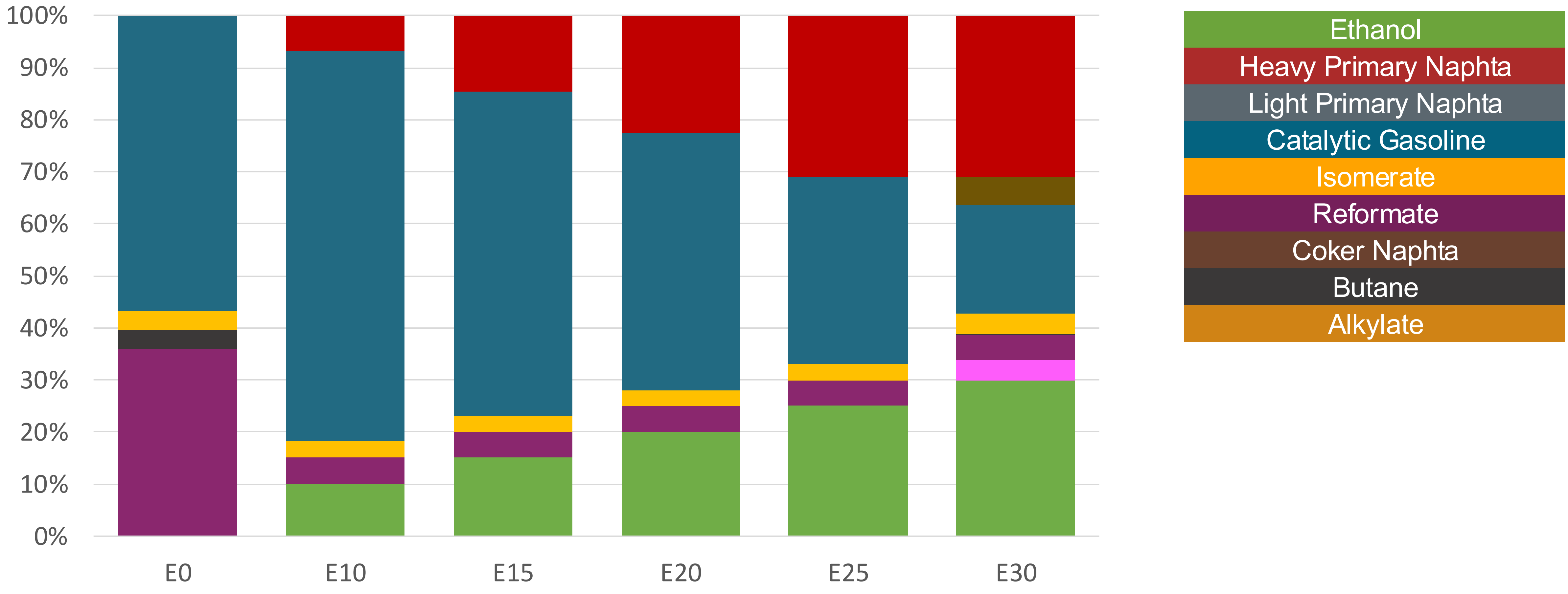
Source: Faro90

Guatemala – Regular – Constant Octane



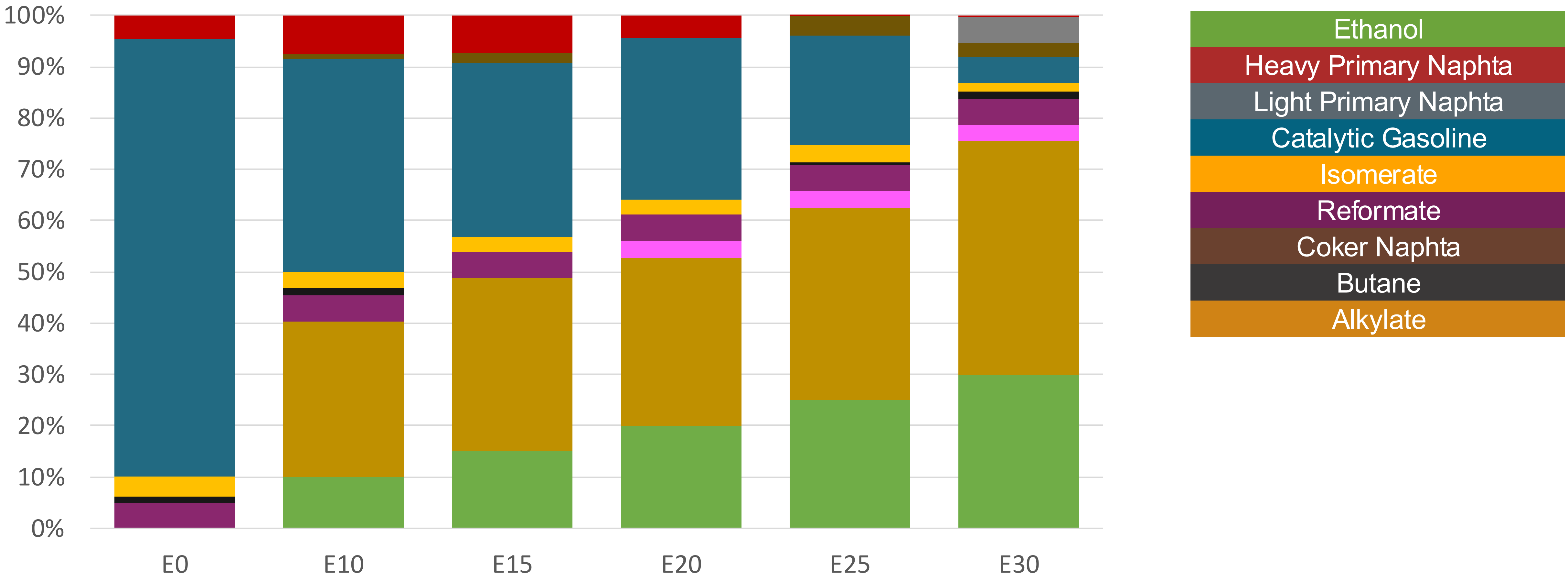
Octane (RON)	91.0	91.0	91.0	91.0	91.0	91.0
Price (USD/gal)	$ 2.336	$. 2.078	$. 1.934	$. 1.787	$. 1.636	$. 1.518

Guatemala – Premium – Constant Octane



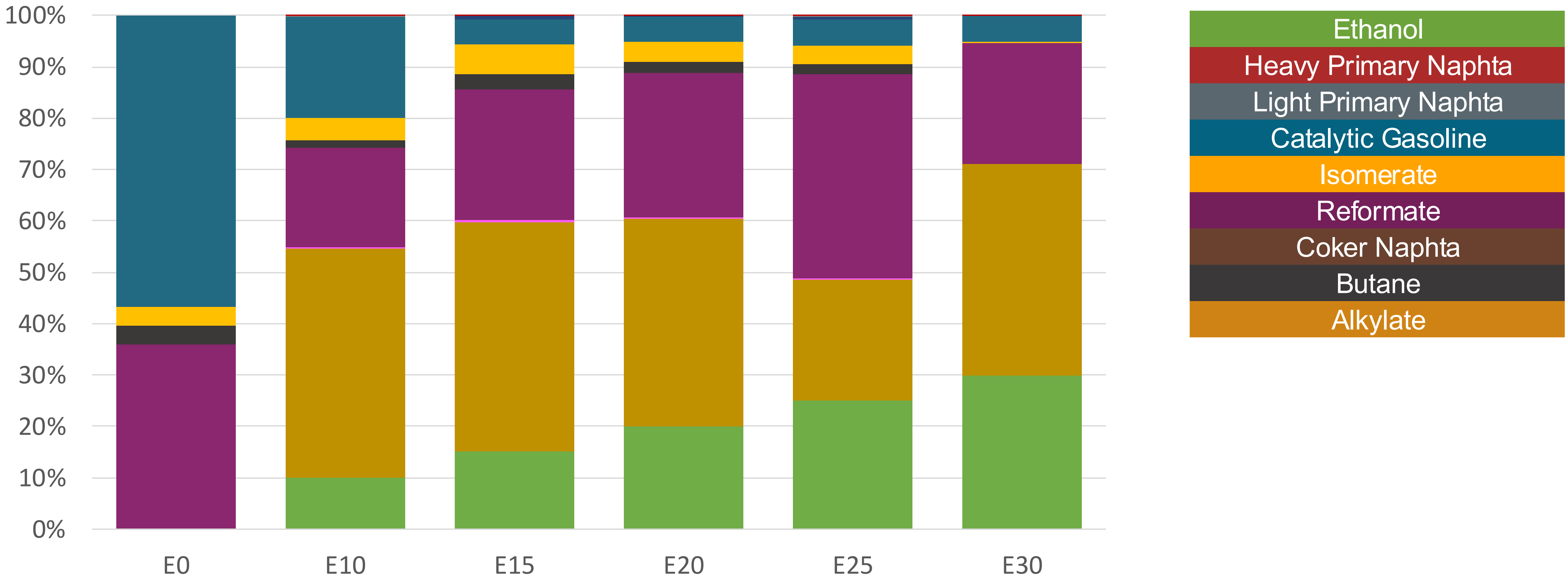
Octane (RON)	95.0	95.0	95.0	95.0	95.0	95.0
Price (USD/gal)	$ 2.586	$ 2.283	$ 2.149	$ 2.007	$ 1.857	$ 1.709

Guatemala – Regular – Increased Octane



Octane (RON)	91.0	94.6	96.6	99.1	101.2	102.4
Price (USD/gal)	$ 2.336	$ 2.336	$ 2.336	$ 2.336	$ 2.336	$ 2.336

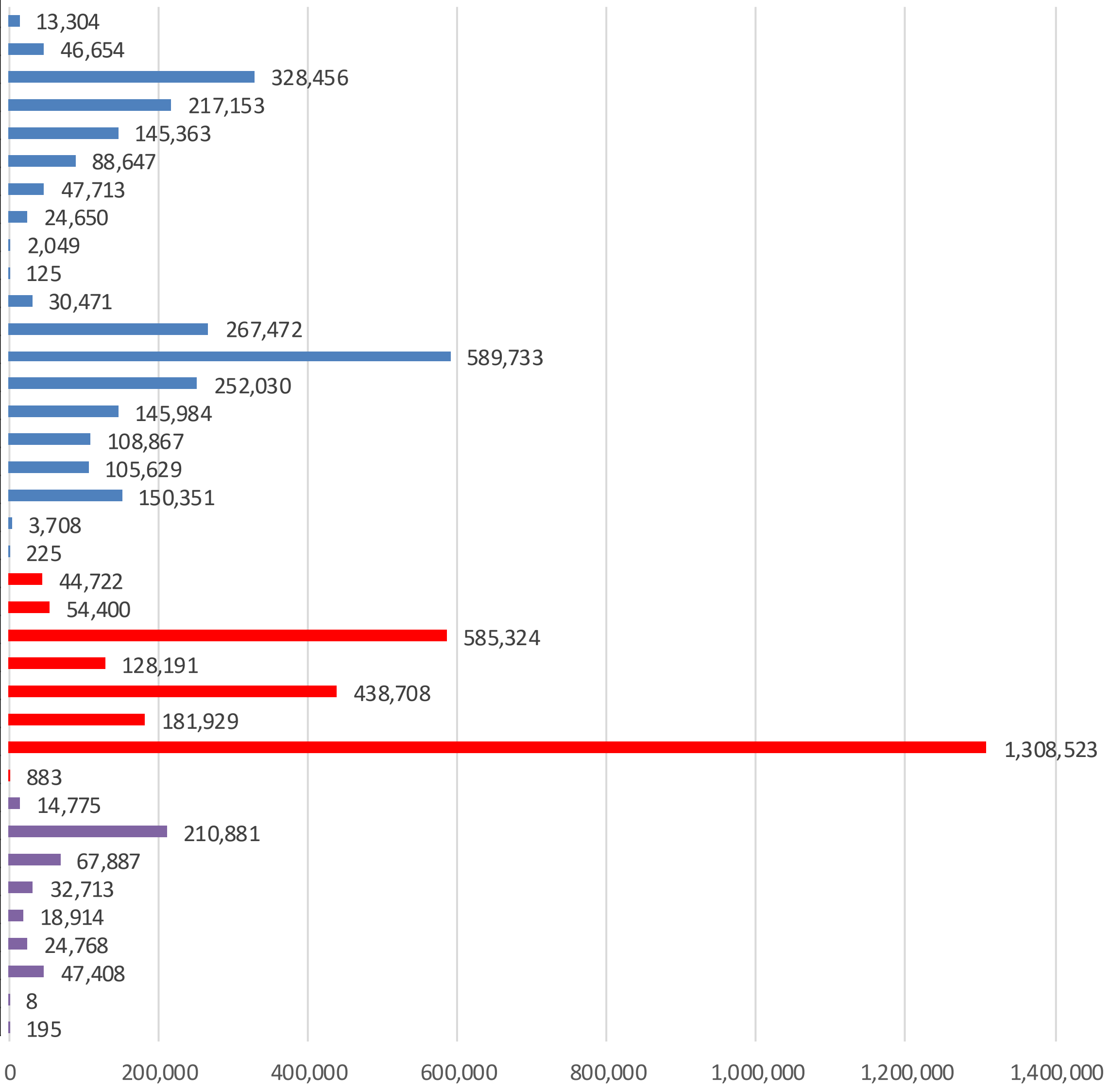
Guatemala – Premium – Increased Octane



	E0	E10	E15	E20	E25	E30
Octane (RON)	95.0	98.0	100.3	102.6	104.9	105.8
Price (USD/gal)	$ 2.586	$ 2.586	$ 2.586	$ 2.586	$ 2.586	$ 2.586

Gasoline Vehicular Fleet - Guatemala

Type	Motor	Exhaust	Recuperator	Age	Number of Vehicles
Auto/Light	Carburator	3-Ways	PCV Tank	<8 years, < 160 mkm	13,304
	Single FI-Pt	3-Ways		<8 years, < 160 mkm	46,654
	Multi FI-Pt	3-Ways		<8 years, < 160 mkm	328,456
		Euro III		<8 years, < 160 mkm	217,153
		Euro IV		<8 years, < 160 mkm	145,363
		Euro V / Euro 6		<8 years, < 160 mkm	88,647
				4-8 years, 80-160 mkm	47,713
				>4years, >80 mkm	24,650
	Hybrid			All	2,049
	Electric			All	125
Auto/Medium	Carburator	3-Ways	PCV Tank	<8 years, < 160 mkm	30,471
	Single FI-Pt	3-Ways		<8 years, < 160 mkm	267,472
	Multi FI-Pt	3-Ways		<8 years, < 160 mkm	589,733
		Euro III		<8 years, < 160 mkm	252,030
		Euro IV		<8 years, < 160 mkm	145,984
		Euro V / Euro 6		<8 years, < 160 mkm	108,867
				4-8 years, 80-160 mkm	105,629
				>4years, >80 mkm	150,351
	Hybrid			All	3,708
	Electric			All	225
Small Engines	FI 4-ciclos	None	PCV	<8 years, < 160 mkm	44,722
		Euro II		<8 years, < 160 mkm	54,400
		Euro III		<8 years, < 160 mkm	585,324
		Euro IV		<8 years, < 160 mkm	128,191
				4-8 years, 80-160 mkm	438,708
		Euro V		4-8 years, 80-160 mkm	181,929
				>4years, >80 mkm	1,308,523
	Eléctricos			All	883
Trucks	Carburator	3-Ways	PCV	<8 years, < 160 mkm	14,775
	FI	3-Ways		<8 years, < 160 mkm	210,881
		Euro III		<8 years, < 160 mkm	67,887
		Euro IV		<8 years, < 160 mkm	32,713
		Euro V / Euro 6		<8 years, < 160 mkm	18,914
				4-8 years, 80-160 mkm	24,768
				>4years, >80 mkm	47,408
	Hybrid			All	8
	Electric			All	195

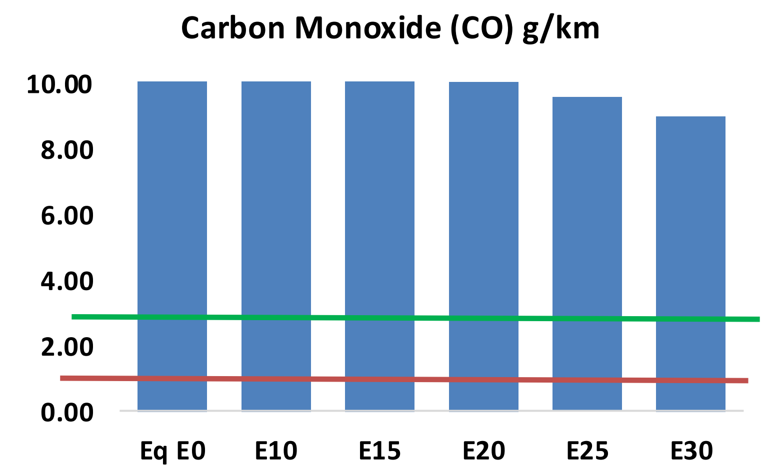


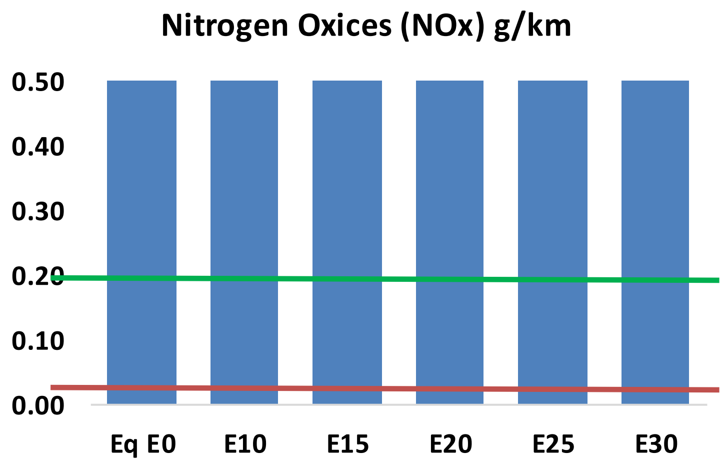
Gasoline Vehicular Fleet: **5,728,610**
Average age: **13 years**
Motorcycles: **48%**

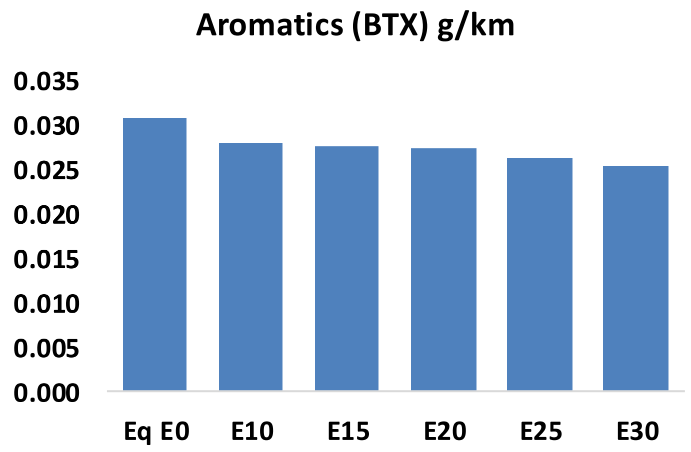
Source: INE, analysis Faro 90

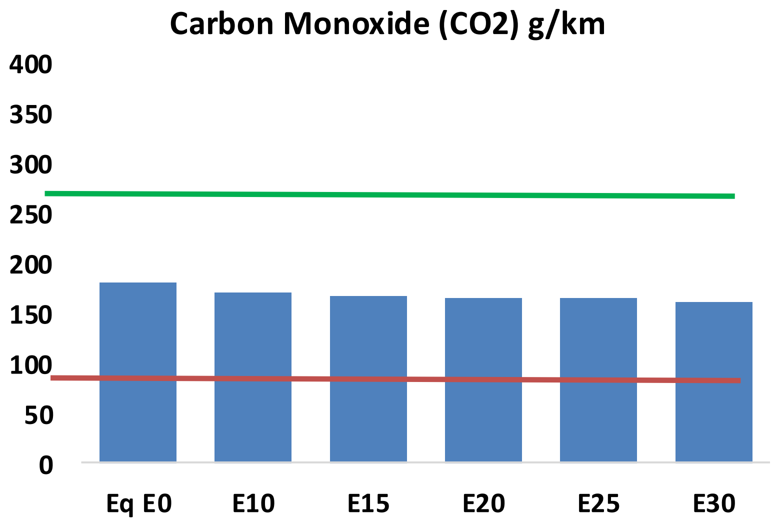
Guatemala – Vehicular Emissions

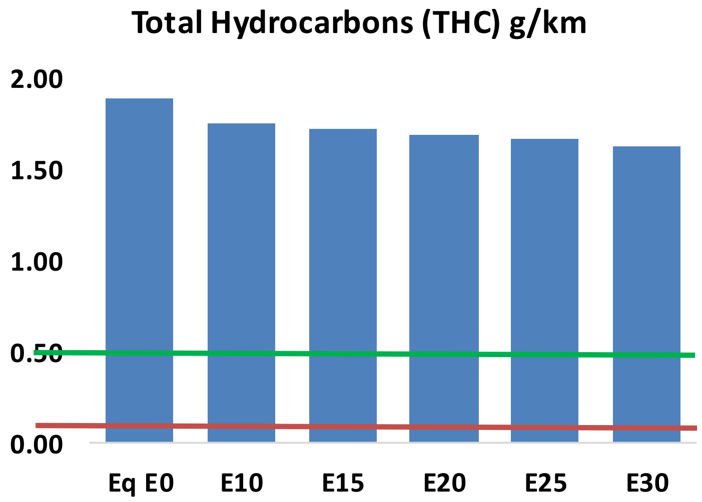
United States
Euro 6

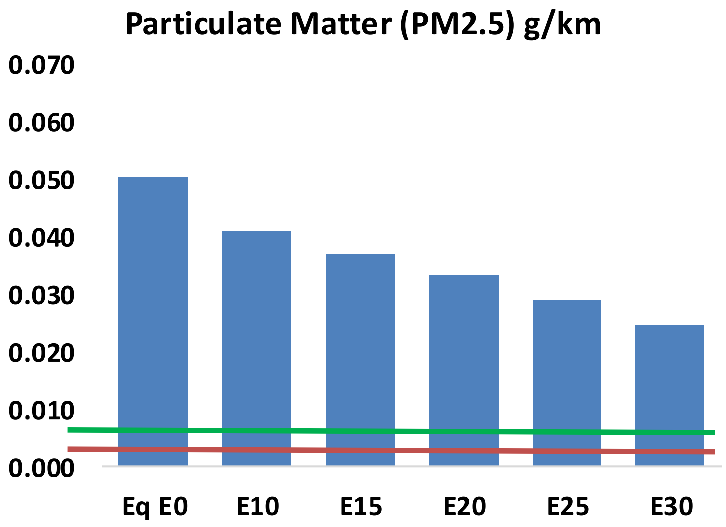








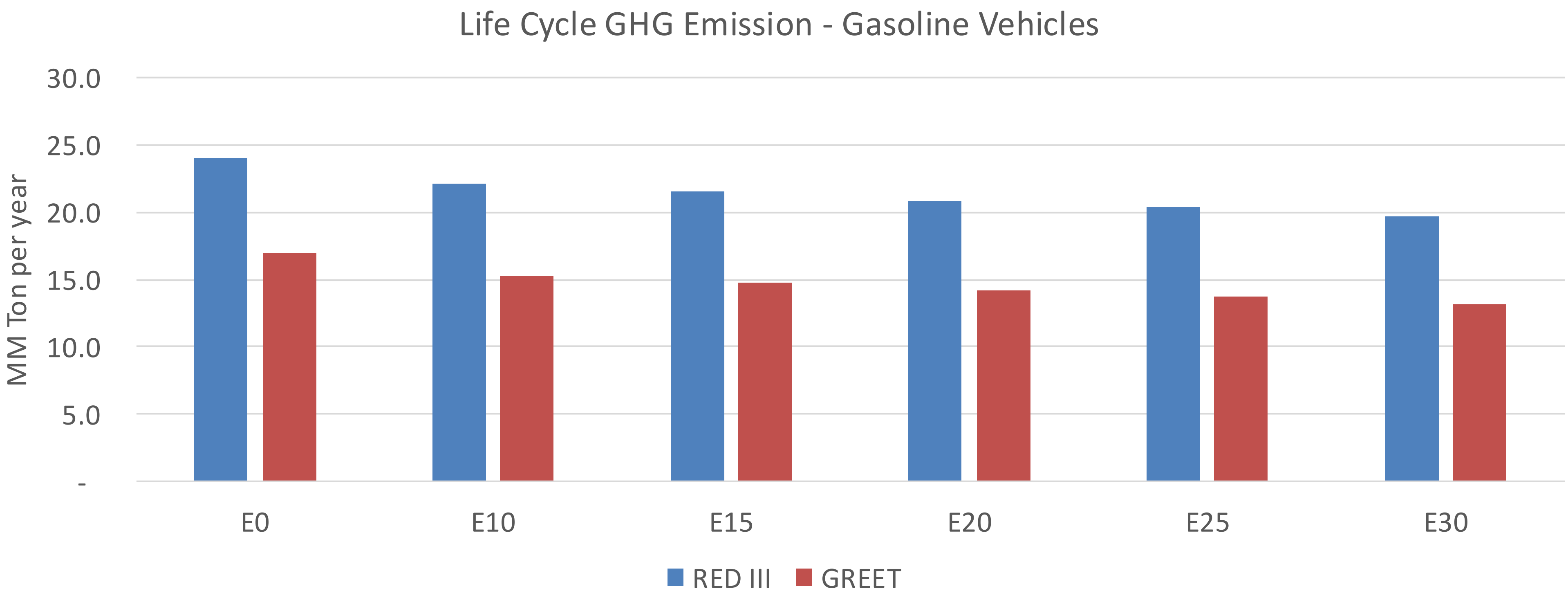




Guatemala – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	1,247,997	1,169,880	1,091,763	1,031,227	971,909	926,781	870,513	-13%	-22%
CO2	17,319,676	16,874,313	16,453,692	16,122,886	15,959,191	15,915,024	15,621,664	-5%	-8%
VOC	182,030	175,687	169,345	165,641	162,600	160,469	156,894	-7%	-11%
NOx	77,890	71,919	67,928	65,950	62,703	60,186	57,205	-13%	-19%
SOx	203	188	172	159	147	133	119	-15%	-28%
NH3	6,836	6,768	6,701	6,733	6,808	6,862	6,906	-2%	0%
N2O	315	314	312	314	320	324	328	-1%	2%
CH4	40,547	40,547	40,547	41,156	42,048	42,818	43,548	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	697	653	609	575	542	517	482	-13%	-22%
Acetaldehyde	1,716	1,969	2,890	4,401	5,996	6,851	8,095	68%	249%
Formaldehyde	6,423	6,244	7,279	8,547	8,954	9,906	10,784	13%	39%
Benzene	2,977	2,861	2,711	2,668	2,645	2,530	2,448	-9%	-11%
PM2.5	1,021	928	835	753	673	588	498	-18%	-34%
PM10	3,821	3,474	3,126	2,821	2,522	2,201	1,865	-18%	-34%

Guatemala – Gasoline Vehicle Fleet GEI Emissions



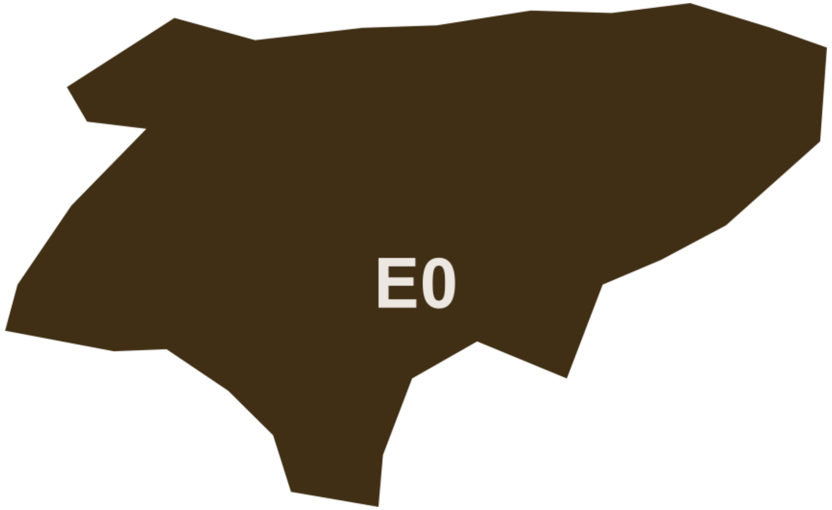
Country Emissions 2024 (MMT/year)	43.5
2035 Target (MMT/year)	21.7
Delta	21.7

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	10.0%	12.8%	16.2%	18.8%	22.3%
RED II/III	0.0%	7.9%	10.4%	13.1%	15.3%	18.1%

% 2035 Target	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	7.8%	10.0%	12.6%	14.7%	17.4%
RED II/III	0.0%	8.8%	11.4%	14.5%	16.9%	20.1%

Source: Greet, RED II/III, climateactiontracker.org, F90

Ethanol Blending in Gasoline - Honduras

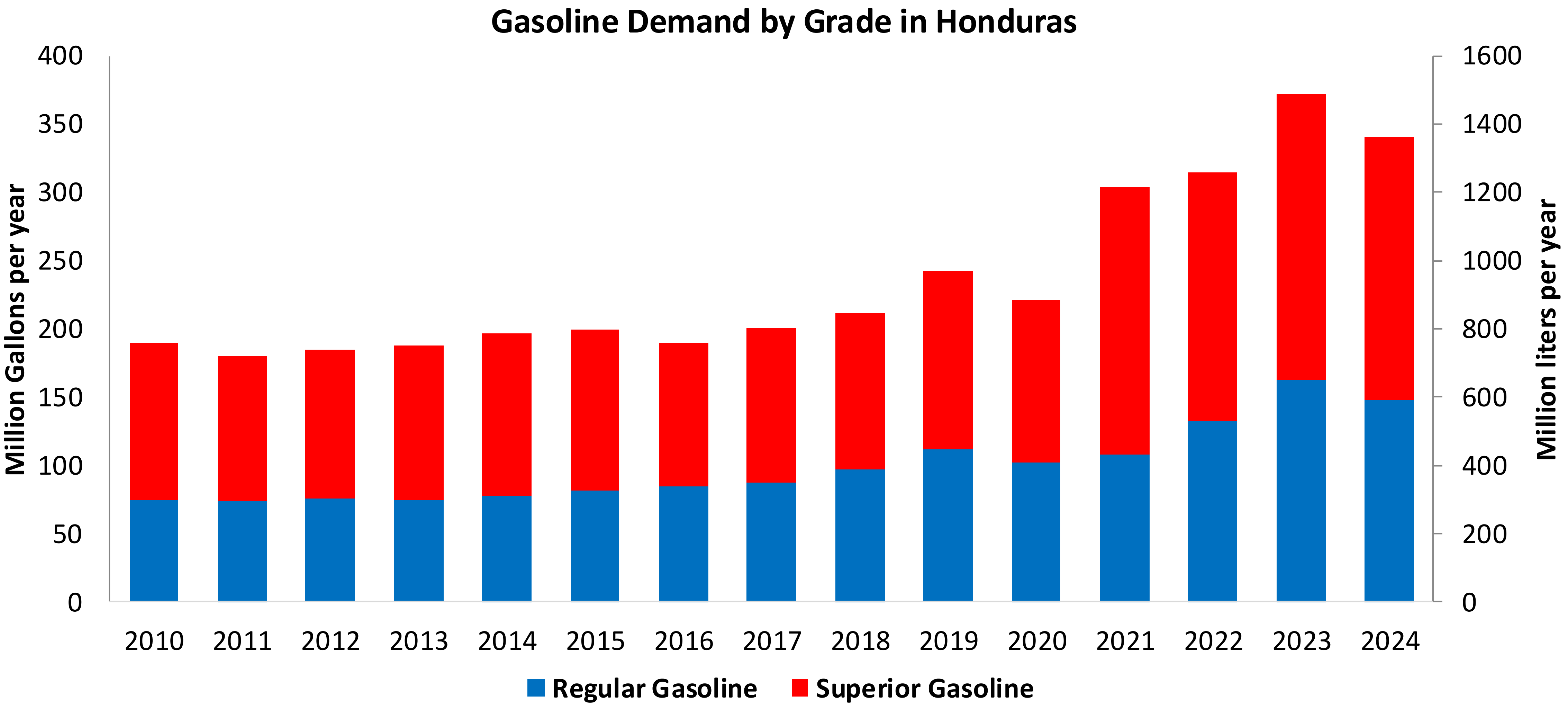


In 2024, gasoline demand was 374 million gallons (1,290 million liters). Regular gasoline RON 91 represented 46% while Superior gasoline RON 95 reached 54% of national demand. Honduras does not have gasoline production capacity and supplies its market with imports mainly from the United States.

Honduras has no national mandate for ethanol blends. A new biofuel's law is under discussion.

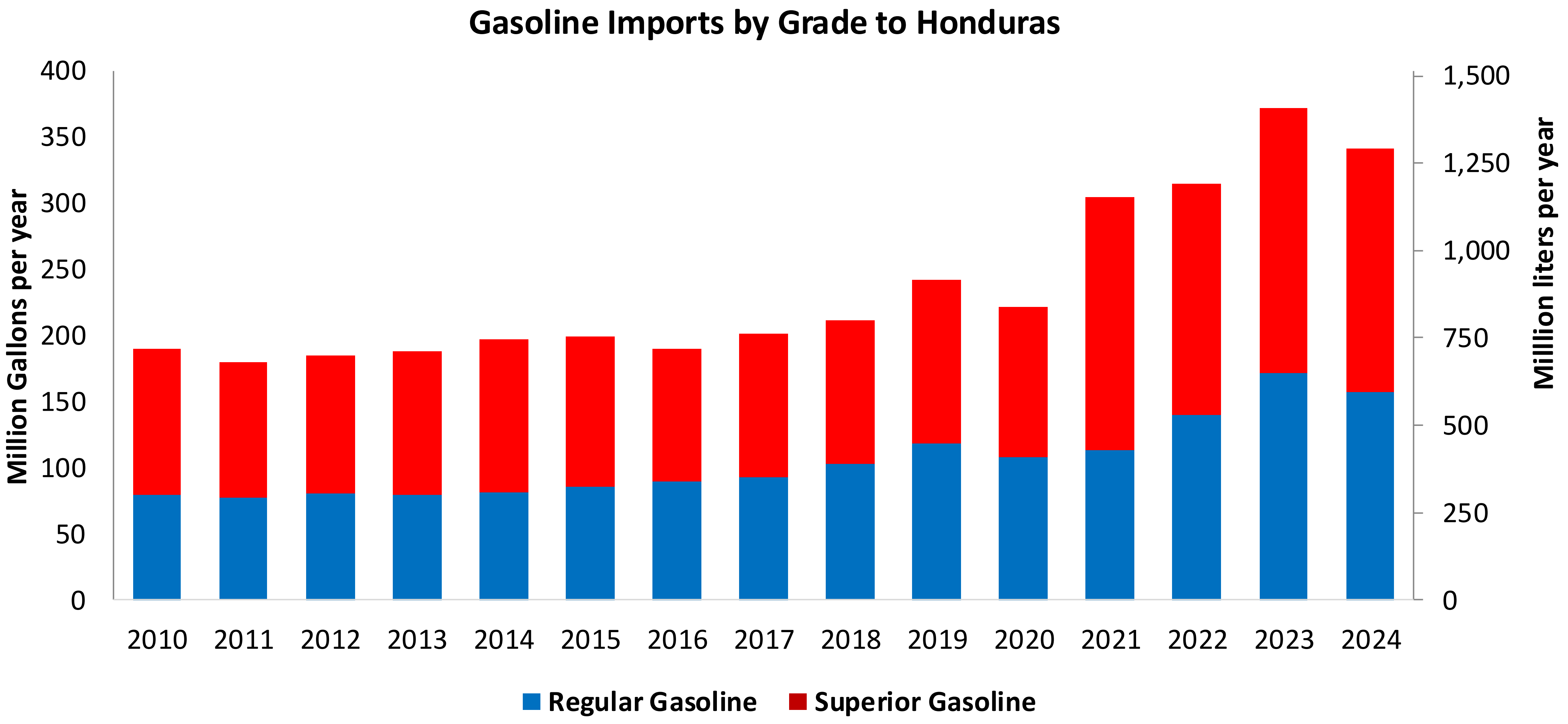
Source: CEPAL, Secretaría de Energía Honduras

Gasoline Demand in Honduras



Source: CEPAL, Secretaría de Energía Honduras

Gasoline Imports in Honduras



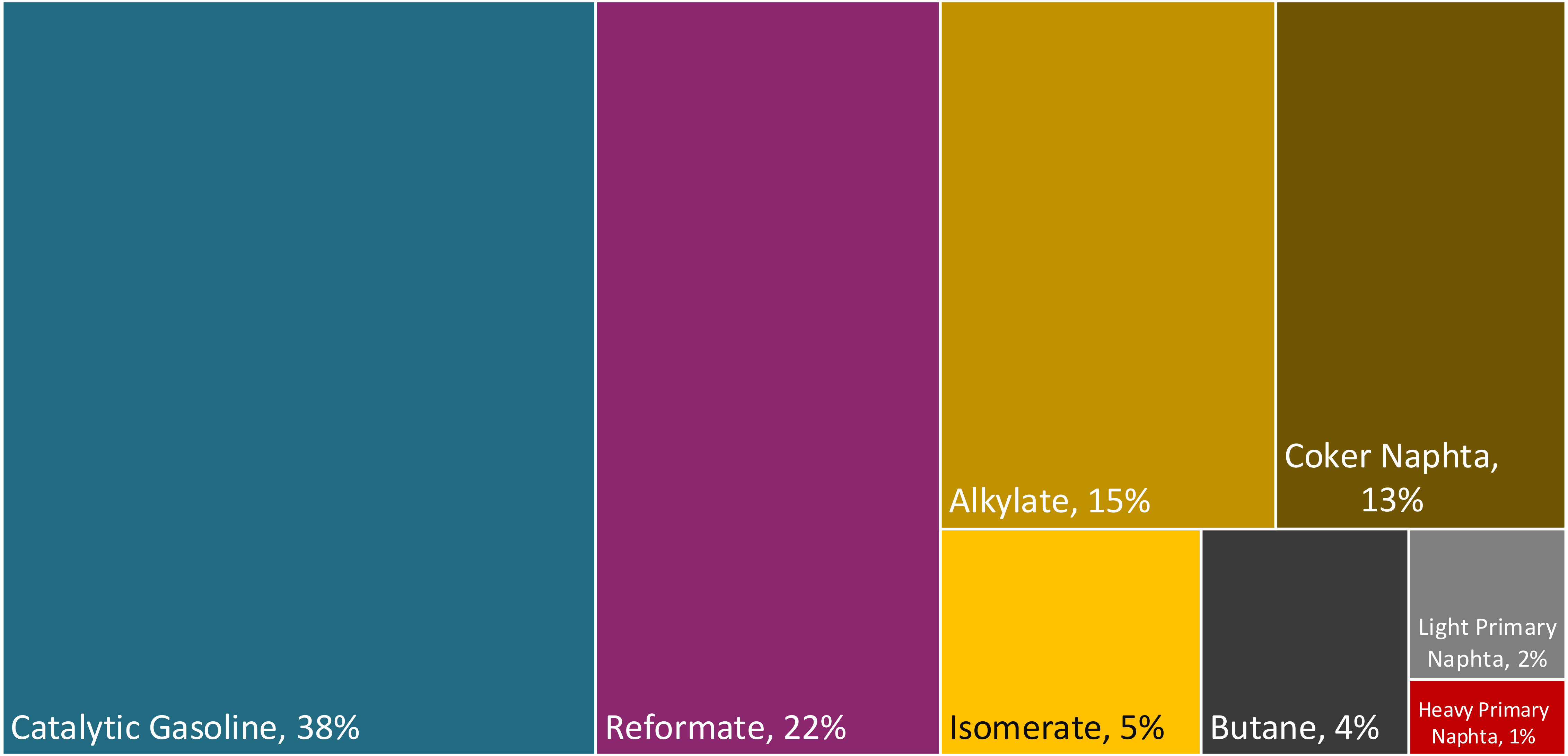
Source: CEPAL, Secretaría de Energía Honduras

Gasoline Quality in Honduras

Name	RTCA 75.01.08:19 p.m.	RTCA 75.01.19:19	EN 228:2012 + A1:2017 (Euro 6 enabling)			
Implementation Date	2021	2021	2017			
Applicability	Whole country	Whole country	All countries			
Selected Grade	Gasoline Premium	Gasoline Regular	RON 95 E5	RON 95 E10	RON 98 E5	RON 98 E10
Benzene Content	2,5% v/v	2,5% v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v
Aromatics	50% v/v	50% v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v
Olefins	30% v/v	30% v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v
Lead Content	< 0,013 g/l	< 0,013 g/l	< 5 mg/l	< 5 mg/l	< 5 mg/l	< 5 mg/l
Manganese	0,25% v/v	0,25% v/v	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l
RON	> 95	> 91	> 95	> 95	> 98	> 98
MON	-	-	> 85	> 88	> 85	> 88
AKI						
Sulfur Content	< 500 mg/kg	< 500 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg
Oxygen Content	2,7% v/v	2,7% v/v	<2,7 % m/m	<3,7 % m/m	<2,7 % m/m	<3,7 % m/m
Ethanol (EtOH)	-	-	<5 %v/v	<10 %v/v	<5 %v/v	<10 %v/v
RVP 37.8°C (Summer)	< 69 kPa	< 69 kPa	<> 60 - 70 kPa *Depends on the country, RVP is regulated in the EU Fuel Quality Directive			
RVP 37.8 °C(Winter)						
RVP 37.8°C (Transition)						
MTBE	-	-	-	-	-	-
Ethers 5 or more C Atoms	-	-	Based on oxygen content	<22 %v/v	Based on oxygen content	<22 %v/v

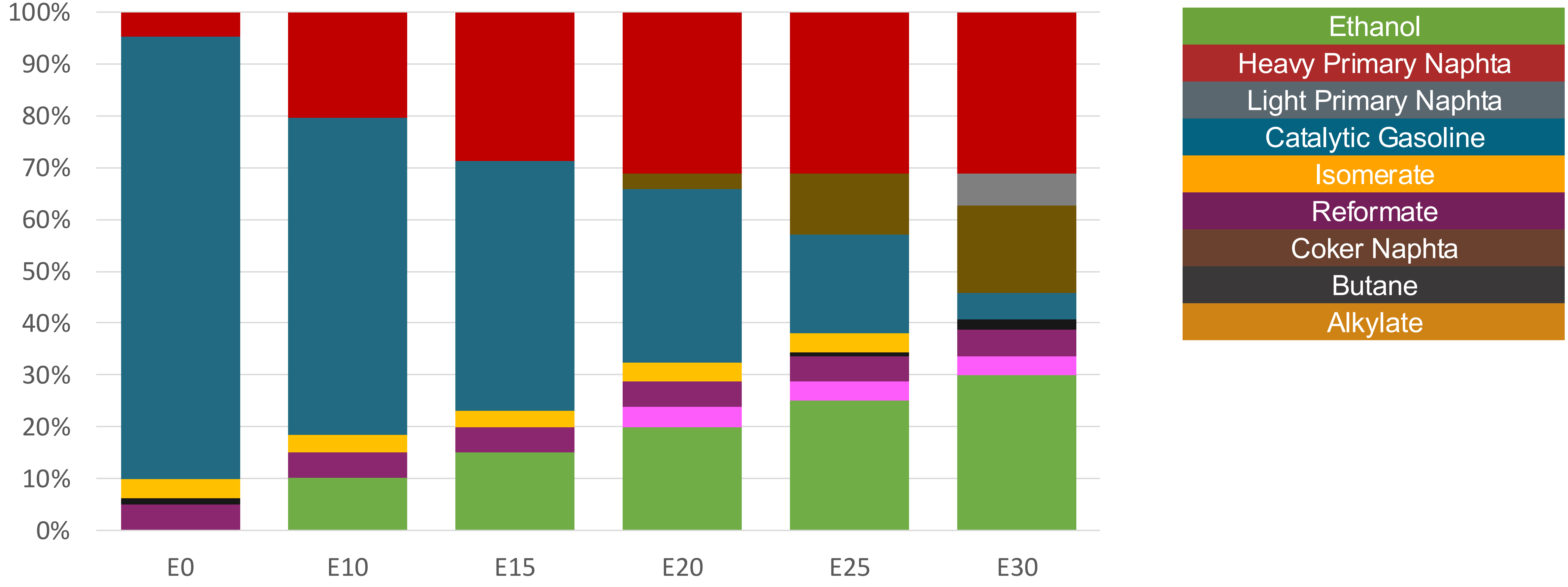
Source:
RTCA

Blending Components - Honduras



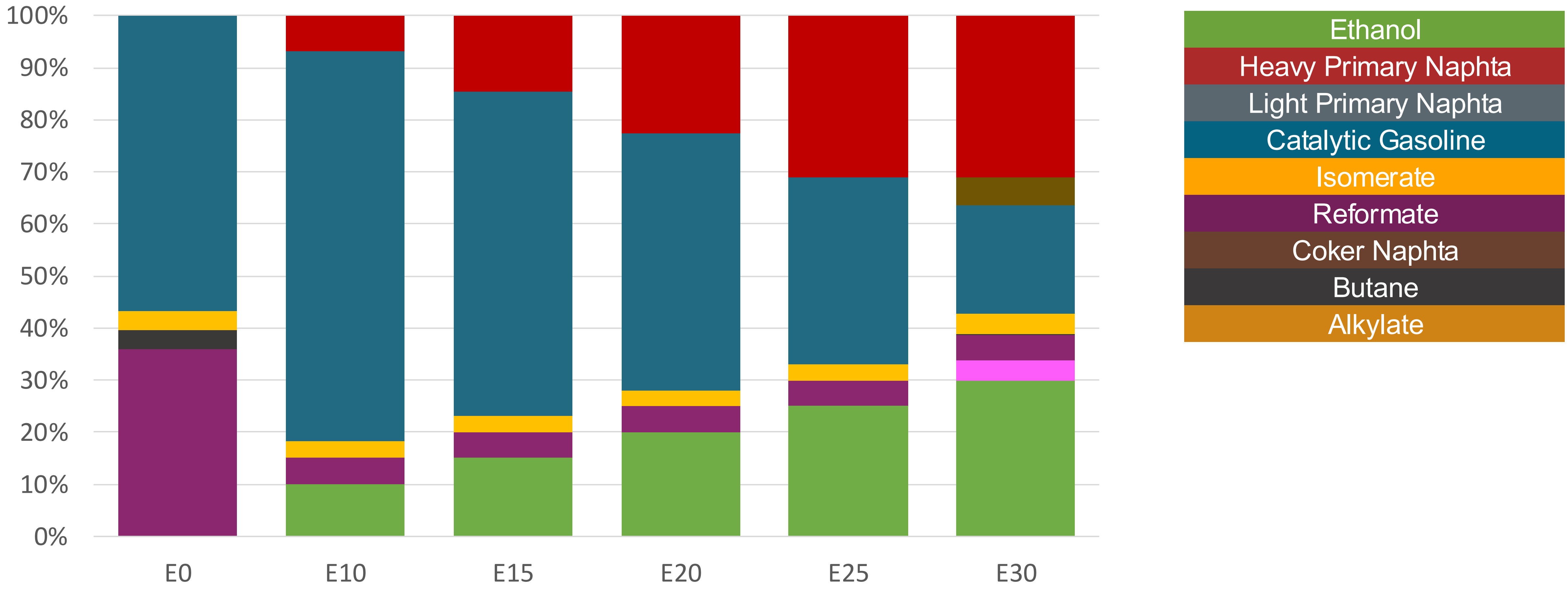
Source: Faro90

Honduras – Regular – Constant Octane



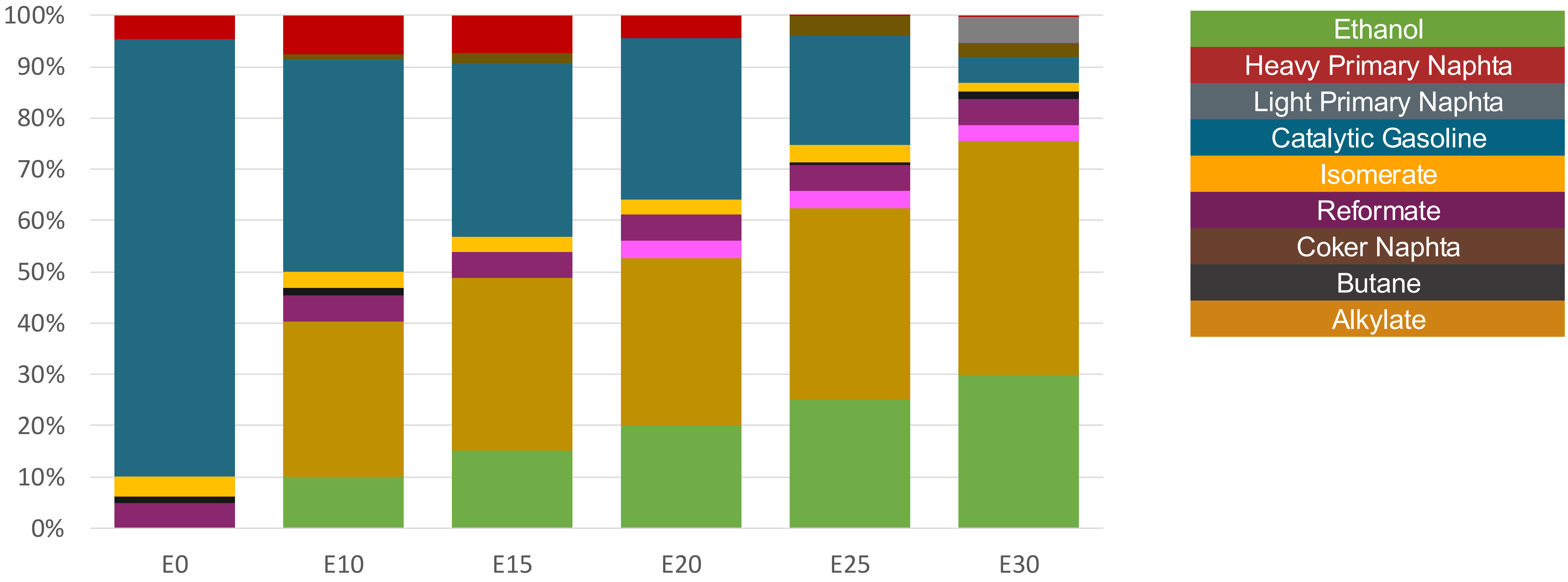
Octane (RON)	91.0	91.0	91.0	91.0	91.0	91.0
Price (USD/gal)	$ 2.156	$ 1.885	$ 1.734	$ 1.592	$ 1.447	$ 1.477

Honduras – Premium – Constant Octane



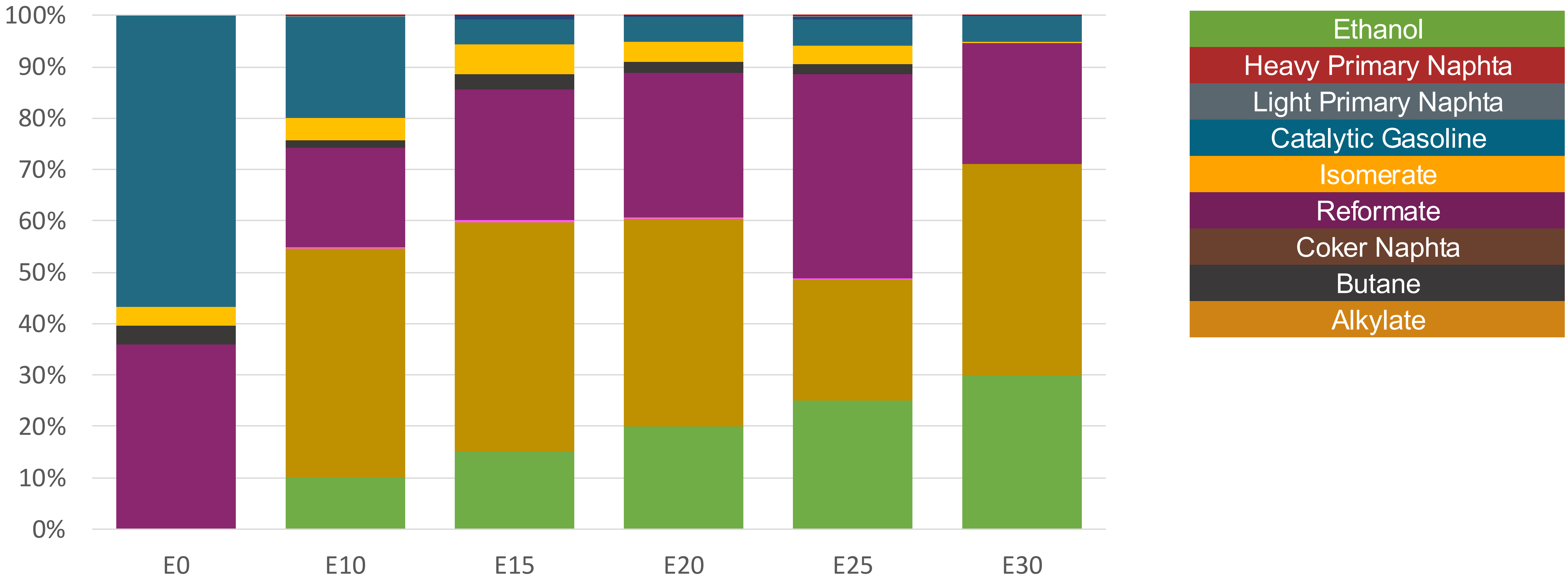
Octane (RON)	95.0	95.0	95.0	95.0	95.0	95.0
Price (USD/gal)	$ 2.536	$ 2.233	$ 2.099	$ 1.957	$ 1.807	$ 1.659

Honduras – Regular – Increased Octane



Octane (RON)	91.0	94.6	96.6	99.1	101.2	102.4
Price (USD/gal)	$ 2.156	$ 2.156	$ 2.156	$ 2.156	$ 2.156	$ 2.156

Honduras – Premium – Increased Octane



Octane (RON)	95.0	98.0	100.3	102.6	104.9	105.8
Price (USD/gal)	$ 2.536	$ 2.536	$ 2.536	$ 2.536	$ 2.536	$ 2.536

Gasoline Vehicular Fleet in Honduras

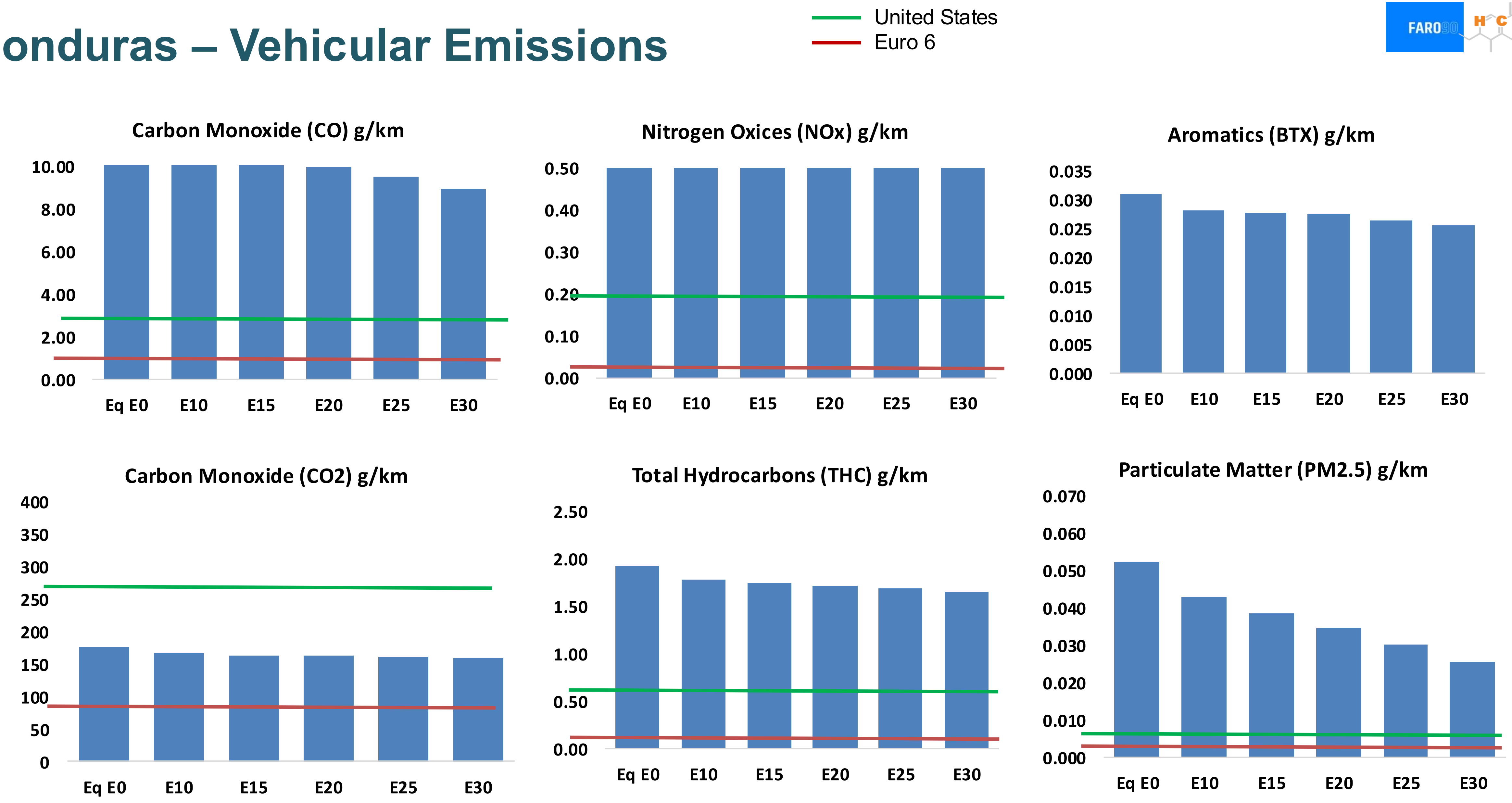
Type	Motor	Exhaust	Recuperator	Age	Vehicle Numbers
Light Vehicles	Carburator	3-Ways	PCV Tank	<8 years, < 160 mkm	7,426
	Single FI-Pt	3-Ways		<8 years, < 160 mkm	26,041
	Multi FI-Pt	3-Ways		<8 years, < 160 mkm	183,333
		Euro III		<8 years, < 160 mkm	121,208
		Euro IV		<8 years, < 160 mkm	81,137
		Euro V / Euro 6		<8 years, < 160 mkm	49,480
				4-8 years, 80-160 mkm	26,632
				>4years, >80 mkm	13,759
Medium Size Vehicles	Carburator	3-Ways	PCV Tank	<8 years, < 160 mkm	18,844
	Single FI-Pt	3-Ways		<8 years, < 160 mkm	165,407
	Multi FI-Pt	3-Ways		<8 years, < 160 mkm	364,696
		Euro III		<8 years, < 160 mkm	155,857
		Euro IV		<8 years, < 160 mkm	90,278
		Euro V / Euro 6		<8 years, < 160 mkm	67,324
				4-8 years, 80-160 mkm	65,322
				>4years, >80 mkm	92,978
Small Engines	FI 4-ciclos	None	PCV	<8 years, < 160 mkm	29,558
		Euro II		<8 years, < 160 mkm	35,954
		Euro III		<8 years, < 160 mkm	386,853
		Euro IV		<8 years, < 160 mkm	84,724
				4-8 years, 80-160 mkm	289,951
		Euro V		4-8 years, 80-160 mkm	120,241
				>4years, >80 mkm	864,831

0 100,000 200,000 300,000 400,000 500,000 600,000 700,000 800,000 900,000 1,000,000

Vehicular Fleet: **3,341,832**
Average Age: **13 years**
Motorcycles: **50%**

Source: INE, analysis Faro 90

Honduras – Vehicular Emissions



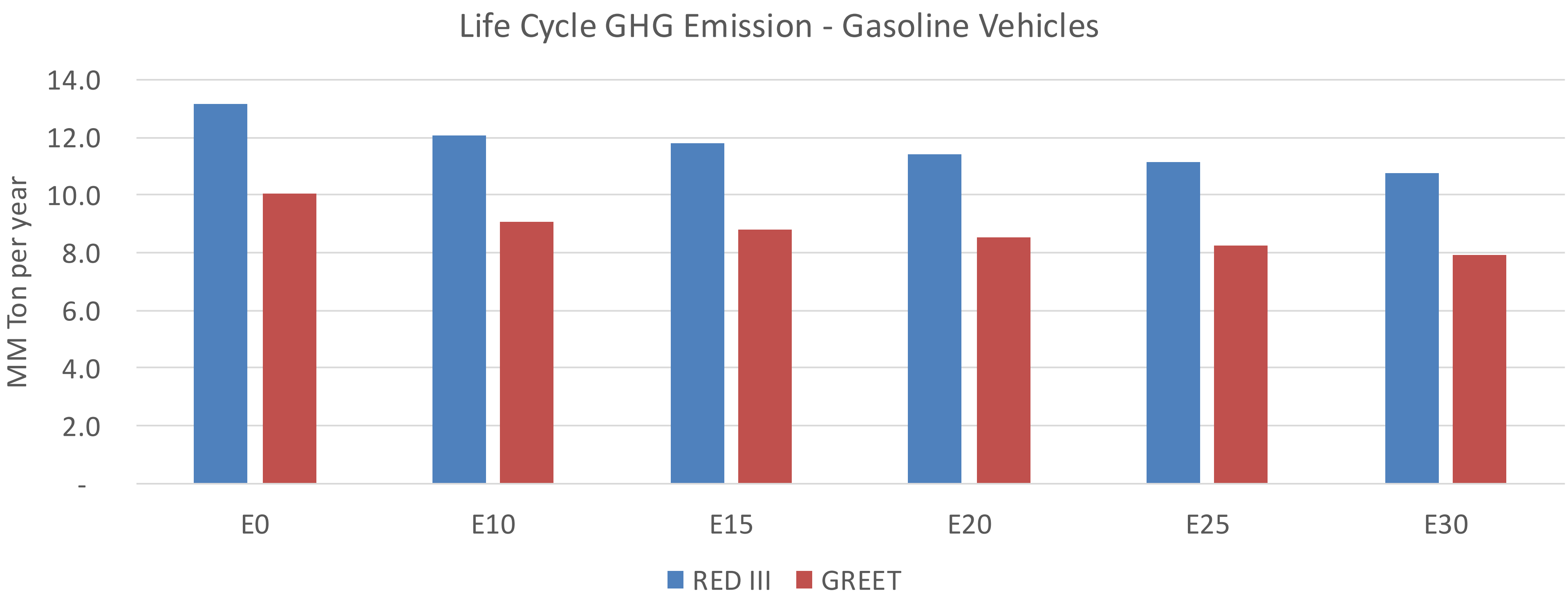
Source: Faro90

Honduras – Gasoline Vehicle Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	769,641	720,753	671,866	633,811	596,439	567,980	532,623	-13%	-23%
CO2	10,458,718	10,189,779	9,935,782	9,736,021	9,637,171	9,612,562	9,435,375	-5%	-8%
VOC	114,270	110,163	106,056	103,603	101,554	100,106	97,726	-7%	-11%
NOx	48,100	44,412	41,946	40,724	38,717	37,162	35,319	-13%	-20%
SOx	123	113	104	96	89	81	72	-15%	-28%
NH3	4,189	4,148	4,106	4,126	4,172	4,205	4,232	-2%	0%
N2O	192	191	190	191	195	197	199	-1%	2%
CH4	25,506	25,506	25,506	25,889	26,450	26,934	27,393	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	434	406	378	356	334	318	295	-13%	-23%
Acetaldehyde	1,087	1,248	1,831	2,788	3,799	4,341	5,129	68%	249%
Formaldehyde	4,114	4,000	4,663	5,475	5,736	6,346	6,908	13%	39%
Benzene	1,849	1,776	1,683	1,657	1,643	1,571	1,520	-9%	-11%
PM2.5	619	563	507	457	409	357	302	-18%	-34%
PM10	2,490	2,264	2,037	1,838	1,643	1,434	1,216	-18%	-34%

Source: Faro90

Honduras – Gasoline Vehicle Fleet GEI Emissions



Country Emissions 2024 (MMT/year)	30.7
2035 Target (MMT/year)	15.3
Delta	15.3

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	9.8%	12.4%	15.6%	18.0%	21.3%
RED II/III	0.0%	8.2%	10.6%	13.3%	15.5%	18.3%

% 2035 Target	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	6.4%	8.2%	10.2%	11.8%	14.0%
RED II/III	0.0%	7.1%	9.1%	11.4%	13.3%	15.7%

Source: Greet, RED II/III, climateactiontracker.org, F90

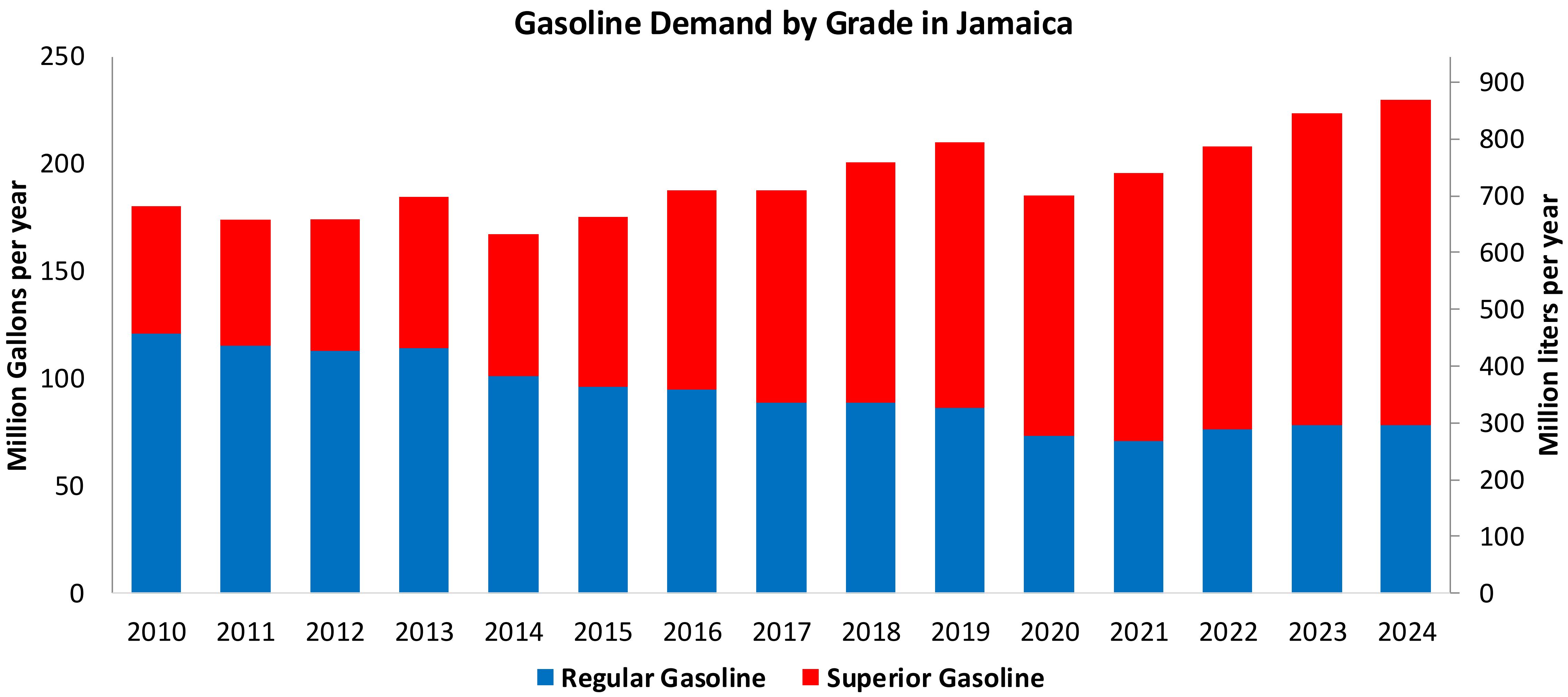
Ethanol Blending in Gasoline - Jamaica

In 2024, gasoline demand was 230 million gallons (869 million liters). Gasoline production was 34 million gallons (130 million liters) and imports 195 million gallons (739 million liters) mainly from United States. Regular grade (AKI 87) had a market share of 34% and Premium grade (AKI 90) 66%.

Current gasoline quality norm mandates E10 (JS 341:2017) . Previously, ethanol was supplied with local production to Petrojam, however, in recent years, ethanol is imported from the United States.

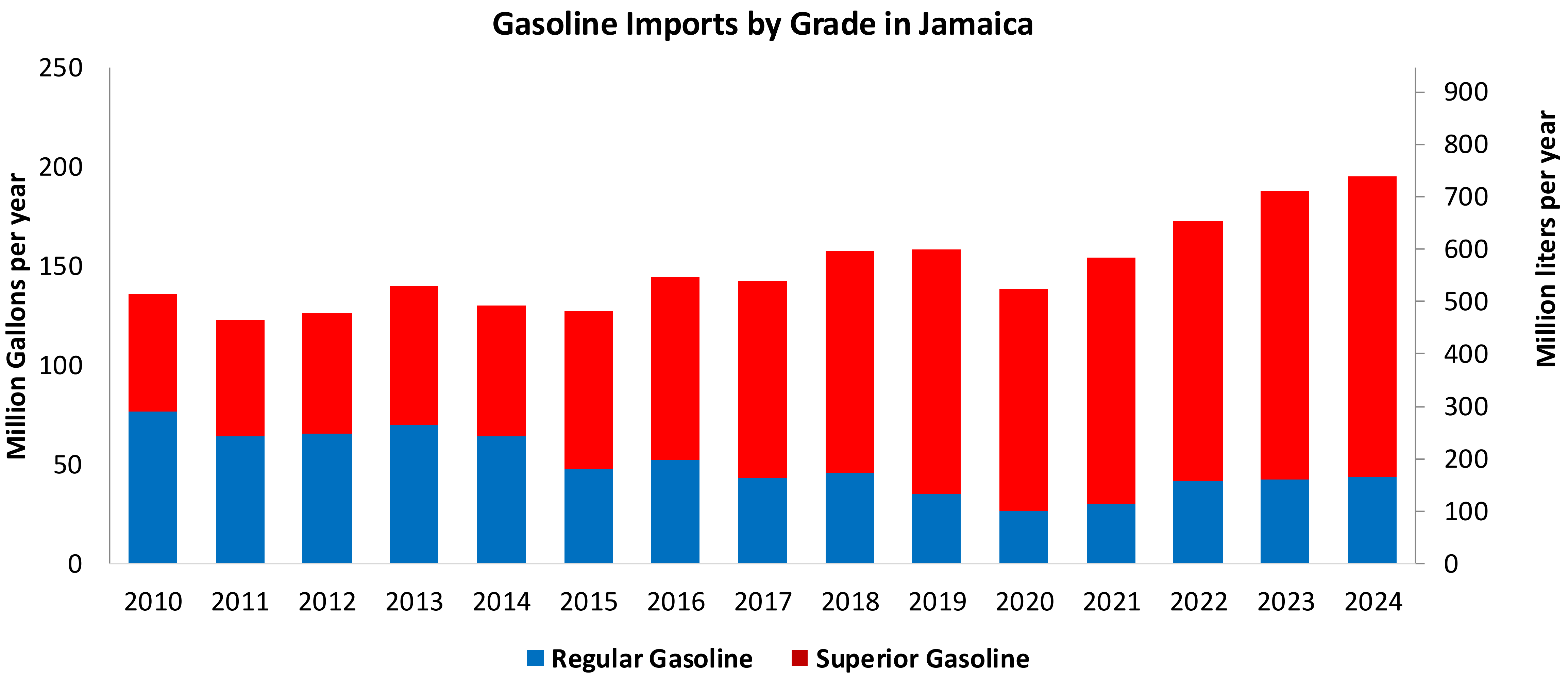
Source: Petrojam, MSET

Gasoline Demand in Jamaica



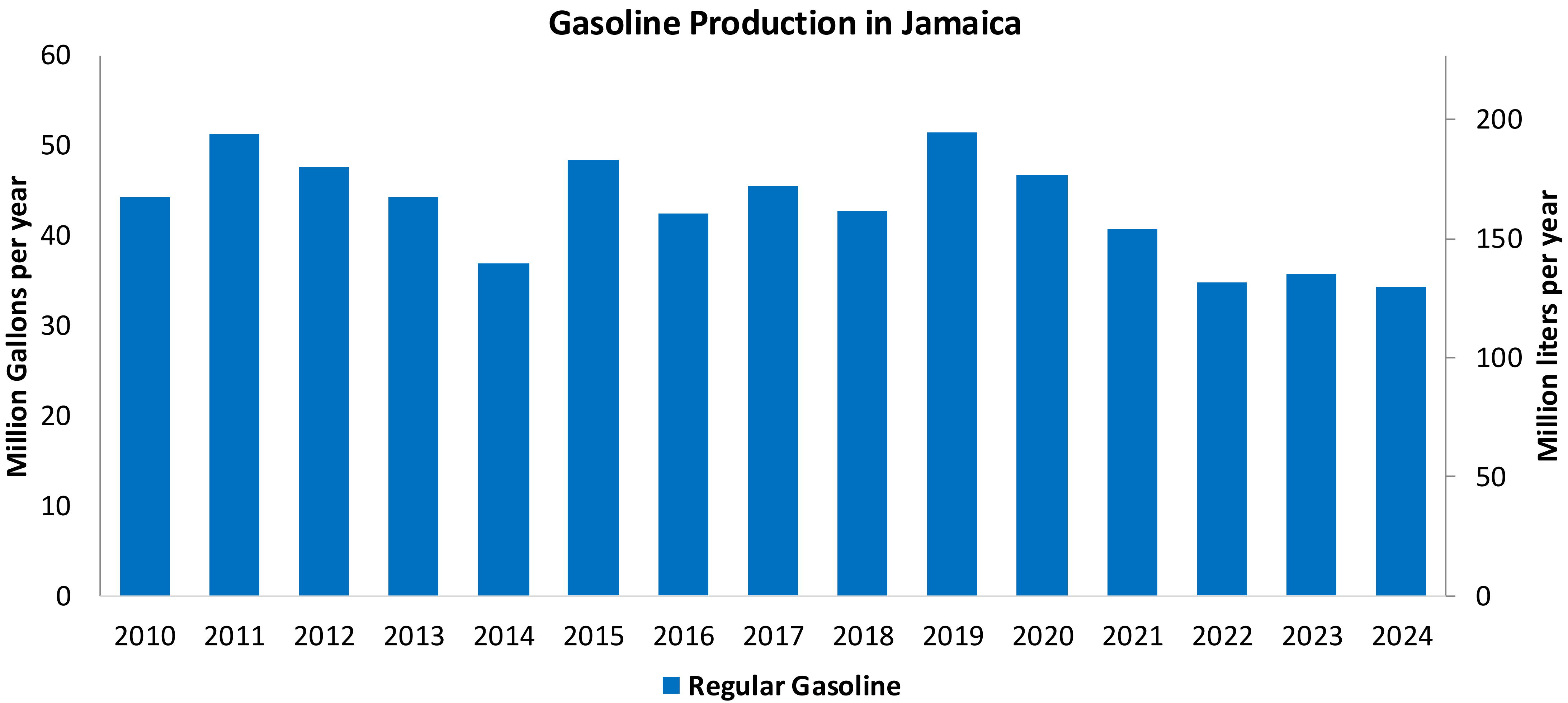
Source: Petrojam, MSET

Gasoline Imports in Jamaica



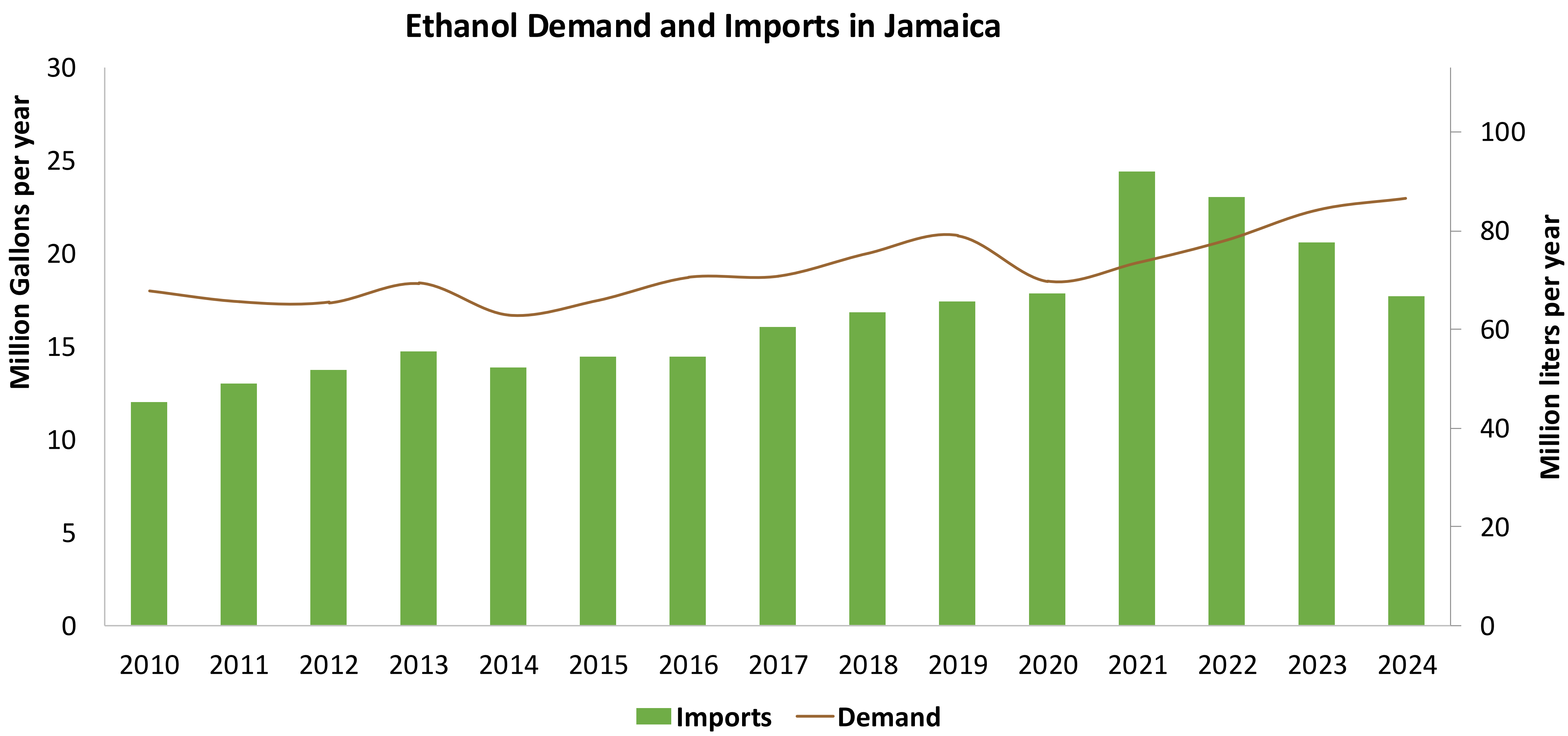
Source: Petrojam, MSET

Gasoline Production in Jamaica



Source: Petrojam, MSET

Ethanol Balance in Jamaica



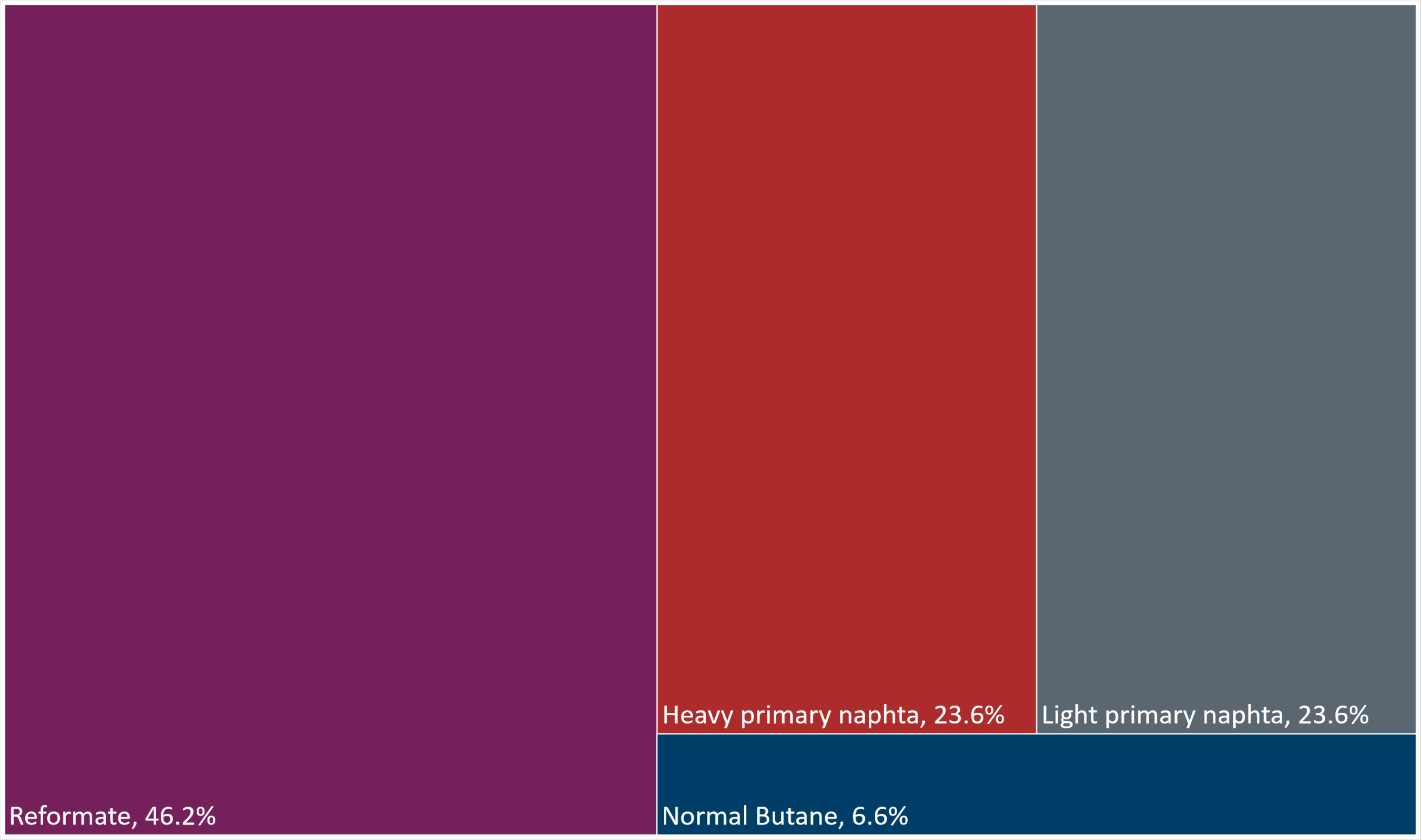
Source: MSET

Gasoline Quality in Jamaica

Name	JS 341:2017		EN 228:2012 + A1:2017 (Euro 6 enabling)			
Implementation Date	2017		2017			
Applicability	Whole country	Whole country	All countries			
Selected Grade	Unleaded Petrol AKI 87 E10	Unleaded Petrol AKI 90 E10	RON 95 E5	RON 95 E10	RON 98 E5	RON 98 E10
Benzene Content	< 5 %v/v	< 5 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v
Aromatics	< 45 %v/v	< 45 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v
Olefins	-	-	< 18 %v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v
Lead Content	< 0,013 g/l	< 0,013 g/l	< 5 mg/l	< 5 mg/l	< 5 mg/l	< 5 mg/l
Manganese	< 18 mg/l	< 18 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l
RON			> 95	> 95	> 98	> 98
MON			> 85	> 88	> 85	> 88
AKI	90	87				
Sulfur Content	< 1.500 mg/kg	< 1.500 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg
Oxygen Content	< 4 %m/m (Gasoline with no ethanol)	< 4 %m/m (Gasoline with no ethanol)	<2,7 % m/m	<3,7 % m/m	<2,7 % m/m	<3,7 % m/m
Ethanol (EtOH)	< 10 %v/v	< 10 %v/v	<5 %v/v	<10 %v/v	<5 %v/v	<10 %v/v
RVP 37.8°C (Summer)	< 61 kPa	< 69 kPa	<> 60 - 70 kPa *Depends on the country, RVP is regulated in the EU Fuel Quality Directive			
RVP 37.8 °C(Winter)						
RVP 37.8°C (Transition)						
MTBE	< 15% v/v for Gasoline with no ethanol	< 15% v/v for Gasoline with no ethanol	-	-	-	-
Ethers 5 or more C Atoms	15% v/v (other oxygenates)	15% v/v (other oxygenates)	Based on oxygen content	<22 %v/v	Based on oxygen content	<22 %v/v

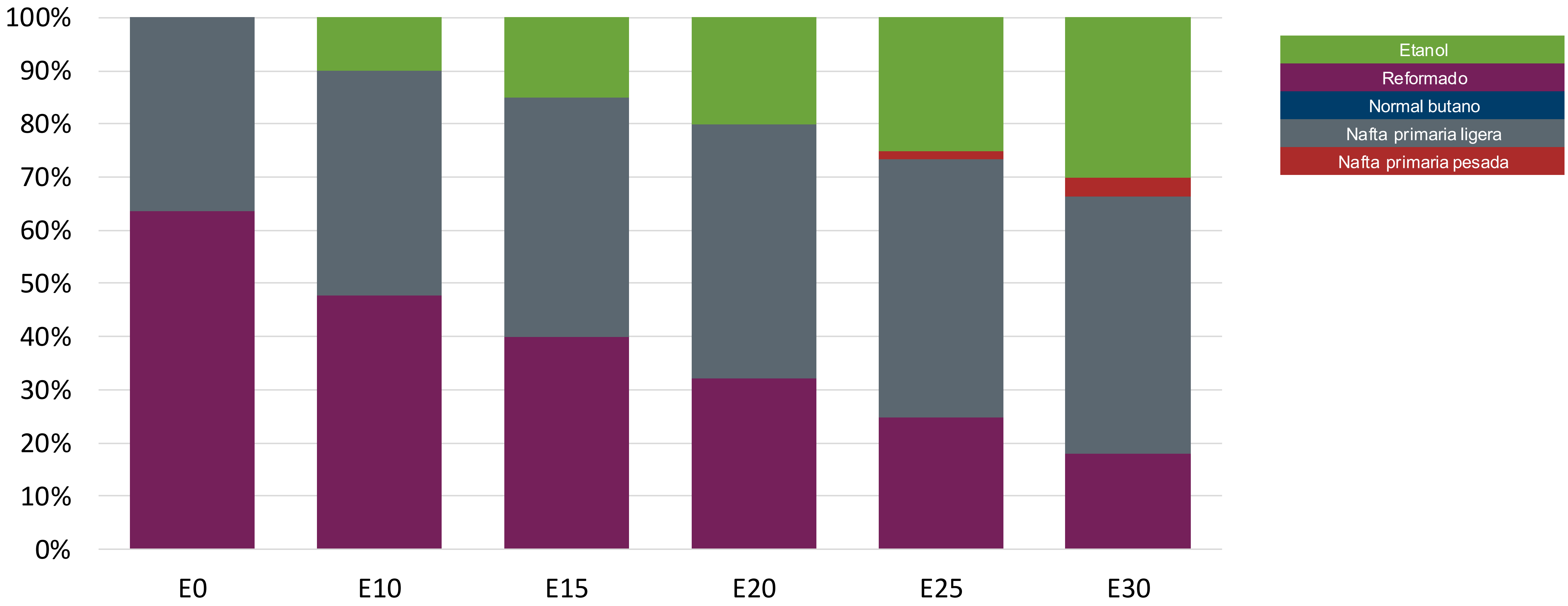
Source: Bureau Standards Jamaica (BSJ)

Available Blending Components



Source: Faro90

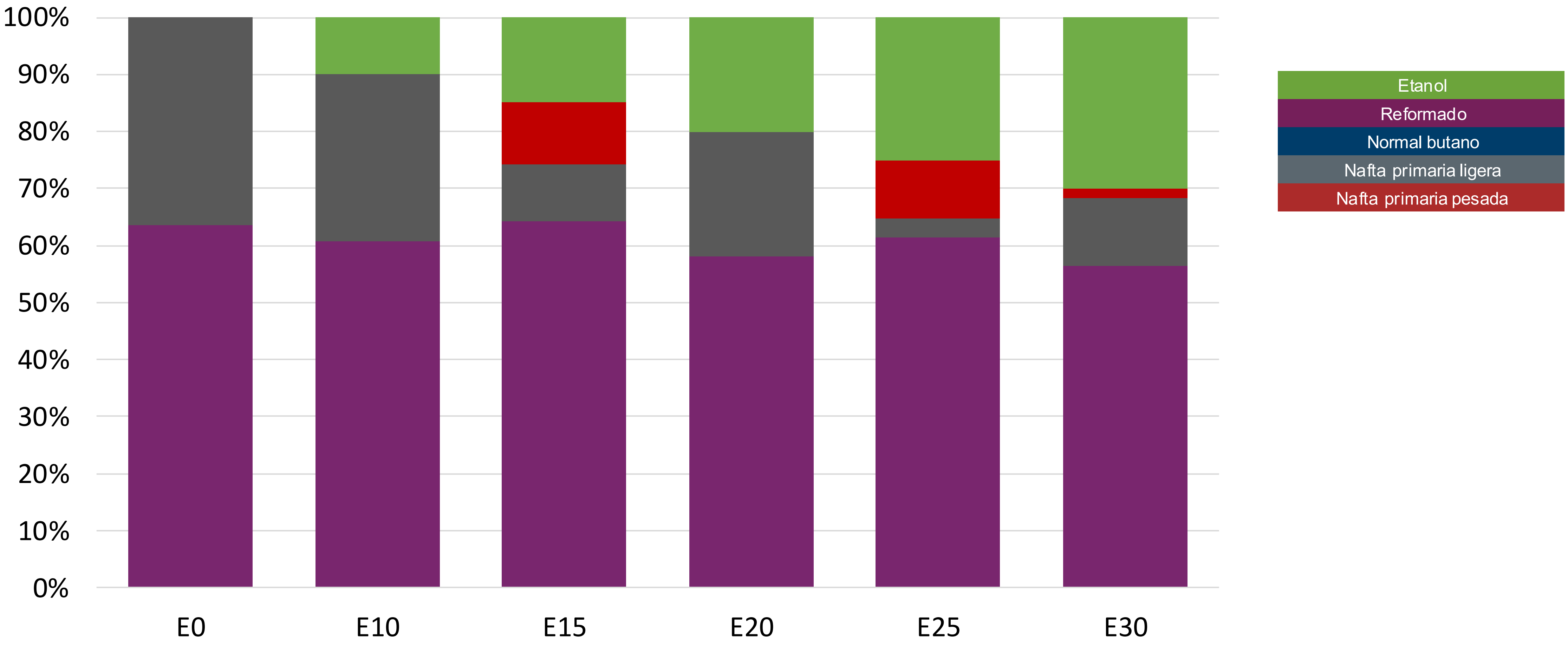
Regular Gasoline – Constant Octane



Octane (AKI)	87.0	87.0	87.0	87.0	87.0	87.0
Price (USD/gal)	$ 2.512	$ 2.314	$ 2.214	$ 2.115	$ 2.016	$ 1.917

Source: Faro90

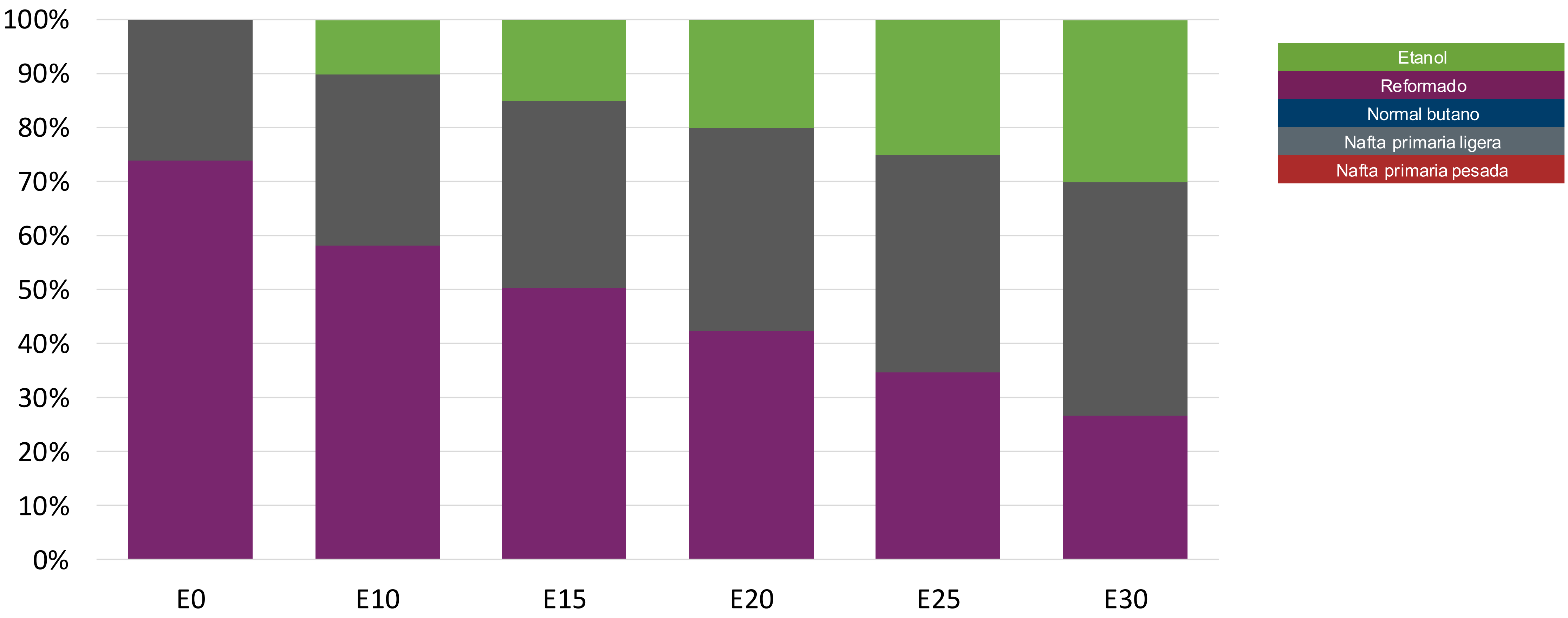
Regular Gasoline – Increased Octane



Octane (AKI)	87.0	90.8	92.6	94.5	96.4	98.3
Price (USD/gal)	$ 2.512	$ 2.512	$ 2.512	$ 2.512	$ 2.512	$ 2.512

Source: Faro90

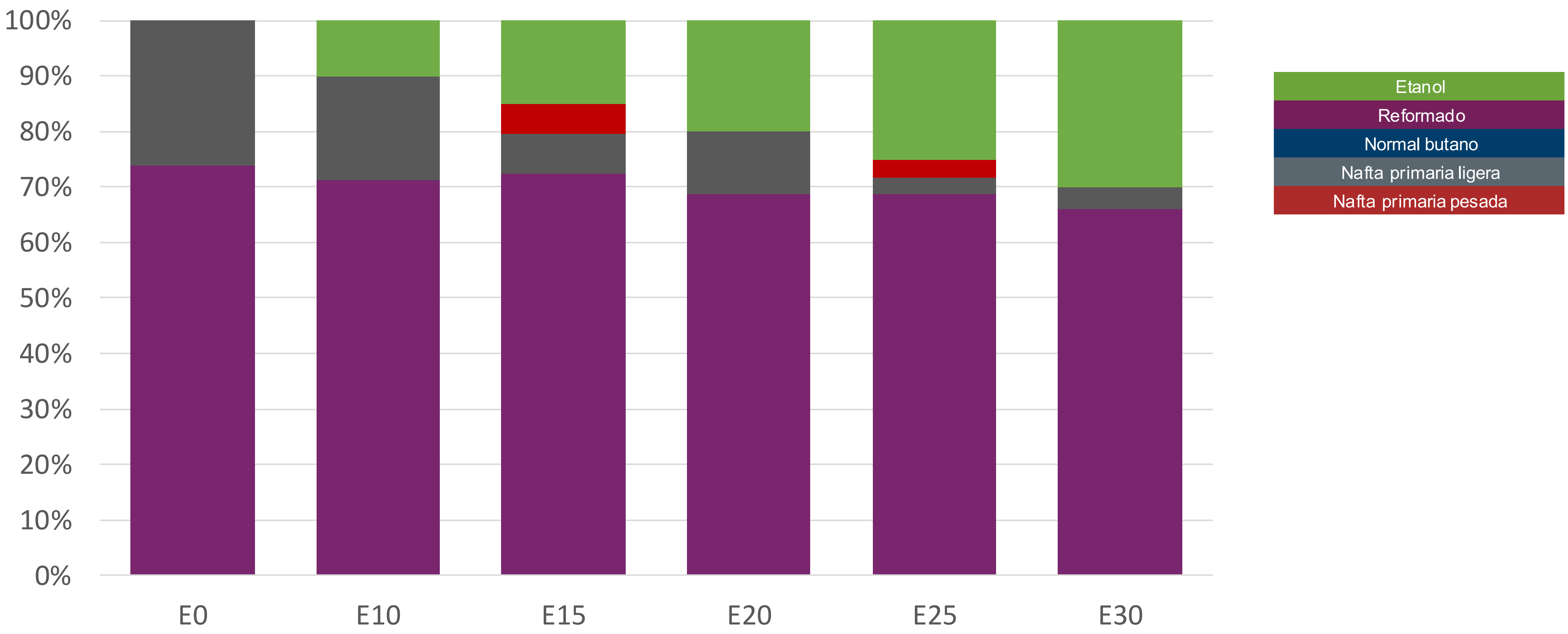
Premium Gasoline – Constant Octane



Octane (AKI)	90.0	90.0	90.0	90.0	90.0	90.0
Price (USD/gal)	$ 2.670	$ 2.472	$ 2.372	$ 2.273	$ 2.174	$ 2.075

Source: Faro90

Premium Gasoline – Octane Increment



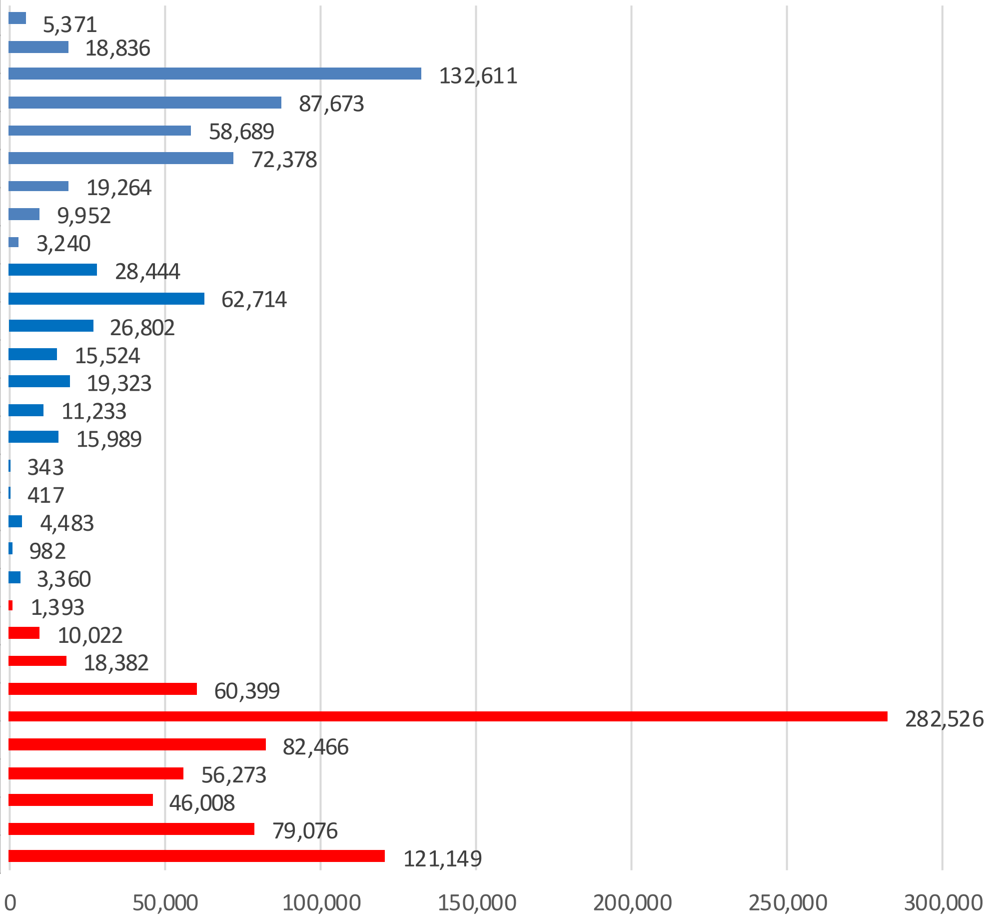
Octane (AKI)	90.0	93.8	95.6	97.5	99.4	101.3
Price (USD/gal)	$ 2.670	$ 2.670	$ 2.670	$ 2.670	$ 2.670	$ 2.670

Source: Faro90

Gasoline Vehicle Fleet - Jamaica

Type	Motor	Exhaust	Recuperator	Age
Auto/Light	Carburator	3-Ways	PCV Tank	<8 years, < 160 mkm
	Single FI-Pt	3-Ways		<8 years, < 160 mkm
	Multi FI-Pt	3-Ways		<8 years, < 160 mkm
		Euro III		<8 years, < 160 mkm
		Euro IV		<8 years, < 160 mkm
		Euro V / Euro 6		<8 years, < 160 mkm
				4-8 years, 80-160 mkm
				>4years, >80 mkm
Auto/Medium	Carburator	3-Ways	PCV Tank	<8 years, < 160 mkm
	Single FI-Pt	3-Ways		<8 years, < 160 mkm
	Multi FI-Pt	3-Ways		<8 years, < 160 mkm
		Euro III		<8 years, < 160 mkm
		Euro IV		<8 years, < 160 mkm
		Euro V / Euro 6		<8 years, < 160 mkm
				4-8 years, 80-160 mkm
				>4years, >80 mkm
Small Engines	FI 4-cycles	None	PCV	<8 years, < 160 mkm
		Euro II		<8 years, < 160 mkm
		Euro III		<8 years, < 160 mkm
		Euro IV		<8 years, < 160 mkm
				4-8 years, 80-160 mkm
		Euro V		4-8 years, 80-160 mkm
				>4years, >80 mkm

Vehicle Numbers



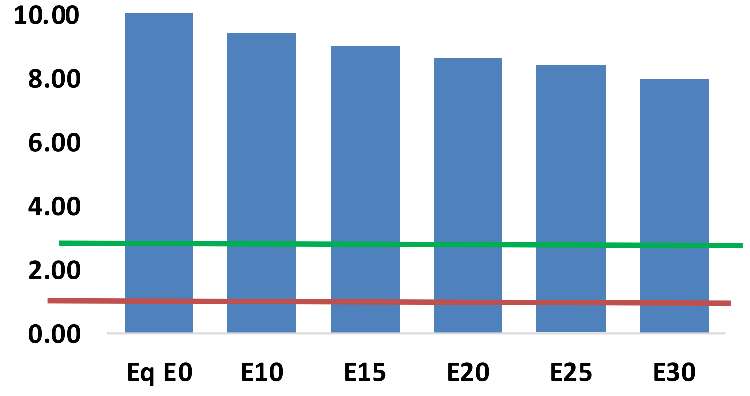
Vehicular Fleet: **609,044**
Average Age: **20 years**
Motorcycles: **2.3%**

Source: MSET, Faro 90

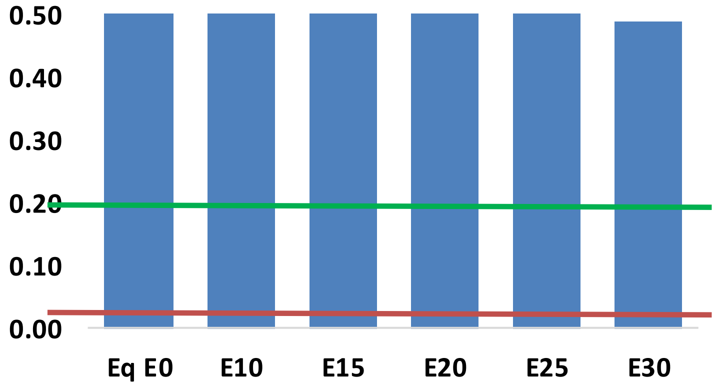
Jamaica – Vehicular Emissions

United States
Euro 6

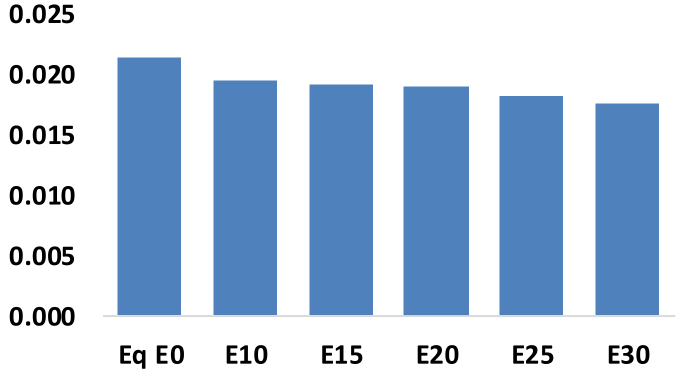
Carbon Monoxide (CO) g/km



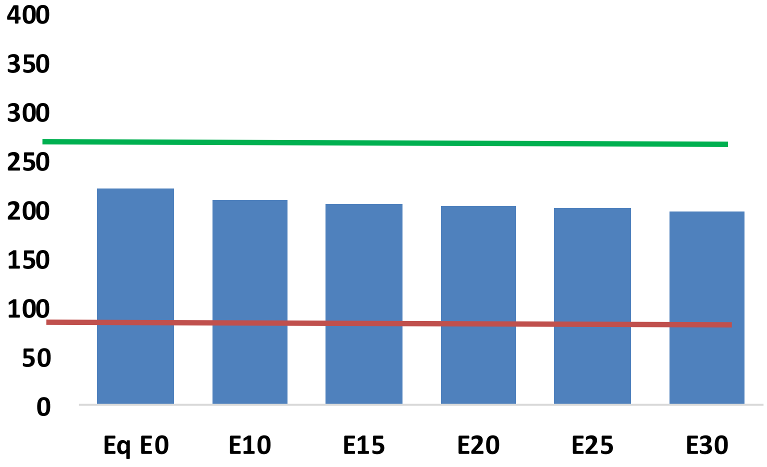
Nitrogen Oxices (NOx) g/km



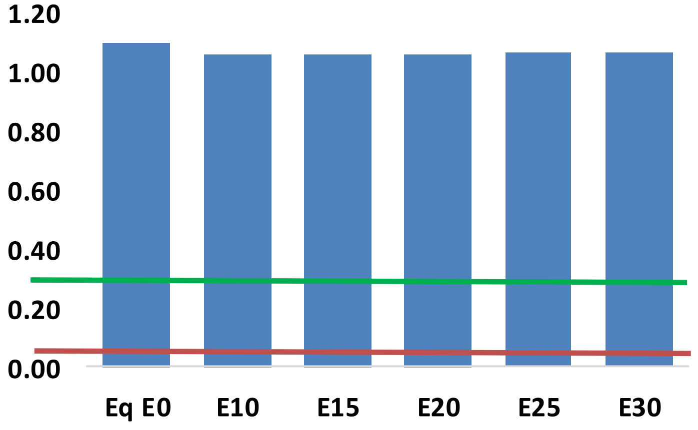
Aromatics (BTX) g/km



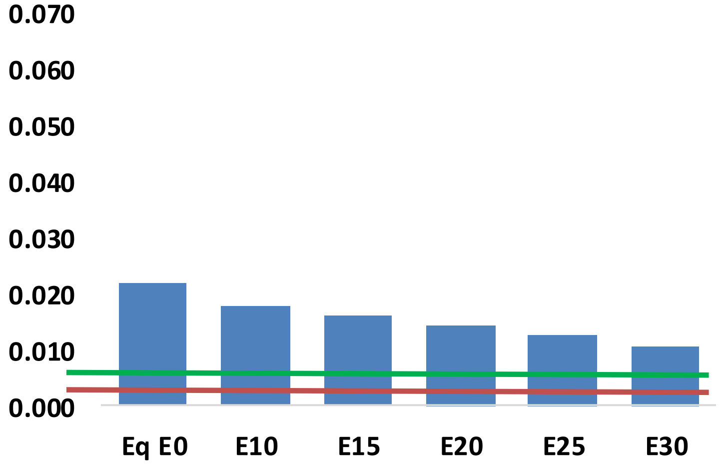
Carbon Monoxide (CO2) g/km



Total Hydrocarbons (THC) g/km



Particulate Matter (PM2.5) g/km



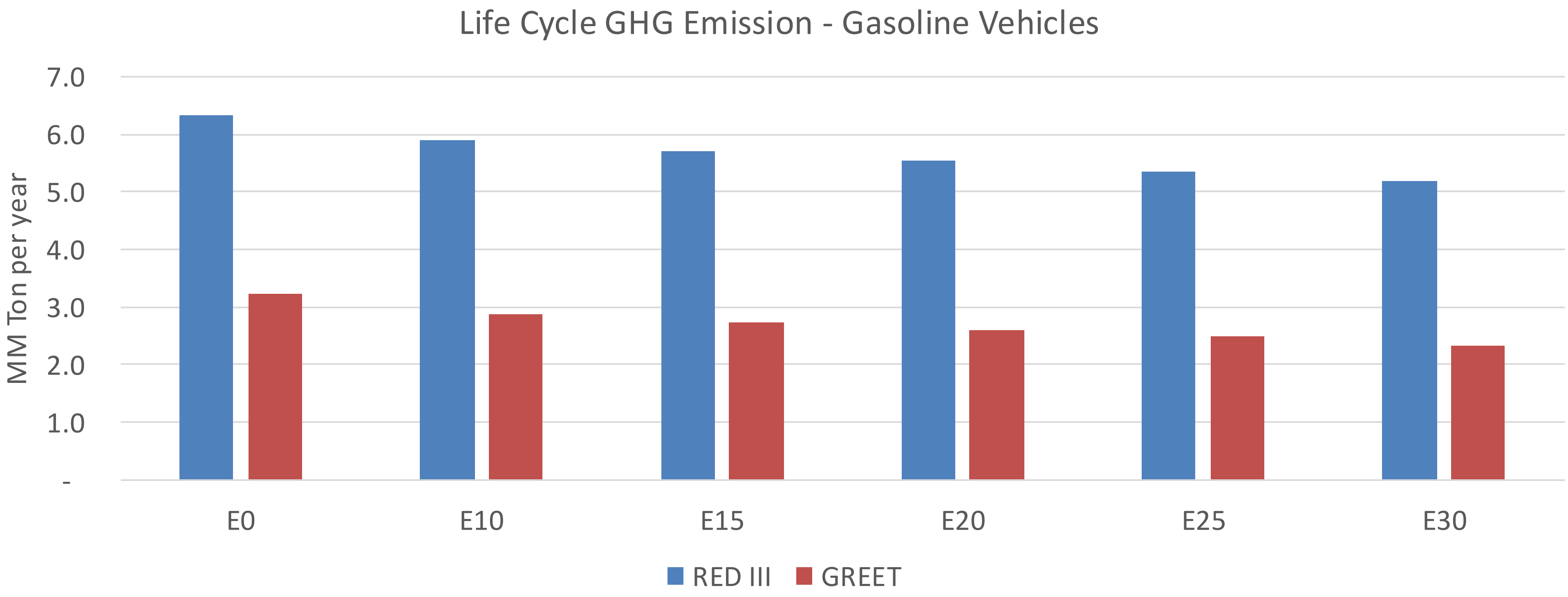
Source: Faro90

Jamaica – Gasoline Vehicle Fleet Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	166,859	158,208	149,556	143,282	137,342	132,889	127,019	-10%	-18%
CO2	3,490,486	3,400,730	3,315,961	3,249,293	3,216,303	3,199,172	3,140,202	-5%	-8%
VOC	17,388	17,100	16,813	16,782	16,845	16,922	16,924	-3%	-3%
NOx	10,587	9,775	9,234	8,968	8,528	8,188	7,785	-13%	-19%
SOx	41	38	35	32	29	27	24	-15%	-28%
NH3	1,181	1,169	1,158	1,163	1,176	1,186	1,193	-2%	0%
N2O	54	54	54	54	55	56	56	-1%	2%
CH4	3,735	3,735	3,735	3,791	3,873	3,944	4,011	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	95	92	89	87	86	86	84	-6%	-9%
Acetaldehyde	175	200	294	448	610	697	824	68%	249%
Formaldehyde	501	487	568	667	698	773	841	13%	39%
Benzene	339	326	308	304	301	288	279	-9%	-11%
PM2.5	200	181	163	147	132	115	97	-18%	-34%
PM10	147	134	120	109	97	85	72	-18%	-34%

Source: Faro90

Jamaica – Gasoline Vehicle Fleet GEI Emissions



Country Emissions 2024 (MMT/year)	9.4
2035 Target (MMT/year)	4.7
Delta	4.7

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	11.0%	15.0%	19.3%	23.2%	27.8%
RED II/III	0.0%	7.2%	10.0%	12.8%	15.5%	18.4%

% 2035 Target	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	7.5%	10.3%	13.2%	15.9%	19.0%
RED II/III	0.0%	9.7%	13.4%	17.3%	20.8%	24.8%

Source: Greet, RED II/III, climateactiontracker.org, F90

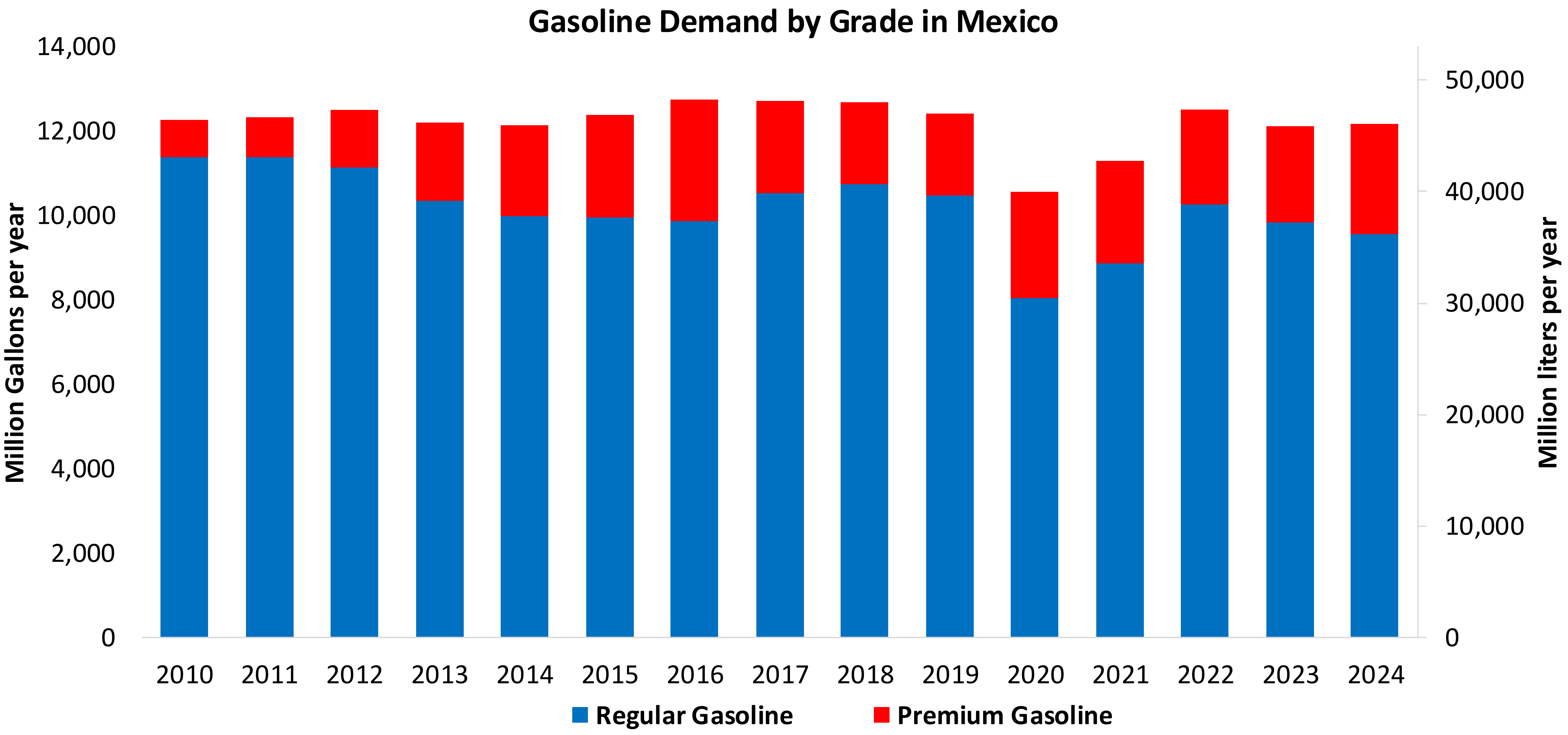
Ethanol Blending in Gasoline - Mexico

In 2024, gasoline consumption was 12,200 million gallons (46,000 million liters). There are two grades of gasoline, Regular (AKI 87) and Premium (AKI 91). Regular gasoline is the main grade in the market, representing 90% of the volume. There are stringent gasoline specifications for major metropolitan areas in Mexico City, Guadalajara and Monterrey. Demand is much higher than production, making necessary to import 40% of gasoline mainly from the United States.

Gasoline quality norm (NOM-016-CRE de 2016) allows blends with a maximum of 5.8% v/v of ethanol with the exception of metropolitan areas of Valle de Mexico, Guadalajara and Monterrey. Current ethanol use is equivalent to an average of 1.06% in the Rest of the Country grade gasoline. In 2025, a new biofuels law was approved that fosters the production with agricultural and other organic residuals. Recently there are new state level efforts to increase ethanol consumption (e.g. Tamaulipas).

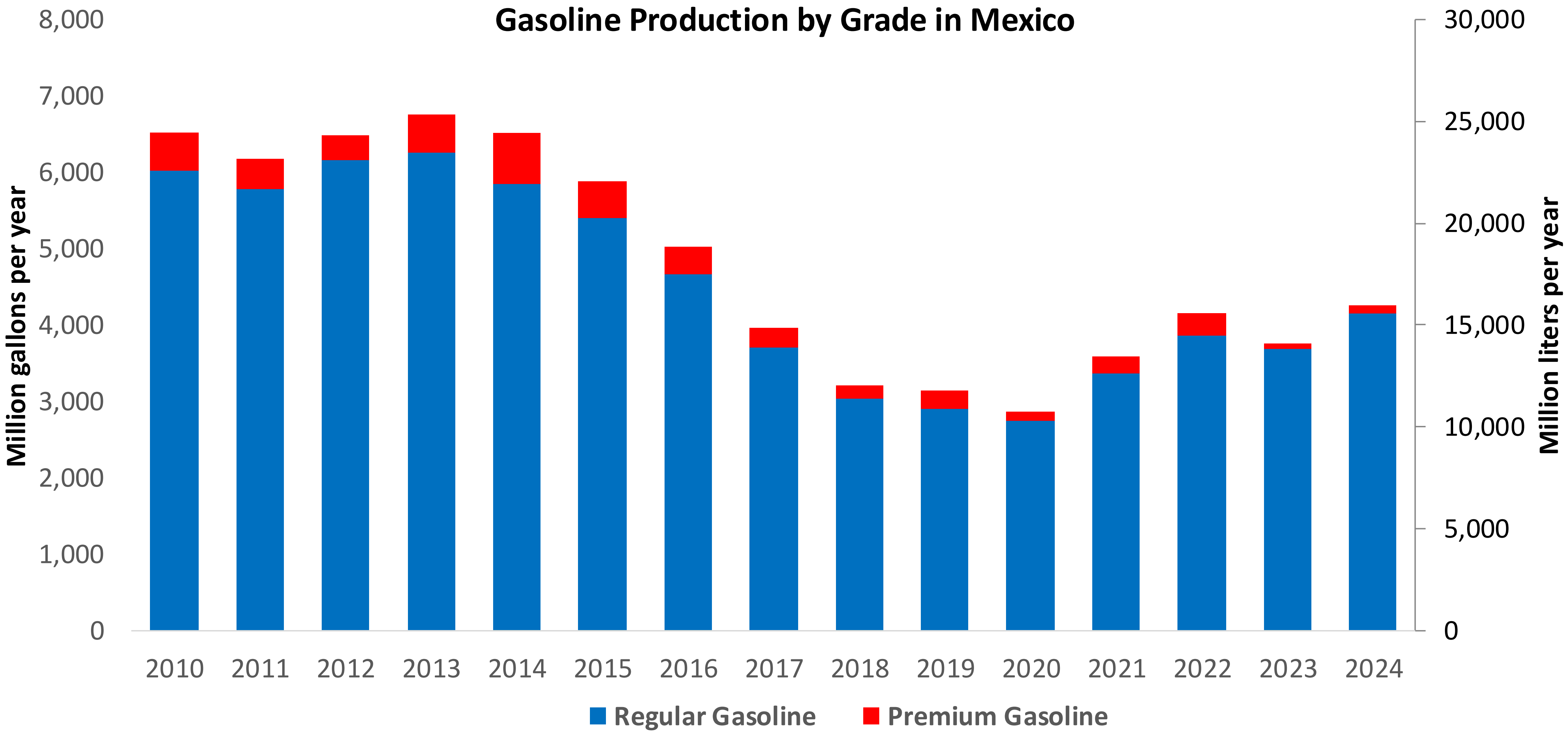
Source: SIE - SENER, 2024

Gasoline Demand in Mexico



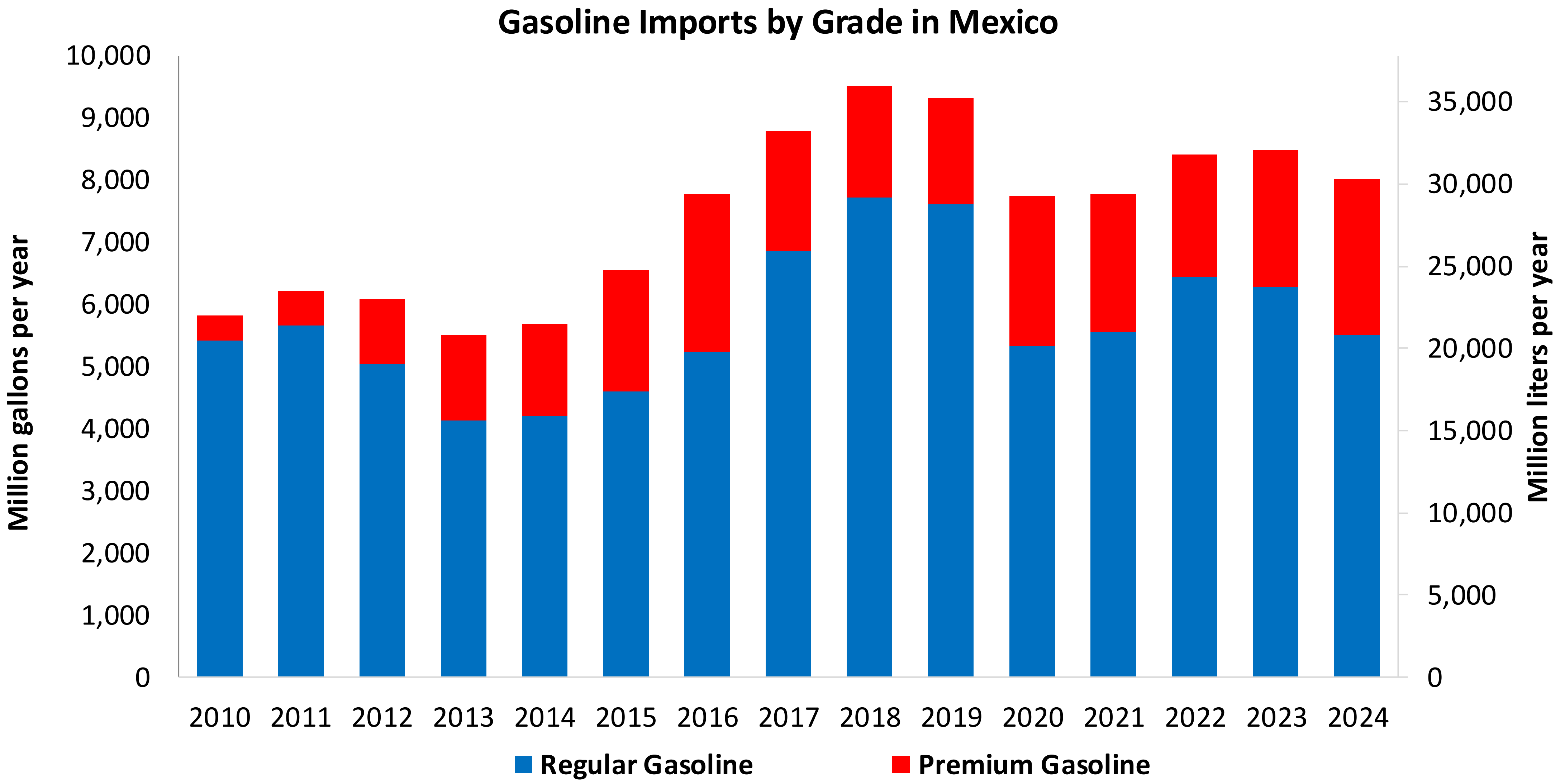
Source: SIE - SENER, 2025

Gasoline Production in Mexico



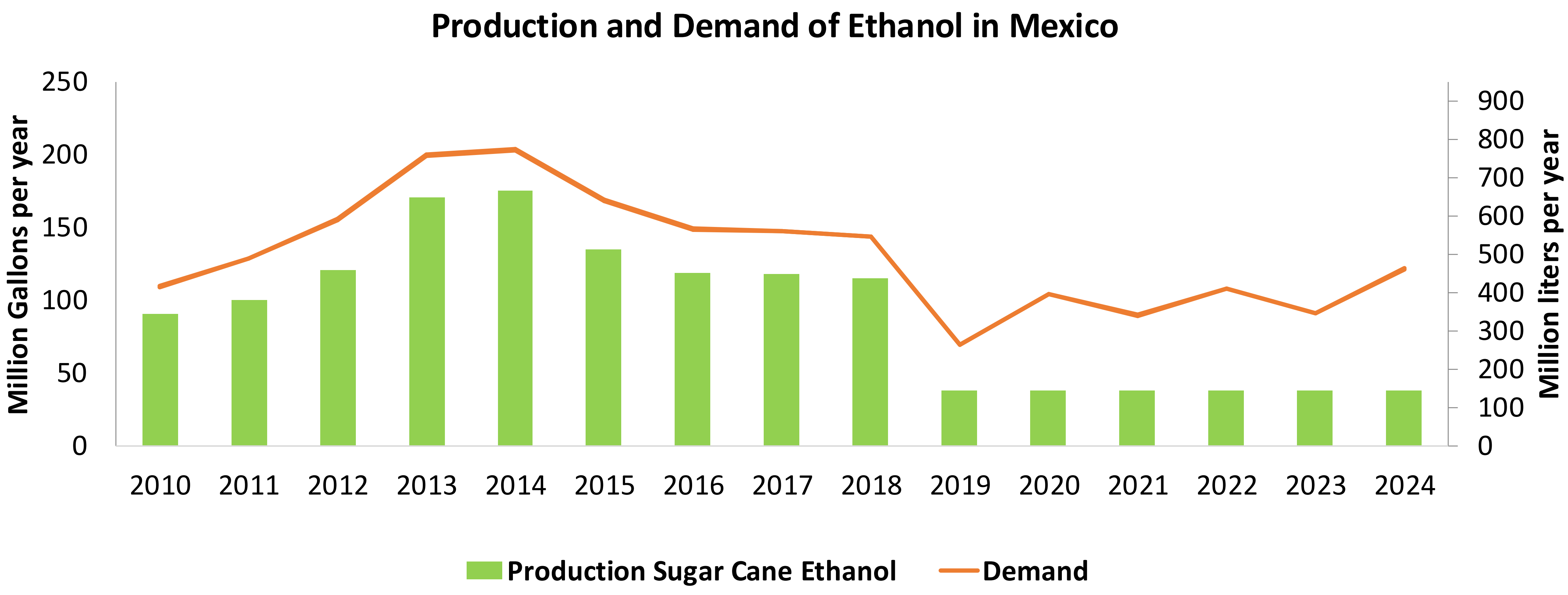
Source: SIE - SENER, 2025

Gasoline Imports in Mexico



Source: SIE - SENER, 2025

Ethanol Balance in Mexico



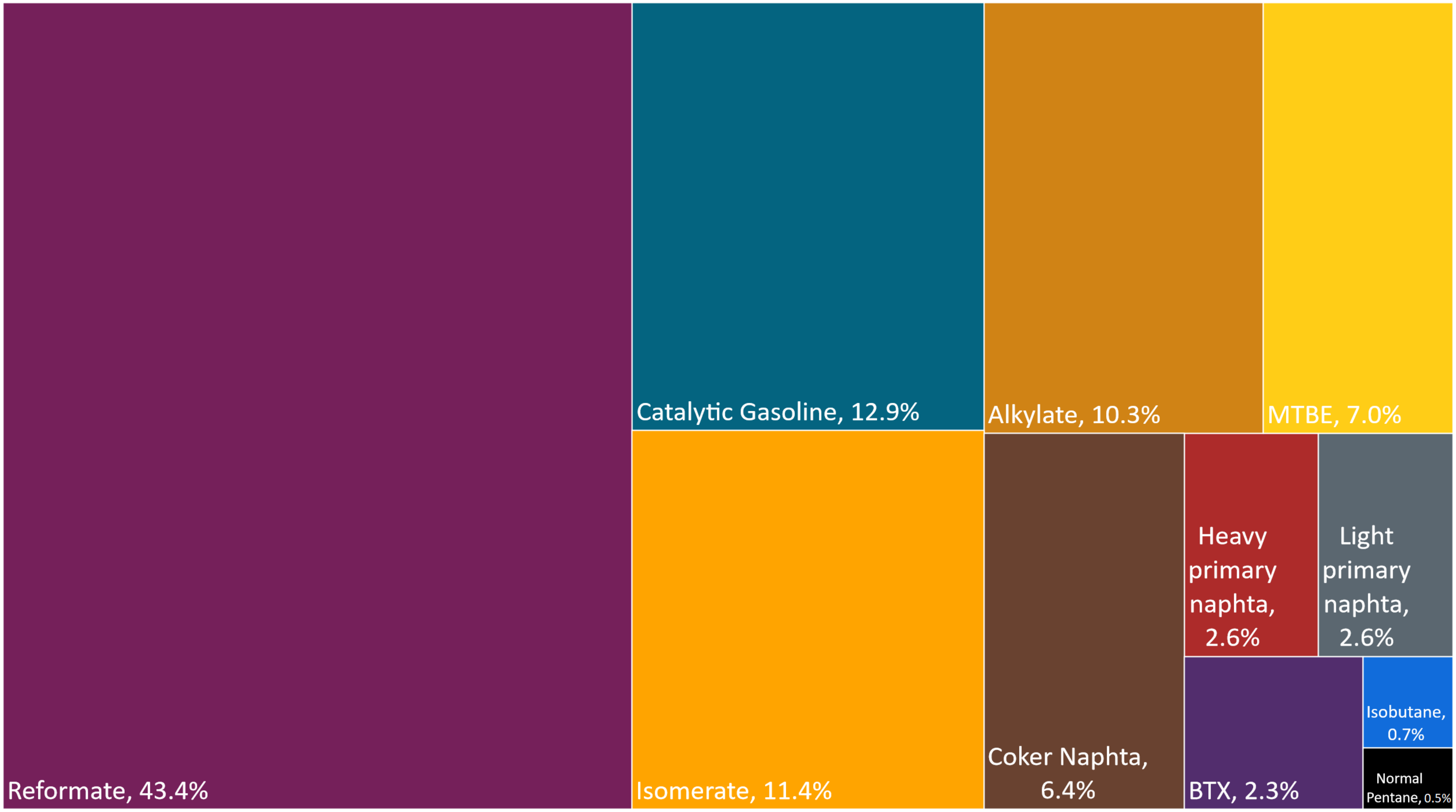
Source: CEDRSSA, 2020; EIA, 2025; Sanchez, 2014.

Gasoline Quality in Mexico

Name	NOM-016-CRE-2016						EN 228:2012 + A1:2017 (Euro 6 enabling)			
Implementation Date	2016						2017			
Applicability	Mexico City	Mexico City	Guadalajara and Monterrey	Guadalajara and Monterrey	Rest of the country	Rest of the country	All countries			
Selected Grade	Gasoline Regular	Gasoline Premium	Gasoline Regular	Gasoline Premium	Gasoline Regular	Gasoline Premium	RON 95 E5	RON 95 E10	RON 98 E5	RON 98 E10
Benzene Content	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 2 %v/v	< 2 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v
Aromatics	< 25 %v/v	< 25 %v/v	< 32 %v/v	22,6% v/v	Report	22,6% v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v
Olefins	< 10 %v/v	< 10 %v/v	< 11,9 %v/v	< 11,9 %v/v	Report	< 2,5 mg/l	< 18 %v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v
Lead Content	-	-	-	-	-	-	< 5 mg/l	< 5 mg/l	< 5 mg/l	< 5 mg/l
Manganese	0 mg/l	0 mg/l	0 mg/l	0 mg/l	0 mg/l	0 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l
RON		94		94		94	> 95	> 95	> 98	> 98
MON	82		82		82		> 85	> 88	> 85	> 88
AKI										
Sulfur Content	< 80 mg/kg	< 80 mg/kg	< 80 mg/kg	< 80 mg/kg	< 80 mg/kg	< 80 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg
Oxygen Content	< 3,7 %m/m	< 2,7 %m/m	< 3,7 %m/m	< 2,7 %m/m	< 3,7 %m/m	< 2,7 %m/m	<2,7 % m/m	<3,7 % m/m	<2,7 % m/m	<3,7 % m/m
Ethanol (EtOH)	Not allowed	Not allowed	Not allowed	Not allowed	> 5,8 %v/v	> 5,8 %v/v	<5 %v/v	<10 %v/v	<5 %v/v	<10 %v/v
RVP 37.8°C (Summer)	< 62 kPa	< 62 kPa	< 62 kPa	< 62 kPa	< 69 kPa North, < 62 kPa Center, South, Pacific	< 69 kPa North, < 62 kPa Center, South, Pacific	<> 60 - 70 kPa *Depends on the country, RVP is regulated in the EU Fuel Quality Directive			
RVP 37.8 °C(Winter)	< 79 kPa	< 79 kPa	< 79 kPa	< 79 kPa	< 79 kPa	< 79 kPa				
RVP 37.8°C (Transition)	< 54 kPa	< 54 kPa	< 54 kPa Guadalajara	< 54 kPa Guadalajara						
MTBE							-	-	-	-
Ethers 5 or more C Atoms	-	-	-	-	-	-	Based on oxygen content	<22 %v/v	Based on oxygen content	<22 %v/v

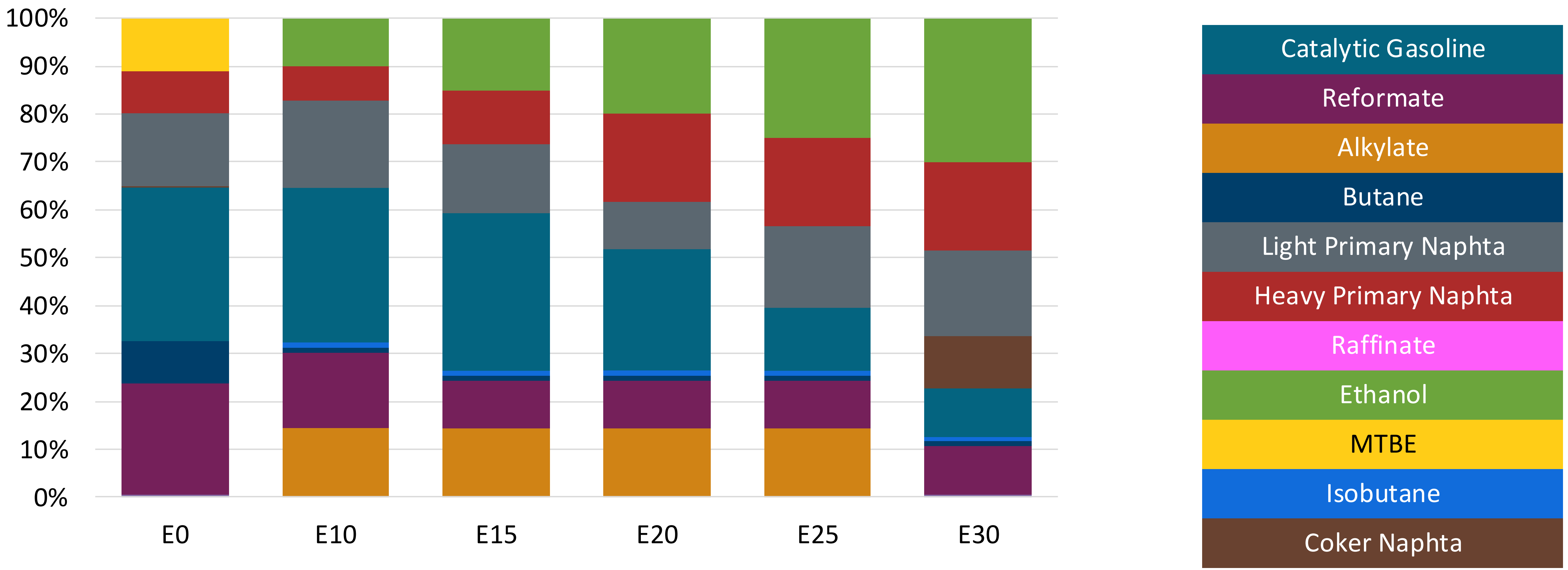
Source: CRE, 2016

Available Blending Components



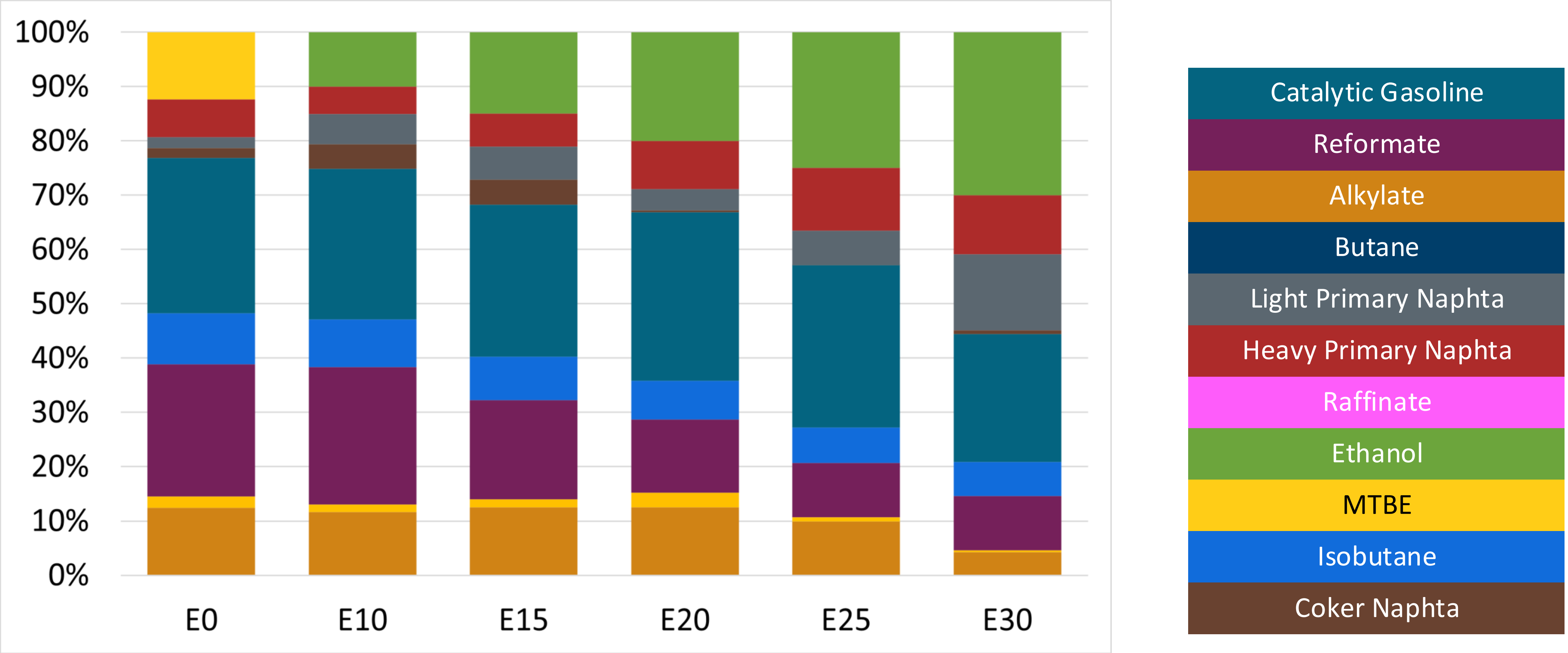
Source: Faro90

Regular Gasoline RP – Constant Octane



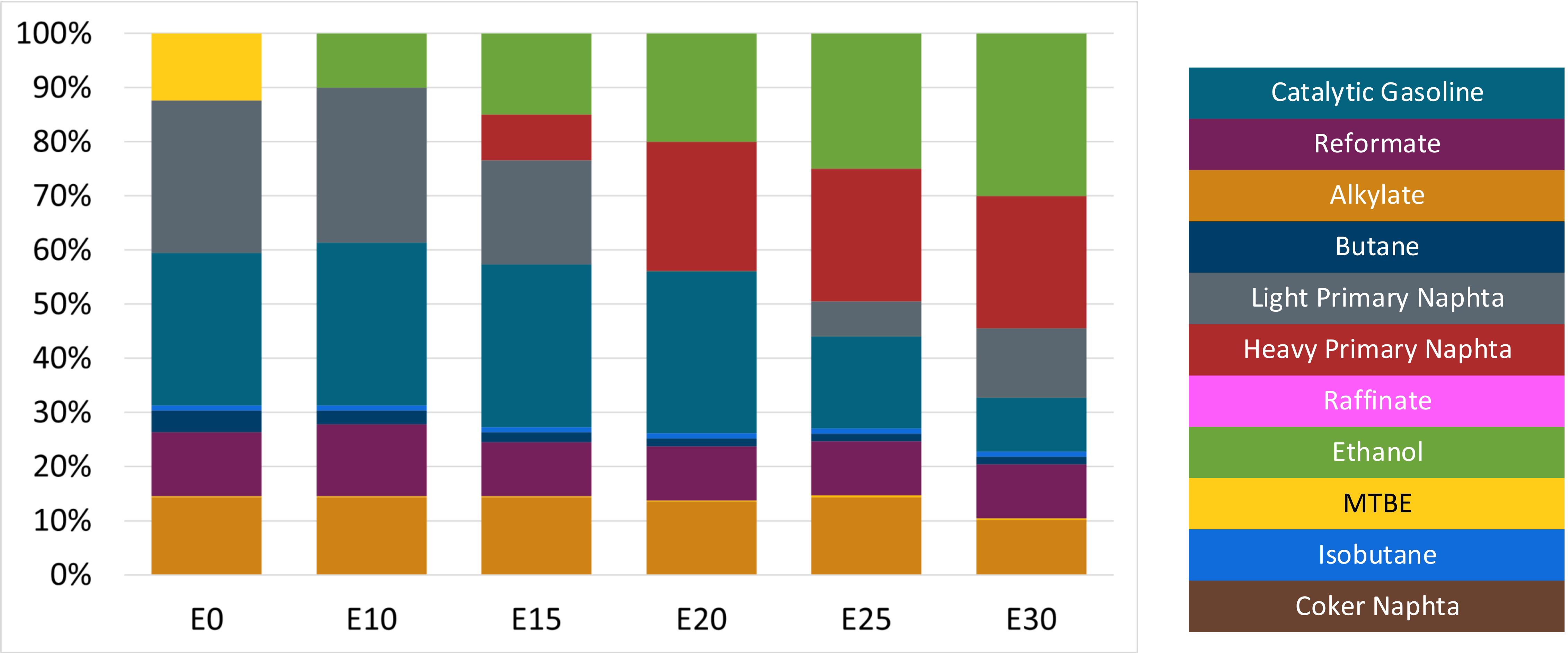
	E0	E10	E15	E20	E25	E30
Octane (AKI)	87.1	87.0	87.0	87.0	87.0	87.0
Price (US$/gal)	$ 2.245	$ 2.286	$ 2.210	$ 2.138	$ 2.067	$ 2.005

Premium Gasoline RP – Constant Octane



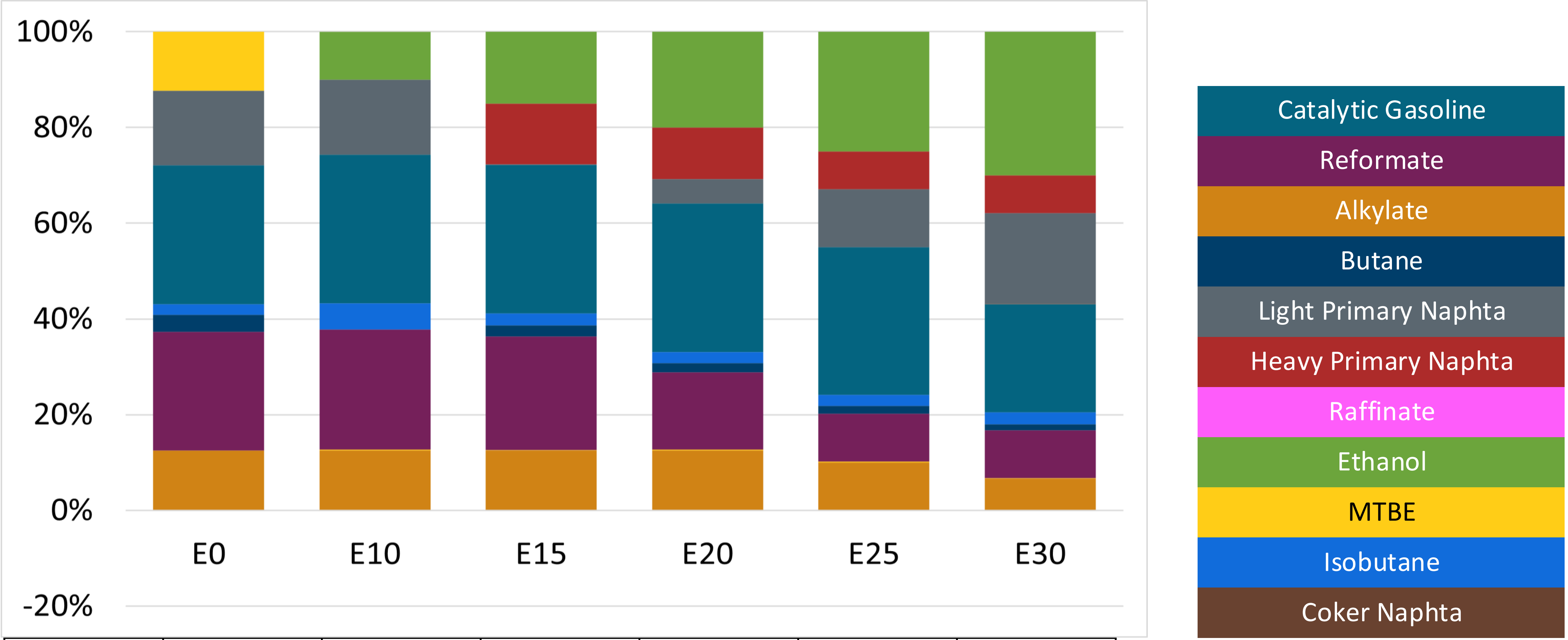
Octane (AKI)	91.0	91.0	91.0	91.0	91.0	91.0
Price (US$/gal)	$ 2.423	$ 2.395	$ 2.331	$ 2.266	$ 2.201	$ 2.134

Regular Gasoline ZM – Constant Octane



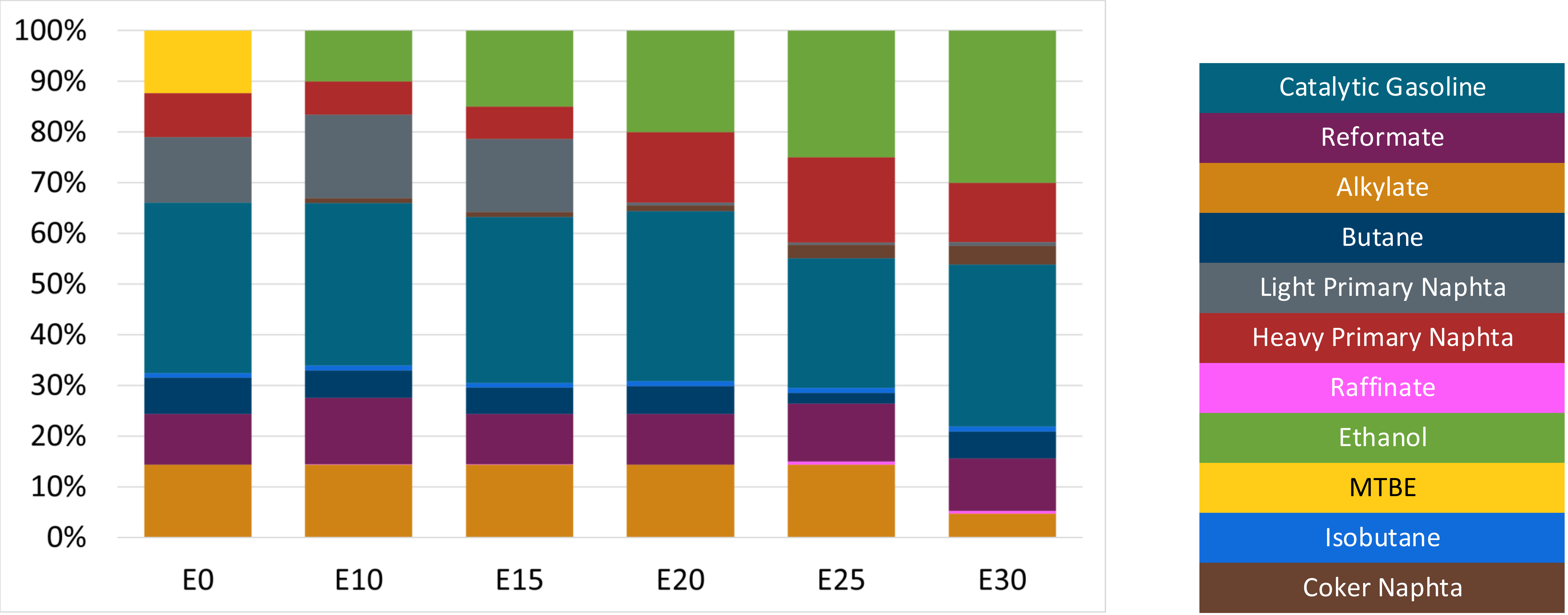
Octane (AKI)	87.0	87.0	87.0	87.0	87.0	87.0
Price (US$/gal)	$ 2.280	$ 2.265	$ 2.199	$ 2.131	$ 2.061	$ 1.990

Premium Gasoline ZM – Constant Octane



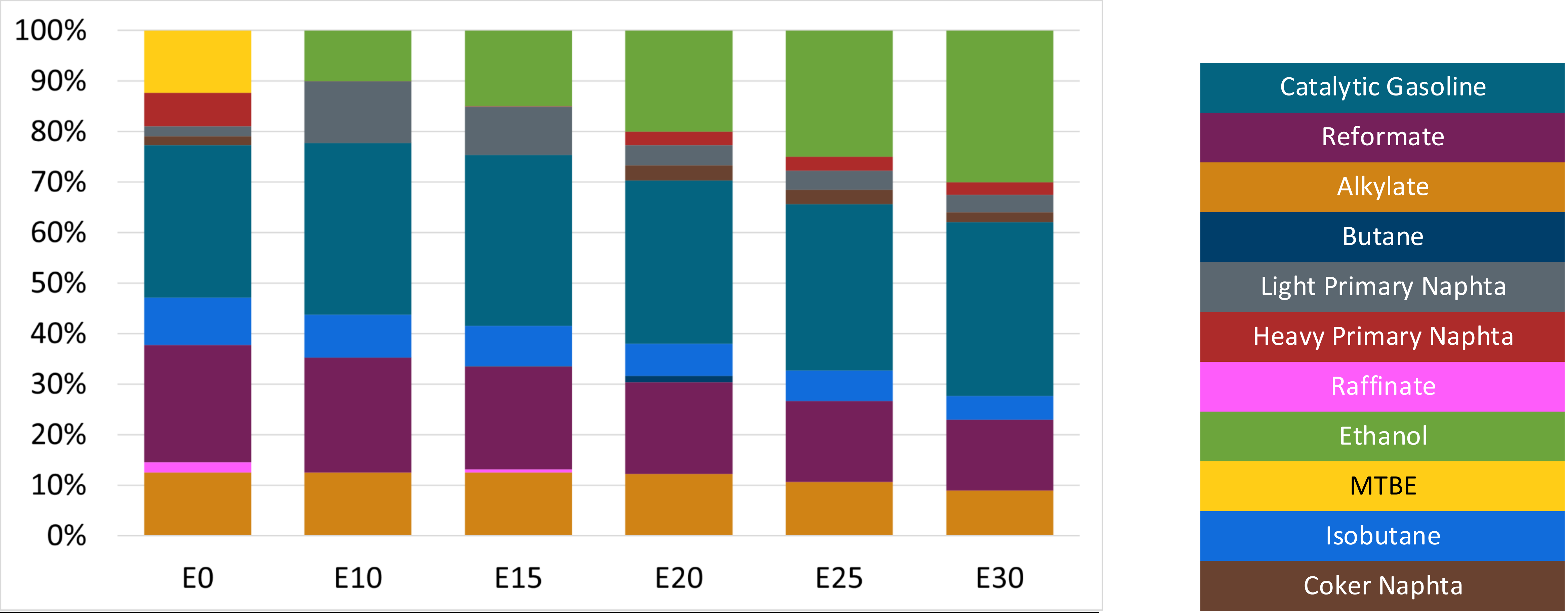
	E0	E10	E15	E20	E25	E30
Octane (AKI)	91.0	91.0	91.0	91.0	91.0	91.0
Price (US$/gal)	$ 2.478	$ 2.442	$ 2.377	$ 2.310	$ 2.242	$ 2.172

Regular Gasoline RP – Increased Octane



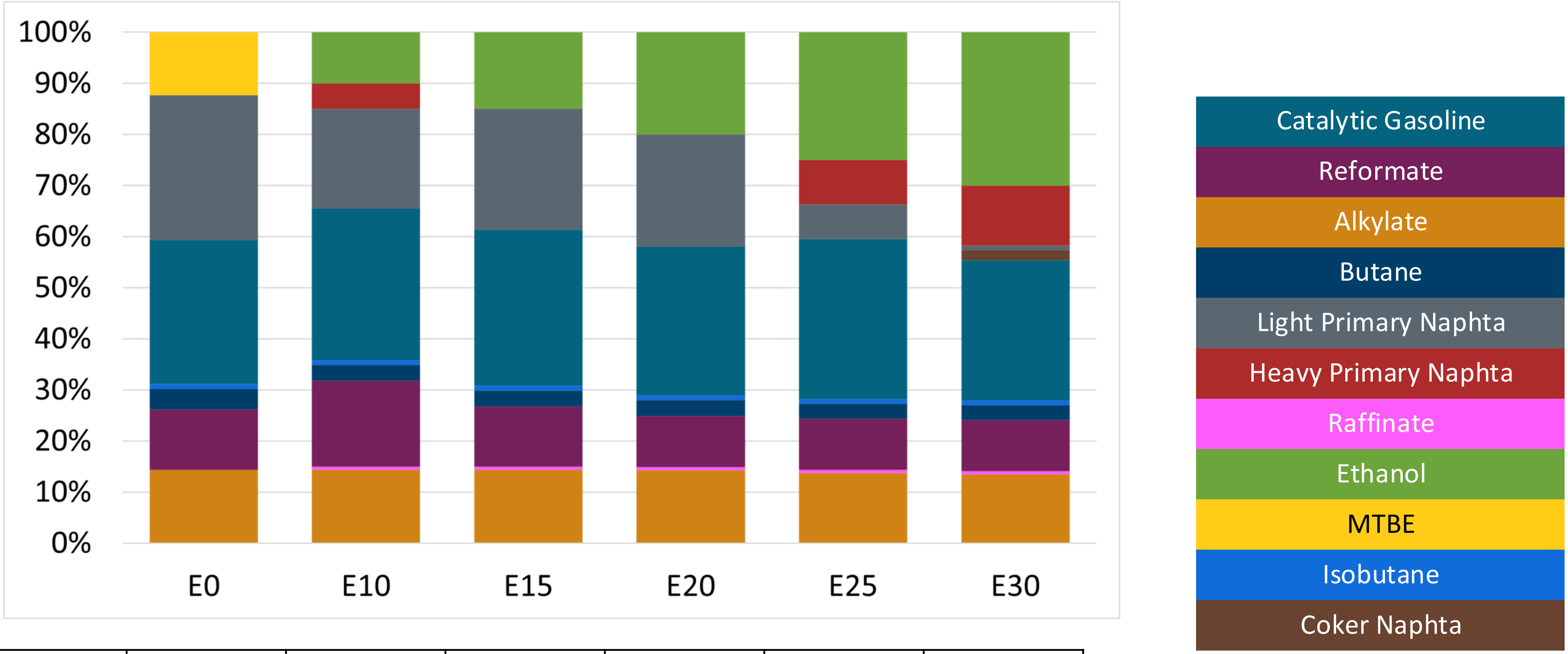
Octane (AKI)	87.0	87.2	88.6	90.1	90.5	92.9
Price (US$/gal)	$ 2.423	$ 2.423	$ 2.423	$ 2.423	$ 2.423	$ 2.423

Premium Gasoline RP – Increased Octane



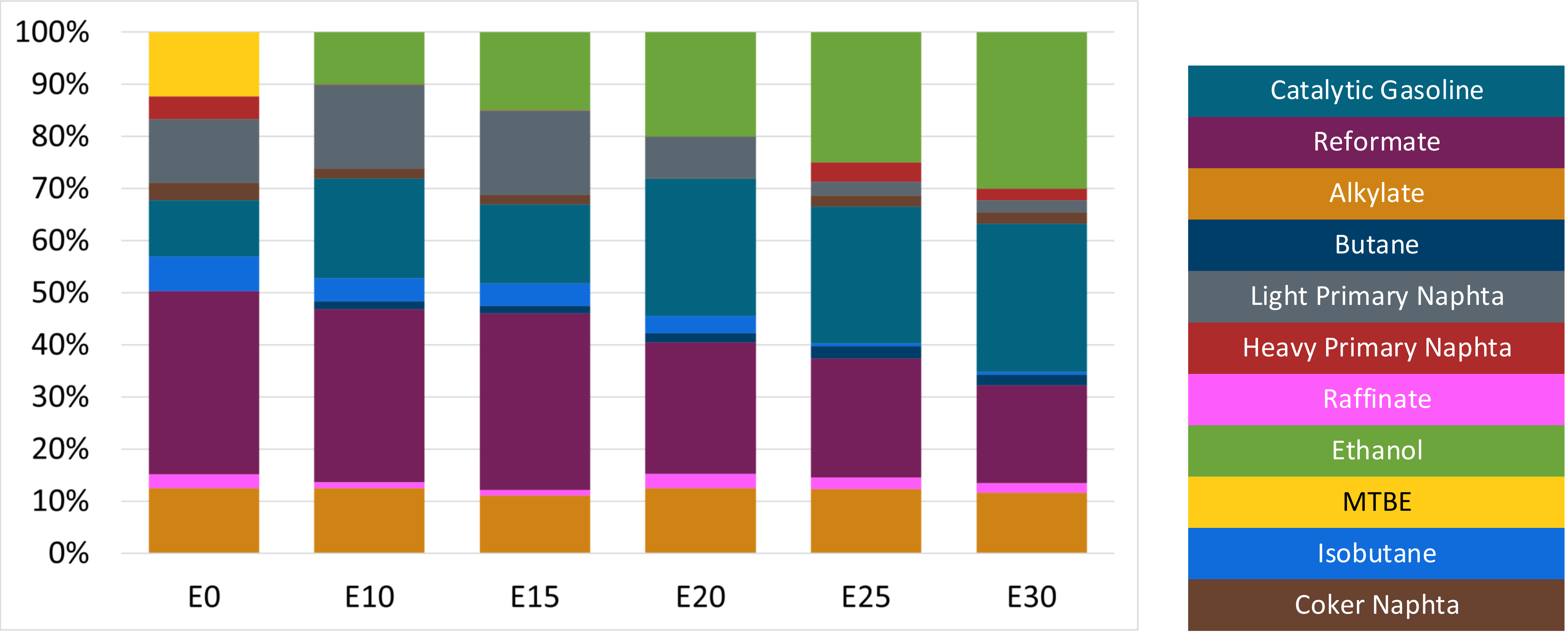
	E0	E10	E15	E20	E25	E30
Octane (AKI)	91.0	91.6	92.8	94.1	95.1	96.2
Price (US$/gal)	$ 2.423	$ 2.423	$ 2.423	$ 2.423	$ 2.423	$ 2.423

Regular Gasoline ZM – Increased Octane



Octane (AKI)	87.0	87.4	88.9	90.4	91.7	93.2
Price (US$/gal)	$ 2.280	$ 2.280	$ 2.280	$ 2.280	$ 2.280	$ 2.280

Premium Gasoline ZM – Increased Octane



Octane (AKI)	91.0	91.6	93.0	94.3	95.1	96.5
Price (US$/gal)	$ 2.475	$ 2.475	$ 2.475	$ 2.475	$ 2.475	$ 2.475

Vehicular Fleet in Mexico

Type	Heavy	Motor	Exhaust	Recuperator	Age	No. Vehículos
Auto/Sml Truck	Light	Carburator	3-Ways	PCV Tank	+8years	14.255
		Single FI	3-Ways		+8years	1.638.045
		Multi FI	3-Ways		+8years	6.559.661
			Euro III		+8years	3.064.588
			Euro IV		+8years	2.425.660
					4-8years	3.611.585
			Euro V		4-8years	855.809
					-4years	2.303.300
	Medium	Carburator	3-Ways	PCV Tank	+8years	2.024
		Single FI	3-Ways		+8years	64.686
		Multi FI	3-Ways		+8years	999.381
			Euro III		+8years	1.065.314
			Euro IV		+8years	770.022
					4-8years	1.274.081
			Euro V		4-8years	386.678
					-4years	1.374.040
Sml Engines	Light	4-Cycle	Euro II	PCV	+8years	2.714.667
					4-8years	1.945.800
			Euro III		-4years	2.797.515
Truck	Heavy	Carburator	3-Ways	PCV	+8years	19.293
		Fuel-Injection	3-Ways		+8years	2.132.687
			Euro III		+8years	1.222.113
			Euro IV		+8years	722.270
			Euro V		4-8years	904.413
					-4years	841.493

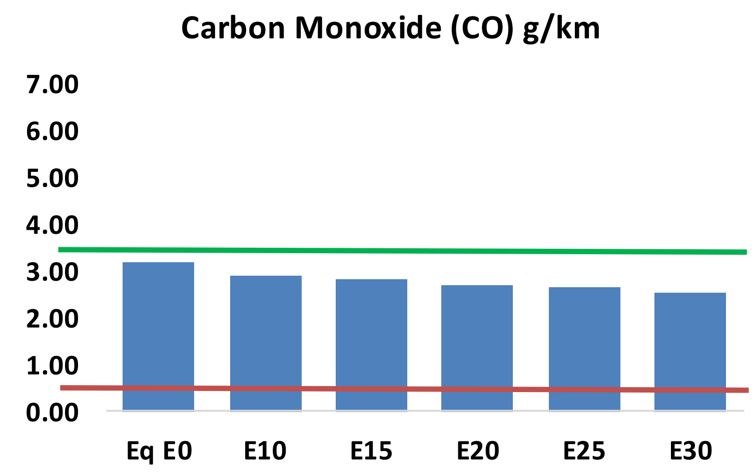
Total = 60,978,233

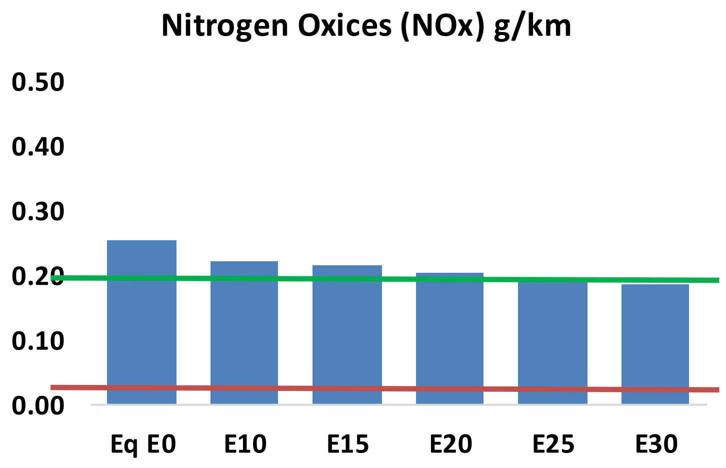
Gasoline Vehicular Fleet: **46,767,837**
Average Age: **10 years**

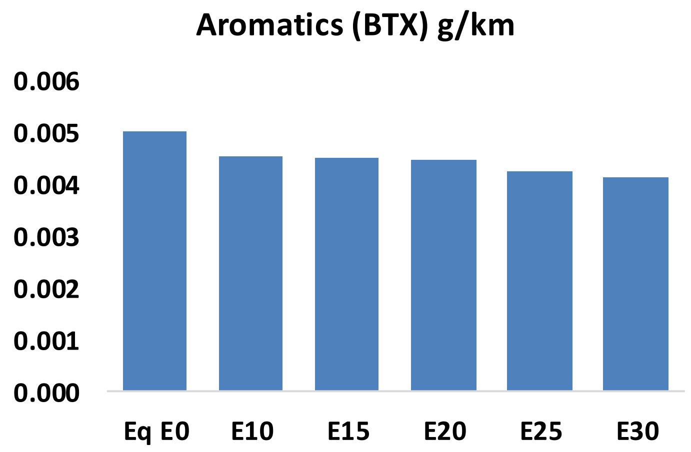
Source: INEGI

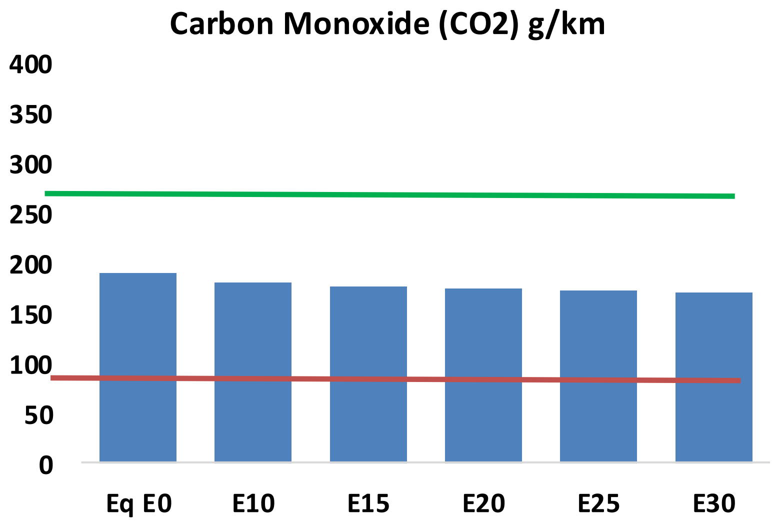
Vehicular Emissions in Mexico

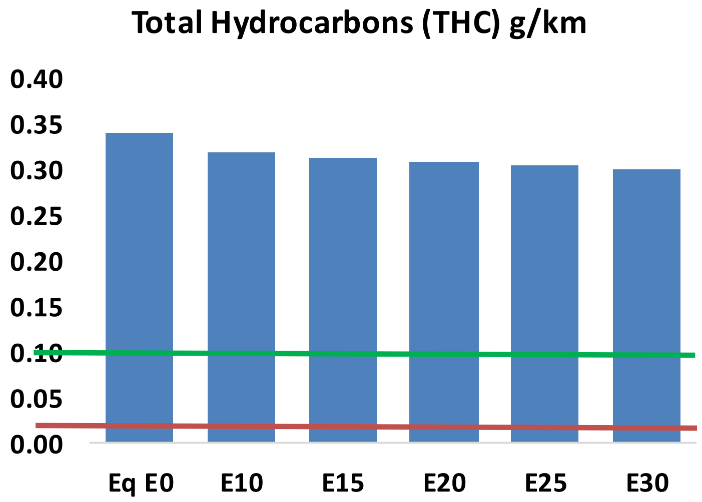
United States
Euro 6

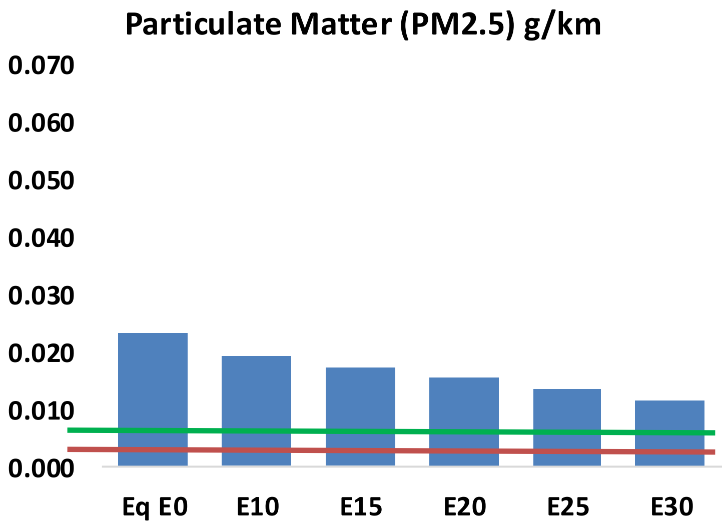












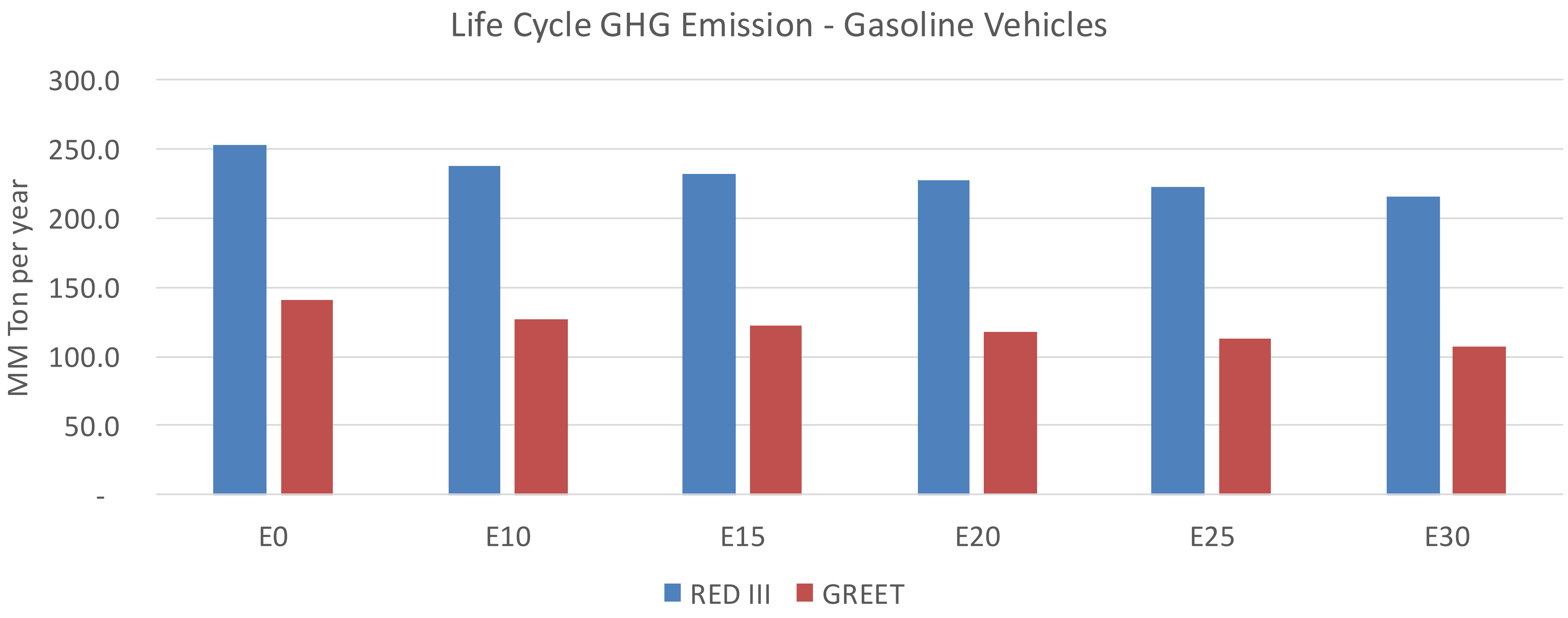
Source: Faro90, 2025.

Mexico – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	3,756,123	3,581,907	3,407,692	3,287,278	3,176,391	3,094,230	2,983,504	-9%	-15%
CO2	222,198,681	216,485,000	211,088,747	206,844,752	204,744,657	203,598,569	199,845,662	-5%	-8%
VOC	398,231	385,700	373,169	366,429	361,270	357,793	351,424	-6%	-9%
NOx	299,828	276,842	260,897	252,685	239,555	229,218	217,074	-13%	-20%
SOx	2,532	2,341	2,150	1,988	1,832	1,664	1,487	-15%	-28%
NH3	74,987	74,245	73,503	73,852	74,682	75,267	75,752	-2%	0%
N2O	3,036	3,019	3,007	3,022	3,084	3,118	3,154	-1%	2%
CH4	100,570	100,570	100,570	102,079	104,291	106,202	108,012	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	3,683	3,602	3,521	3,494	3,485	3,484	3,463	-4%	-5%
Acetaldehyde	8,141	9,343	13,709	20,877	28,443	32,502	38,405	68%	249%
Formaldehyde	26,810	26,065	30,385	35,678	37,378	41,351	45,015	13%	39%
Benzene	5,887	5,658	5,361	5,276	5,232	5,004	4,842	-9%	-11%
PM2.5	13,538	12,307	11,076	9,993	8,934	7,798	6,609	-18%	-34%
PM10	13,699	12,454	11,209	10,113	9,040	7,891	6,688	-18%	-34%

Source: Faro90

Mexico – Gasoline Vehicle Fleet GEI Emissions



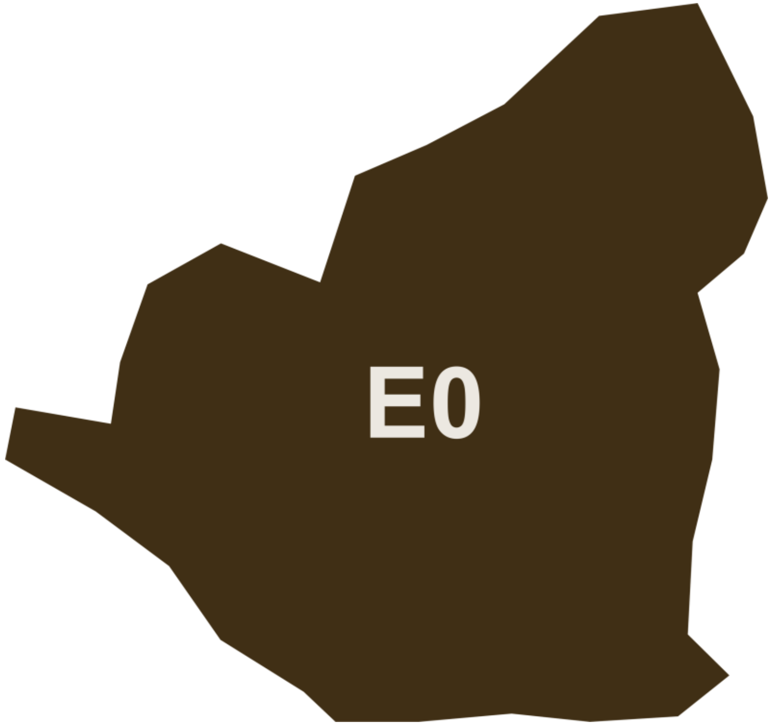
Country Emissions 2024 (MMT/year)	751.4
2035 Target (MMT/year)	500.4
Delta	251.0

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	9.4%	13.0%	16.5%	19.5%	23.5%
RED II/III	0.0%	5.9%	8.2%	10.3%	12.3%	14.6%

% 2035 Target	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	5.3%	7.3%	9.2%	10.9%	13.1%
RED II/III	0.0%	6.0%	8.2%	10.4%	12.4%	14.8%

Source: Greet, RED II/III, climateactiontracker.org, F90

Ethanol Blending in Gasoline - Nicaragua

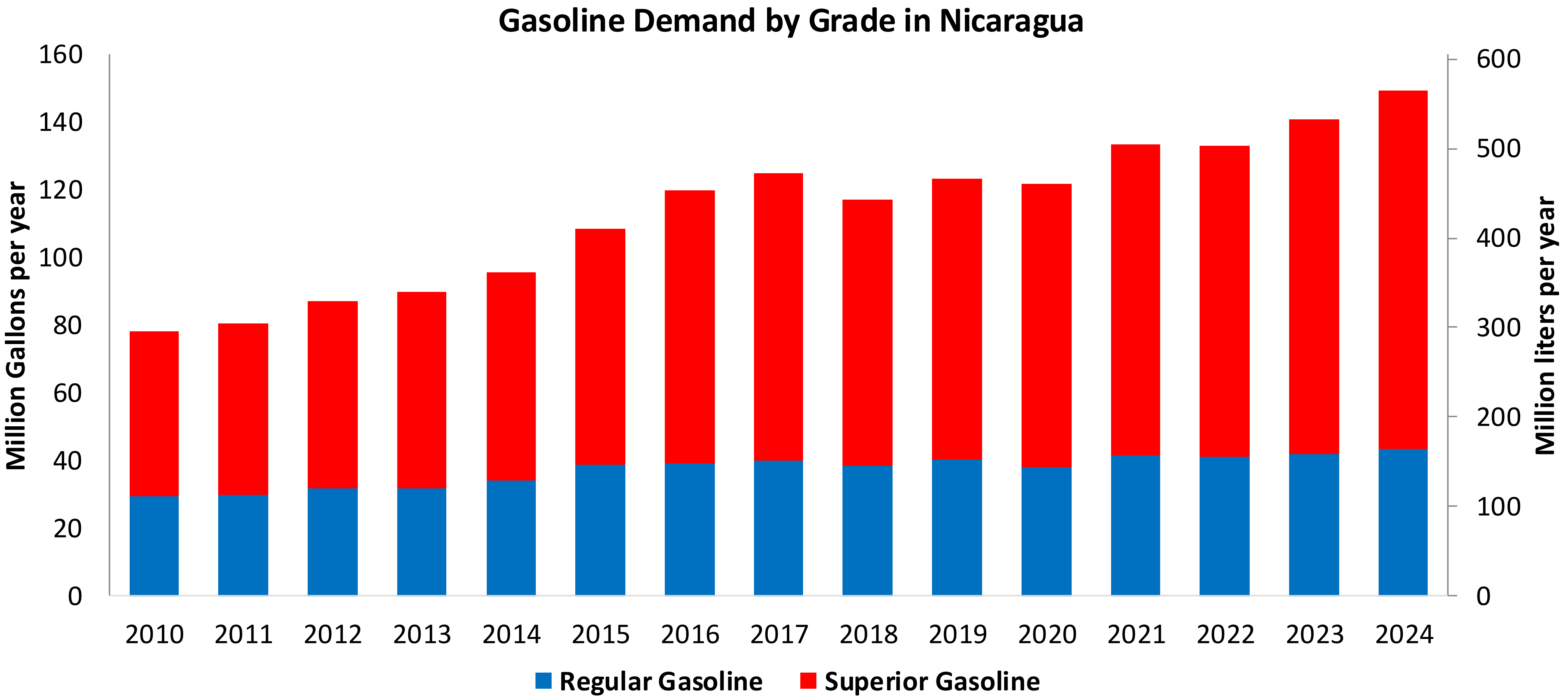


In 2024, gasoline demand was 149 million gallons (565 million liters). Nicaragua is the only country in Central America that has active refinery capacity, which produced 33% of gasoline national demand. The rest of demand is imported, previously from Venezuela and since 2016 mainly from United States. There are two gasoline grades in the country, regular with 29% and premium with 71% of market share.

According to the Ministry of Energy and RTCA, Nicaragua has no ethanol mandate.

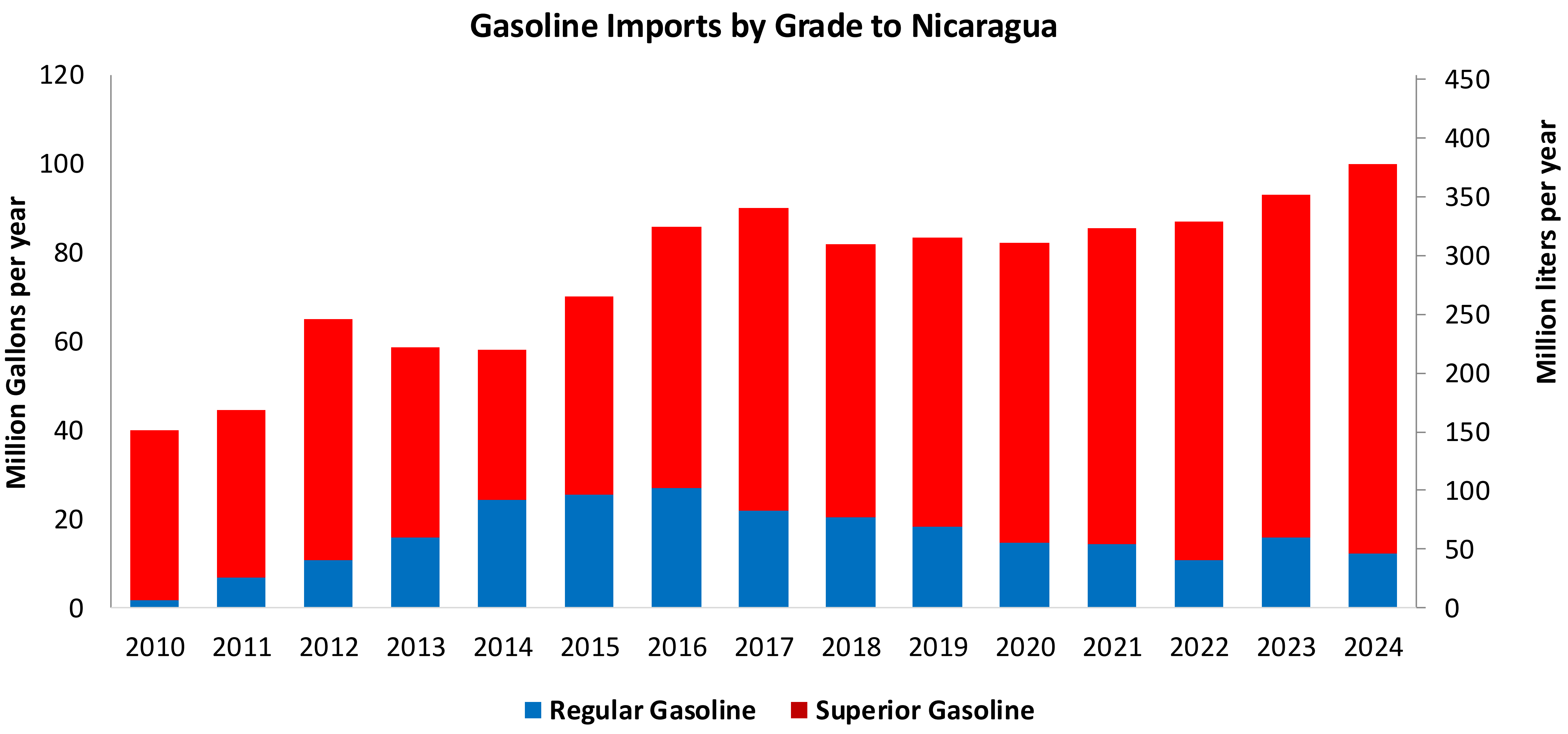
Source: MEM Nicaragua, RTCA

Gasoline Demand in Nicaragua



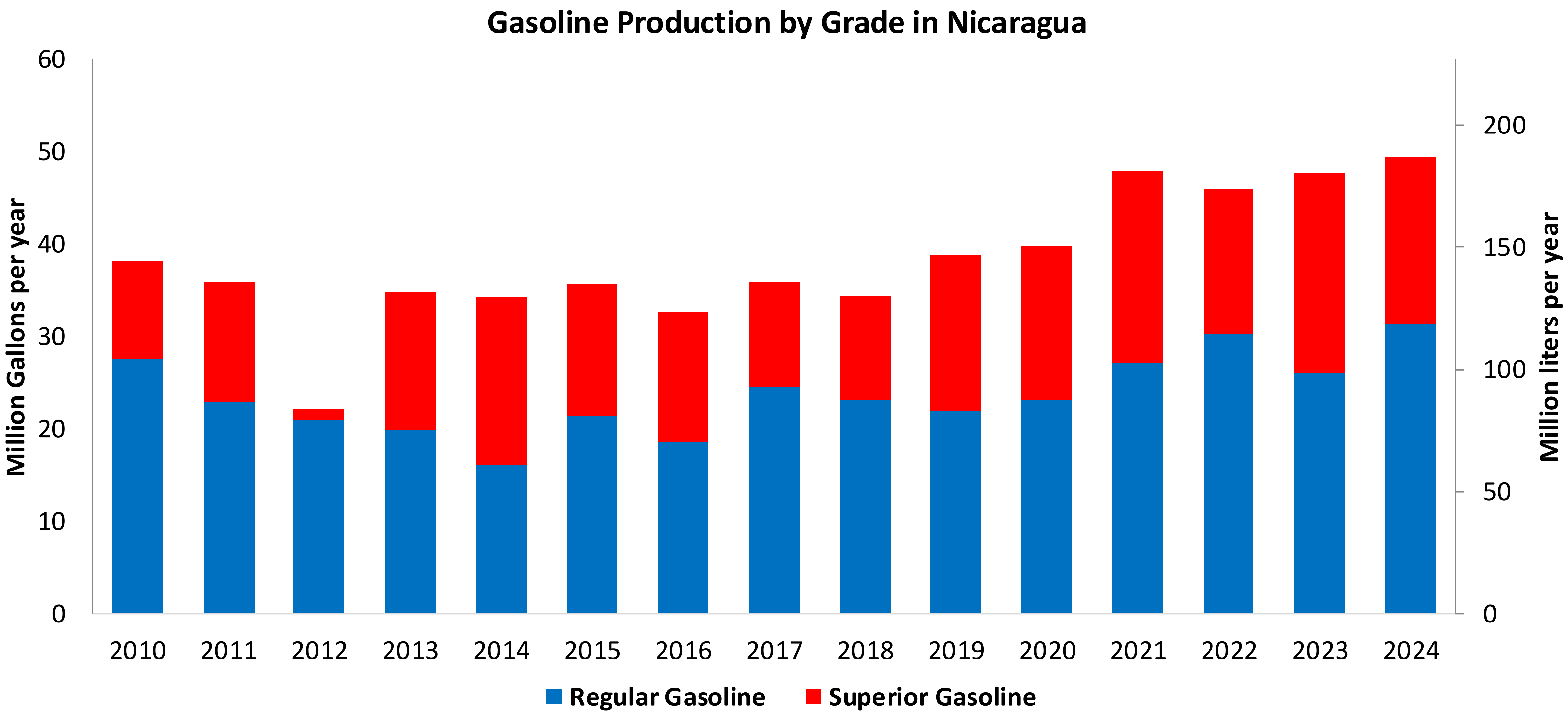
Source: MEM Nicaragua

Gasoline Imports in Nicaragua



Source: MEM Nicaragua

Gasoline Production in Nicaragua



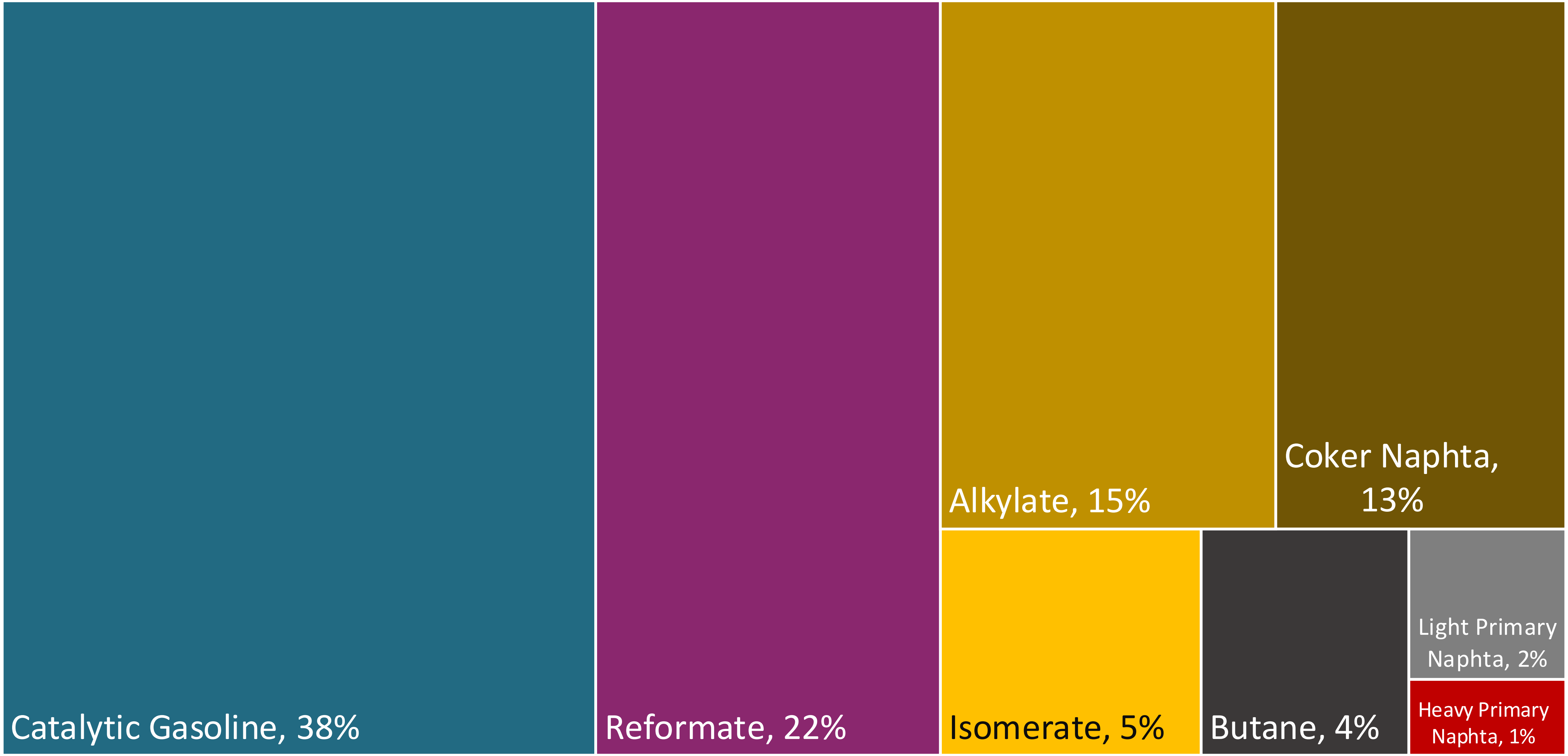
Source: MEM Nicaragua

Gasoline Quality in Nicaragua

Name	RTCA 75.01.20:19	RTCA 75.01.19:19	EN 228:2012 + A1:2017 (Euro 6 enabling)			
Implementation Date	2021	2021	2017			
Applicability	Whole country	Whole country	All countries			
Selected Grade	Gasoline Premium	Gasoline Regular	RON 95 E5	RON 95 E10	RON 98 E5	RON 98 E10
Benzene Content	5% v/v	5% v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v
Aromatics	50% v/v	50% v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v
Olefins	30% v/v	30% v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v
Lead Content	< 0,013 g/l	< 0,013 g/l	< 5 mg/l	< 5 mg/l	< 5 mg/l	< 5 mg/l
Manganese	Report	Report	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l
RON	> 95	> 91	> 95	> 95	> 98	> 98
MON	-	-	> 85	> 88	> 85	> 88
AKI						
Sulfur Content	< 500 mg/kg	< 500 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg
Oxygen Content	2,7% v/v	2,7% v/v	<2,7 % m/m	<3,7 % m/m	<2,7 % m/m	<3,7 % m/m
Ethanol (EtOH)	-	-	<5 %v/v	<10 %v/v	<5 %v/v	<10 %v/v
RVP 37.8°C (Summer)	< 69 kPa	< 69 kPa	<> 60 - 70 kPa *Depends on the country, RVP is regulated in the EU Fuel Quality Directive			
RVP 37.8 °C(Winter)						
RVP 37.8°C (Transition)						
MTBE	-	-	-	-	-	-
Ethers 5 or more C Atoms	-	-	Based on oxygen content	<22 %v/v	Based on oxygen content	<22 %v/v

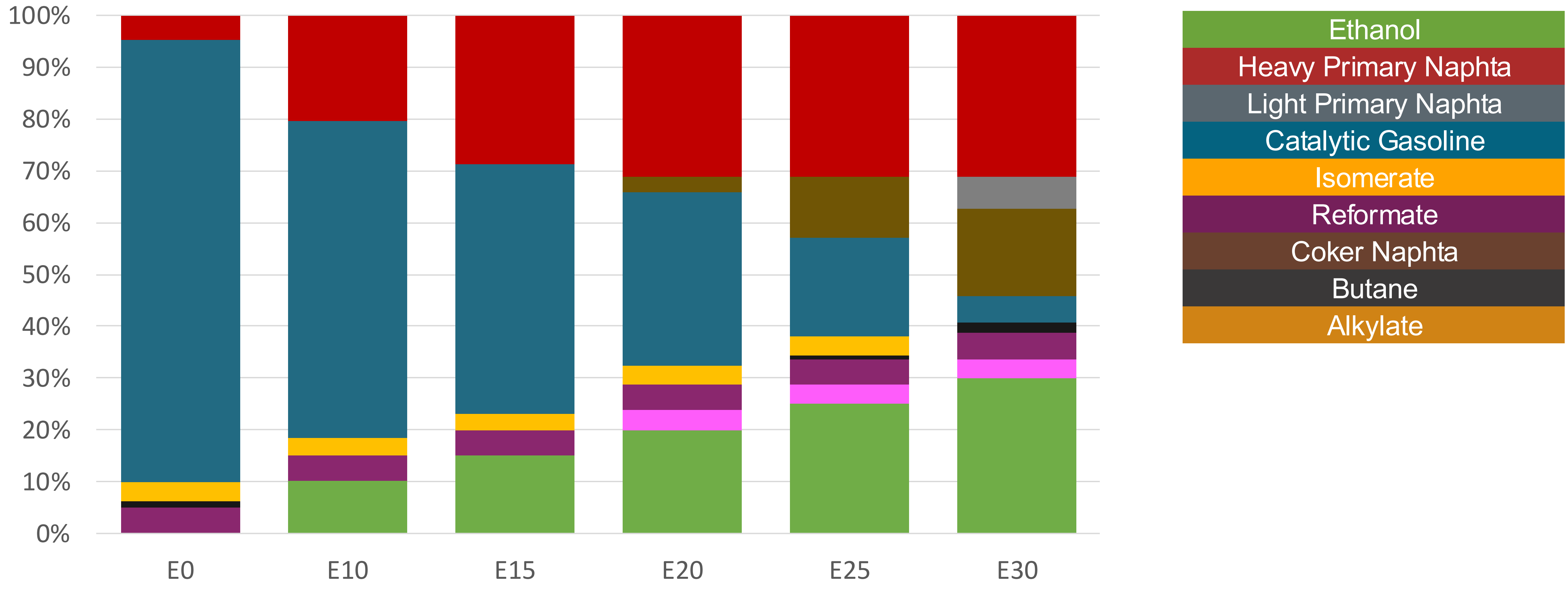
Source: RTCA

Blending Components - Nicaragua



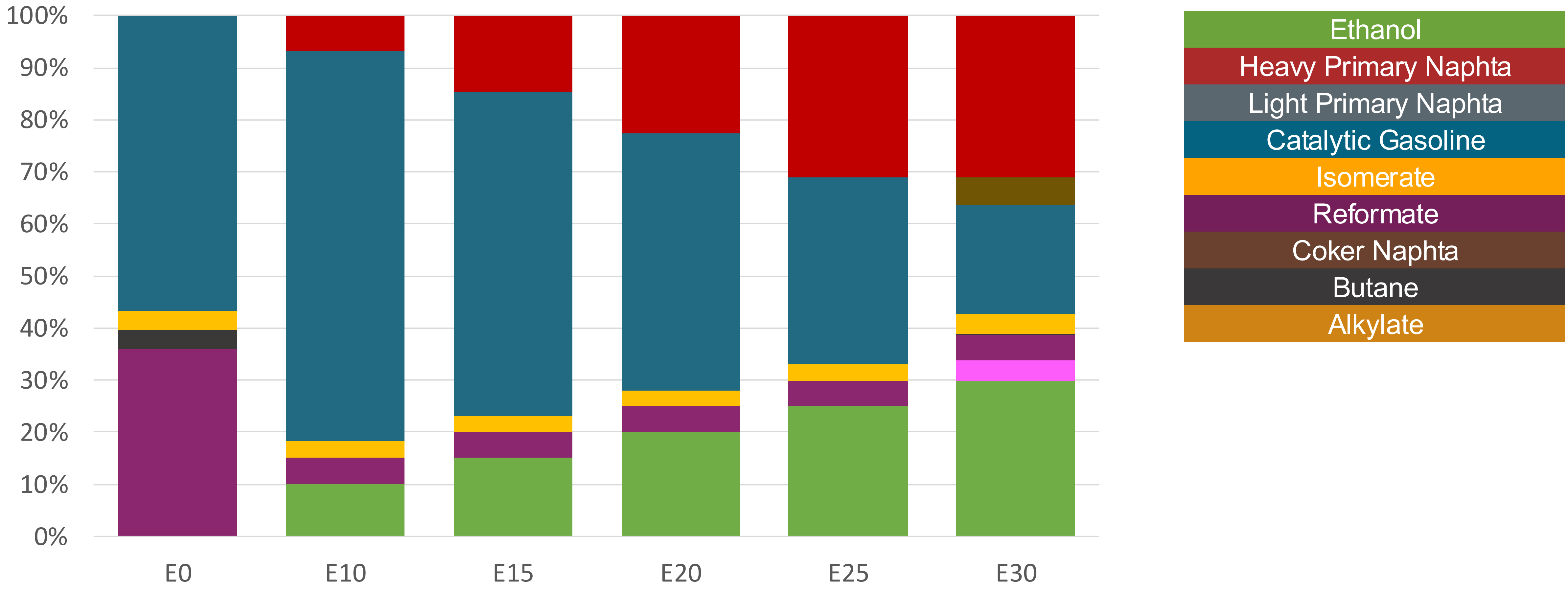
Source: Faro90

Nicaragua – Regular – Constant Octane



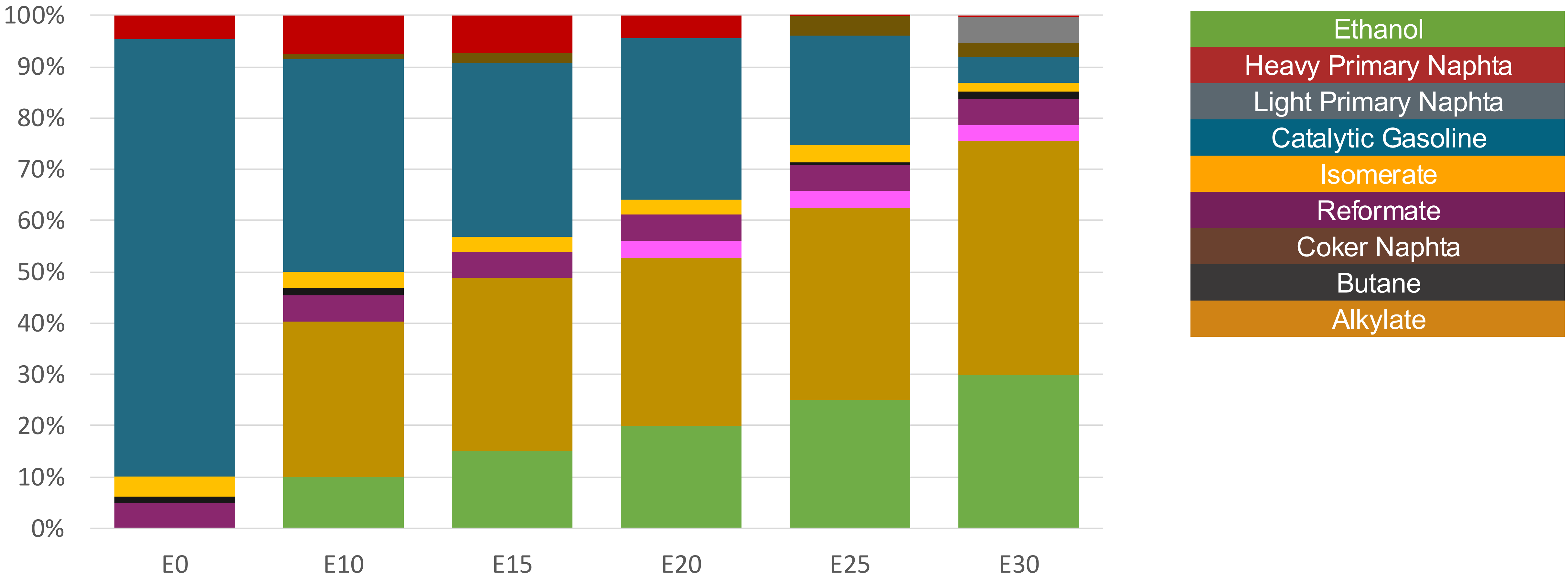
Octane (RON)	91.0	91.0	91.0	91.0	91.0	91.0
Price (USD/gal)	$ 2.156	$ 1.885	$ 1.734	$ 1.592	$ 1.447	$ 1.477

Nicaragua – Premium – Constant Octane



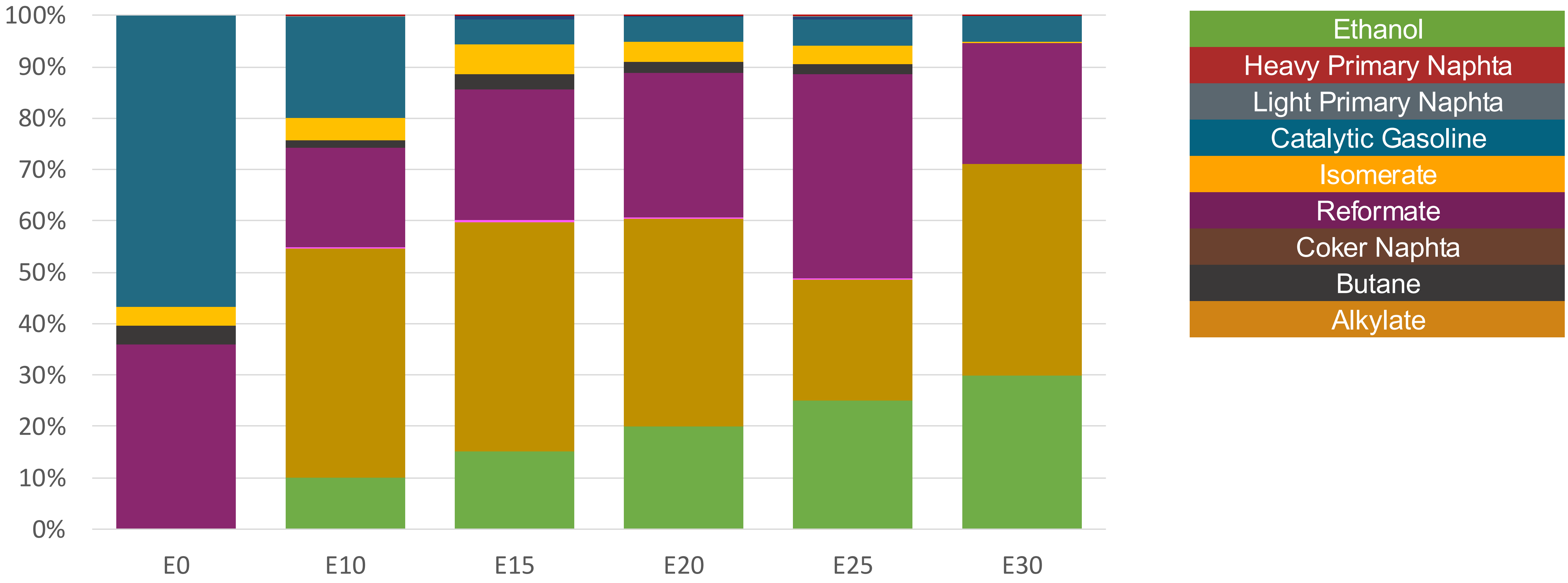
	E0	E10	E15	E20	E25	E30
Octane (RON)	95.0	95.0	95.0	95.0	95.0	95.0
Price (USD/gal)	$ 2.536	$ 2.233	$ 2.099	$ 1.957	$ 1.807	$ 1.659

Nicaragua – Regular – Increased Octane



Octane (RON)	91.0	94.6	96.6	99.1	101.2	102.4
Price (USD/gal)	$ 2.156	$ 2.156	$ 2.156	$ 2.156	$ 2.156	$ 2.156

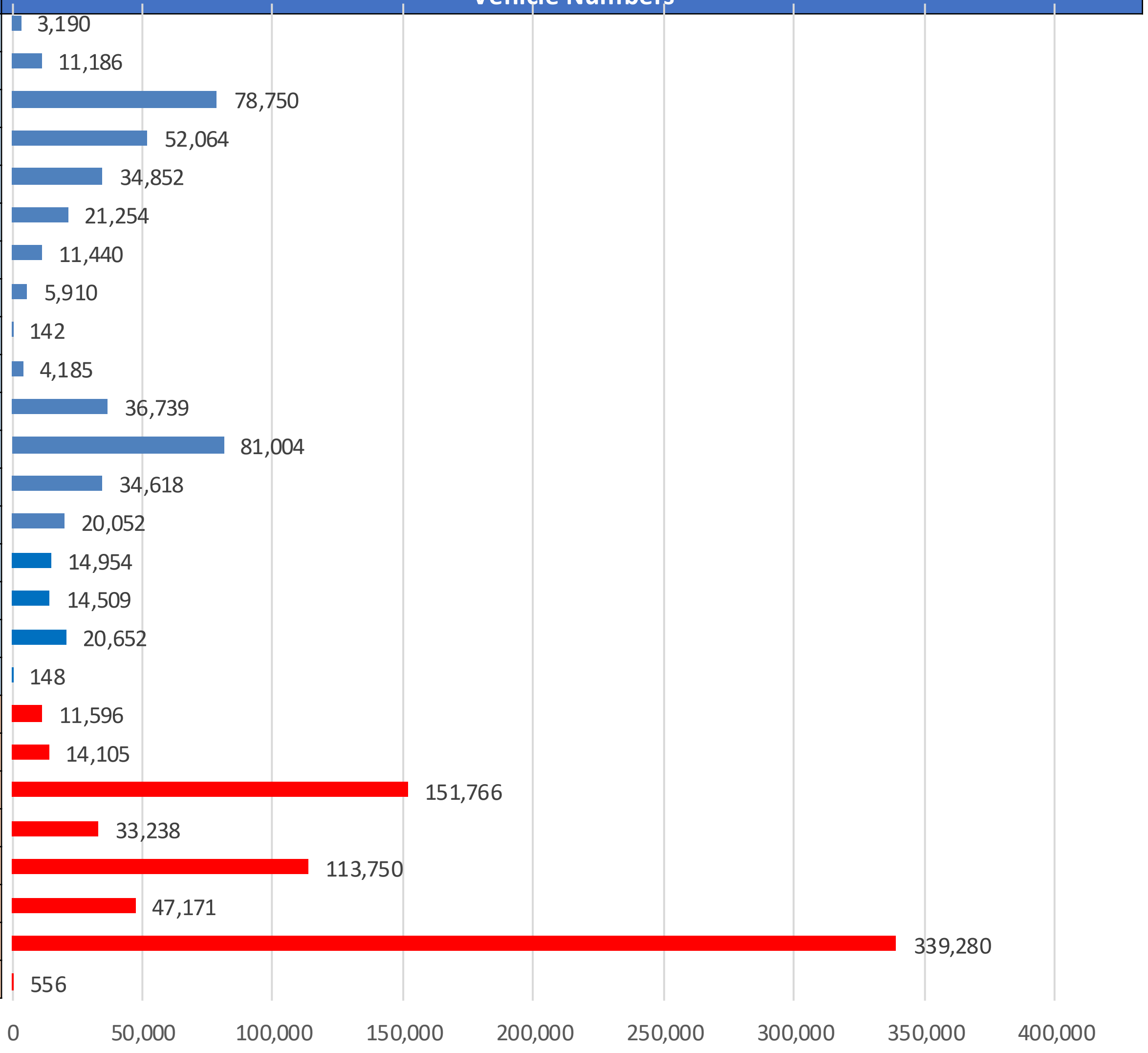
Nicaragua – Premium – Increased Octane



Octane (RON)	95.0	98.0	100.3	102.6	104.9	105.8
Price (USD/gal)	$ 2.536	$ 2.536	$ 2.536	$ 2.536	$ 2.536	$ 2.536

Vehicular Fleet a gasolina en Nicaragua

Type	Motor	Exhaust	Recuperator	Age	Vehicle Numbers
Auto/Light	Carburator	3-Ways	PCV Tank	<8 years, < 160 mkm	3,190
	Single FI-Pt	3-Ways		<8 years, < 160 mkm	11,186
	Multi FI-Pt	3-Ways		<8 years, < 160 mkm	78,750
		Euro III		<8 years, < 160 mkm	52,064
		Euro IV		<8 years, < 160 mkm	34,852
		Euro V / Euro 6		<8 years, < 160 mkm	21,254
				4-8 years, 80-160 mkm	11,440
				>4years, >80 mkm	5,910
	Electric			All	142
Auto/Medium	Carburator	3-Ways	PCV Tank	<8 years, < 160 mkm	4,185
	Single FI-Pt	3-Ways		<8 years, < 160 mkm	36,739
	Multi FI-Pt	3-Ways		<8 years, < 160 mkm	81,004
		Euro III		<8 years, < 160 mkm	34,618
		Euro IV		<8 years, < 160 mkm	20,052
		Euro V / Euro 6		<8 years, < 160 mkm	14,954
				4-8 years, 80-160 mkm	14,509
				>4years, >80 mkm	20,652
	Electric			All	148
Small Engines	FI 4-cycles	None	PCV	<8 years, < 160 mkm	11,596
		Euro II		<8 years, < 160 mkm	14,105
		Euro III		<8 years, < 160 mkm	151,766
		Euro IV		<8 years, < 160 mkm	33,238
				4-8 years, 80-160 mkm	113,750
		Euro V		4-8 years, 80-160 mkm	47,171
				>4years, >80 mkm	339,280
	Electric			All	556



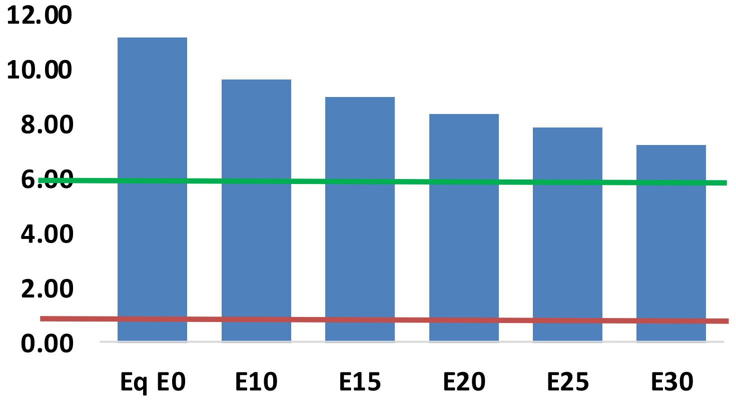
Vehicular Fleet: **1,156,265**
Average Age: **11 years**
Motorcycles: **56%**

Source: ine.gob.hn, analysis Faro 90

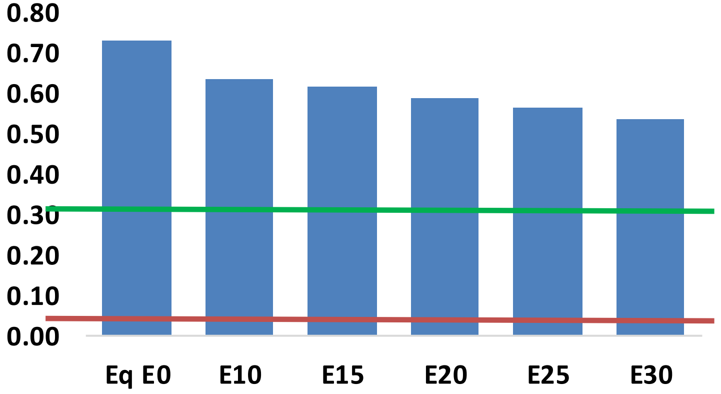
Nicaragua - Vehicular Emissions

United States
Euro 6

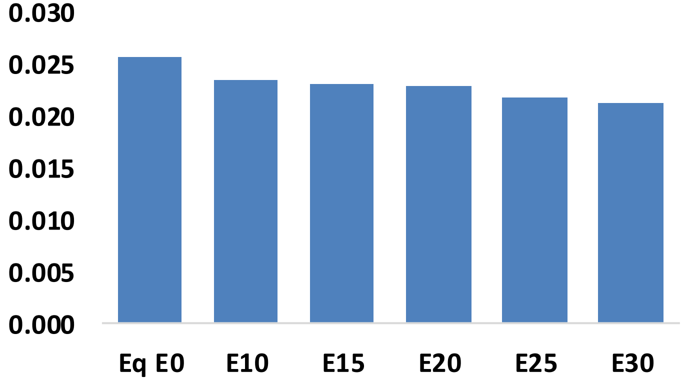
Carbon Monoxide (CO) g/km



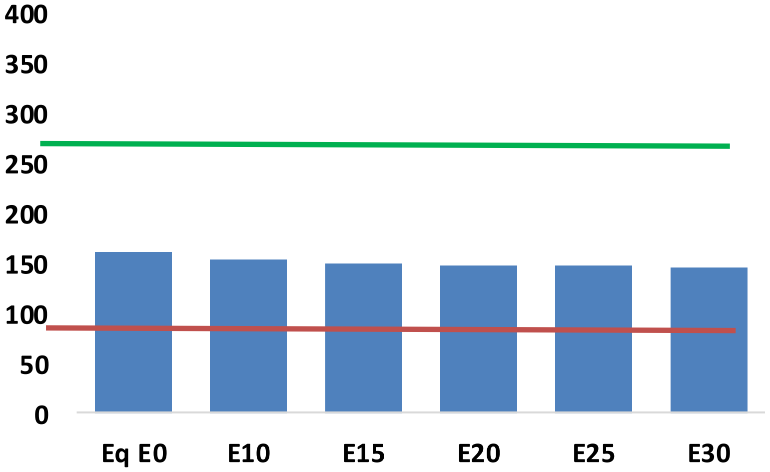
Nitrogen Oxices (NOx) g/km



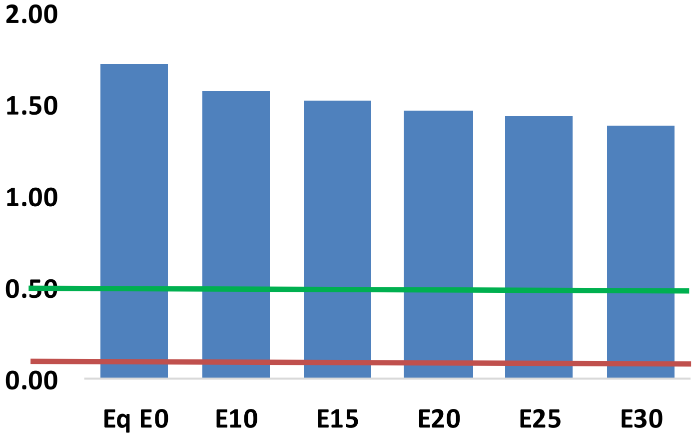
Aromatics (BTX) g/km



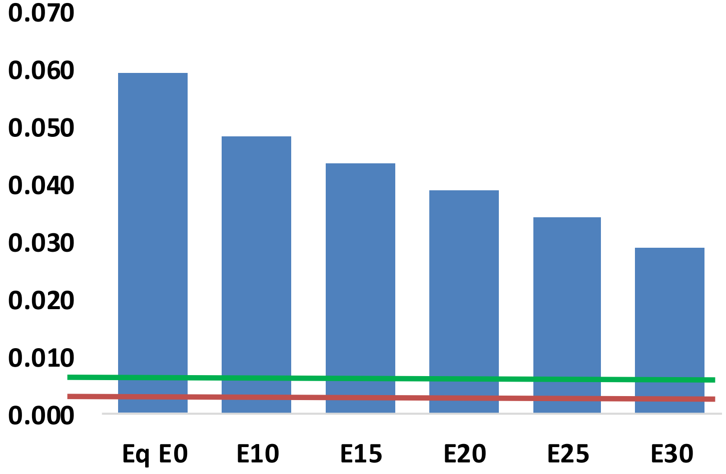
Carbon Monoxide (CO2) g/km



Total Hydrocarbons (THC) g/km



Particulate Matter (PM2.5) g/km



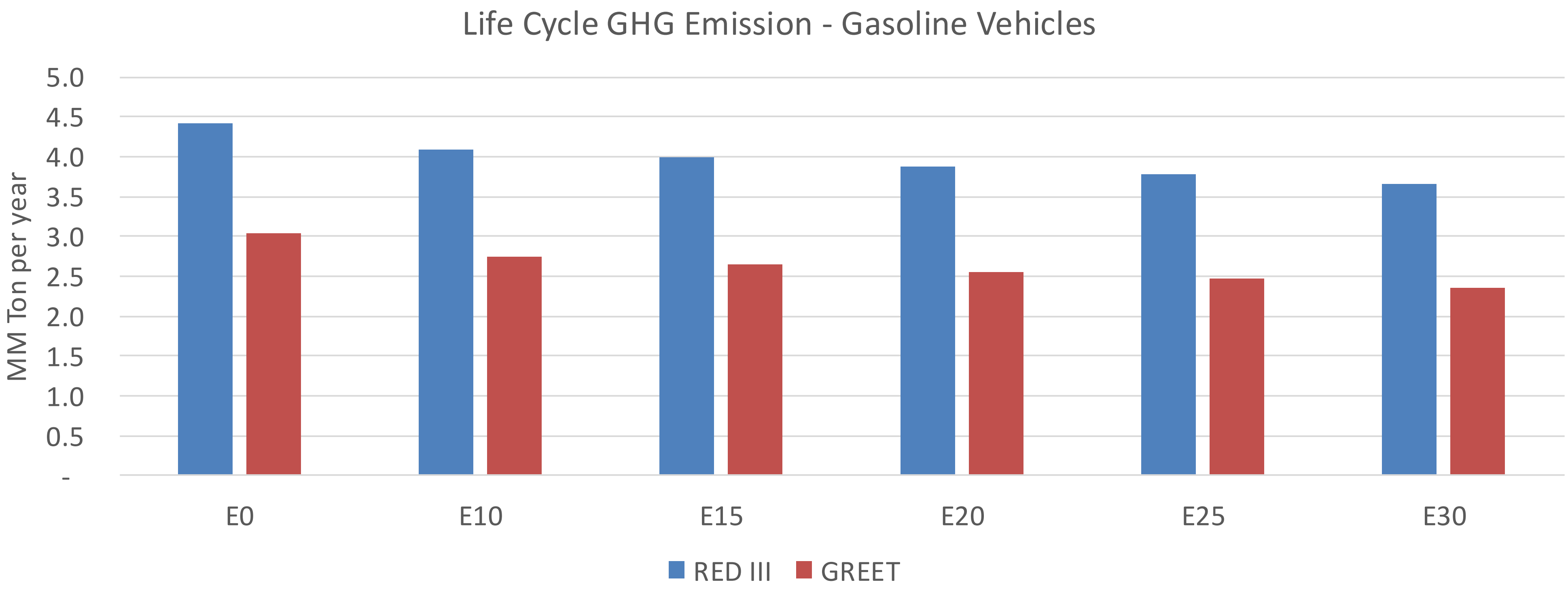
Source: Faro90

Nicaragua - Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	214,932	199,644	184,355	172,068	159,814	150,425	139,047	-14%	-26%
CO2	3,079,570	3,000,381	2,925,592	2,866,772	2,837,666	2,831,064	2,778,880	-5%	-8%
VOC	33,116	31,642	30,168	29,170	28,261	27,591	26,595	-9%	-15%
NOx	14,004	12,931	12,209	11,848	11,259	10,802	10,260	-13%	-20%
SOx	35	33	30	28	26	23	21	-15%	-28%
NH3	1,316	1,303	1,290	1,296	1,311	1,321	1,329	-2%	0%
N2O	52	52	52	52	53	54	54	-1%	2%
CH4	7,490	7,490	7,490	7,602	7,767	7,909	8,044	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	148	138	128	121	113	107	99	-13%	-24%
Acetaldehyde	395	454	665	1,013	1,381	1,578	1,864	68%	249%
Formaldehyde	1,528	1,486	1,732	2,034	2,131	2,357	2,566	13%	39%
Benzene	494	475	450	443	439	420	406	-9%	-11%
PM2.5	191	173	156	141	126	110	93	-18%	-34%
PM10	945	859	773	697	623	544	461	-18%	-34%

Source: Faro90

Nicaragua – Gasoline Vehicle Fleet GEI Emissions



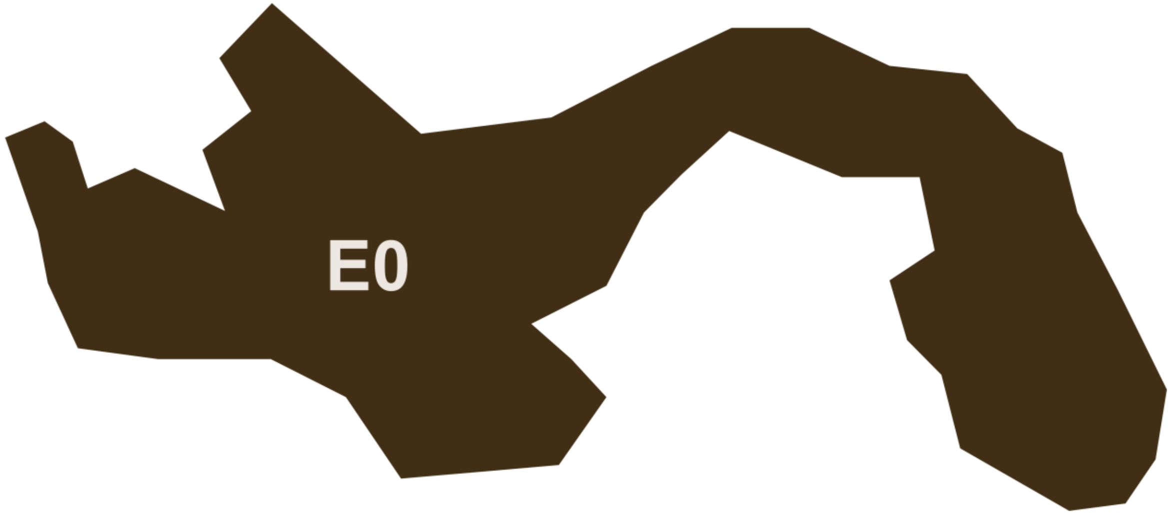
Country Emissions 2024 (MMT/year)	41.1
2035 Target (MMT/year)	28.8
Delta	12.3

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	10.0%	12.9%	16.3%	18.9%	22.5%
RED II/III	0.0%	7.5%	9.8%	12.4%	14.5%	17.1%

% 2035 Target	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	2.5%	3.2%	4.0%	4.7%	5.5%
RED II/III	0.0%	2.7%	3.5%	4.4%	5.2%	6.1%

Source: Greet, RED II/III, climateactiontracker.org, F90

Ethanol Blending in Gasoline - Panama

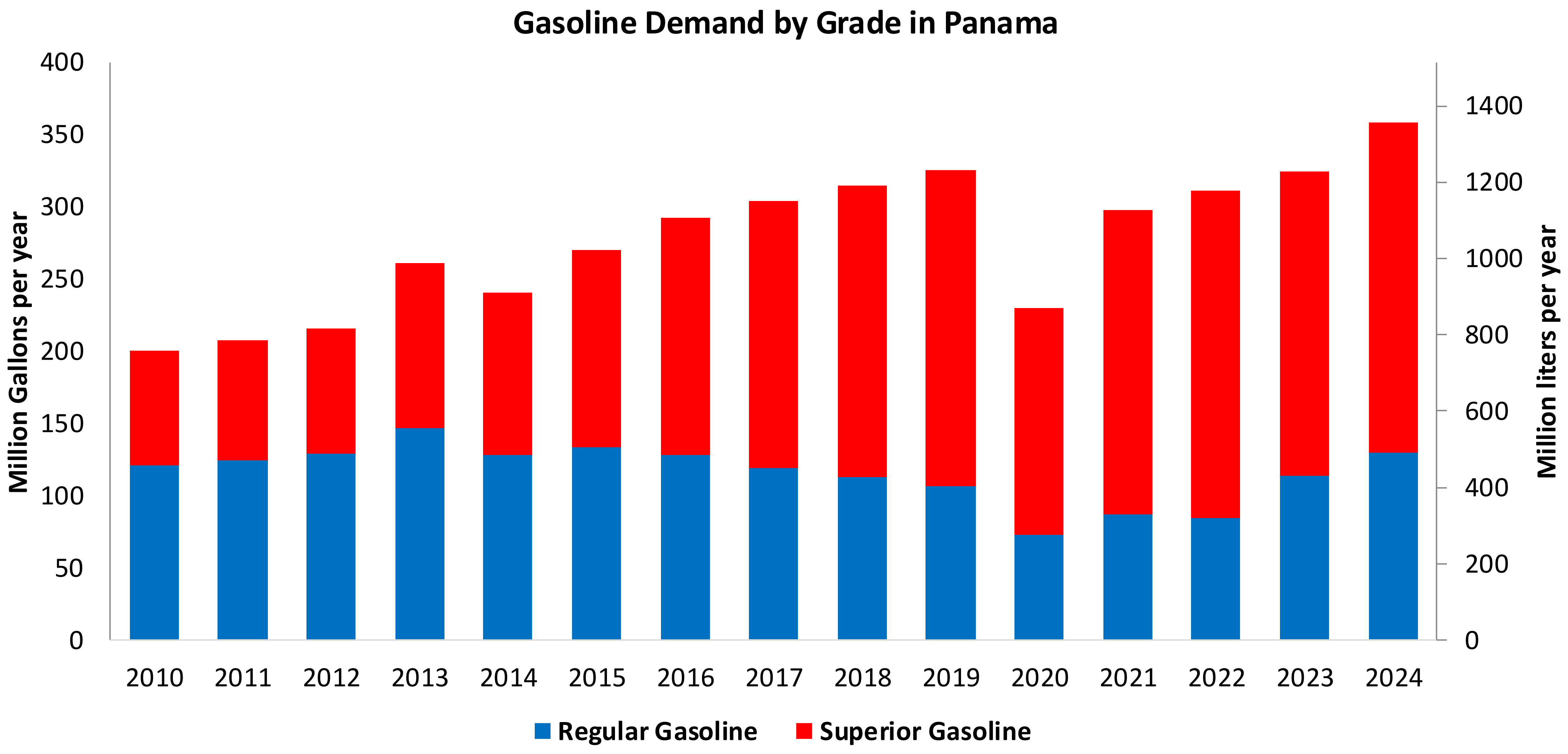


En 2024 demand was 358 million gallons (1356 million liters). Regular gasoline (RON 91) represented 36% of total demand and Superior gasoline (RON 95) 64%. Gasoline is supplied only with imports, mainly from United States, the Netherlands and Germany.

In 2011, 2% v/v ethanol blends with gasoline were authorized, with a planned gradual increase until reaching 10% v/v by 2016. However, in 2013 it was limited to 5% v/v and by 2014 blending stopped because the only national producer announced its production cease due to political dispute. A new mandate implementation has been approved for E10 and is expected to be implemented between 2026 and 2027.

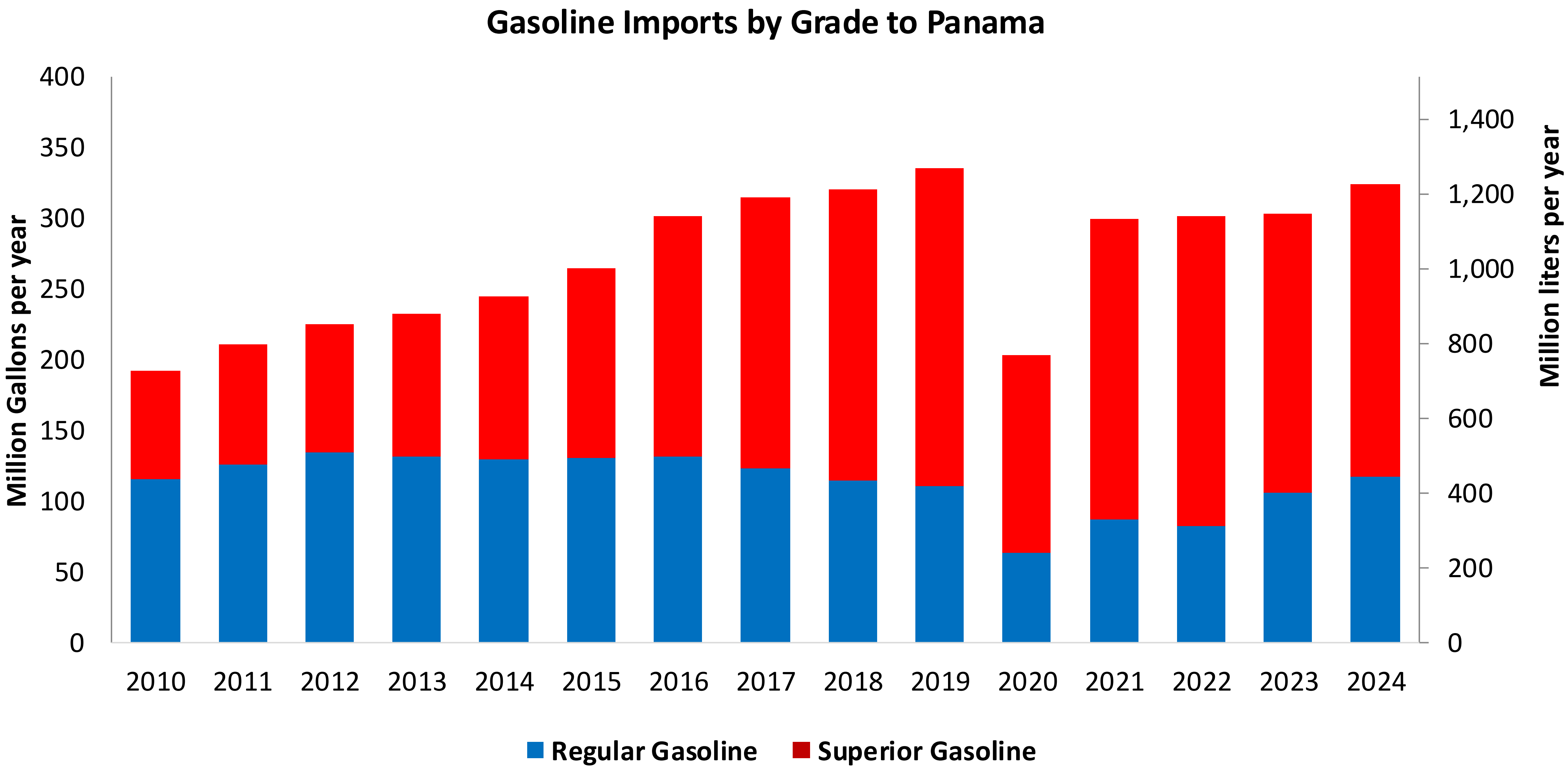
Source: Secretaría de Energía - Panama

Gasoline Demand in Panama



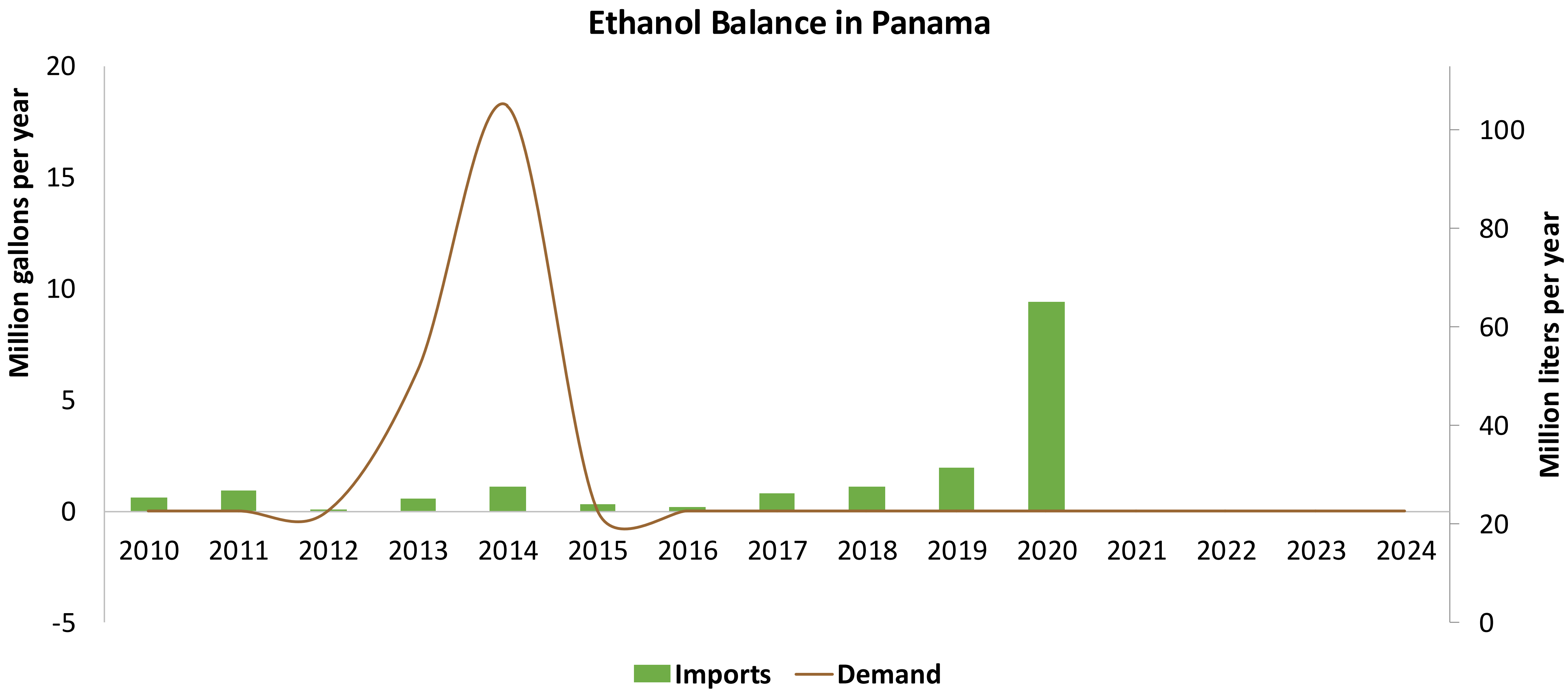
Source: Secretaría de Energía - Panama

Gasoline Imports to Supply Demand in Panama



Source: Secretaría de Energía - Panama

Ethanol Balance in Panama



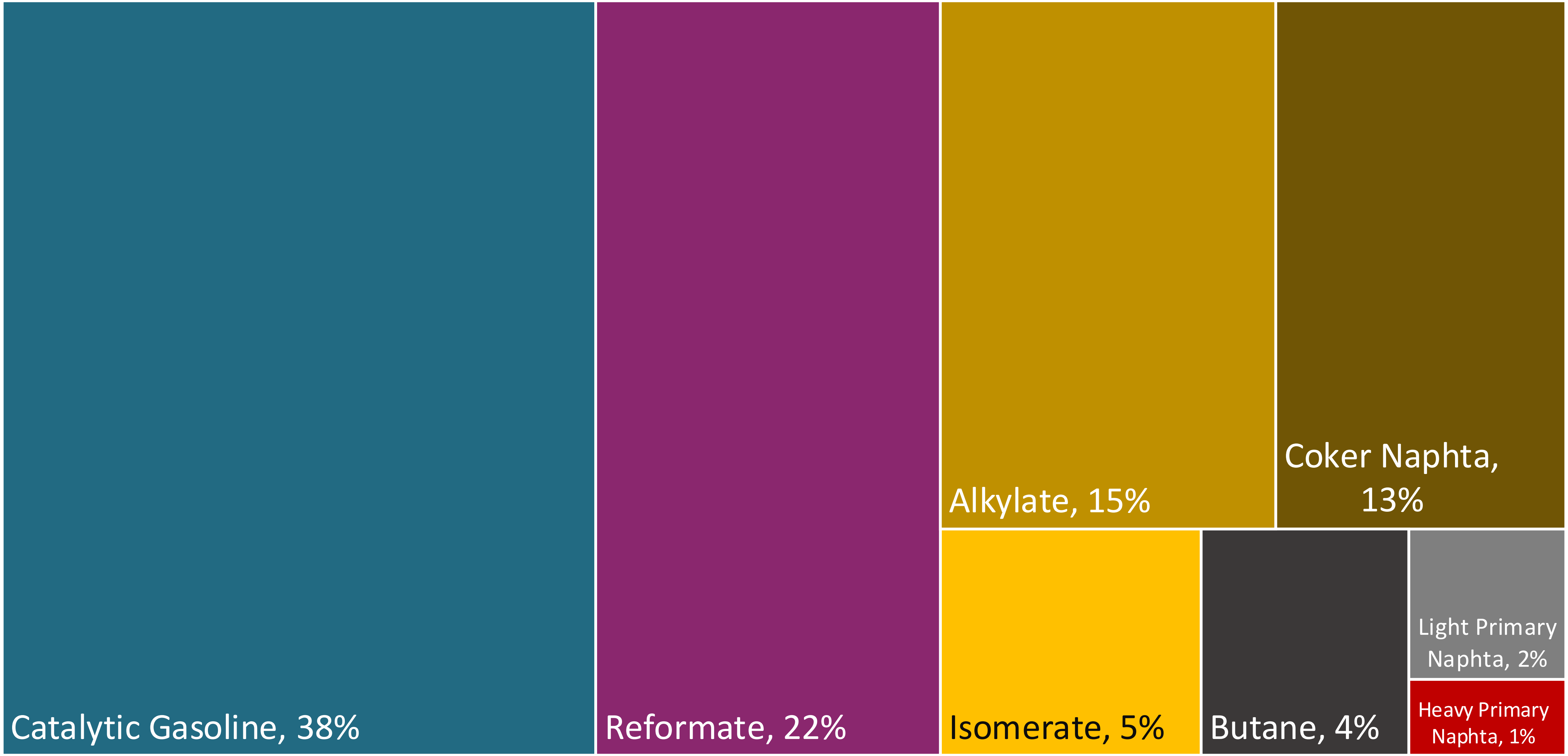
Source: Secretaría de Energía - Panama

Gasoline Quality in Panama

Name	DGNTI-COPANIT 83-2013/ Resolution 425/2020		EN 228:2012 + A1:2017 (Euro 6 enabling)			
Implementation Date	2014/2021		2017			
Applicability	Whole country	Whole country	All countries			
Selected Grade	RON 91	RON 95	RON 95 E5	RON 95 E10	RON 98 E5	RON 98 E10
Benzene Content	< 5% v/v / <1,5% v/v	< 5% v/v / <1,5% v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v
Aromatics	< 50% v/v	< 50% v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v
Olefins	< 30% v/v	< 30% v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v
Lead Content	< 0,013 g/l	< 0,013 g/l	< 5 mg/l	< 5 mg/l	< 5 mg/l	< 5 mg/l
Manganese	-	-	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l
RON	> 91	> 95	> 95	> 95	> 98	> 98
MON	-	-	> 85	> 88	> 85	> 88
AKI						
Sulfur Content	< 500 mg/kg / < 150 mg/kg	< 500 mg/kg / < 150 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg
Oxygen Content	-/ < 0,7% v/v	-/ < 0,7% v/v	<2,7 % m/m	<3,7 % m/m	<2,7 % m/m	<3,7 % m/m
Ethanol (EtOH)	10% v/v	10% v/v	<5 %v/v	<10 %v/v	<5 %v/v	<10 %v/v
RVP 37.8°C (Summer)	< 69 kPa (< 76 kPa if ethanol is added)	< 69 kPa (< 76 kPa if ethanol is added)	<> 60 - 70 kPa *Depends on the country, RVP is regulated in the EU Fuel Quality Directive			
RVP 37.8 °C(Winter)						
RVP 37.8°C (Transition)						
MTBE	0% v/v if ethanol is added	0% v/v if ethanol is added	-	-	-	-
Ethers 5 or more C Atoms	-	-	Based on oxygen content	<22 %v/v	Based on oxygen content	<22 %v/v

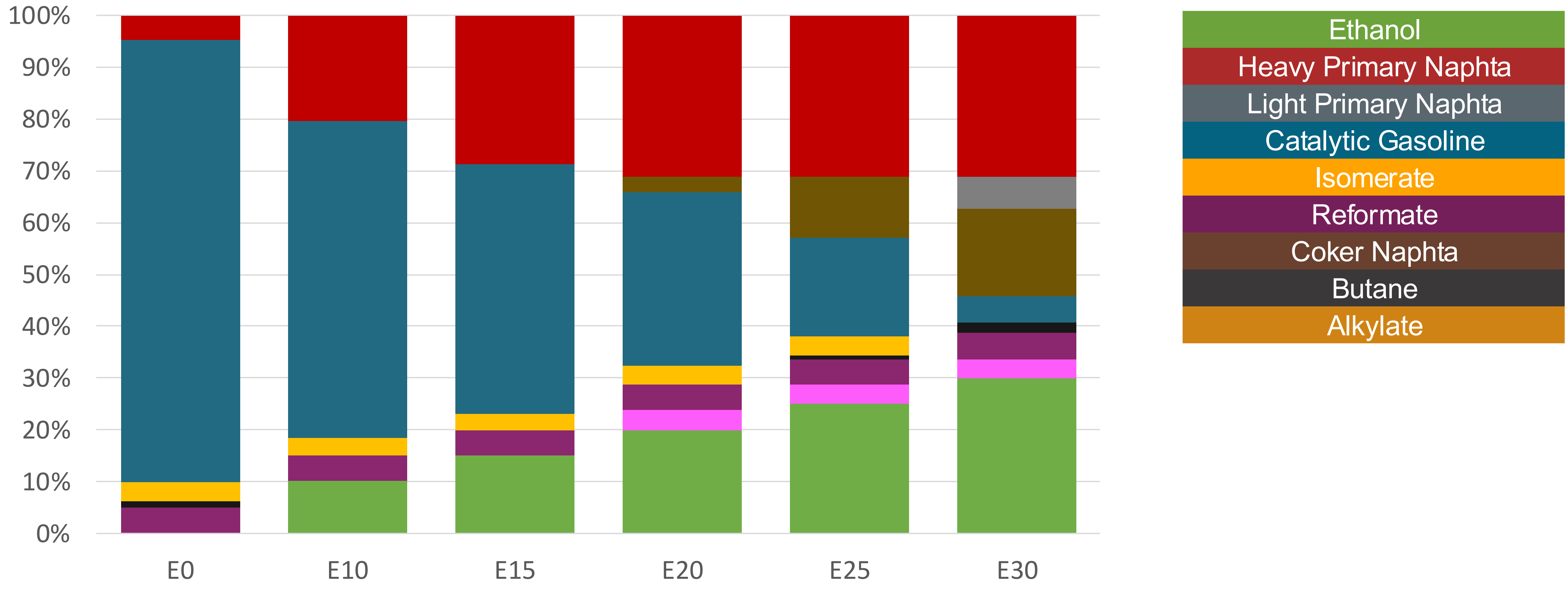
Source: RTCA

Blending Components - Panama



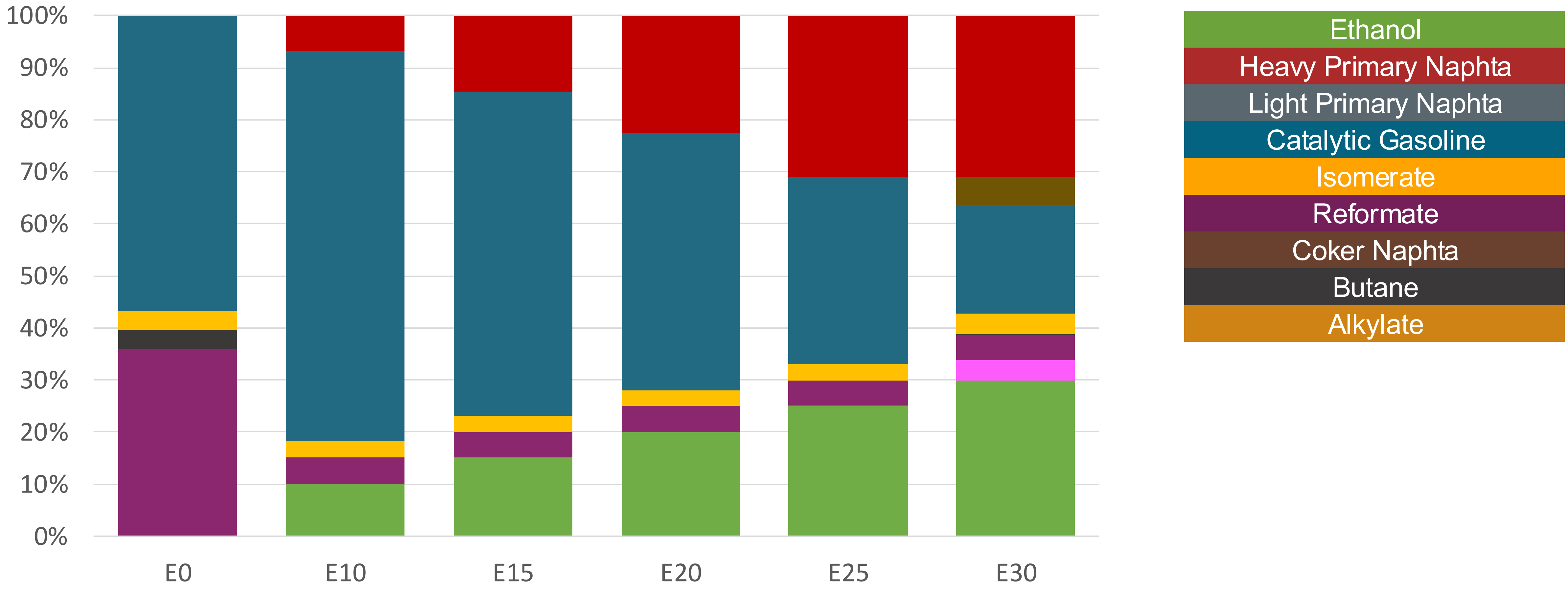
Source: Faro90

Panama – Regular – Constant Octane



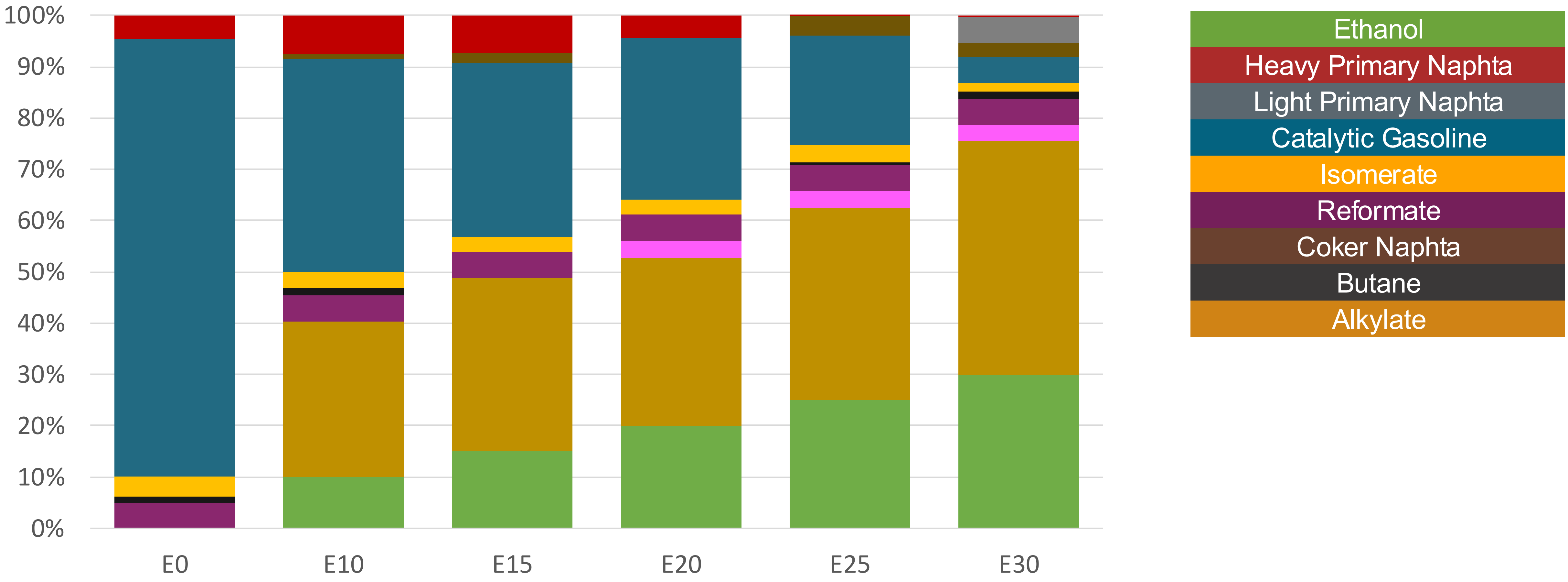
Octane (RON)	91.0	91.0	91.0	91.0	91.0	91.0
Price (USD/gal)	$ 2.286	$ 2.028	$ 1.884	$ 1.737	$ 1.586	$ 1.468

Panama – Premium – Constant Octane



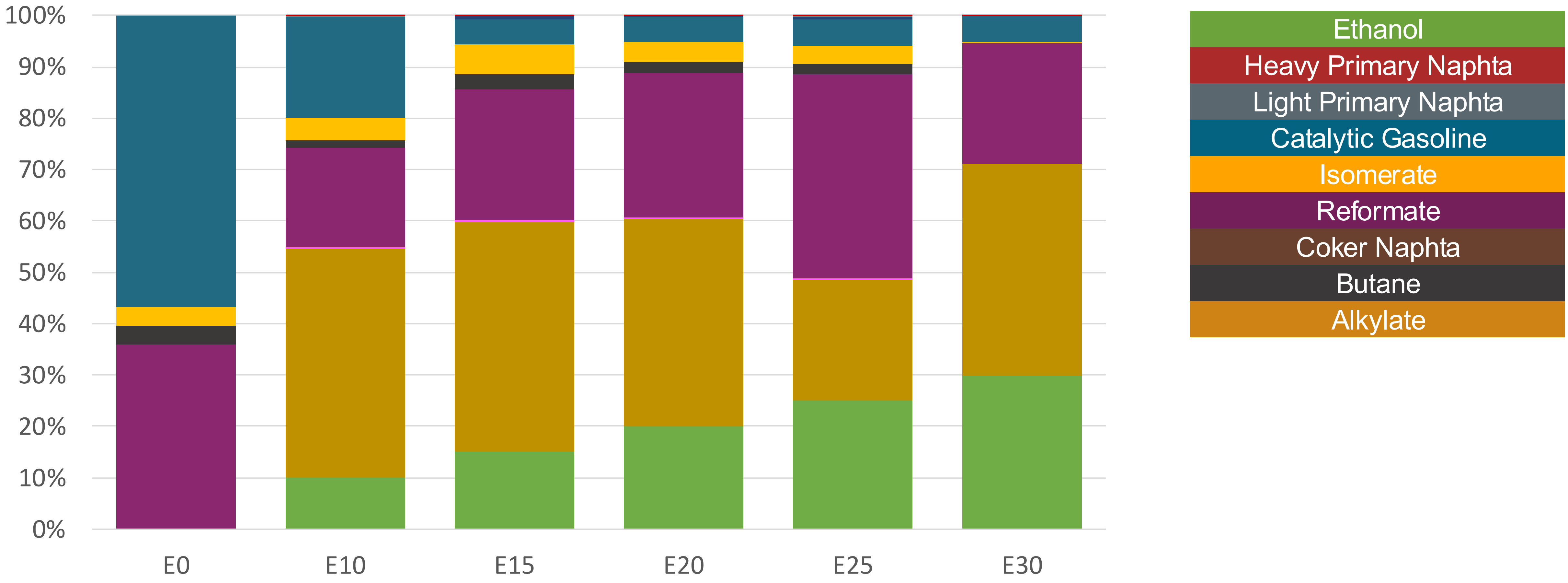
Octane (RON)	95.0	95.0	95.0	95.0	95.0	95.0
Price (USD/gal)	$ 2.536	$ 2.233	$ 2.099	$ 1.957	$ 1.807	$ 1.659

Panama – Regular – Increased Octane



Octane (RON)	91.0	94.6	96.6	99.1	101.2	102.4
Price (USD/gal)	$ 2.286	$ 2.286	$ 2.286	$ 2.286	$ 2.286	$ 2.286

Panama – Premium – Increased Octane



Octane (RON)	95.0	98.0	100.3	102.6	104.9	105.8
Price (USD/gal)	$ 2.536	$ 2.536	$ 2.536	$ 2.536	$ 2.536	$ 2.536

Gasoline Vehicular Fleet in Panama

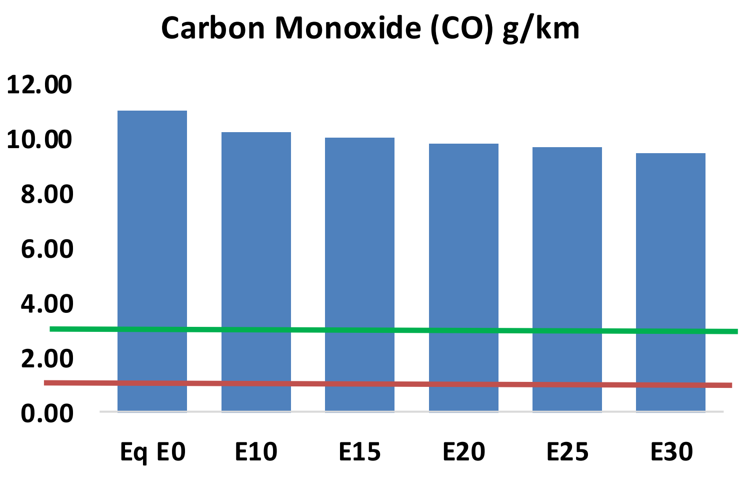
Type	Motor	Exhaust	Recuperator	Age	Number of Vehicles
Small and Medium Cars	Multi FI-Pt	Euro V/VI	PCV Tank	>4years, >80 mkm	116,145
				4-8 years, 80-160 mkm	132,236
				<8 years, < 160 mkm	88,408
		Euro IV		<8 years, < 160 mkm	86,492
		Euro III		<8 years, < 160 mkm	82,088
		3-Ways		<8 years, < 160 mkm	182,260
Small Engines	FI 4-ciclos	Euro III	PCV	>4years, >80 mkm	7,393
				4-8 years, 80-160 mkm	4,208
		Euro II		4-8 years, 80-160 mkm	4,208
				<8 years, < 160 mkm	27,957
Trucks	FI	Euro V	PCV	>4years, >80 mkm	20,467
				4-8 years, 80-160 mkm	23,303
				<8 years, < 160 mkm	15,579
		Euro IV		<8 years, < 160 mkm	15,241
		Euro III		<8 years, < 160 mkm	14,465
		3-Ways		<8 years, < 160 mkm	32,117
				Total	**852,566**

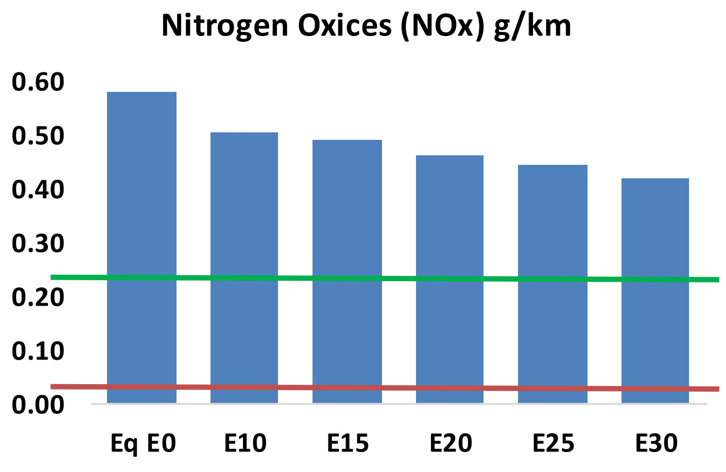
Vehicle Fleet: **852,566**
Average Age: **11.5 years**
Motorcycles: **5%**

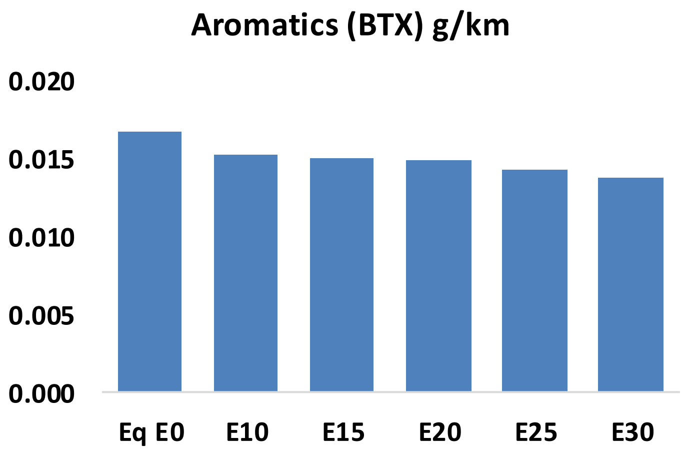
Source: INEC, analysis Faro 90

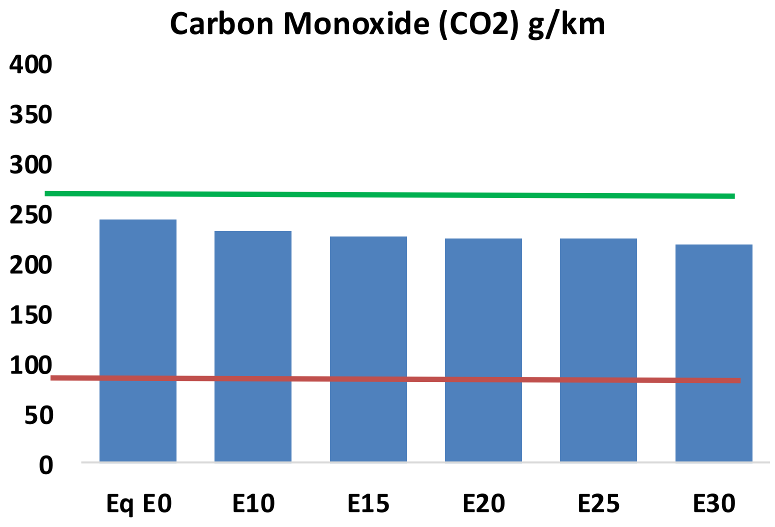
Vehicle Emissions in Panama

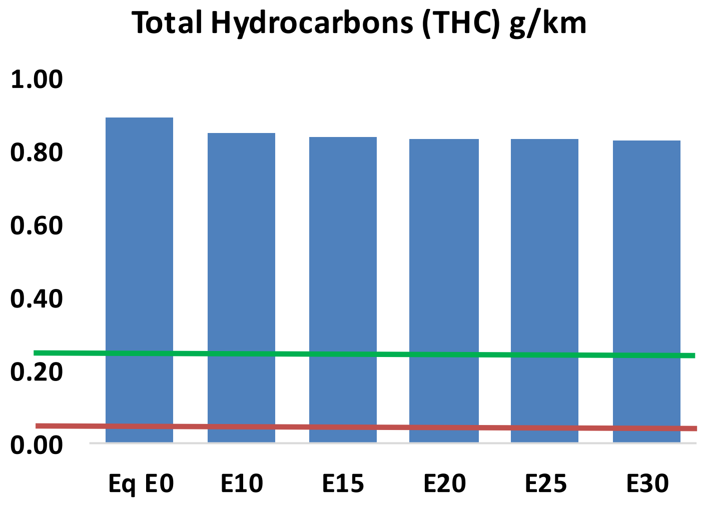
United States
Euro 6 Standard

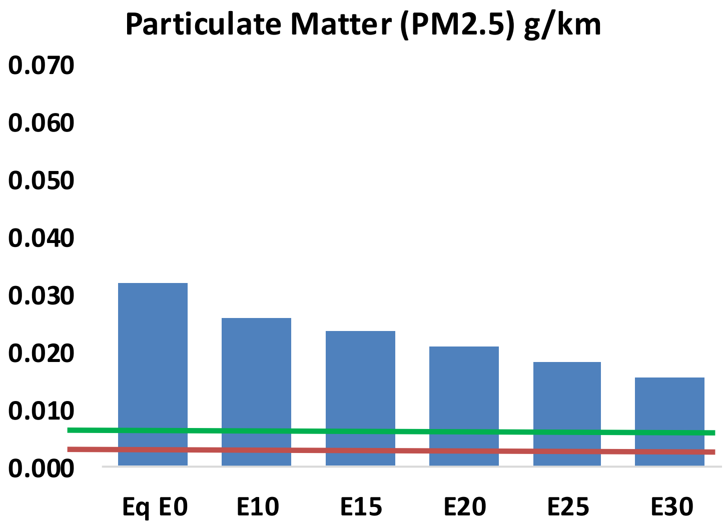












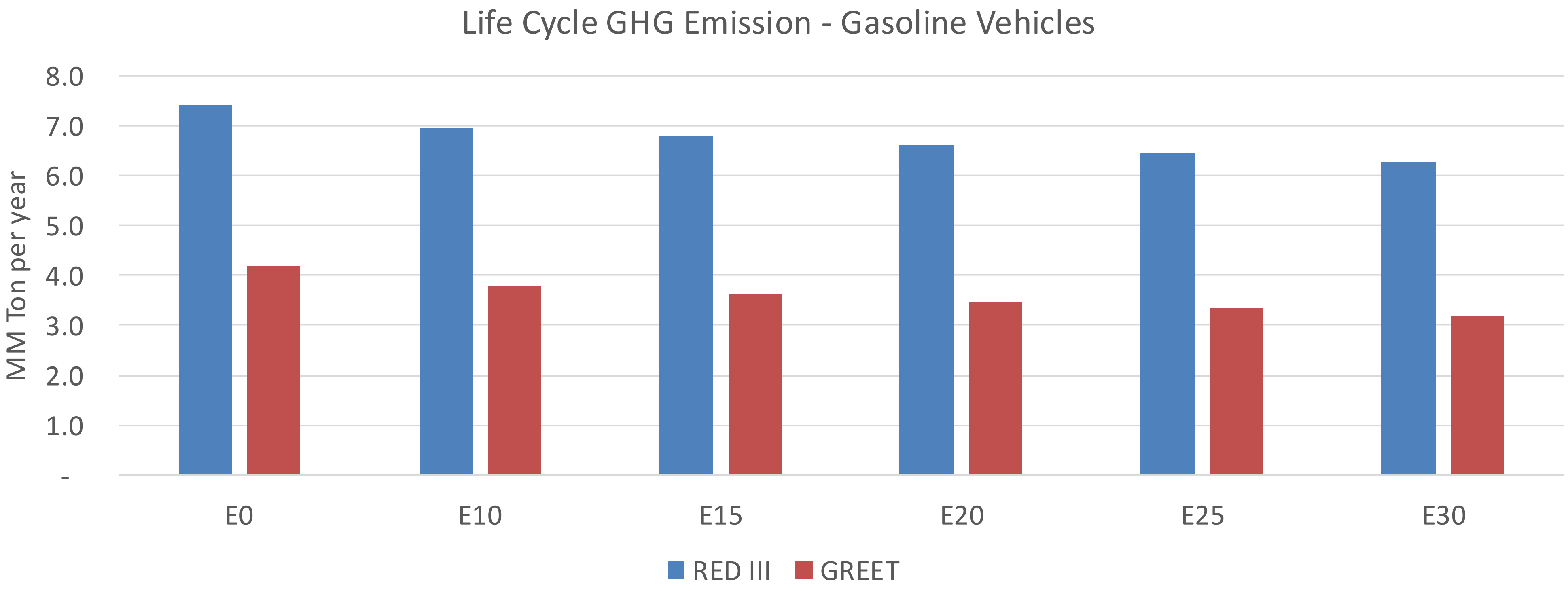
Source: Faro90

Panama – Vehicle Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	240,410	231,736	223,062	217,868	213,522	210,446	205,730	-7%	-11%
CO2	5,305,431	5,169,006	5,040,160	4,938,826	4,888,682	4,867,636	4,777,912	-5%	-8%
VOC	19,385	18,916	18,446	18,261	18,166	18,121	17,963	-5%	-6%
NOx	12,636	11,667	10,993	10,645	10,090	9,652	9,138	-13%	-20%
SOx	65	60	55	51	47	43	38	-15%	-28%
NH3	1,503	1,488	1,473	1,480	1,497	1,509	1,519	-2%	0%
N2O	189	188	187	188	192	194	196	-1%	2%
CH4	4,279	4,279	4,279	4,343	4,437	4,519	4,596	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	119	116	113	112	112	112	111	-5%	-6%
Acetaldehyde	229	262	385	586	799	913	1,079	68%	249%
Formaldehyde	755	734	856	1,005	1,053	1,165	1,268	13%	39%
Benzene	364	350	332	327	324	310	300	-9%	-11%
PM2.5	377	343	309	279	249	217	184	-18%	-34%
PM10	313	285	256	231	207	180	153	-18%	-34%

Source: Faro90

Panama – Gasoline Vehicle Fleet GEI Emissions



Country Emissions 2024 (MMT/year)	23.8
2035 Target (MMT/year)	16.7
Delta	7.1

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	9.7%	13.2%	16.7%	19.8%	23.7%
RED II/III	0.0%	6.3%	8.6%	11.0%	13.1%	15.6%

% 2035 Target	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	5.7%	7.7%	9.8%	11.6%	13.9%
RED II/III	0.0%	6.6%	9.0%	11.5%	13.6%	16.2%

Source: Greet, RED II/III, climateactiontracker.org, F90

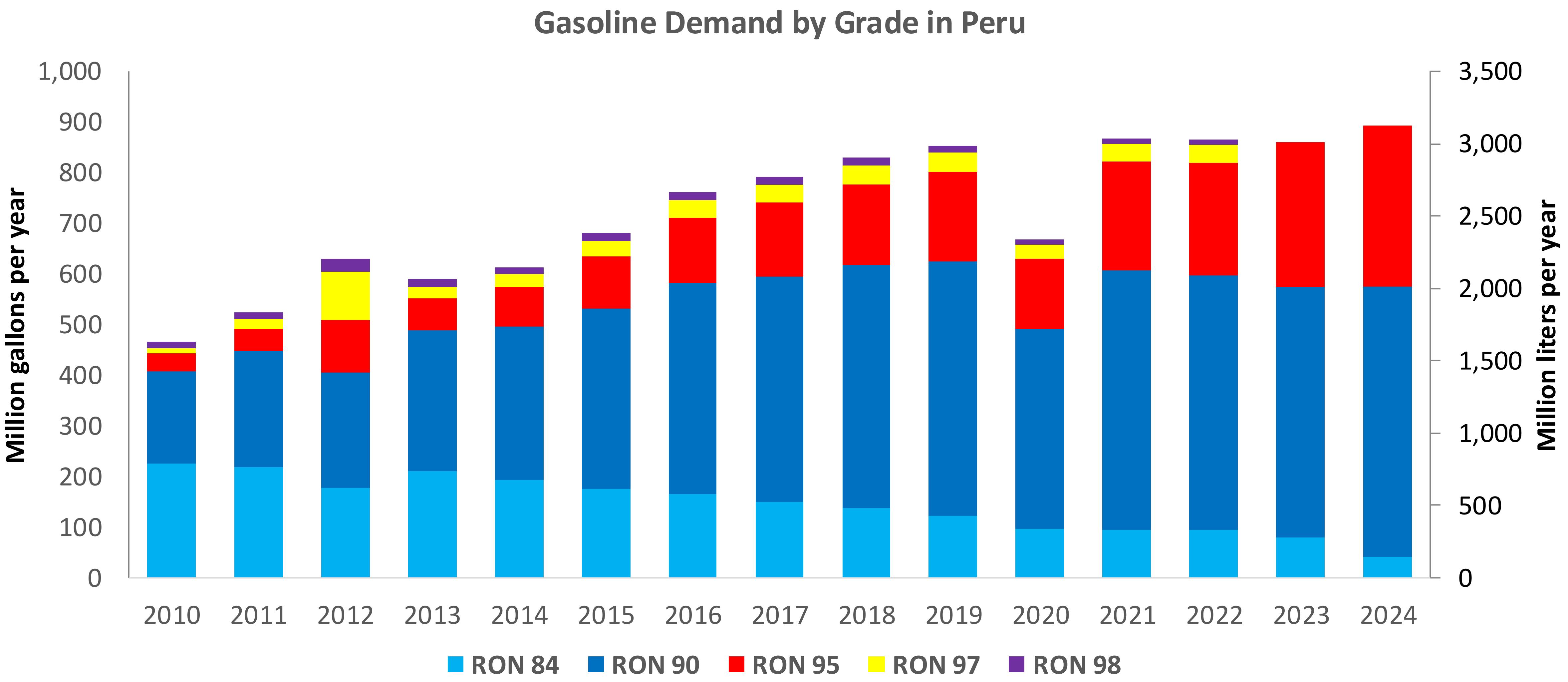
Ethanol Blending in Gasoline - Peru

In 2024, Peru gasoline consumption surpassed 900 million gallons (3,000 millions liters). Starting in 2023 a program to increase octane in gasoline was implemented; RON84 and RON 90 gasoline are commercialized as Regular gasoline (RON 91) with a 70% market participation. RON 95, RON 97 and RON 98 gasolines are now marketed as Premium gasoline (RON 96) with a 30% market share. Gasoline with ethanol (gasohol) is required to meet one additional octane number (RON 91 and RON 96 respectively). Local gasoline supply is done with production from Petroperú, Repsol and Primax. Gasoline imports account for 10 to 15% of demand, and are supplied by local markets, United states and Europe.

Blends with 7.8% v/v are allowed (Decreto Supremo N.° 021-2007-EM: MINEM). Peru produces ethanol from sugarcane; however, its destination is mainly the EU market. Ethanol imports from the United States and to a lesser degree from Brazil are used for the national blend.

Source: MEM, 2024

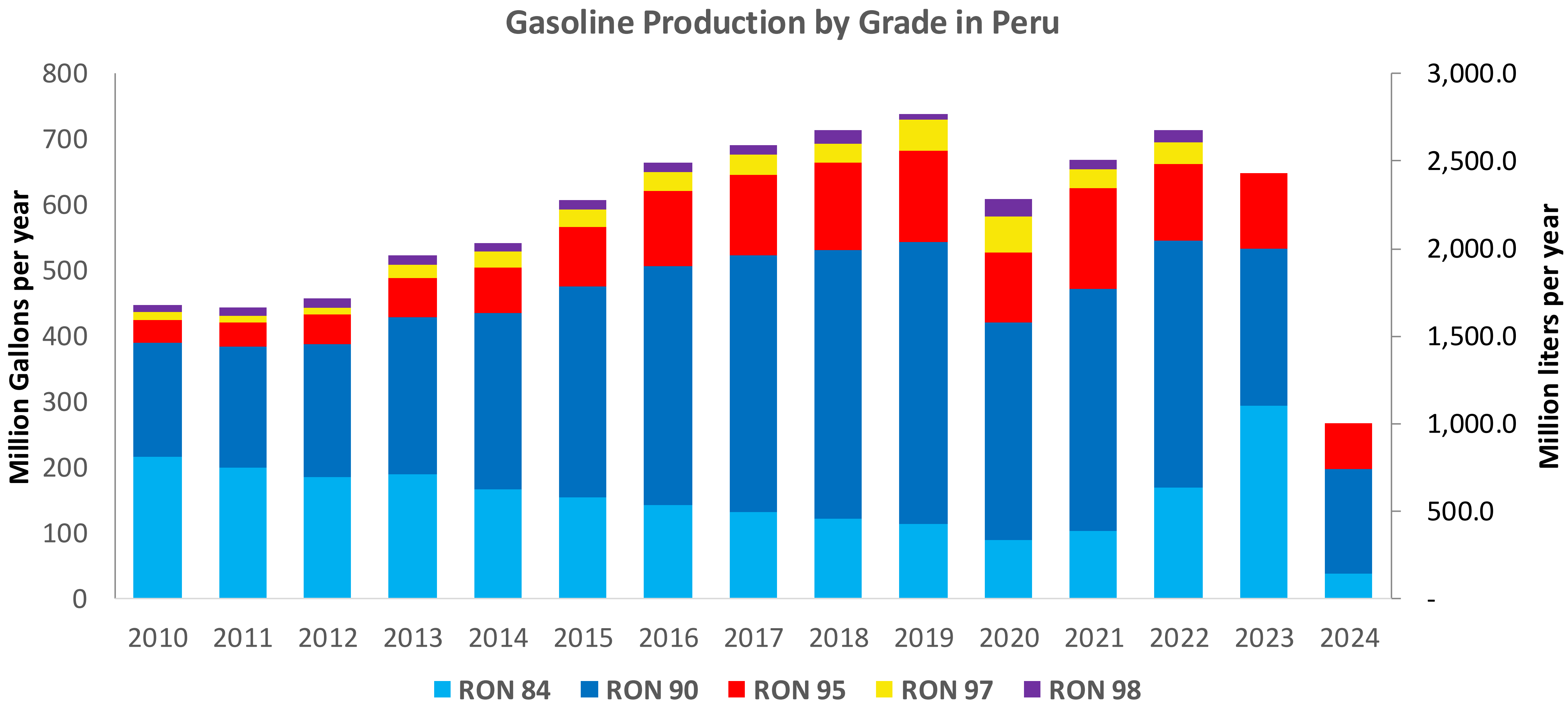
Gasoline Demand in Peru



Since 2023, RON 90 is commercialized as Regular Gasoline (RON 90-91) and RON 95, RON 97 and RON 98 are now marketed as Premium Gasoline (RON 95-96).

Source: MINEM, 2024

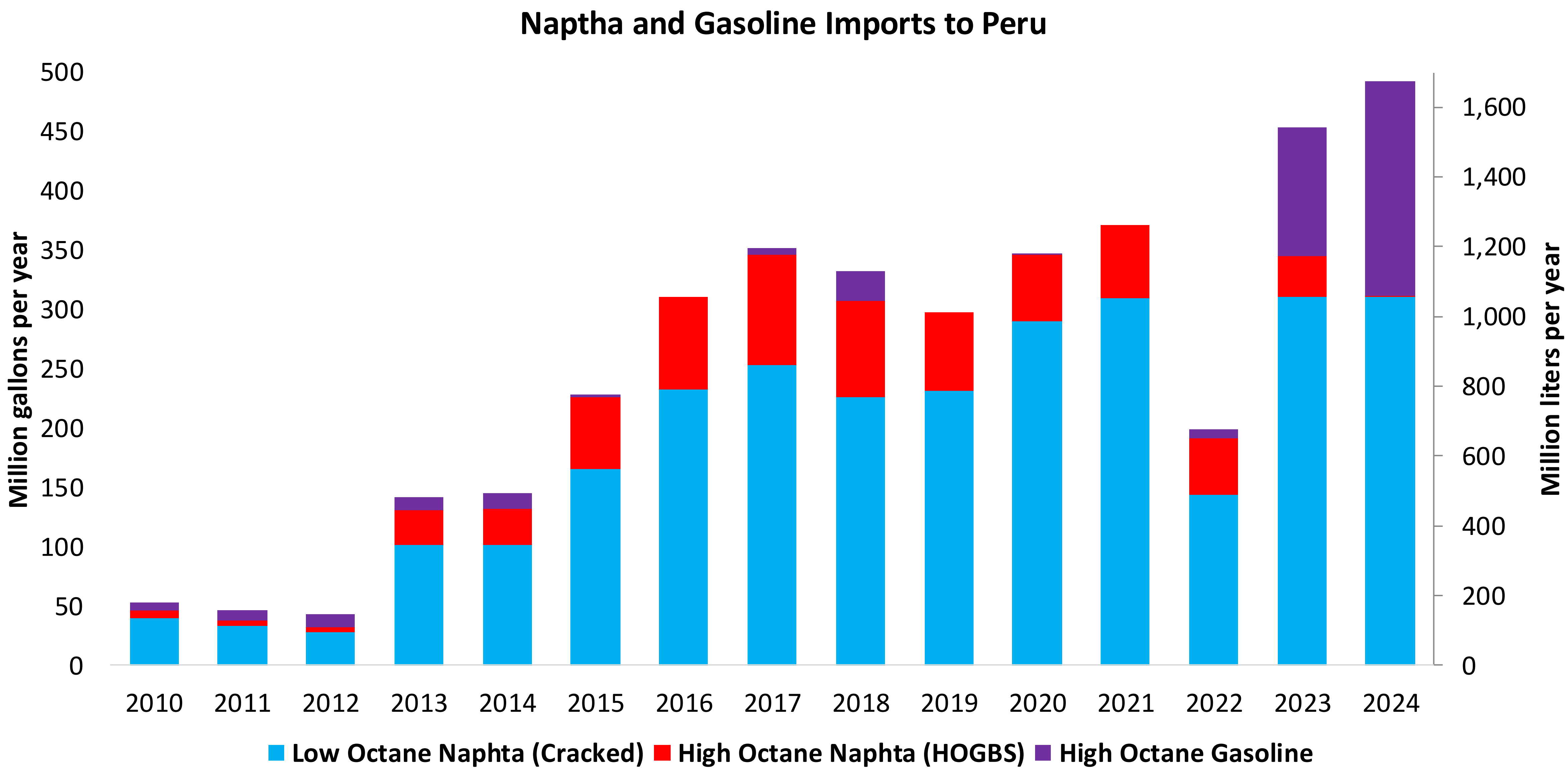
Gasoline Production in Peru



Since 2023, RON 90 is commercialized as Regular Gasoline (RON 90-91) and RON 95, RON 97 and RON 98 are now marketed as Premium Gasoline (RON 95-96).

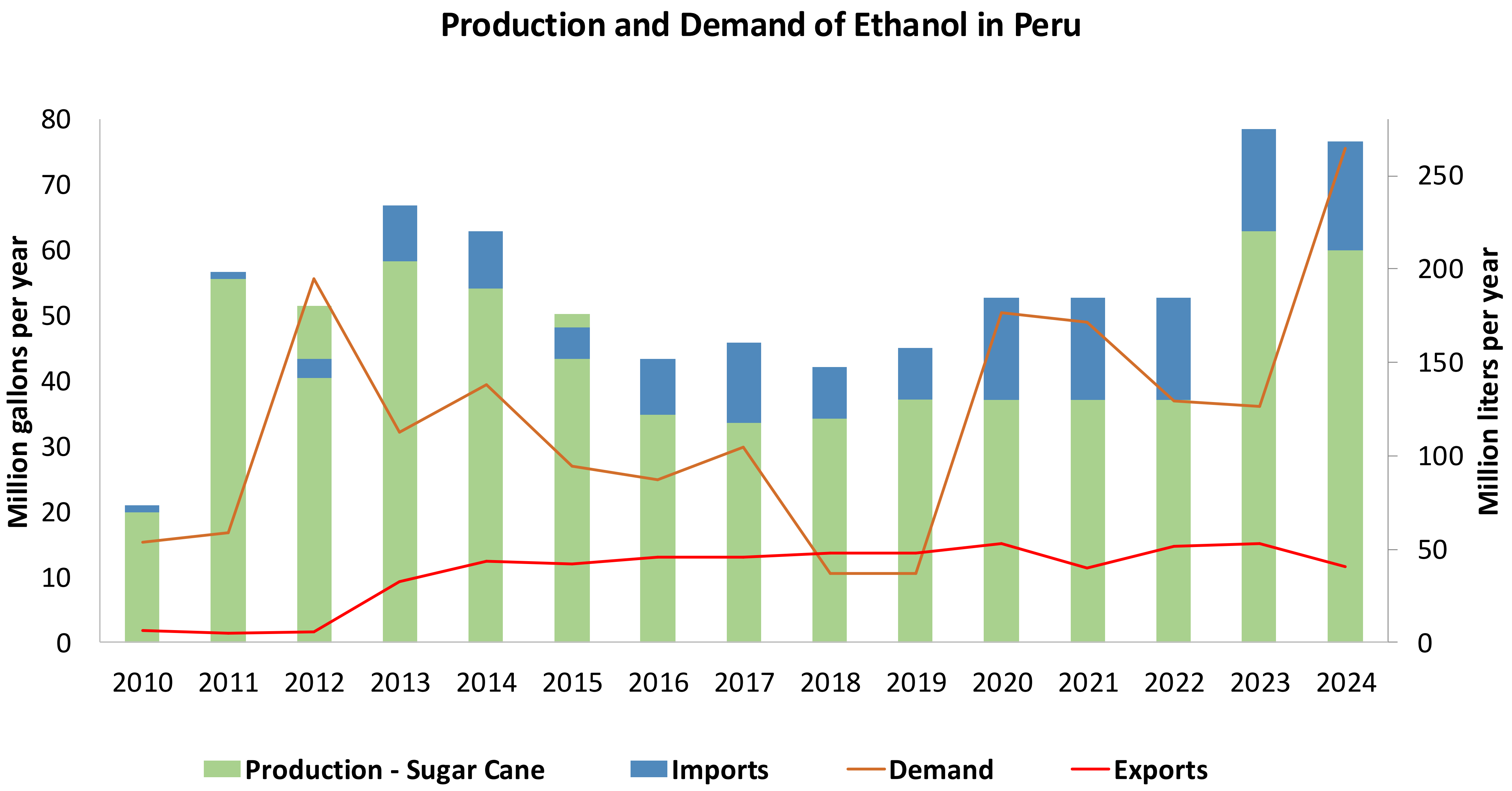
Source: MINEM, 2024

Gasoline Imports in Peru



Source: MINEM, 2024

Ethanol Balance in Peru



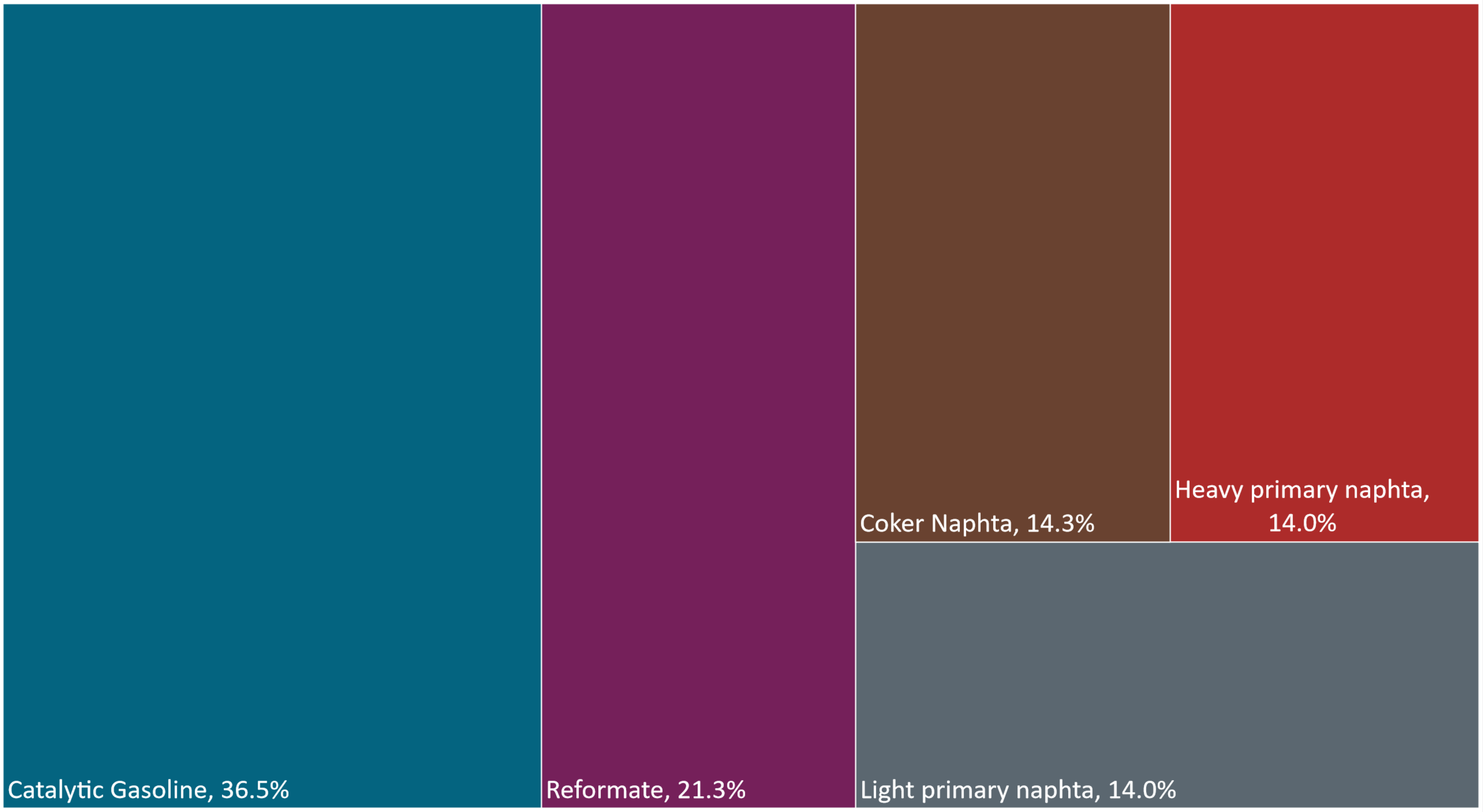
Source: MINEM, 2024

Gasoline Quality in Peru

Name	Ministerial Resolution Nº 469-2021-MINEM/DM				EN 228:2012 + A1:2017 (Euro 6 enabling)			
Implementation Date	2021				2017			
Applicability	Whole contry	Whole contry	Whole contry	Whole contry	All countries			
Selected Grade	Gasolina Regular	Gasolina Premium	Gasohol Regular	Gasohol Premium	RON 95 E5	RON 95 E10	RON 98 E5	RON 98 E10
Benzene Content	< 2% v/v	< 2% v/v	< 2% v/v	< 2% v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v
Aromatics	< 45% v/v	< 45% v/v	< 45% v/v	< 45% v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v
Olefins	< 25% v/v	< 25% v/v	< 25% v/v	< 25% v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v
Lead Content	< 0,013 g/l	< 0,013 g/l	< 0,013 g/l	< 0,013 g/l	< 5 mg/l	< 5 mg/l	< 5 mg/l	< 5 mg/l
Manganese	< 0,25 mg/l	< 0,25 mg/l	< 0,25 mg/l	< 0,25 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l
RON	> 90	> 95	> 91	> 96	> 95	> 95	> 98	> 98
MON	-	-	-	-	> 85	> 88	> 85	> 88
AKI								
Sulfur Content	< 50 mg/kg	< 50 mg/kg	< 50 mg/kg	< 50 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg
Oxygen Content	Reportar	Reportar	< 3,45 %m/m	< 3,45 %m/m	<2,7 % m/m	<3,7 % m/m	<2,7 % m/m	<3,7 % m/m
Ethanol (EtOH)			< 7,8 %v/v	< 7,8 %v/v	<5 %v/v	<10 %v/v	<5 %v/v	<10 %v/v
RVP 37.8°C (Summer)	< 76 kPa	< 76 kPa	< 76 kPa	< 76 kPa	<> 60 - 70 kPa *Depends on the country, RVP is regulated in the EU Fuel Quality Directive			
RVP 37.8 °C(Winter)								
RVP 37.8°C (Transition)								
MTBE					-	-	-	-
Ethers 5 or more C Atoms	-	-	-	-	Based on oxygen content	<22 %v/v	Based on oxygen content	<22 %v/v

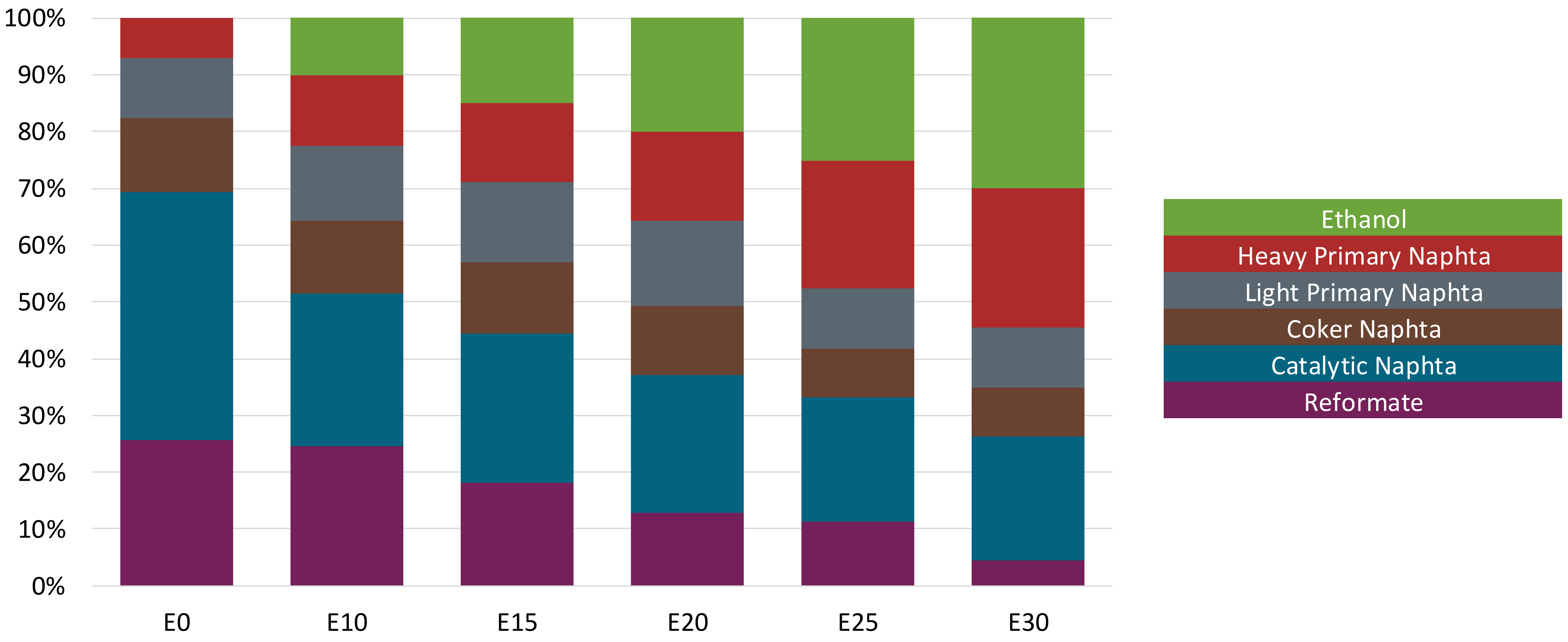
Source: MINEM, 2023

Available Blending Components



Source: Faro90

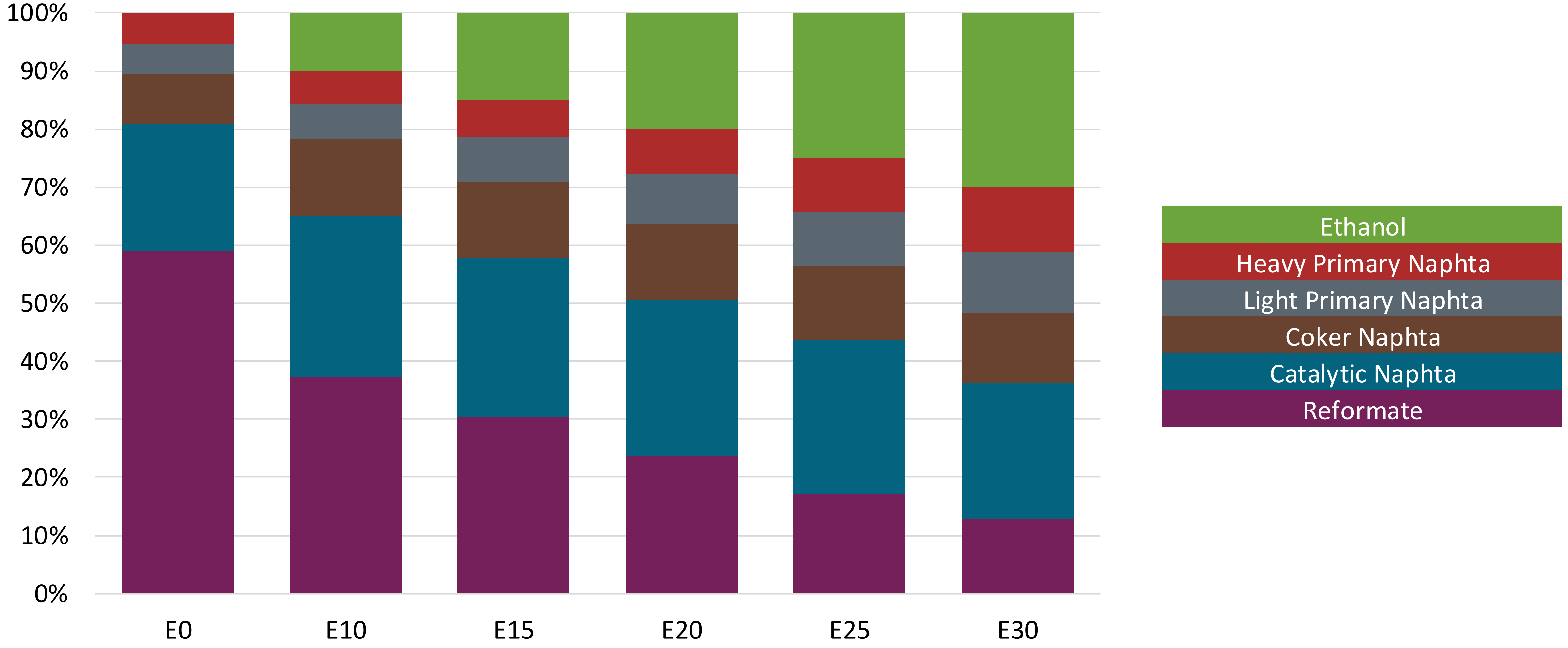
Peru – Regular – Constant Octane



	E0	E10	E15	E20	E25	E30
Octane (RON)	90.0	90.0	90.0	90.0	90.0	90.0
Price (US$/gal)	$ 2.412	$ 2.242	$ 2.155	$ 2.070	$ 1.995	$ 1.908

Source: Faro90

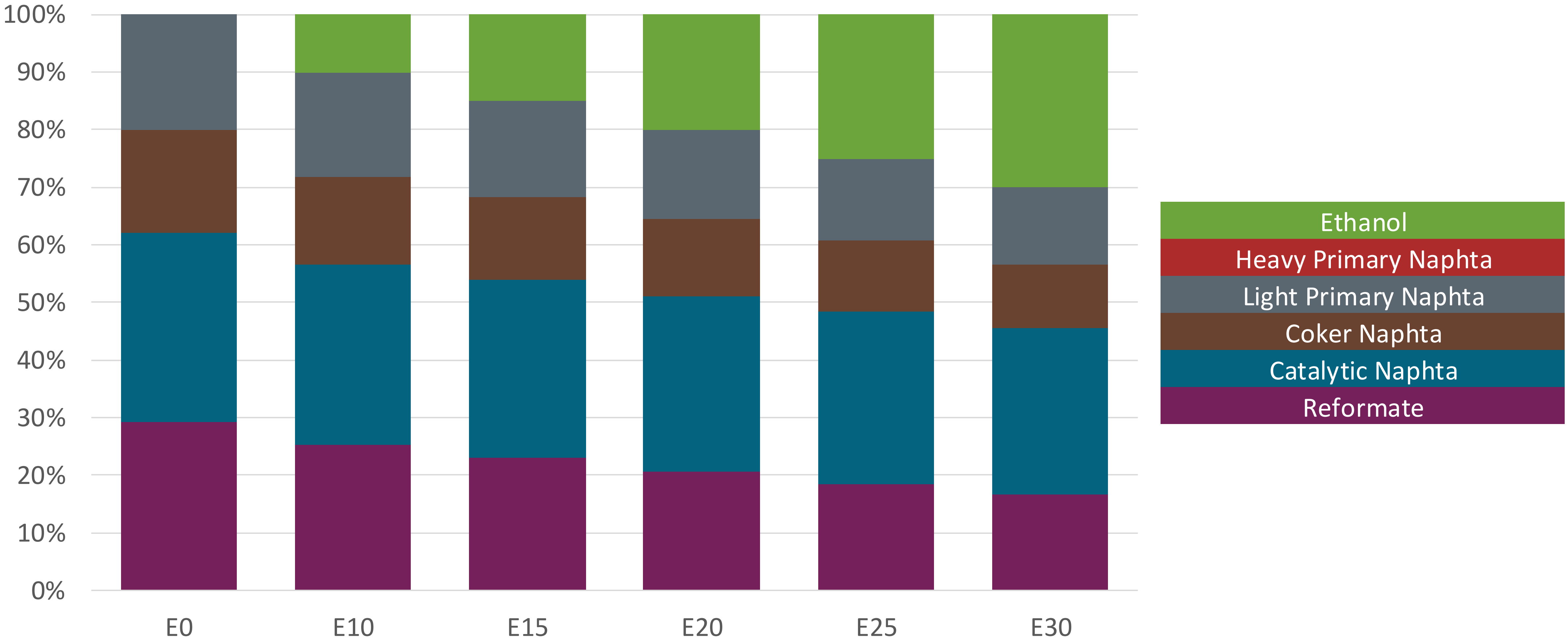
Peru – Premium – Constant Octane



	E0	E10	E15	E20	E25	E30
Octane (RON)	95.0	95.0	95.0	95.0	95.0	95.0
Price (US$/gal)	$ 2.672	$ 2.483	$ 2.395	$ 2.309	$ 2.223	$ 2.138

Source: Faro90

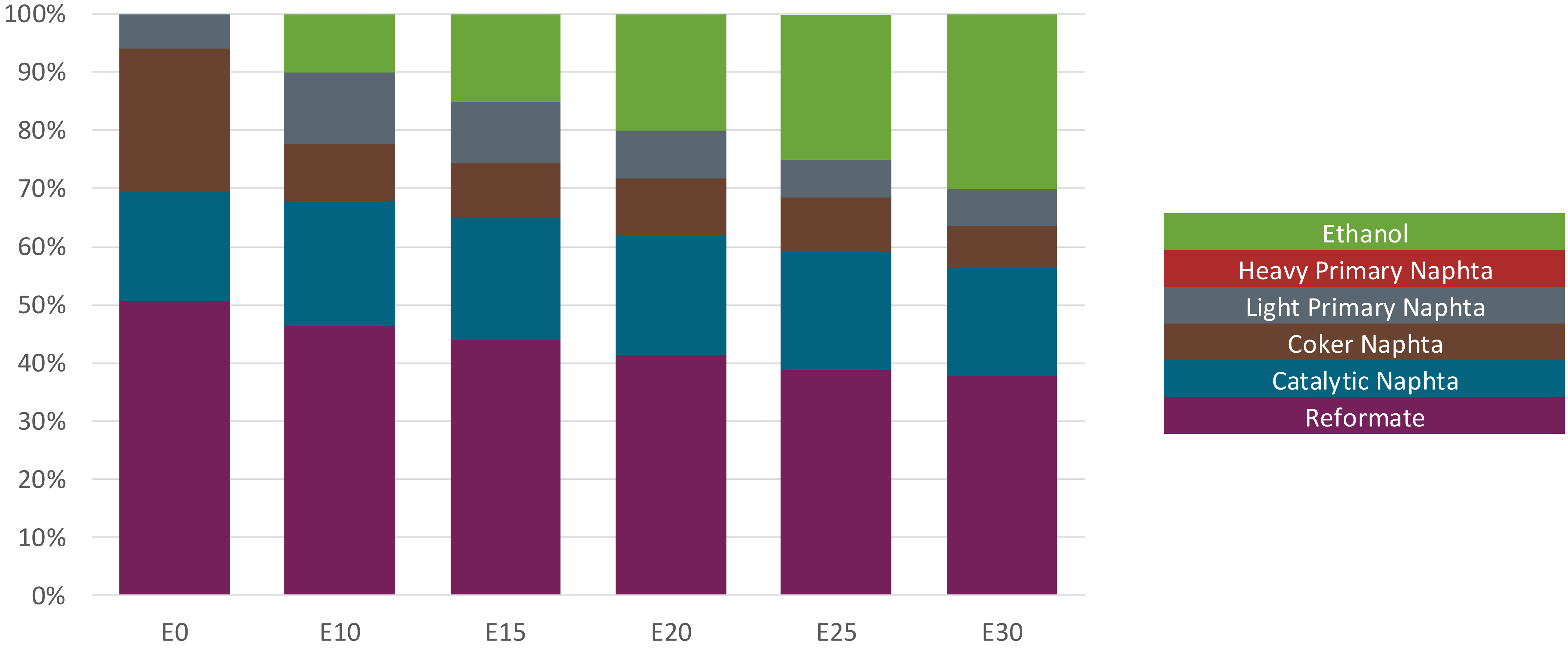
Peru – Regular – Increased Octane



	E0	E10	E15	E20	E25	E30
Octane (RON)	90.0	93.5	95.2	96.9	98.4	105.4
Price (US$/gal)	$ 2.410	$ 2.412	$ 2.412	$ 2.412	$ 2.401	$ 2.412

Source: Faro90

Peru – Premium – Increased Octane

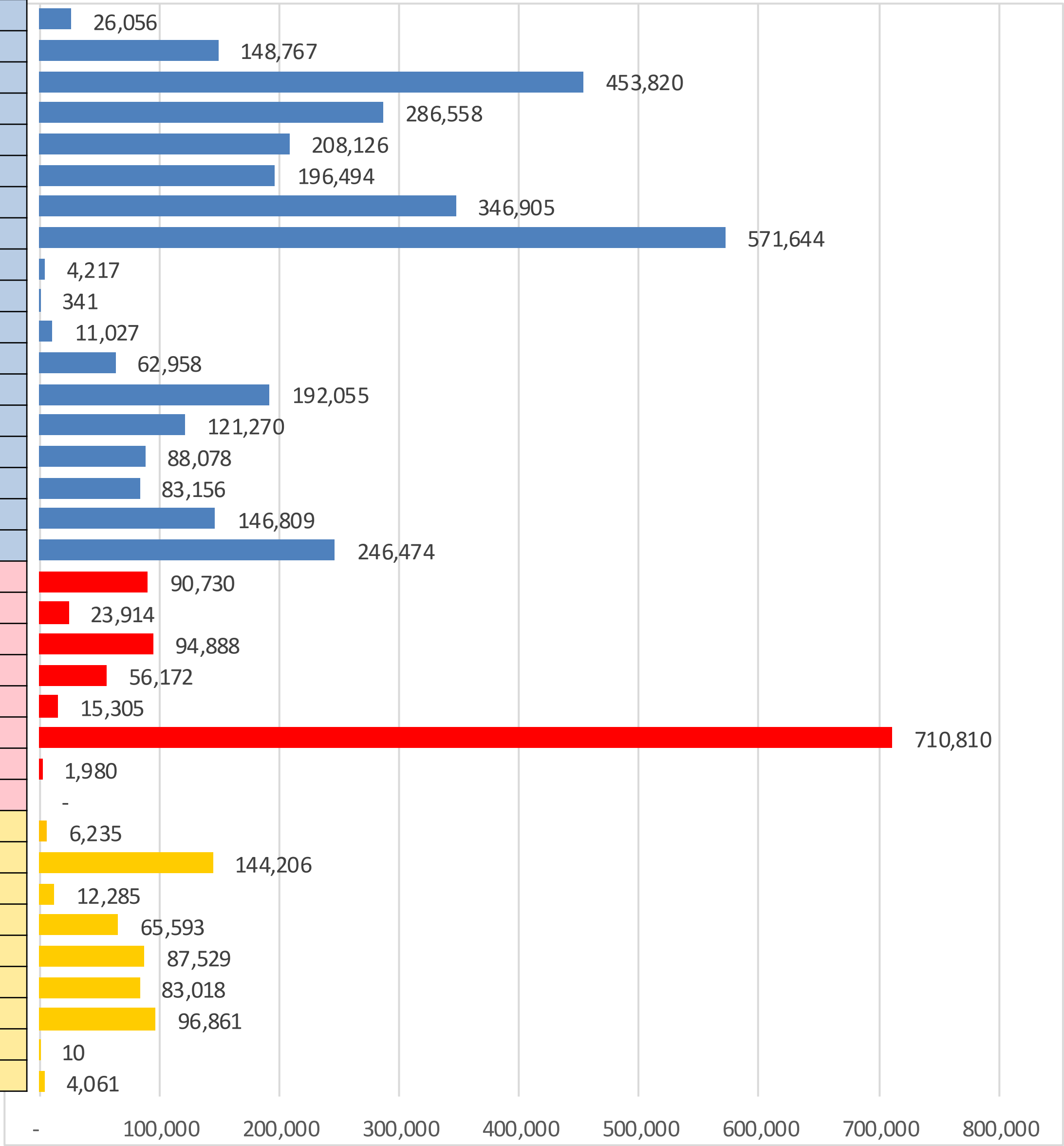


Octane (RON)	95.0	96.4	97.8	99.4	100.8	102.1
Price (US$/gal)	$ 2.623	$ 2.561	$ 2.545	$ 2.531	$ 2.515	$ 2.496

Source: Faro90

Peru – Gasoline Vehicle Fleet

Type	Weight	Motor	Exhaust	Recuperator	Age
Light / Medium	Light	Carburator	Not specified	PCV	+8years
		Single FI	Not specified	PCV	+8years
		Multi FI	Not specified	PCV	+8years
			Euro II	PCV Tank	+8years
			Euro III		+8years
					4-8years
			Euro IV		4-8years
					-4years
			Hybrids		
			Electric		
	Medium	Carburator	Not specified	PCV	+8years
		Single FI	Not specified	PCV	+8years
		Multi FI	Not specified	PCV	+8years
			Euro II	PCV Tank	+8years
			Euro III		+8years
					4-8years
			Euro IV		4-8years
					-4years
Small Engines	Light	4-Cycle FI	Not specified	PCV	+8years
			Euro II		+8years
			Euro III		+8years
					4-8years
			Euro IV		4-8years
					-4years
			Hybrids		
			Electric		
Heavy	Heavy	Carburator	Not specified	PCV	+8years
		Fuel-Injection	Not specified		+8years
			Euro II		+8years
			Euro III		+8years
					4-8years
			Euro IV		4-8years
					-4years
			Hybrids		
			Electric		
				Total	4,682,300

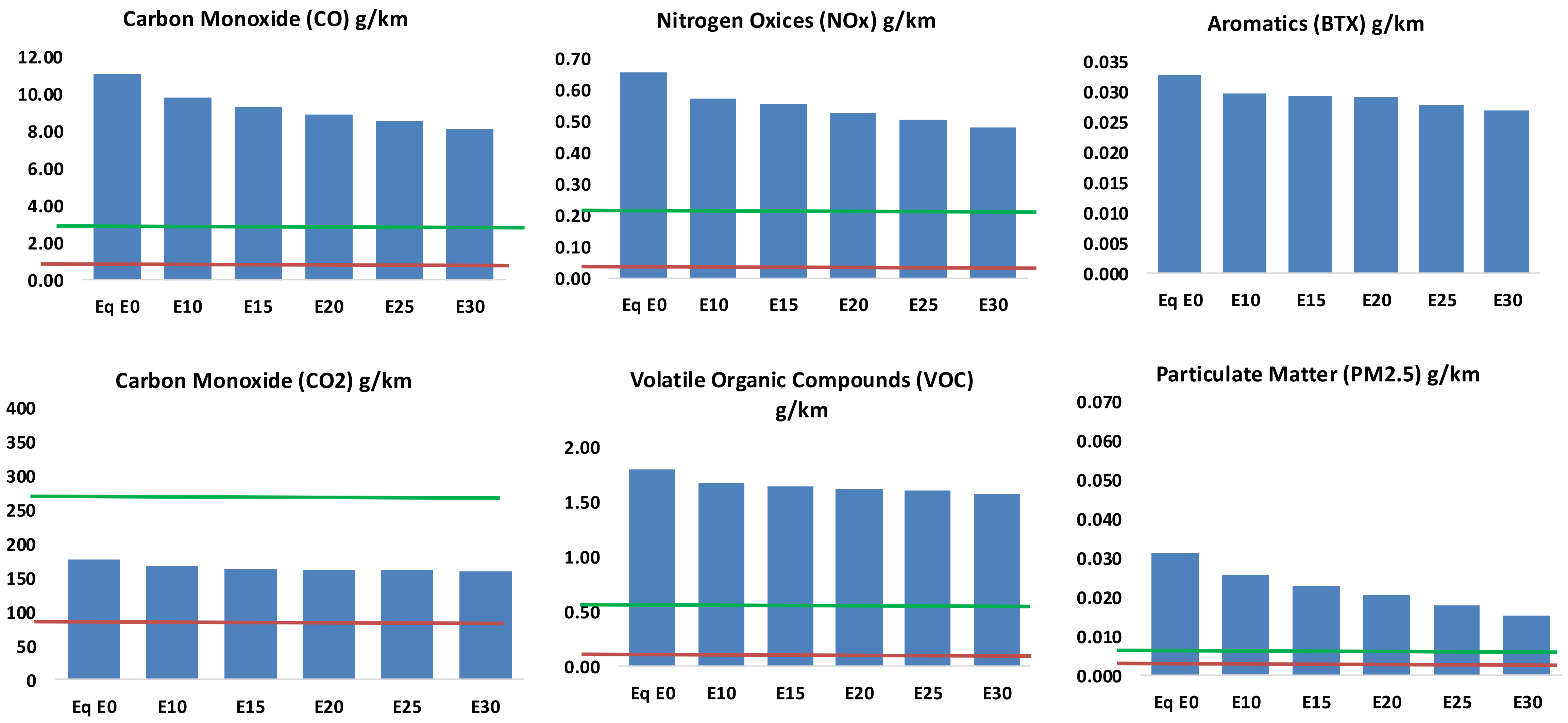


Gasoline vehicles: **4,182.016**
Average Age: **13.2 years**
Motorcycles: 21%

Source: Ministerio de Transporte y Comunicaciones, 2024

Peru – Emissions

United States
Euro 6



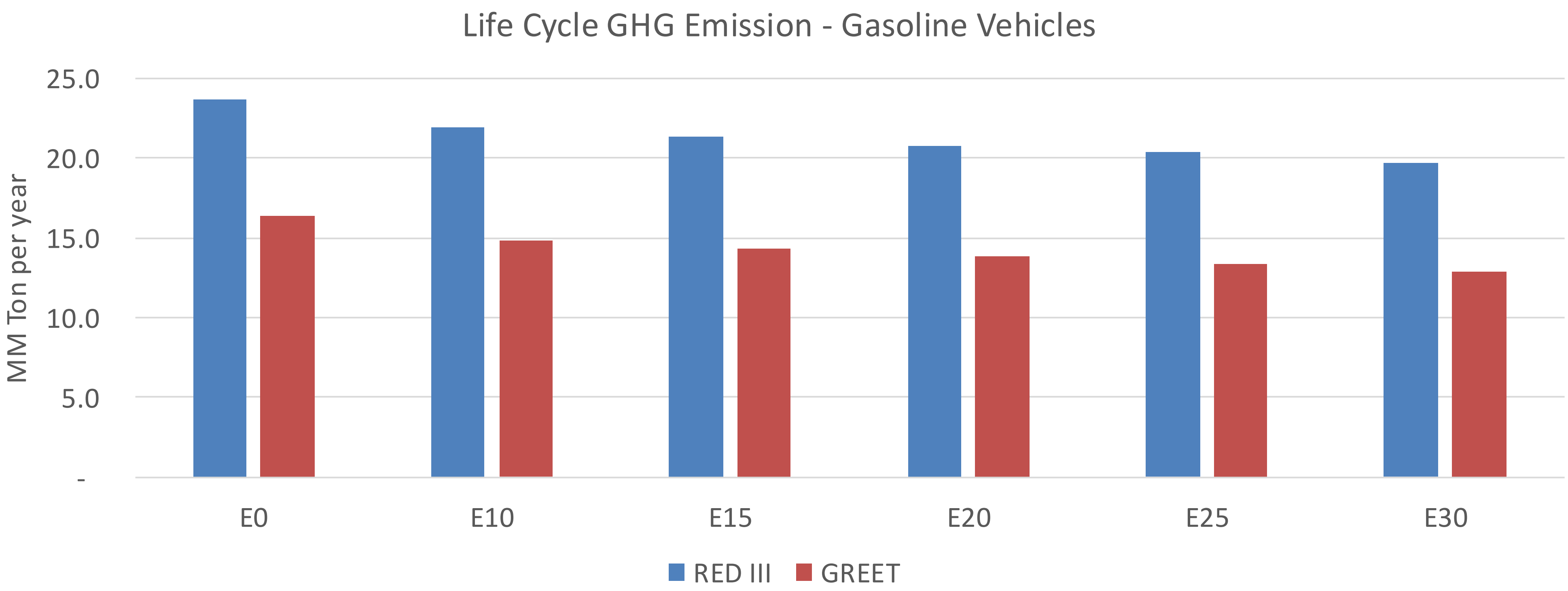
Source: Faro90

Peru – Gasoline Vehicle Fleet Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	1,170,451	1,104,009	1,037,566	987,711	939,656	903,341	857,311	-11%	-20%
CO2	18,524,884	18,048,530	17,598,640	17,244,815	17,069,728	17,005,418	16,691,959	-5%	-8%
VOC	189,345	183,067	176,788	173,258	170,449	168,514	165,139	-7%	-10%
NOx	69,155	63,853	60,317	58,570	55,695	53,468	50,830	-13%	-19%
SOx	213	197	181	167	154	140	125	-15%	-28%
NH3	6,875	6,807	6,739	6,771	6,847	6,901	6,946	-2%	0%
N2O	190	189	188	189	193	195	198	-1%	2%
CH4	41,845	41,845	41,845	42,472	43,393	44,188	44,941	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	604	579	555	539	525	515	499	-8%	-13%
Acetaldehyde	1,308	1,501	2,202	3,353	4,569	5,221	6,169	68%	249%
Formaldehyde	4,502	4,377	5,102	5,991	6,277	6,944	7,559	13%	39%
Benzene	3,455	3,320	3,146	3,096	3,070	2,936	2,842	-9%	-11%
PM2.5	1,172	1,065	959	865	773	675	572	-18%	-34%
PM10	2,101	1,910	1,719	1,551	1,386	1,210	1,025	-18%	-34%

Source: Faro90

Peru – Gasoline Vehicle Fleet GEI Emissions



Country Emissions 2024 (MMT/year)	111.2
2035 Target (MMT/year)	96.1
Delta	15.1

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	9.6%	12.5%	15.6%	18.2%	21.6%
RED II/III	0.0%	7.3%	9.5%	12.0%	14.0%	16.5%

% 2035 Target	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	10.4%	13.5%	16.9%	19.7%	23.4%
RED II/III	0.0%	11.4%	14.8%	18.7%	21.8%	25.8%

Source: Greet, RED II/III, climateactiontracker.org, F90

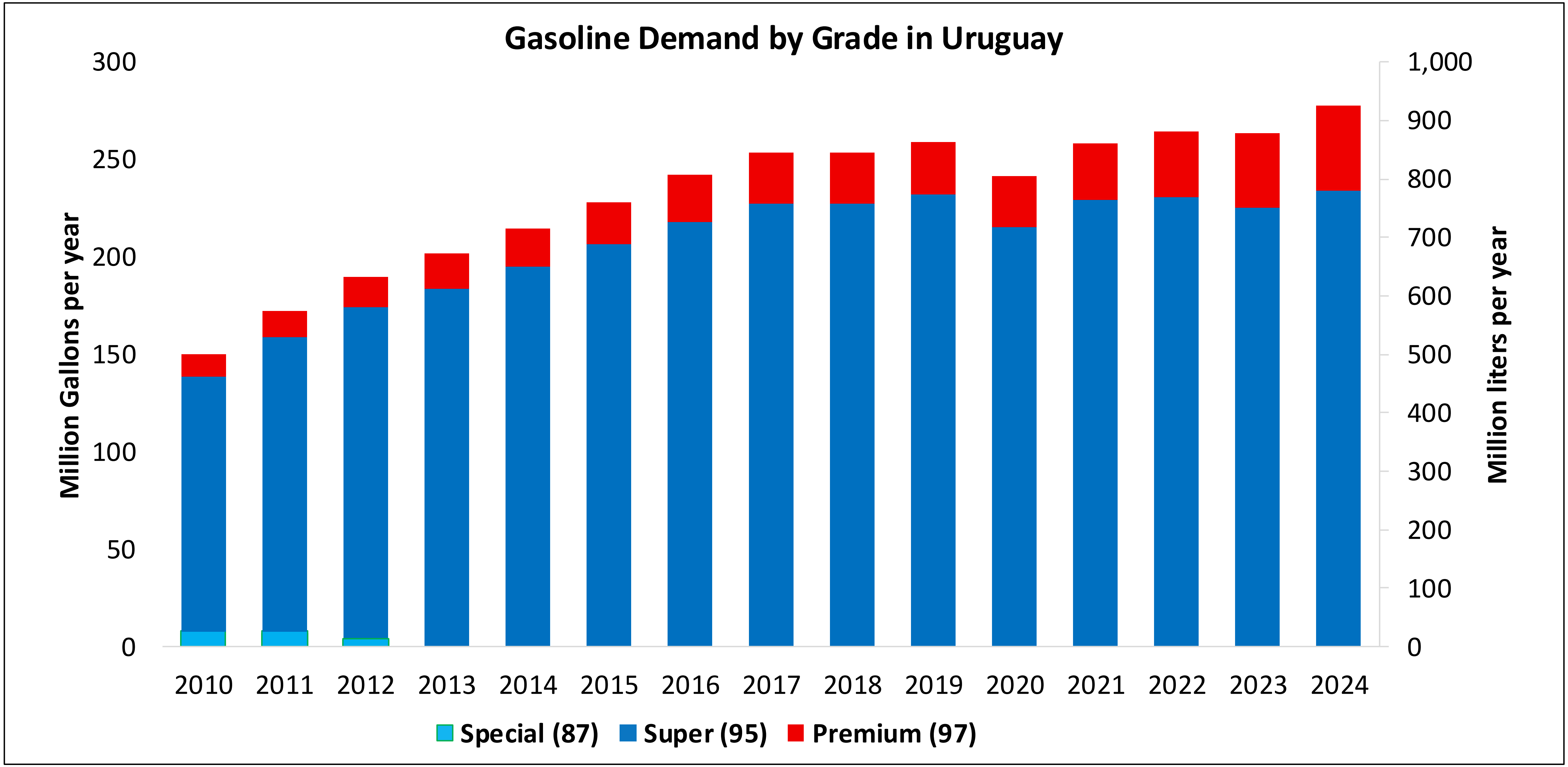
Ethanol Blending in Gasoline - Uruguay

In 2024, gasoline demand was 245 million gallons (926 million liters). Super Gasoline with RON 95 represented 84% of total demand and Premium Gasoline with RON 97 the rest. Gasoline production was 125 million gallons (471 million liters) and covers 64% of local demand.

Gasoline Blends with up to 10% v/v are allowed.. In Uruguay there are two ethanol plants with an overall capacity of 100 million liters per year, raw material commonly used are sugar cane, sorghum, and corn. In 2024, ethanol consumption was 24 million gallons (91 million liters).

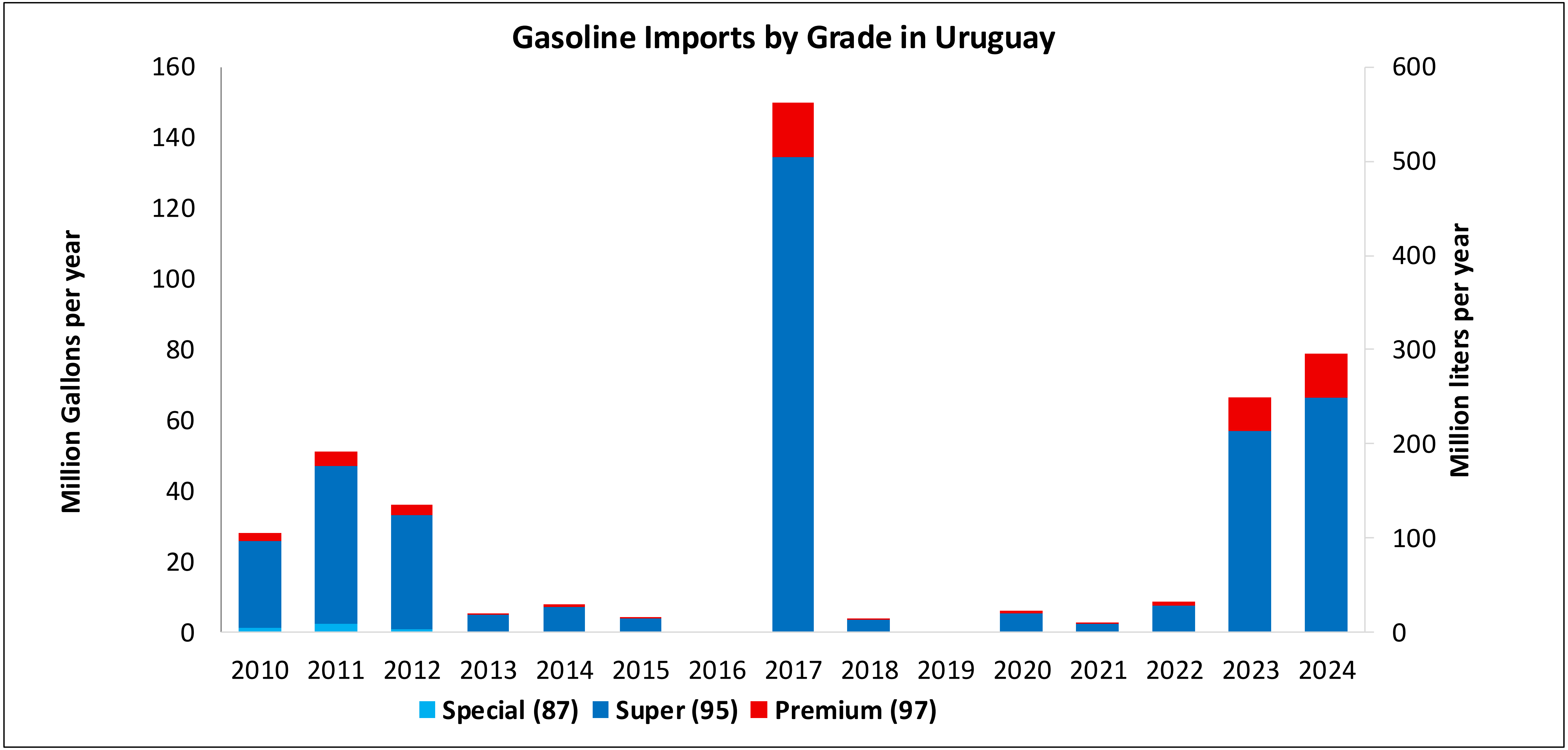
Source: ANCAP

Gasoline Demand in Uruguay



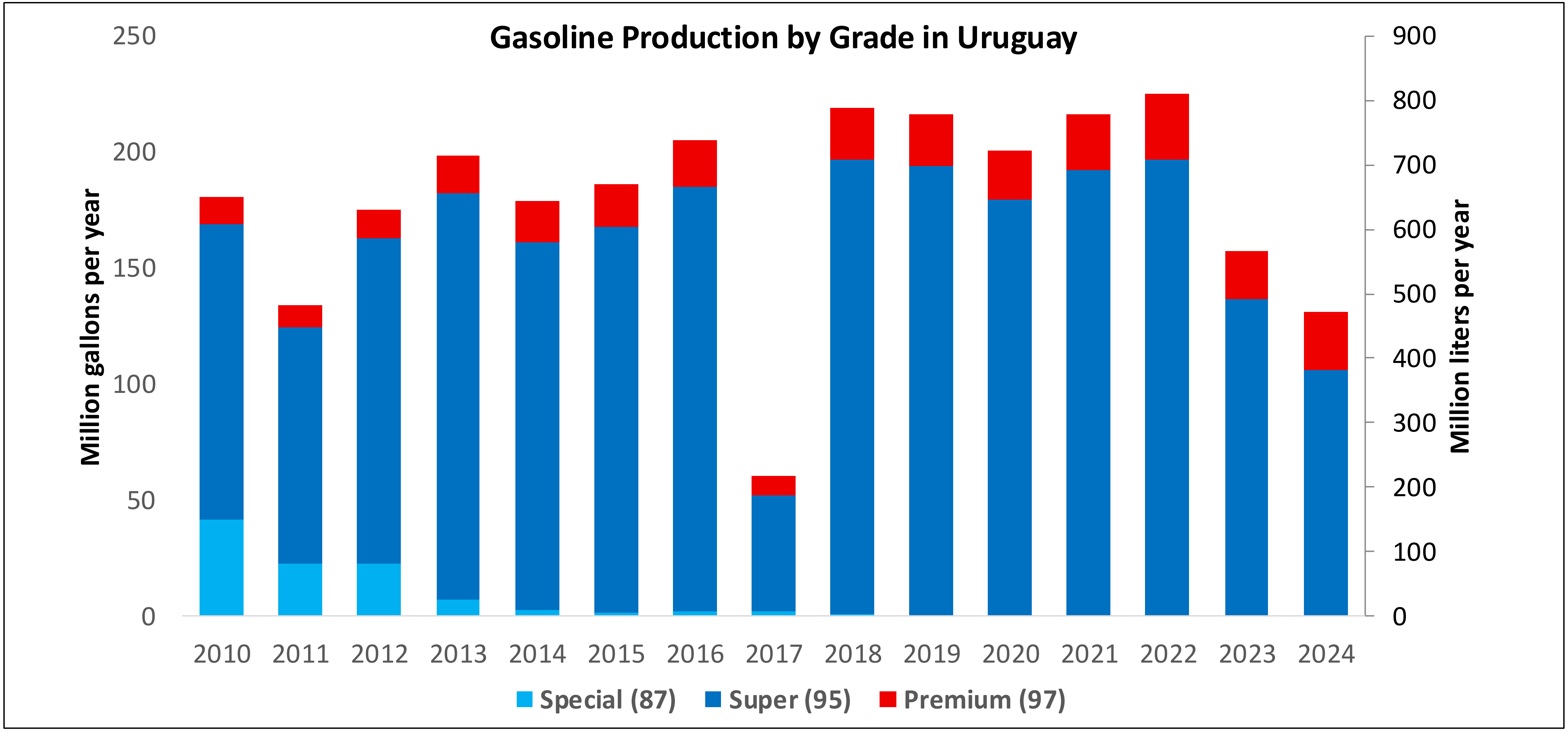
Source: ANCAP

Gasoline Imports in Uruguay



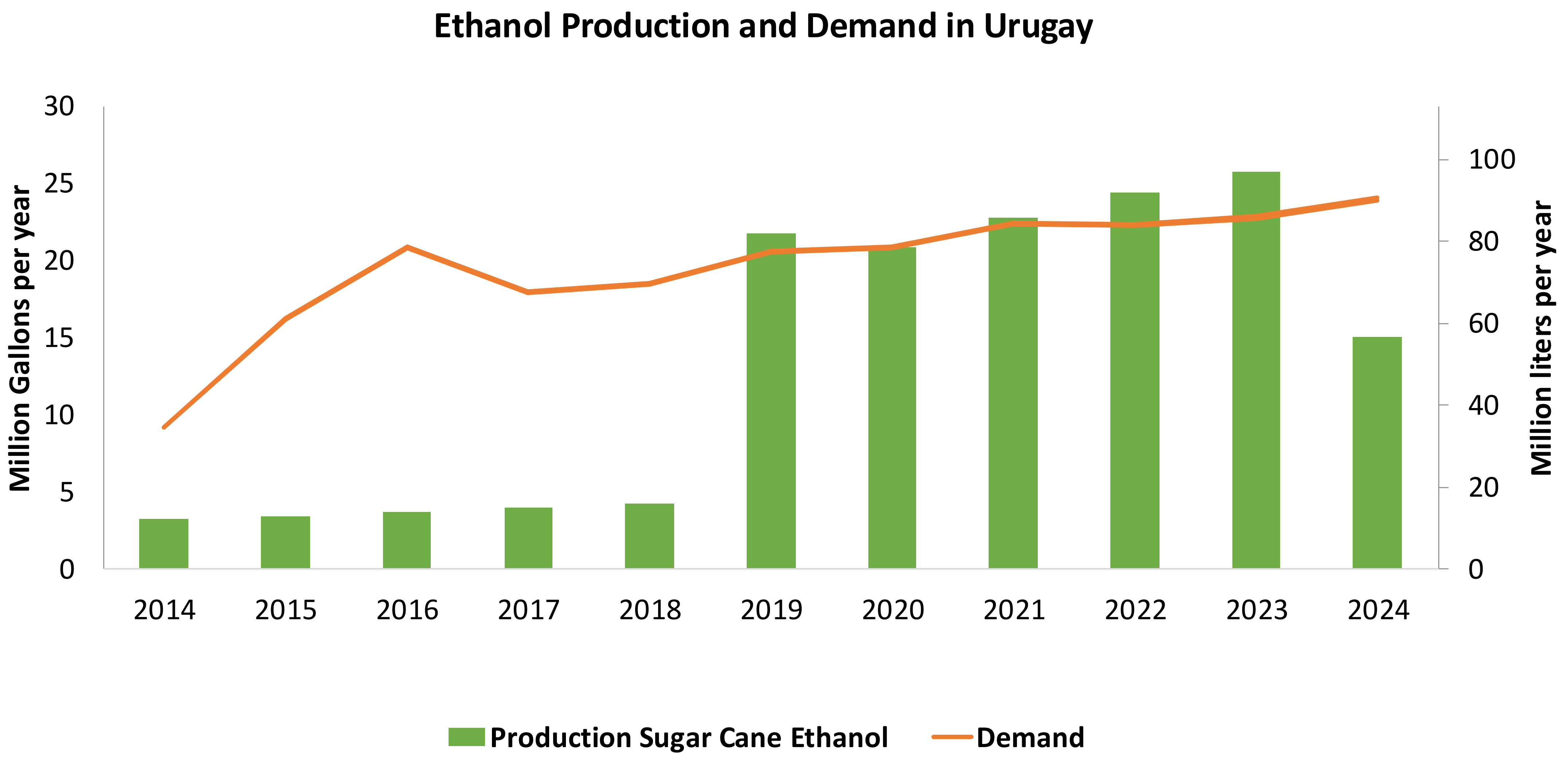
Source: ANCAP

Gasoline Production in Uruguay



Source: ANCAP

Ethanol Demand in Uruguay



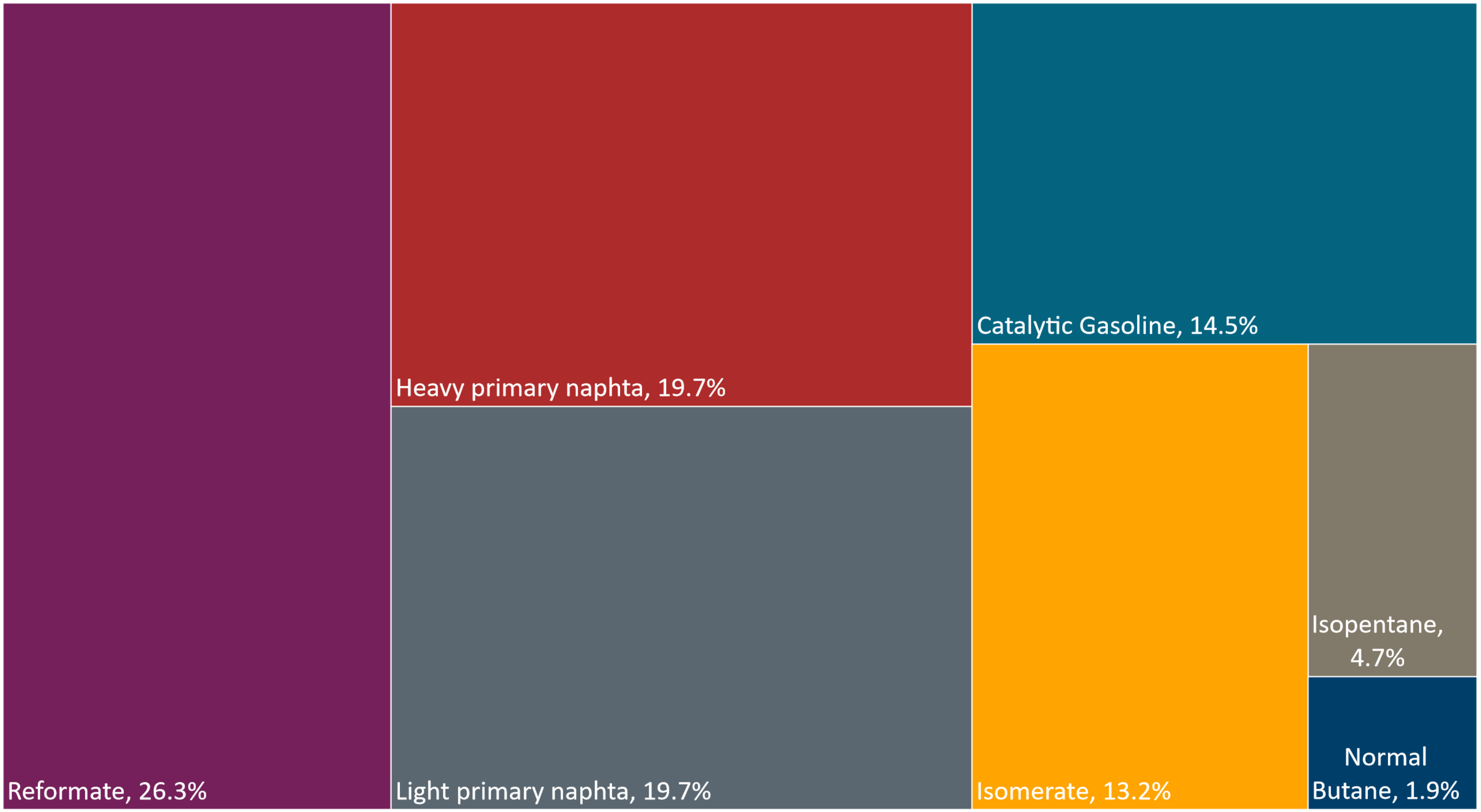
Source: ANCAP

Gasoline Quality in Uruguay

Name	Resolution 110/2014		ANCAP		EN 228:2012 + A1:2017 (Euro 6 enabling)			
Implementation Date	2019		2020		2017			
Applicability	Whole country	Whole country	Whole country	Whole country	All countries			
Selected Grade	Super 95 30-S	Premium 97 30-S	Super 95 30-S	Premium 97 30-S	RON 95 E5	RON 95 E10	RON 98 E5	RON 98 E10
Benzene Content	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v
Aromatics	< 40 %v/v	< 40 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v
Olefins	< 20 %v/v	< 20 %v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v
Lead Content	< 0,005 g/l	< 0,005 g/l	< 0,005 g/l	< 0,005 g/l	< 5 mg/l	< 5 mg/l	< 5 mg/l	< 5 mg/l
Manganese	< 2,5 mg/l	< 2,5 mg/l	< 2,5 mg/l	< 2,5 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l
RON	> 95	> 97	> 95	> 97	> 95	> 95	> 98	> 98
MON	> 82	> 84	> 85	> 85	> 85	> 88	> 85	> 88
AKI								
Sulfur Content	< 30 mg/kg	< 30 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg
Oxygen Content	< 2,7 %m/m	< 2,7 %m/m	< 2,7 %m/m	< 2,7 %m/m	<2,7 % m/m	<3,7 % m/m	<2,7 % m/m	<3,7 % m/m
Ethanol (EtOH)	< > 10 %v/v	< > 10 %v/v			<5 %v/v	<10 %v/v	<5 %v/v	<10 %v/v
RVP 37.8°C (Summer)	<> 72 kPa	<> 72 kPa	<> 50-80 kPa	<> 50-80 kPa	<> 60 - 70 kPa *Depends on the country, RVP is regulated in the EU Fuel Quality Directive			
RVP 37.8 °C(Winter)	<> 83 kPa	<> 83 kPa	<> 45-67 kPa	<> 45-67 kPa				
RVP 37.8°C (Transition)								
MTBE					-	-	-	-
Ethers 5 or more C Atoms	-	-	-	-	Based on oxygen content	<22 %v/v	Based on oxygen content	<22 %v/v

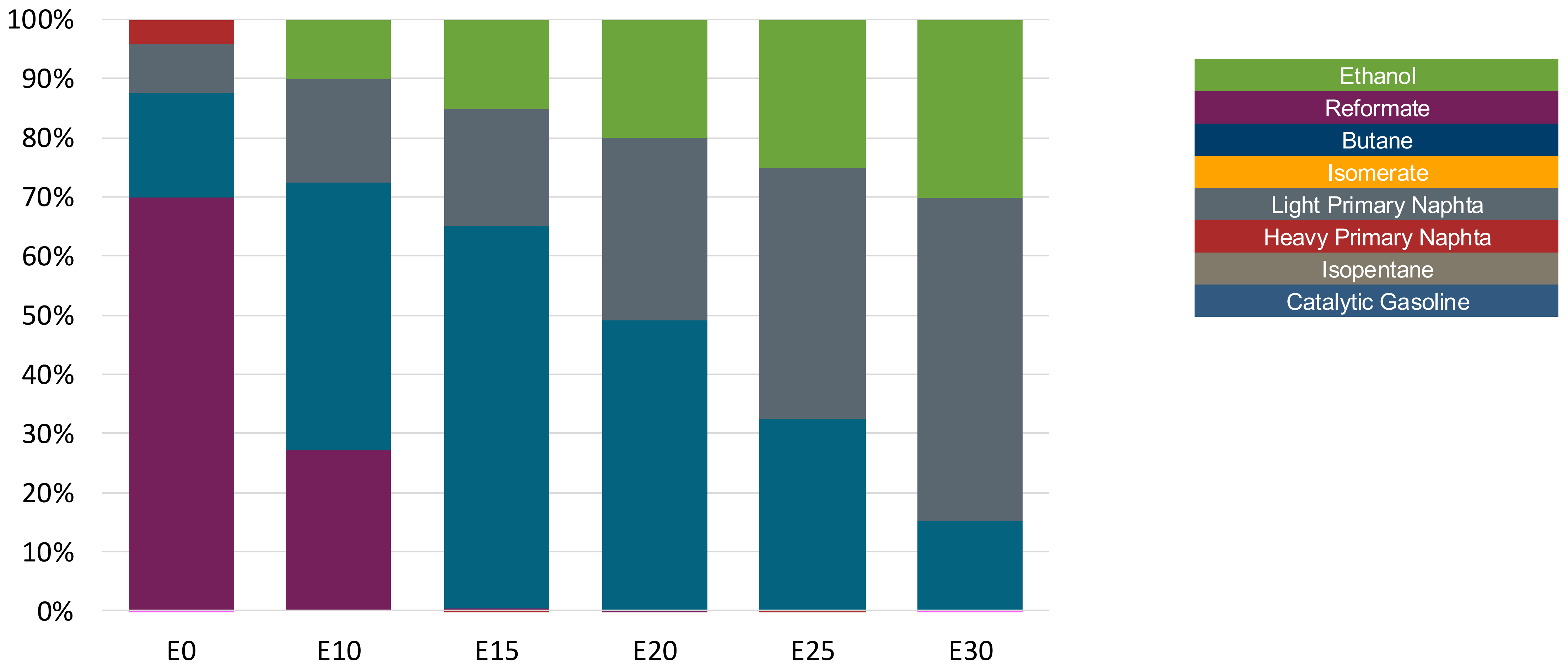
Source: ANCAP

Available Blending Components



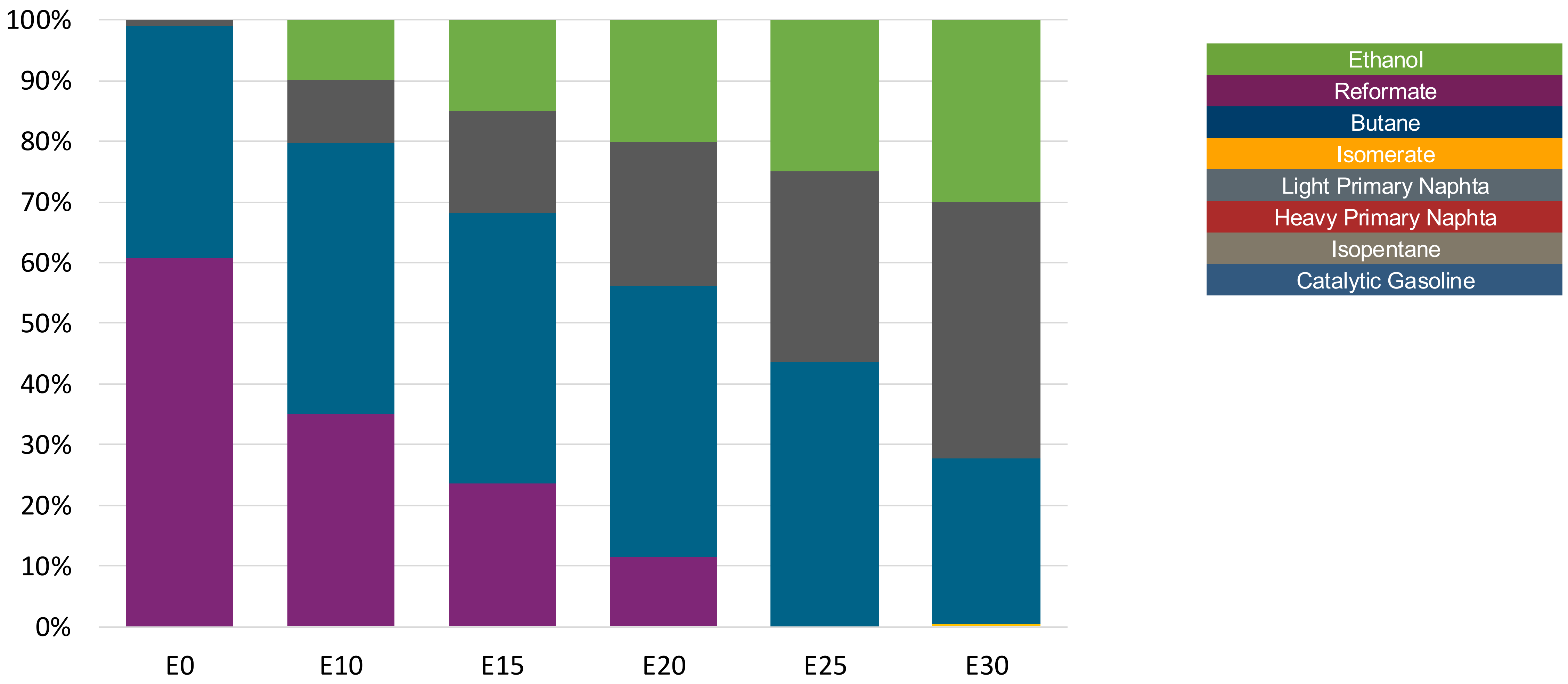
Source: Faro90

Super Gasoline – Constant Octane



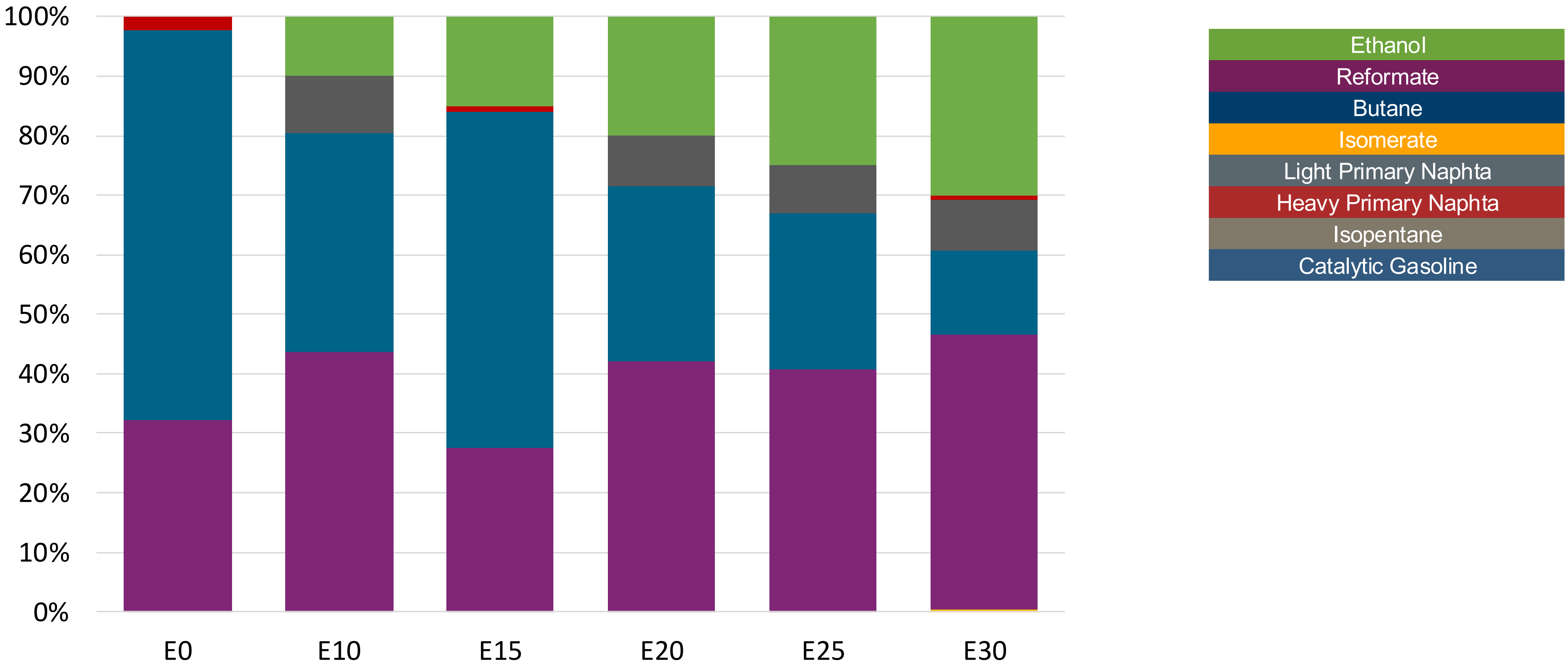
Octane (RON)	94.7	95.5	95.5	95.5	95.5	95.5
Price (US$/gal)	$ 2.433	$ 2.239	$ 2.109	$ 2.012	$ 1.912	$ 1.807

Premium Gasoline – Constant Octane



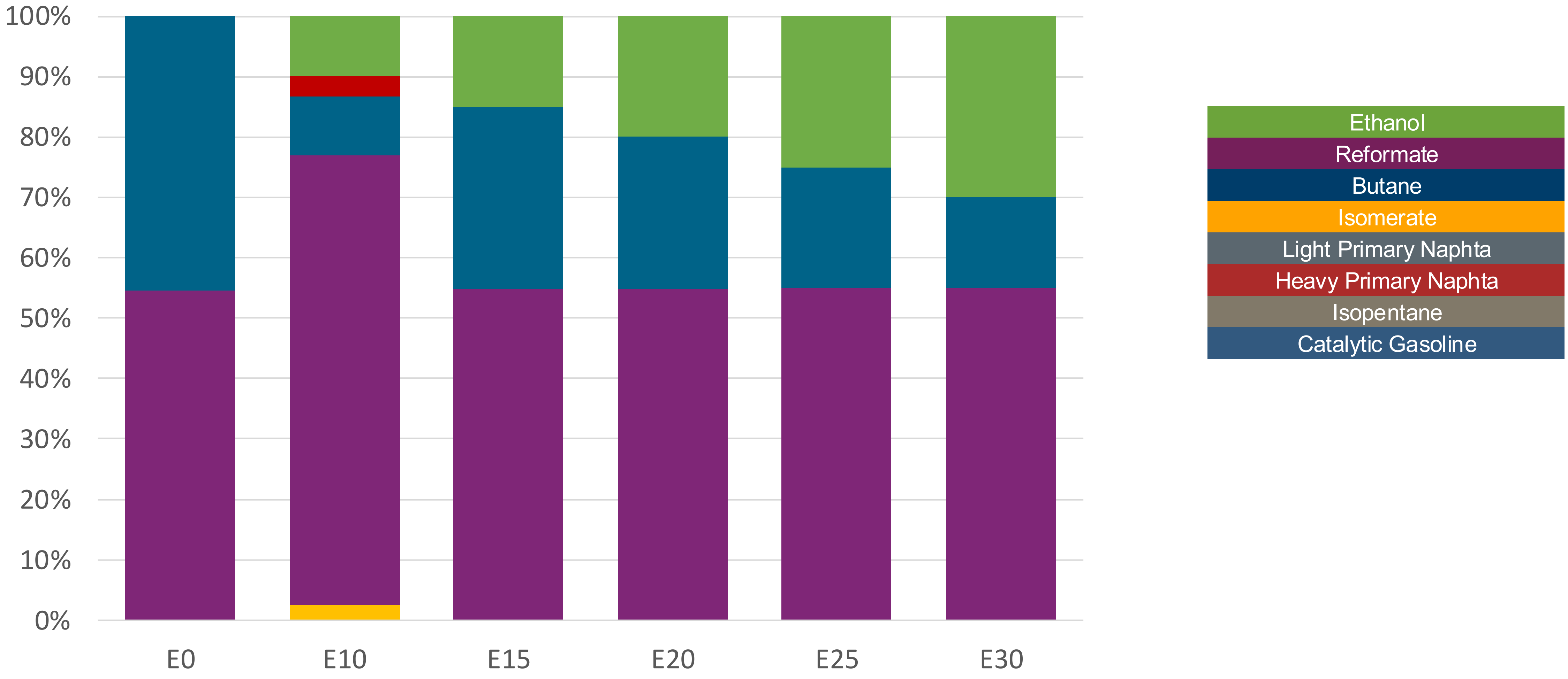
	E0	E10	E15	E20	E25	E30
Octane (RON)	97.0	97.5	97.5	97.5	97.5	97.5
Price (US$/gal)	$ 2.519	$ 2.333	$ 2.229	$ 2.120	$ 2.008	$ 1.907

Super Gasoline – Increased Octane



Octane (RON)	94.5	98.2	100.8	102.1	103.9	105.5
Price (US$/gal)	$ 2.375	$ 2.375	$ 2.375	$ 2.375	$ 2.375	$ 2.375

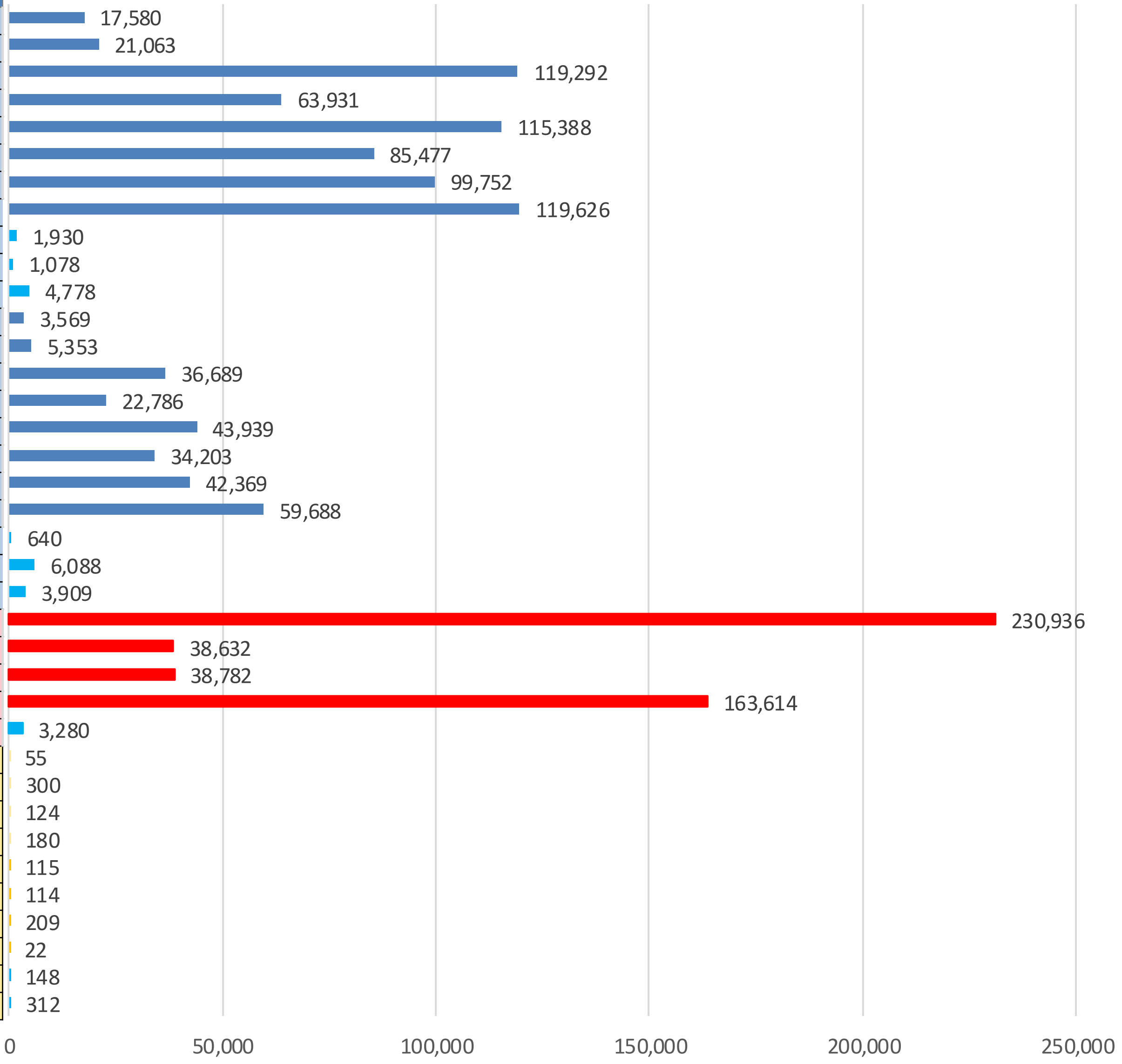
Premium Gasoline – Increased Octane



	E0	E10	E15	E20	E25	E30
Octane (RON)	96.8	100.5	102.7	104.5	106.2	107.7
Price (US$/gal)	$ 2.501	$ 2.501	$ 2.501	$ 2.501	$ 2.501	$ 2.501

Gasoline Vehicular Fleet in Uruguay

Type	Weight	Motor	Exhaust	Recuperator	Age
Auto/Sml Truck	Light	Carburator	3-Ways	PCV Tank	+8years
		Single FI	3-Ways		+8years
		Multi FI	3-Ways		+8years
			Euro III		+8years
			Euro IV		+8years
			Euro V		+8years
					4-8years
					-4years
		Hybrid	Hybrid		4-8years
					-4years
		Electric			
	Medium	Carburator	3-Ways	PCV Tank	+8years
		Single FI	3-Ways		+8years
		Multi FI	3-Ways		+8years
			Euro III		+8years
			Euro IV		+8years
			Euro V		+8years
					4-8years
					-4years
		Hybrid	Hybrid		4-8years
					-4years
		Electric			
Sml Engines	Light	4-Cycle	Euro II	PCV	+8years
					4-8years
			Euro III		4-8years
					-4years
		Eléctricas			-8years
Truck	Heavy	Carburator	3-Ways	PCV	+8years
		Fuel-Injection	3-Ways		+8years
			Euro III		+8years
			Euro IV		+8years
			Euro V		+8years
					4-8years
					-4years
		Hybrid	Hybrid		4-8years
					4years
		Electric			

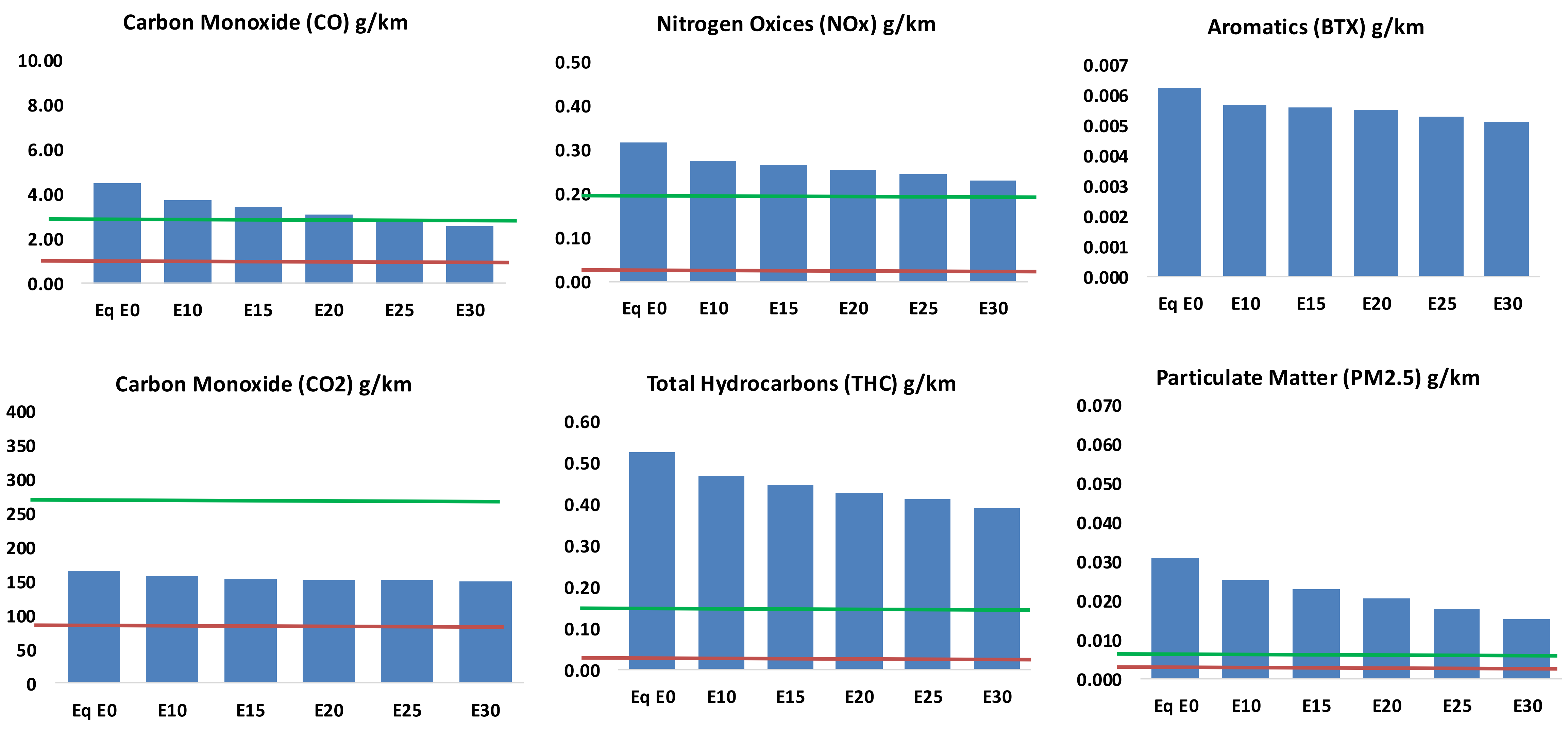


Gasoline Vehicular Fleet: **1,373,674**
Average Age: **8 years**

Source: MIEM, 2025.

Vehicular Emissions in Uruguay

United States
Euro 6



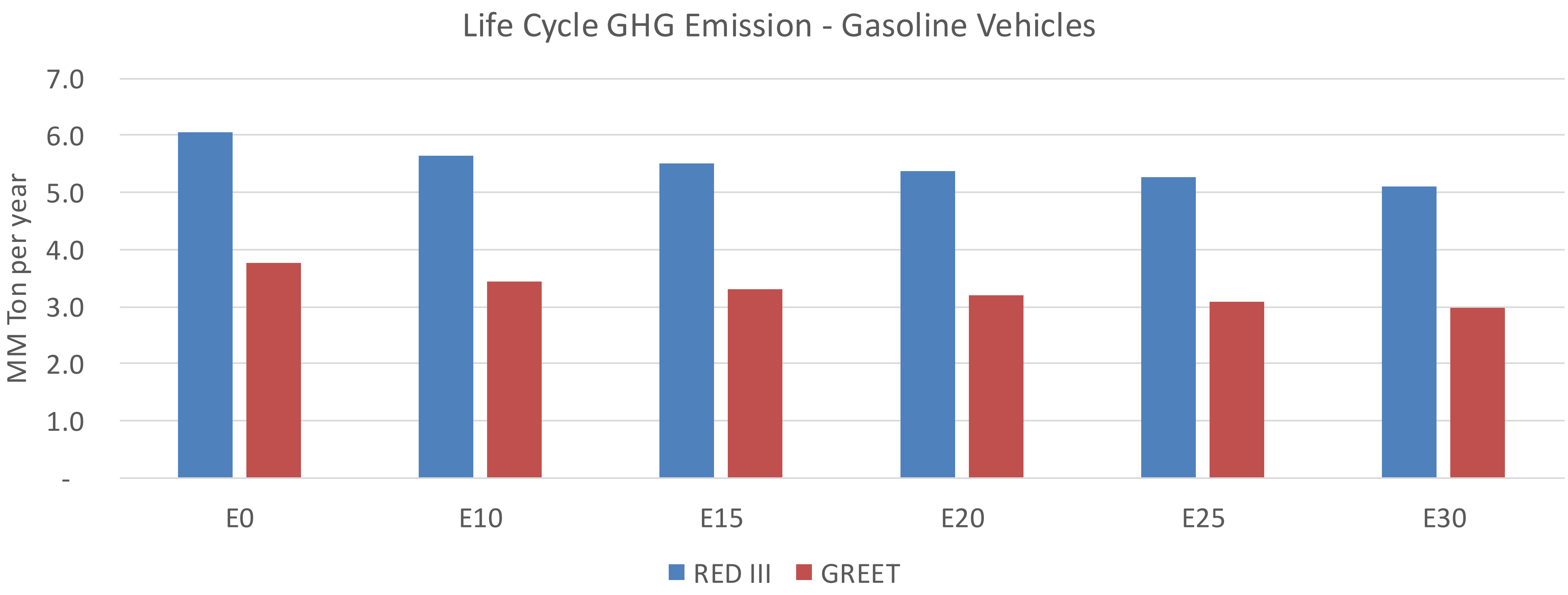
Source: Faro90, 2025.

Vehicular Emissions in Uruguay

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	153,217	140,183	127,148	116,222	105,114	96,539	87,128	-17%	-31%
CO2	5,613,763	5,469,409	5,333,075	5,225,852	5,172,794	5,145,685	5,050,835	-5%	-8%
VOC	17,828	16,857	15,886	15,169	14,483	13,966	13,238	-11%	-19%
NOx	10,733	9,910	9,356	9,079	8,627	8,275	7,859	-13%	-20%
SOx	63	59	54	50	46	42	37	-15%	-28%
NH3	1,923	1,904	1,885	1,894	1,915	1,930	1,943	-2%	0%
N2O	85	85	84	85	87	88	89	-1%	2%
CH4	4,530	4,530	4,530	4,597	4,697	4,783	4,865	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	137	131	125	121	118	115	111	-9%	-14%
Acetaldehyde	343	394	578	880	1,199	1,371	1,620	68%	249%
Formaldehyde	1,226	1,192	1,389	1,631	1,709	1,890	2,058	13%	39%
Benzene	212	204	193	190	188	180	174	-9%	-11%
PM2.5	353	321	289	261	233	203	172	-18%	-34%
PM10	695	632	568	513	458	400	339	-18%	-34%

Source: Faro 90, 2025.

Uruguay – Gasoline Vehicle Fleet GEI Emissions



Country Emissions 2024 (MMT/year)	44.6
2035 Target (MMT/year)	31.2
Delta	13.4

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	9.1%	12.1%	15.2%	17.8%	21.3%
RED II/III	0.0%	6.5%	8.9%	11.2%	13.1%	15.6%

% 2035 Target	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	2.5%	3.4%	4.3%	5.0%	6.0%
RED II/III	0.0%	3.0%	4.0%	5.0%	5.9%	7.0%

Source: Greet, RED II/III, climateactiontracker.org, F90